

ELECTRICAL CIRCUIT ANALYSIS (20A02301T)

LECTURE NOTES

II-B.TechI-Semester

Prepared by

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ELECTRICAL AND ELECTRONICS ENGINEERING

Course Code	ELECTRICAL CIRCUIT ANALYSIS	L T P C
20A02301T		3 0 0 3

UNIT - I Locus Diagrams & Resonance 8 Hrs

Series R-L, R-C, R-L-C and Parallel Combination with Variation of Various Parameters - Resonance-Series, Parallel Circuits, Frequency Response, Concept of Bandwidth and Q Factor.

UNIT - II Two Port Networks 9 Hrs

Two Port Network Parameters – Impedance – Admittance - Transmission and Hybrid Parameters and their Relations - Concept of Transformed Network - Two Port Network Parameters Using Transformed Variables.

UNIT - III Transient Analysis 12 Hrs

D.C Transient Analysis: Transient Response of R-L, R-C, R-L-C Series Circuits for D.C Excitation - Initial Conditions in network - Initial Conditions in elements - Solution Method Using Differential Equation and Laplace Transforms - Response of R-L & R-C Networks to Pulse Excitation.

A.C Transient Analysis: Transient Response of R-L, R-C, R-L-C Series Circuits for Sinusoidal Excitations - Solution Method Using Differential Equations and Laplace Transforms.

UNIT - IV Fourier Transforms 10 Hrs

Fourier Theorem - Trigonometric Form and Exponential Form of Fourier series – Conditions of Symmetry - Line Spectra and Phase Angle Spectra - Analysis of Electrical Circuits to Non Sinusoidal Periodic Waveforms. Fourier Integrals and Fourier Transforms – Properties of Fourier Transforms and Application to Electrical Circuits.

UNIT - V Filters 9 Hrs

Filters – Low Pass – High Pass, Band Pass and Band Stop– RC, RL filters– derived filters and composite filters design – Attenuators – Principle of Equalizers – Series and Shunt Equalizers – L Type - T type and Bridged – T and Lattice Equalizers.

Textbooks:

1. William Hayt, Jack E. Kemmerly and Jamie Phillips, “Engineering Circuit Analysis”, Mc Graw Hill, 9th Edition, 2019.
2. A. Chakrabarti, “Circuit Theory: Analysis & Synthesis”, Dhanpat Rai & Sons, 2008.

Reference Books:

1. M.E. Van Valkenberg, “Network Analysis”, 3rd Edition, Prentice Hall (India), 1980.
2. V. Del Toro, “Electrical Engineering Fundamentals”, Prentice Hall International, 2009.
3. Charles K. Alexander and Matthew. N. O. Sadiku, “Fundamentals of Electric Circuits” Mc Graw Hill, 5th Edition, 2013.
4. MahamoodNahvi and Joseph Edminister, “Electric Circuits” Schaum’s Series, 6th Edition, 2013.
5. John Bird, Routledge, “Electrical Circuit Theory and Technology”, Taylor & Francis, 5th Edition, 2014.

Online Learning Resources:

- https://onlinecourses.nptel.ac.in/noc21_ee99/preview
- https://onlinecourses.nptel.ac.in/noc21_ee14/preview

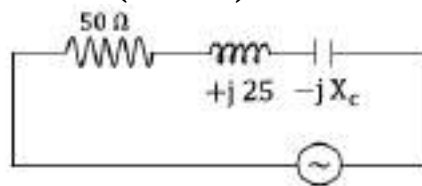
VEU INSTITUTE OF TECHNOLOGY:P.KOTHAKOTA

Department of EEE

ECA

UNIT-1 LOCUS DAIGRAMS

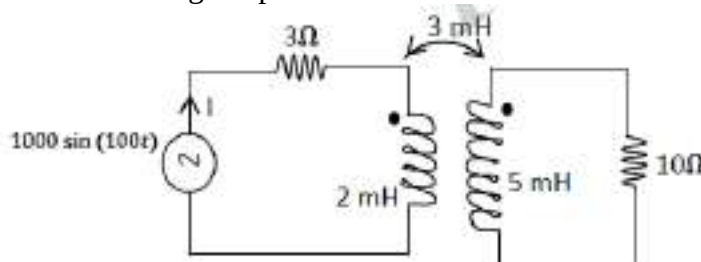
1. Define resonance?
2. Define anti resonance?
3. Define bandwidth and quality factor?
4. Define quality factor and sensitivity of a series RLC circuit.(MAY 18)
5. Define cut off frequency. Obtain the expression of cut off frequency in series resonance?
6. Differentiate series and parallel resonance?
7. Write the relation between bandwidth, quality factor and cut off frequency?
8. For the circuit shown below, determine the value of capacitive reactance and impedance at resonance.(DEC 16)



9. What is the current in the series R-L-C circuit at resonance? What is the reason for that?(NOV 17)
10. Define Selectivity?
11. Define Locus Diagram?

ESSAY QUESTIONS:

1. (a) What is the relation between bandwidth and resonant frequency for series RLC Circuit?
(b) An inductance of 0.5H a resistance of 5 Ω and a capacitance of 8 μ F are in series across a 220 V AC supply. Calculate the frequency at which the circuit resonates. Find the current at resonance bandwidth, half power frequencies and the voltage across capacitance of resonance. (MAY 18)
2. (a) Show that the resonant frequency ω_0 of an RLC series circuit is the geometric mean of ω_1 and ω_2 , the lower and upper half power frequencies respectively.
(b) Given a series RLC circuit with $R = 100$ ohms, $L = 0.5$ H and $C = 40\mu$ F. Calculate the resonant, lower and upper half – power frequencies.(MAY 2017)
3. A series circuit with $R=10\Omega$, $L=0.1$ H and $C=50\mu$ F has an applied voltage $V=50\angle 0^\circ$ V with a variable frequency. Find the resonant frequency & the value of frequency at which maximum voltage occurs across the capacitor & inductor.(DEC 16)
4. Determine the quality factor of a coil for the series circuit consisting of $R=10\Omega$, $L=0.1$ H and $C=10\mu$ F.(DEC 16)
5. In the following coupled circuit determine the current supplied by the source I.(JUNE 16)



6. A coil of impedance $R + j XL$ is connected in series with a 50 Ω resistor and a capacitor $-j XC$ ohms. The combination is connected across a 100 V variable frequency source. At a frequency of 200 Hz, maximum current of 0.7 amps flows in the circuit and the voltage

across the capacitor is 200 volts at this frequency. Find R , X_L , X_C and band width of the circuit. **(JUNE 16)**

(b) An inductance of 0.5 H, a resistance of 5Ω , and a capacitance of $8 \mu\text{F}$ are in series across a 220 V AC supply. Calculate the frequency at which the circuit resonates. Find the current at resonance, Band width, half power frequencies and the voltage across capacitance at resonance. **(NOV 17)**

7. (a) Derive the expression for Q-factor in a series R-L-C circuit.

(b) The Q-factor of a RLC circuit is 5 at its resonance frequency of 1 kHz. Assuming the power dissipation of 250 W when the current drawn is 1 A, find the circuit parameters.

Determine the BW of the circuit. **(NOV 17)**

8. Derive the Expression for Resonant Frequency for Series RLC Circuit?

9. Derive the Expression for Resonant Frequency for Parallel RLC Circuit?

10. What is the relation between bandwidth and resonant frequency for parallel RLC Circuit?

① construct the Locus diagram for Series R-L circuit with Variable R & constant X_L ?

② construct the Locus diagram for series R-L circuit with constant R and variable X_L case ?

③ Draw the Locus diagram for Series R-C circuit with constant R and variable X_C ?

④ Draw the Locus diagram for Series R-C circuit with variable R and constant X_C ?

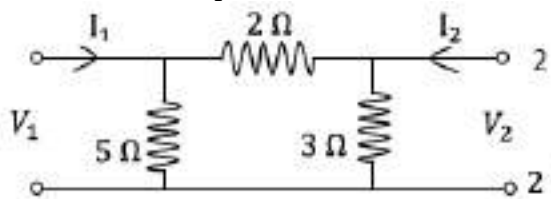
⑤ A circuit consisting of a choke coil with resistance of 2Ω and reactance of 1Ω at 50 Hz . It is connected in series with other choke coil with negligible resistance and variable inductive reactance. Draw the locus diagram of the current drawn from a 150 V , 50 Hz supply. If the inductive reactance is allowed to vary from 0 to 3Ω . calculate the min & max values of current.

⑥ construct the locus diagram for Parallel LC circuit for (i) varying X_L (ii) varying X_C .

UNIT-2 TWO PORT PARAMETERS

SHORT QUESTIONS:

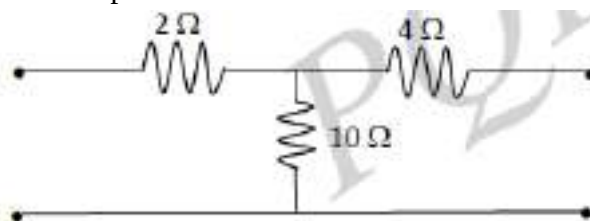
1. Define two-port network?
2. Define Z-parameters?
3. Define Y-parameters?
4. Define h-parameters?(MAY 17, DEC 16)
5. Define transmission or ABCD or chain or general parameters?(NOV 17)
6. Write the symmetry condition for Z, Y, h and ABCD parameters?
7. Write the reciprocity condition for Z, Y, h and ABCD parameters?
8. Draw the parallel connection of 2 two-port networks?
9. Find the mutual impedance of the circuit shown below.(DEC 16)



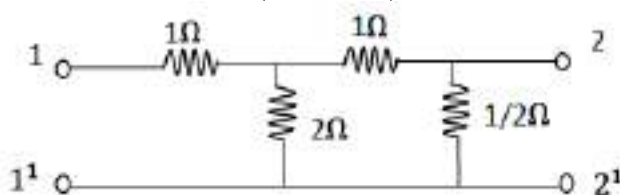
10. What are called transmission parameters? Why are they called so?(DEC 16)
11. When 10 V is applied to a two port resistive network, on no load, the voltage at the open circuit end is 8 V. When 1 amp load is connected, the voltage at the load end is 6 V. Find the voltage at the load end for a load current of 2 amp. The supply voltage is same in all the cases.

ESSAY QUESTIONS:

1. (b) Explain ABCD parameters in two port network.(MAY 18)
2. Find the h-parameters for the network shown in figure below.(MAY 18)

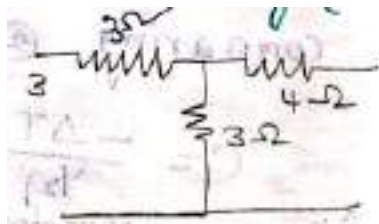


3. (a) Define and explain short circuit parameters by taking one example.
5. Derive the relation between transmission and impedance parameters.
6. Express the Z parameters of a two port network in terms of Y parameters.(DEC 16)
9. Find the Y and Z parameters of the following two port resistive network. Verify the relation between them. .(JUNE 16)



10. (a) Derive the relation between impedance and admittance parameters.
11. (a) Derive the relation between hybrid and impedance parameters.
12. Express the Z parameters of a two port network in terms of h and ABCD parameters?

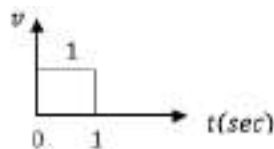
13. Express the Y parameters of a two port network in terms of Z, h, ABCD parameters
14. Express the h parameters of a two port network in terms of Z, Y and ABCD parameters
15. Express the ABCD parameters of a two port network in terms of Y and ABCD parameters
16. Define and explain “h” parameters by taking one example
17. Define and explain ABCD parameters by taking one example
18. Find Z, Y, h and ABCD parameters for the following problem



UNIT-3 TRANSIENT ANALYSIS

SHORT QUESTIONS:

1. Define transients?
2. Define time constant. What is the time constant for series RL and series RC circuit?
3. Define natural and forced responses?
4. Write the natural response equation of series RL and RC circuit?
5. Write the forced response equation of series RL and RC circuit?
6. Write the transient response equation in series RLC circuit?
7. What is excitation?
8. Define time constant of RL circuit. **(MAY 2018)**
9. When an inductor is connected in a circuit, what is the status of inductor under steady state condition? Draw the equivalent circuit? **(MAY 2017)**
10. When a capacitor is connected in a circuit, what is the status of capacitor under steady state condition? Draw the equivalent circuit? **(MAY 2017)**
11. Write down the equation of current for an RC circuit when it is supplied by a DC source. **(DEC 16)**
12. What is the time constant of an RC circuit excited by a DC source? **(DEC 16)**
13. A unit pulse of width 1 sec as shown is applied to an R – L series circuit with $R=1\Omega$ and $L = 1H$. Sketch the current response $i(t)$ in the circuit. **(JUNE 16)**

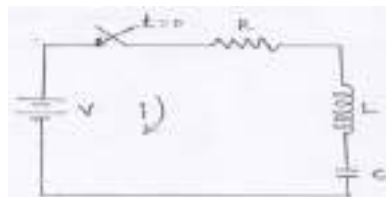


14. A DC source of 1V is suddenly applied to a series R-L-C circuit with $R=2\Omega$, $L = 1H$ and $C = \frac{1}{2} F$. Sketch the current response in the circuit. **(JUNE 16)**
15. When an inductor is connected in a circuit, what is the status of inductor under transient state condition? Draw the equivalent circuit. **(NOV 17)**
16. When a capacitor is connected in a circuit, what is the status of capacitor under transient state condition? Draw the equivalent circuit. **(NOV 17)**

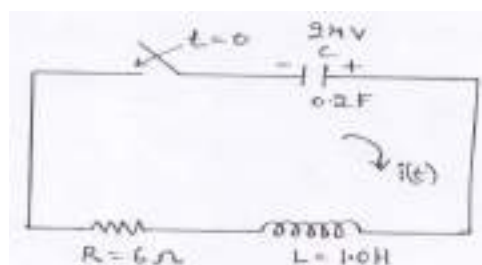
ESSAY QUESTIONS:

1. Obtain the Current Expression for Transient Response of Series RL Circuit excited by DC using Differential Evaluation Method?
2. Obtain the Current Expression for Transient Response of Series RL Circuit excited by DC using Laplace Transform Method?
3. Obtain the Current Expression for Transient Response of Series RL Circuit excited by AC using Differential Evaluation Method?

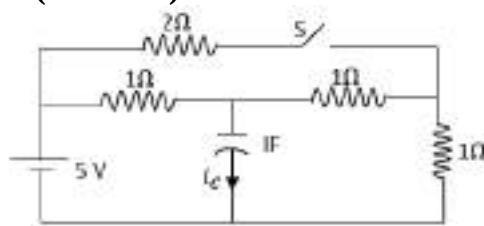
4. Obtain the Current Expression for Transient Response of Series RL Circuit excited by AC using Laplace Transform Method?
5. Obtain the Current Expression for Transient Response of Series RC Circuit excited by DC using Differential Evaluation Method?
6. Obtain the Current Expression for Transient Response of Series RC Circuit excited by DC using Laplace Transform Method?
7. Obtain the Current Expression for Transient Response of Series RC Circuit excited by AC using Differential Evaluation Method?
8. Obtain the Current Expression for Transient Response of Series RC Circuit excited by AC using Laplace Transform Method?
9. Obtain the Current Expression for Transient Response of Series RLC Circuit excited by DC using Differential Evaluation Method?
10. Obtain the Current Expression for Transient Response of Series RLC Circuit excited by DC using Laplace Transform Method?
11. Obtain the Current Expression for Transient Response of Series RLC Circuit excited by AC using Differential Evaluation Method?
12. Obtain the Current Expression for Transient Response of Series RLC Circuit excited by AC using Laplace Transform Method?
13. A series RLC circuit with $R = 300 \text{ ohms}$, $L = 1\text{H}$ and $C = 100\mu\text{F}$ has a constant voltage of 50 V applied to it at $t = 0$. Find the maximum current value. Assume zero initial conditions. **(MAY 2018)**
14. In the circuit shown below, $V = 10 \text{ V}$, $R = 10 \Omega$, $L = 1\text{H}$ and $C = 10\mu\text{F}$. The capacitor is initially uncharged. The switch is closed at $t = 0$. Determine $i(0^+)$, $\frac{di}{dt}(0^+)$ and $\frac{d^2i}{dt^2}(0^+)$. And also derive the formulae used. **(MAY 2017)**



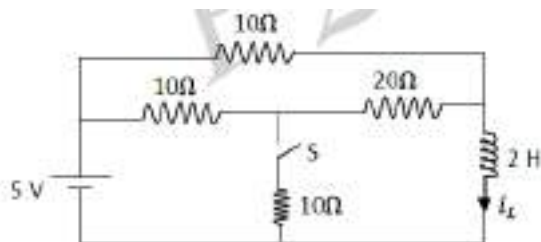
15. In the circuit shown below, $L = 1\text{H}$, $R = 6 \Omega$ and $C = 0.2\text{F}$. The capacitor is initially charged to 24 V and the switch is closed at $t = 0$. Determine the expression for $i(t)$ and the value of current at one second after the switch is closed. And also derive the formulae used.



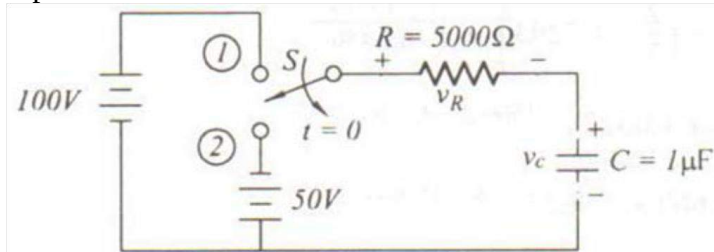
16. The following network is in steady state with S open. At $t = 0$, S is closed. Find $V_c(t)$ for $t > 0$. **(JUNE 16)**



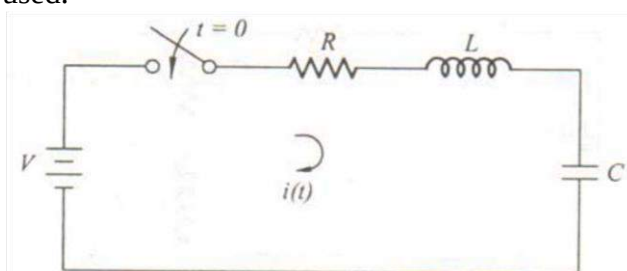
17. In the following network, the switch S is open and steady state is reached. At $t = 0$, S is closed. Find $i_L(t)$ for $t > 0$. **(JUNE 16)**



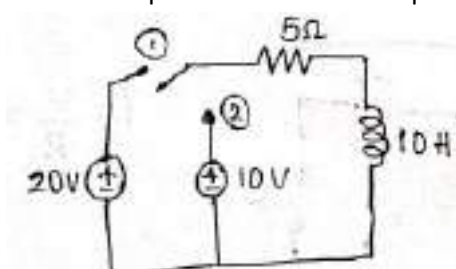
18. The switch in the circuit shown below is moved from position (1) to (2) at $t = 0$. Find the expression for V_C and V_R for $t > 0$. And also derive the formulae used.



19. For the network shown in figure below, determine the expression for $i(t)$. The switch is closed at $t = 0$. Take $V = 1\text{ V}$, $R = 2\ \Omega$, $L = 1\text{ H}$ and $C = 0.5\text{ F}$. And also derive the formulae used.



20. In the circuit shown in the figure, switch is at position 1 and steady state is reached. At $t=0$, switch is moved to position 2. Find the expression for current?



UNIT-4 FOURIER TRANSFORMS

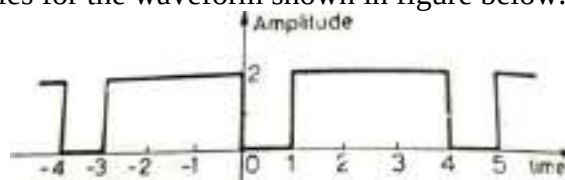
2 Marks Questions

SHORT QUESTIONS:

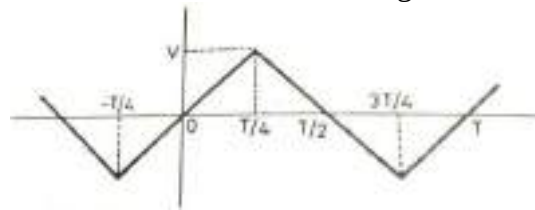
1. Write the any two properties of Fourier transform
2. Write the exponential form of Fourier series
3. Write the Trigonometric form of Fourier series
4. Write the condition for odd and even symmetry
5. Define fourier transforms?
6. Define fourier series?
7. Write the different forms of fourier series?

SHORT QUESTIONS:

1. Obtain the Fourier Series for the waveform shown in figure below.



2. Obtain the Fourier Series for the waveform shown in figure below.

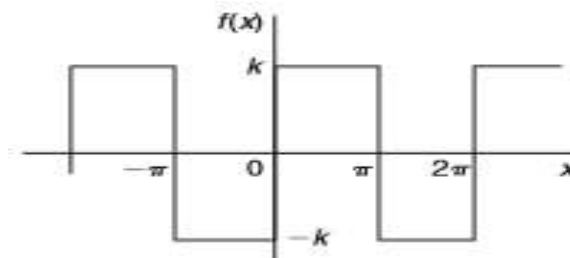


3. Obtain a Fourier series for the periodic function $f(x)$ defined as:

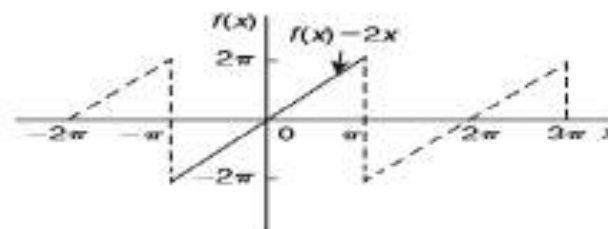
$$f(x) = -k, \text{ when } -\pi < x < 0$$

$$+k, \text{ when } 0 < x < \pi$$

The function is periodic outside of this range with period 2π

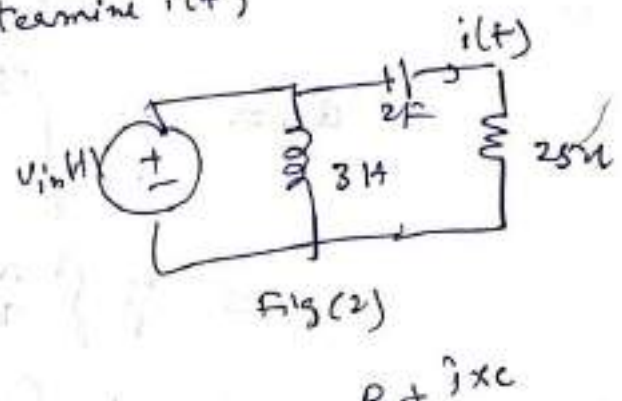
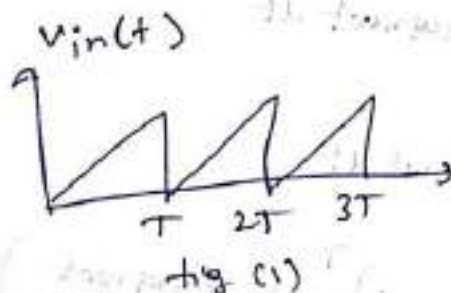


4. Determine the Fourier series to represent the function $f(x) = 2x$ in the range $-\pi$ to $+\pi$

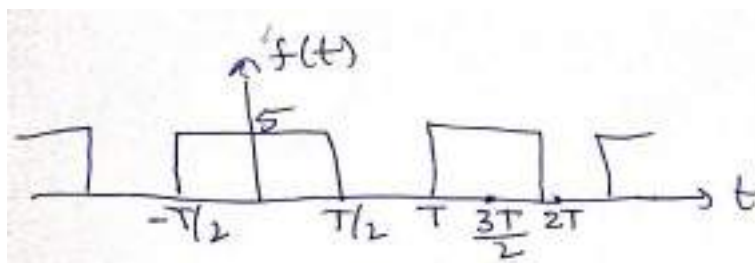


5. Derive the fourier series for both sine and cosine function?
6. Derive the fourier transforms for both sine and cosine function?
7. Find the relation between trigonometry and exponential fourier series?
8. Find the fourier transform of
 - a) Unit step function
 - b) Ramp function
9. Find the fourier series of two sided signal of e^{At} ?
10. Explain the Properties of Fourier Transforms?
- 11.

The voltage having the waveform shown in fig(1) is applied to ckt shown in fig(2) - determine $i(t)$



12. Obtain the expressions for Trigonometric Form and Exponential For of Fourier Series?
13. Develop Fourier series for the following signal?



UNIT-5 FILTERS

1. Define filter?
2. Define constant-K type filter?
3. Define Neper.(MAY 18)
4. Deduce the relation between Neper and Decibel?
5. Define propagation constant, Attenuation constant and phase constant?
6. Write the pass band and stop band conditions for a filter?
7. Write the characteristic equations of T and Π network?
8. Write the applications of different types of filter.(MAY 18)
9. Draw constant-k low pass filter (proto type).(MAY 17)
10. Draw constant-k high pass filter (proto type).(NOV 17)
11. Draw the frequency response characteristics of a band pass filter and indicate the important frequencies in it.

ESSAY TYPE QUESTIONS

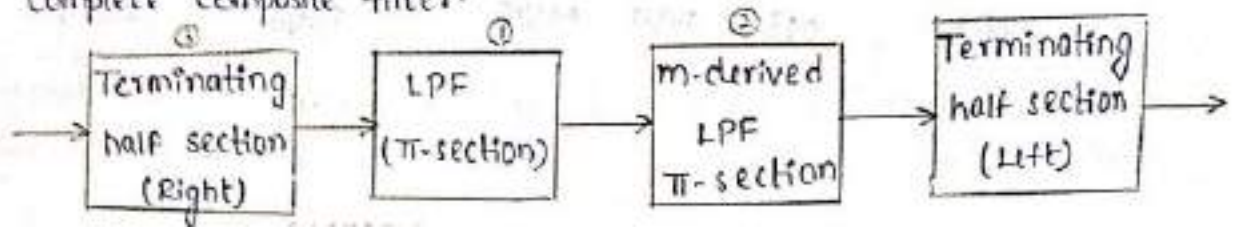
1. Design a constant-K low pass filter, both T and π sections having a cut-off frequency of 2 kHz to operate with a terminated load resistance of 500 Ω . (MAY 18)
2. A T-section low pass filter has an inductance of 30mH in each of the series arms and a shunt arm capacitance of 0.25 μ F. Calculate the cut off frequency, characteristic impedance, ratio of input, output voltages and phase shift at: (i) 1 kHz. (ii) 5 kHz.(MAY 17)
3. Analyze constant K low pass filters.(DEC 16)
4. Analyze constant K High pass filters
5. Analyze constant Band pass filters
6. A prototype high pass filter has a cut off frequency of 8 kHz and a nominal impedance of 600 Ω . Calculate the values of inductance and capacitance used in the filter.(NOV 17)
7. Design a prototype band pass filter with cut off frequencies 1.5 kHz and 5 kHz and a design impedance of 500 Ω .(NOV 17)
8. A low pass section filter consists of an inductance of 25mH in the series arm and two capacitors of 0.2 μ F in the shunt arms. Calculate the cut off frequency, design impedance, attenuation at 5 kHz and phase shift at 2 kHz. Also find the characteristic impedance at 2 kHz.(MAY 17)
9. Analyze m Derived low pass filters
10. Analyze m Derived High pass filters
11. Analyze Composite filters?
- 12.

Design m-derived high pass filter to work into load of 600 Ω with cut off frequency of $1000/\pi$ Hz and peak attenuation frequency at 300 Hz.

13.

Design a complete Low pass filter π section to work into 600Ω if cut-off frequency of 2 kHz and very high attenuation of 2.2 kHz also terminate filter properly.

Complete composite filter.



Dr. A. Hema Sekhar

Electrical circuit Analysis ()

Unit-1: Locus Diagrams and Resonance

Series R-L, R-C, R-L-C and Parallel combination with variation of various parameters - Resonance - Series, Parallel circuits, Frequency Response, concept of bandwidth and Q Factor

Introduction :

For a Particular circuit like R-L, R-C, R-L-C. If any one of the element is variable then depending upon the value of the variable element circuit characteristic changes then circuit parameters like voltage, current and power consumed by the element is also changes.

Def of Locus diagram:

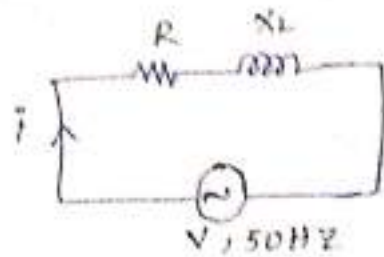
"It is defined as the Locus of the current obtained for various values of the variable element."

classification

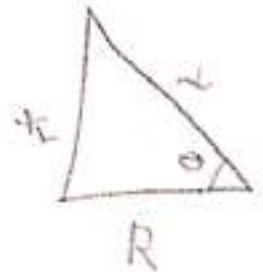
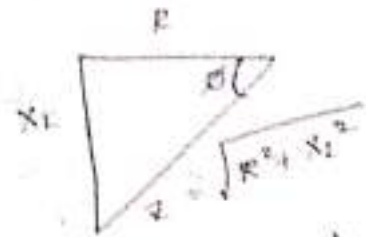
Locus diagrams are classified into two types

- ① Series R-L, R-C and R-L-C circuits.
- ② Parallel combination of circuits.

Locus diagram of R-L circuit



Impedance diagram



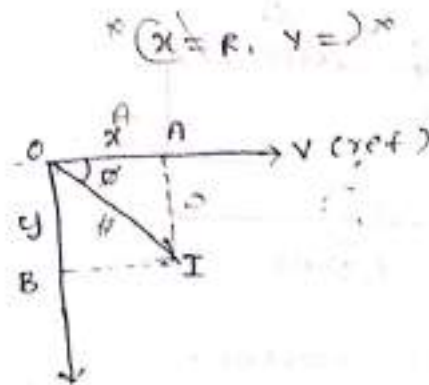
According to KVL

$$V = I(R + jX_L)$$

$$V = IZ, \quad Z = R + jX_L$$

$$I = \frac{V}{Z} \longrightarrow (1)$$

Now, phasor diagram for R-L circuit is



From the phasor diagram

$$\cos \phi = \frac{x}{I}$$

$$x = I \cos \phi \longrightarrow (2)$$

$$\sin \phi = \frac{OB}{I} = \frac{y}{-I}$$

$$y = -I \sin \phi \longrightarrow (3)$$

$$x^2 + y^2 = I^2 \cos^2 \phi + I^2 \sin^2 \phi$$

$$x^2 + y^2 = I^2 (\cos^2 \phi + \sin^2 \phi)$$

$$x^2 + y^2 = I^2$$

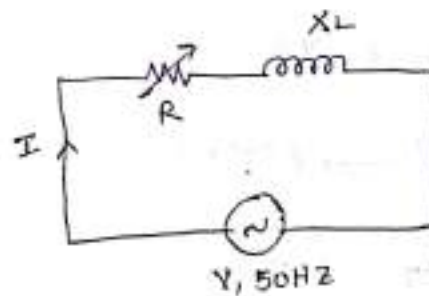
$$x^2 + y^2 = \left(\frac{V}{Z}\right)^2$$

$$x^2 + y^2 = \frac{V^2}{Z^2}$$

$$x^2 + y^2 = \frac{V^2}{(\sqrt{R^2 + X_L^2})^2}$$

$$\boxed{\therefore x^2 + y^2 = \frac{V^2}{R^2 + X_L^2}} \longrightarrow (4)$$

Case (i):- Variable R , constant X_L



Here X_L is constant

$$y = -I \sin \phi$$

$$= -\frac{V}{Z} \times \frac{X_L}{Z}$$

$$= -\frac{V X_L}{R^2 + X_L^2}$$

$$y = -X_L \times \frac{V}{R^2 + X_L^2} \longrightarrow (5)$$

$$\frac{V}{R^2 + X_L^2} = \frac{-y}{X_L} \longrightarrow (6)$$

Substitute Eqn (6) in Eqn (4)

$$x^2 + y^2 = V \times \frac{V}{R^2 + x_L^2}$$

$$x^2 + y^2 = V \times \frac{-y}{x_L}$$

$$x^2 + y^2 + 2 \cdot y \cdot \frac{V}{2x_L} + \left(\frac{V}{2x_L}\right)^2 - \left(\frac{V}{2x_L}\right)^2 = 0$$

$$x^2 + \left(y + \frac{V}{2x_L}\right)^2 = \left(\frac{V}{2x_L}\right)^2$$

$$\therefore x^2 + \left(y + \frac{V}{2x_L}\right)^2 = \left(\frac{V}{2x_L}\right)^2 \longrightarrow (7)$$

Now we know the circle equation

$$(x - x_1)^2 + (y - y_1)^2 = r^2 \longrightarrow (8)$$

comparing eqn's (7) & (8), we get,

$$x_1 = 0,$$

$$y_1 = \frac{-V}{2x_L}$$

$$r = \frac{V}{2x_L}$$

$$\therefore \text{Centre} = (x_1, y_1) = \left(0, \frac{-V}{2x_L}\right)$$

$$\text{radius } (r) = \frac{V}{2x_L}$$

Construction of locus diagram :-

$$\cos \phi = \frac{R}{Z} \quad \& \quad I = \frac{V}{Z} \rightarrow I = \frac{V}{\sqrt{R^2 + x_L^2}}$$

Suppose if $\phi = 30^\circ$ then $\cos 30^\circ = 0.866$

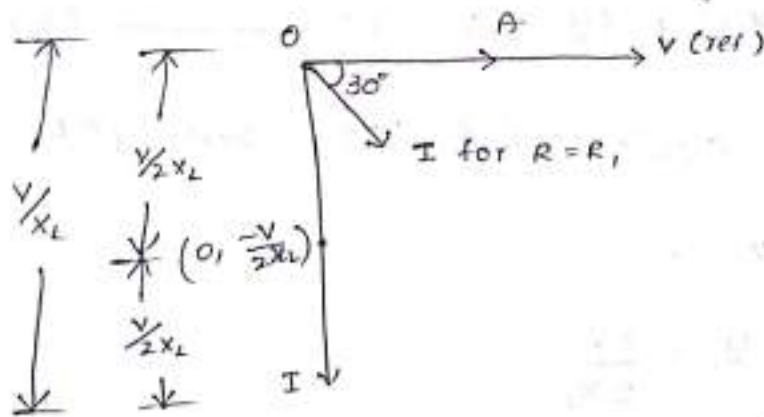
$\phi = 45^\circ$ then $\cos 45^\circ = 0.707$

$\phi = 60^\circ$ then $\cos \phi = 0.5$

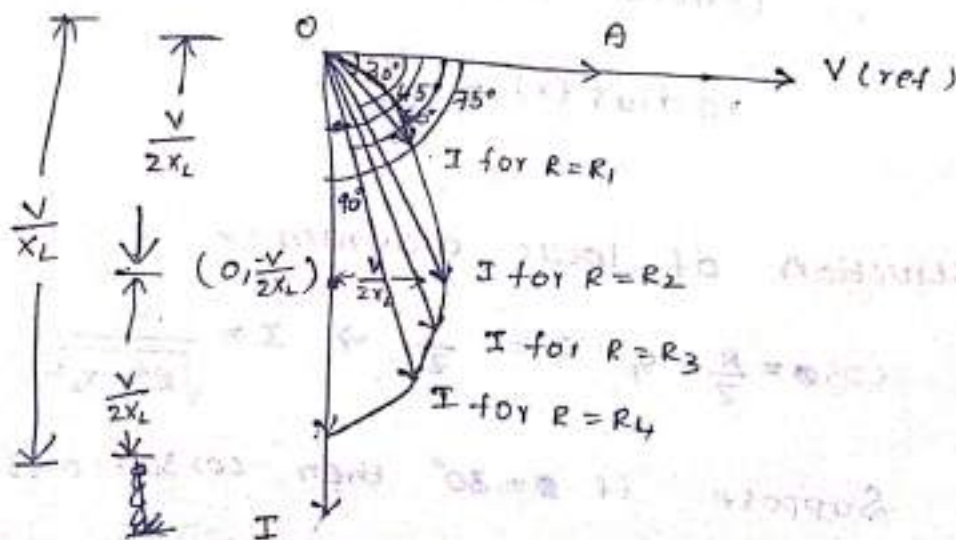
$\phi = 75^\circ$ then $\cos 75^\circ = 0.25$

$\phi = 90^\circ$ then $\cos 90^\circ = 0$

If $\cos 30^\circ = 0.866$ and compared due to other angle it is high value and we know that R is directly proportional to $\cos \phi$ and ' R ' value is high then if we put high value of ' R ' in current eqn then the magnitude of current is less. that means at 30° magnitude of current is less.



If angle increases ' R ' value decreases then current values increases.



We know

$$x^2 + y^2 = \frac{V^2}{R^2 + x_L^2} \longrightarrow (4)$$

Here constant 'R'.

$$x = I \cos \theta$$

$$= \frac{V}{\sqrt{R^2 + x_L^2}} \times \frac{R}{\sqrt{R^2 + x_L^2}}$$

$$x = \frac{V \cdot R}{R^2 + x_L^2}$$

$$\frac{V}{R^2 + x_L^2} = \frac{x}{R} \longrightarrow (5)$$

Sub Eqn (5) in Eqn (4)

$$x^2 + y^2 = \frac{Vx}{R}$$

$$x^2 + y^2 - \frac{Vx}{R} = 0$$

$$x^2 + y^2 - 2x \cdot \frac{V}{2R} = 0$$

$$x^2 + y^2 - 2x \cdot \frac{V}{2R} + \left(\frac{V}{2R}\right)^2 - \left(\frac{V}{2R}\right)^2 = 0$$

$$\left(x - \frac{V}{2R}\right)^2 + y^2 - \left(\frac{V}{2R}\right)^2 = 0$$

$$\left(x - \frac{V}{2R}\right)^2 + y^2 = \left(\frac{V}{2R}\right)^2 \longrightarrow (6)$$

Now we know the circle equation

$$(x - x_1)^2 + (y - y_1)^2 = r^2 \longrightarrow (7)$$

Comparing Eqn's (6) & (7), we get

$$\text{centre} = (x_1, y_1) = \left(\frac{V}{2R}, 0 \right)$$

$$\text{Radius } (r) = \frac{V}{2R}$$

Here also one of the element in the centre is zero. So locus diagram is a semicircle.

Construction of locus diagram:-

$$\sin \phi = \frac{x_L}{\sqrt{R^2 + x_L^2}}, \quad I = \frac{V}{\sqrt{R^2 + x_L^2}}$$

Case (i):- Constant R , Variable x_L

$$\text{Centre} = \left(\frac{V}{2R}, 0 \right)$$

$$\text{radius} = \frac{V}{2R}$$

here variable is x_L , so

$$\sin \phi = \frac{x_L}{Z}, \quad I = \frac{V}{\sqrt{R^2 + x_L^2}}$$

$$\phi = 30^\circ, \quad \sin 30 = 0.5$$

$$\phi = 45^\circ, \quad \sin 45^\circ = \frac{1}{\sqrt{2}} = 0.707$$

$$\phi = 60^\circ, \quad \sin 60^\circ = 0.866$$

$$\phi = 75^\circ, \quad \sin 75^\circ = 0.965$$

$$\phi = 90^\circ, \quad \sin 90^\circ = 1$$

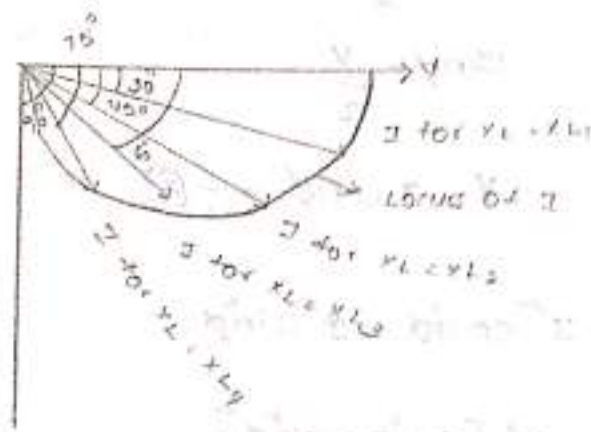
$\phi = 30^\circ$, x_L is small, I is large

$\phi = 45^\circ$, x_L is high, I is small

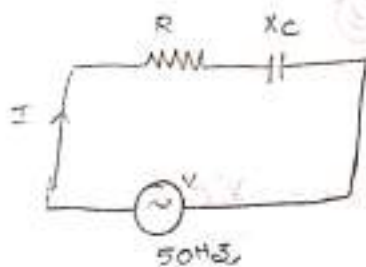
$\phi = 60^\circ$, x_L is very high, I is very small

$\phi = 75^\circ$, x_L is vv high, I is vv small

$\phi = 90^\circ$



* Locus diagram of R-C circuit:-

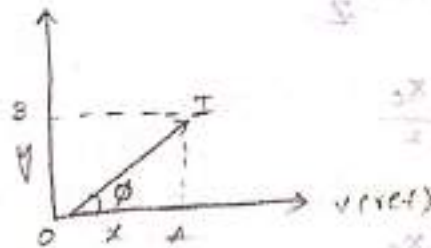


$$V = I(R + jX_c)$$

$$V = IZ$$

$$I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + X_c^2}}$$

Phasor diagram:-



$$\cos \phi = \frac{OA}{OI} = \frac{x}{I}$$

$$x = I \cos \phi \quad \text{--- (1)}$$

$$\cos \phi = \frac{OB}{I}$$

$$\cos \phi = \frac{Y}{I}$$

$$Y = I \cos \phi \quad \text{--- (2)}$$

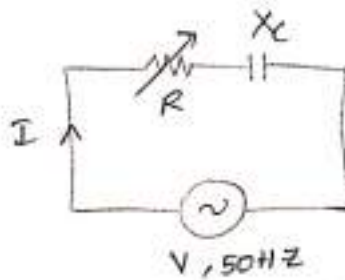
$$X^2 + Y^2 = I^2 \cos^2 \phi + I^2 \sin^2 \phi$$

$$= I^2 [\cos^2 \phi + \sin^2 \phi]$$

$$= I^2 (1)$$

$$\boxed{X^2 + Y^2 = \frac{V^2}{R^2 + X_c^2}} \quad \text{--- (3)}$$

Case (i) :- Variable R or constant X_c :-



Here X_c is constant

$$Y = I \sin \phi$$

$$Y = \frac{V}{Z} \times \frac{X_c}{Z}$$

$$Y = \frac{V X_c}{Z^2}$$

$$Y = \frac{V X_c}{R^2 + X_c^2}$$

$$Y = X_c \times \frac{\frac{V}{I}}{\frac{R^2 + X_c^2}{I^2}} \quad \text{--- (4)}$$

$$\frac{V}{R^2 + x_c^2} = \frac{y}{x_c} \longrightarrow (5)$$

Sub Eqn (5) in Eqn (3)

$$x^2 + y^2 = V \times \frac{y}{x_c}$$

$$x^2 + y^2 - V \times \frac{y}{x_c} = 0$$

$$x^2 + y^2 - 2V \times \frac{y}{2x_c} + \left(\frac{V}{2x_c}\right)^2 - \left(\frac{V}{2x_c}\right)^2 = 0$$

$$x^2 + y^2 - 2y \times \frac{V}{2x_c} + \left(\frac{V}{2x_c}\right)^2 - \left(\frac{V}{2x_c}\right)^2 = 0$$

$$x^2 + \left(y - \left(\frac{V}{2x_c}\right)\right)^2 = \left(\frac{V}{2x_c}\right)^2 \longrightarrow (6)$$

Now we know the circle eqn

$$(x - x_1)^2 + (y - y_1)^2 = r^2 \longrightarrow (7)$$

comparing eqn's (6) & (7)

$$x_1 = 0$$

$$y_1 = \frac{V}{2x_c}$$

$$r = \frac{V}{2x_c}$$

$$\therefore \text{Center} = (x_1, y_1) = \left(0, \frac{V}{2x_c}\right)$$

$$\text{radius } (r) = \frac{V}{2x_c}$$

Construction of locus diagram:-

$$\cos \phi = \frac{R}{Z}, \quad I = \frac{V}{Z} \Rightarrow I = \frac{V}{\sqrt{x^2 + x_c^2}}$$

Suppose

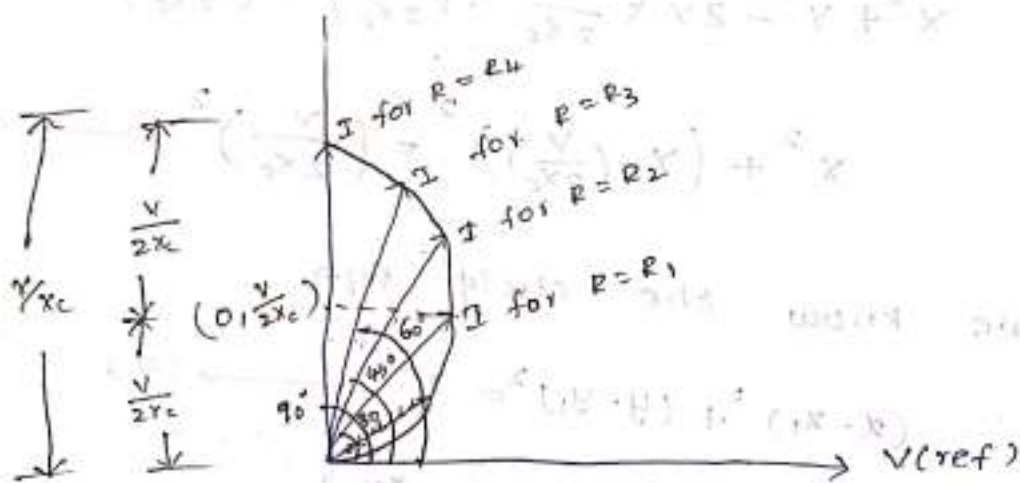
$$\phi = 30^\circ, \text{ Then } \cos \phi = 0.866$$

$$\phi = 45^\circ, \text{ Then } \cos \phi = 0.707$$

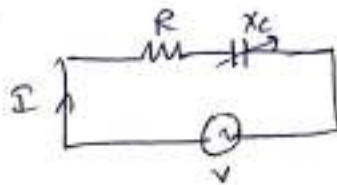
$$\phi = 60^\circ, \text{ Then } \cos \phi = 0.5$$

$$\phi = 75^\circ, \text{ Then } \cos \phi = 0.25$$

$$\phi = 90^\circ, \text{ Then } \cos \phi = 0$$



Case (i) :- constant R, variable X_c



$$x = I \cos \phi$$

$$x = \frac{V}{Z} \times \frac{R}{Z}$$

$$x = \frac{VR}{R^2 + X_c^2}$$

$$\frac{V}{R^2 + X_c^2} = \frac{x}{R} \rightarrow (8)$$

Sub Eqn (8) in Eqn (3)

$$x^2 + y^2 = V \times \frac{x}{R}$$

$$x^2 + y^2 - V \frac{x}{R} = 0$$

$$x^2 + y^2 - 2xy \frac{V}{2R} + \left(\frac{V}{2R}\right)^2 - \left(\frac{V}{2R}\right)^2 = 0$$

$$y^2 + \left(x - \frac{V}{2R}\right)^2 = \left(\frac{V}{2R}\right)^2$$

comparing above eqn w.r.t. $(x-x_1)^2 + (y-y_1)^2 = r^2$

$$x_1 = \frac{V}{2R}, y_1 = 0, r = \frac{V}{2R}$$

$$\left(\frac{V}{2R}, 0\right) \cdot \frac{V}{2R}$$

construction of locus diagram

$$\sin \phi = \frac{x_c}{\sqrt{R^2 + x_c^2}}, \quad \phi = \frac{V}{\sqrt{R^2 + x_c^2}}$$

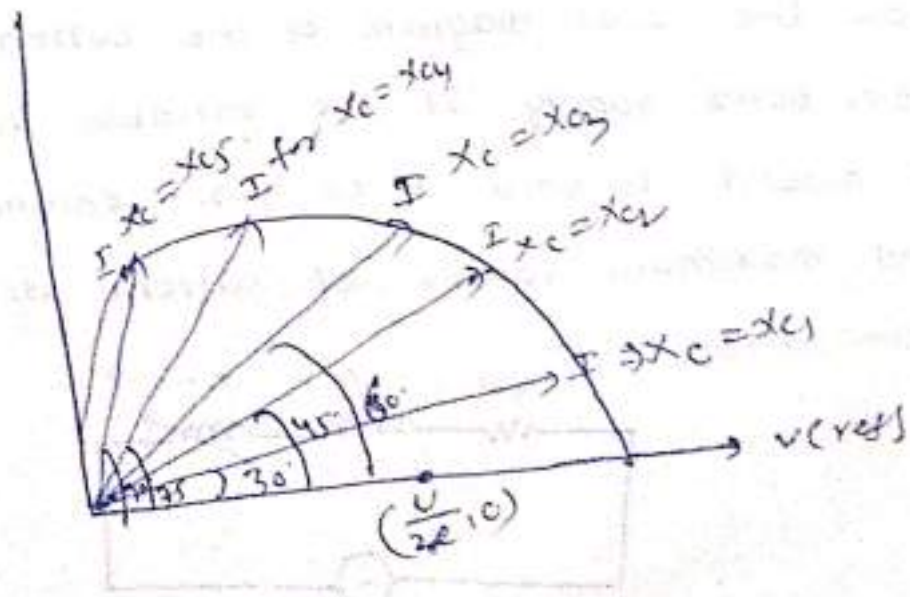
$$\phi = 30^\circ \text{ then } \sin \phi = 0.5$$

$$\phi = 45^\circ \text{ then } \sin \phi = 0.707$$

$$\phi = 60^\circ \text{ then } \sin \phi = 0.866$$

$$\phi = 75^\circ \text{ then } \sin \phi = 0.965$$

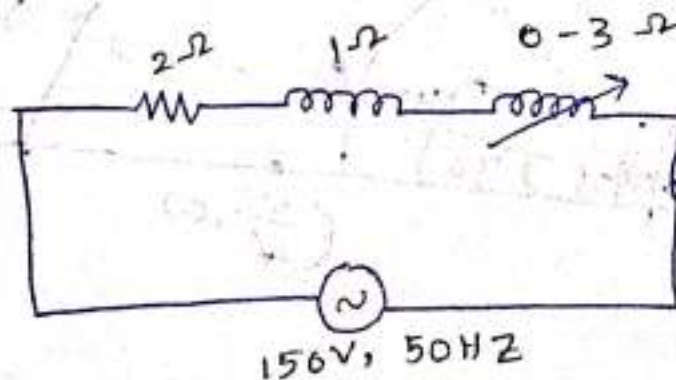
$$\phi = 90^\circ \text{ then } \sin \phi = 1$$



PROBLEMS :-

1. A circuit consisting of a choke coil with resistance of 2Ω and reactance of 1Ω at 50Hz . It is connected in series with other choke coil with negligible resistance and variable inductive reactance. Draw the locus diagram of the current drawn from a 150V , 50Hz supply. If the variable inductive reactance is allowed to vary 0 to 3Ω , calculate the minimum and maximum values of current and corresponding power factor.

Sol:-



In this problem,

variable reactance (X_L) & constant R

$$\text{radius} = \frac{V}{2R} = \frac{150}{2 \times 2}$$

$$\boxed{\text{radius} = 37.5 \text{ m}}$$

$$\text{centre} \left(\frac{V}{2R}, 0 \right) = (37.5, 0)$$

X_L is varies from 1 to 1+3

1 to 4 Ω

case(i):- Put $X_L = 0 \Omega$

$$\text{Total reactance } (X_L) = 1+0 = 1 \Omega$$

$$R = 2 \Omega$$

$$I = \frac{V}{\sqrt{R^2 + X_L^2}} = \frac{150}{\sqrt{2^2 + 1^2}}$$

$$\boxed{I = 67.08 \text{ A}}$$

$$\tan \phi = \frac{X_L}{R}$$

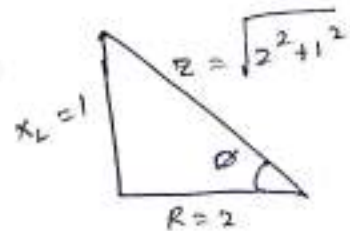
$$\phi_{\min} = \tan^{-1} \left(\frac{X_L}{R} \right)$$

$$= \tan^{-1} \left(\frac{1}{2} \right)$$

$$\phi_{\min} = 26.56^\circ$$

$$\cos \phi_{\min} = \cos 26.56$$

$$\boxed{\cos \phi_{\min} = 0.894}$$



case (ii):- $X_L = 3\Omega$

$$R = 2\Omega$$

$$\text{Total } (X_L) = 1 + 3 = 4\Omega$$

$$I = \frac{V}{\sqrt{R^2 + X_L^2}} = \frac{150}{\sqrt{2^2 + 4^2}}$$

$$I = 33.5 \text{ A}$$

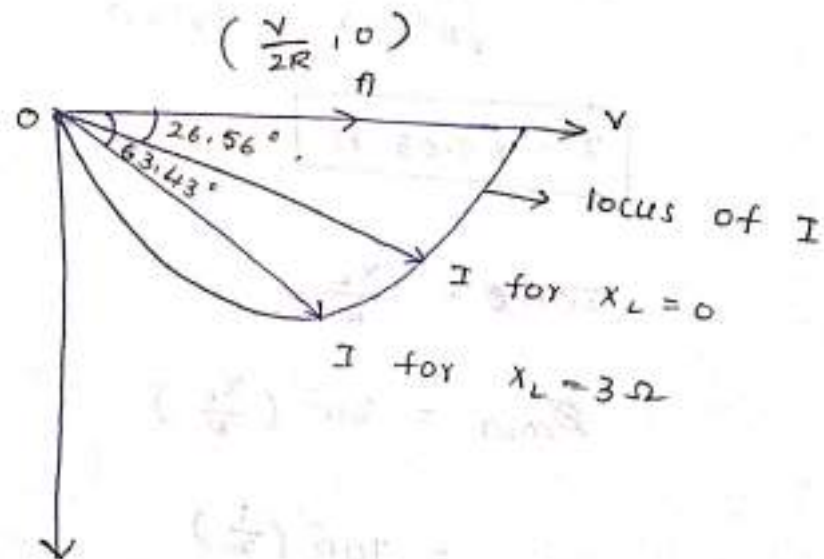
$$\phi_{\max} = \tan^{-1} \left(\frac{X_L}{R} \right)$$

$$= \tan^{-1} \left(\frac{4}{2} \right)$$

$$\phi_{\max} = 63.43$$

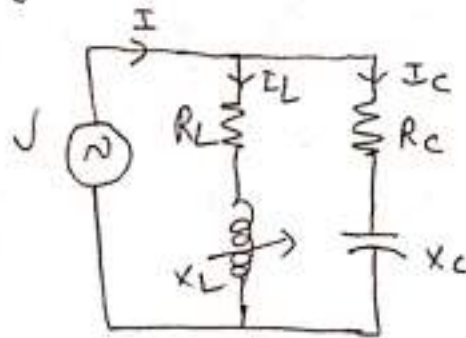
$$\cos \phi_{\max} = \cos (63.43)$$

$$\boxed{\cos \phi_{\max} = 0.447}$$



Parallel RLC circuit

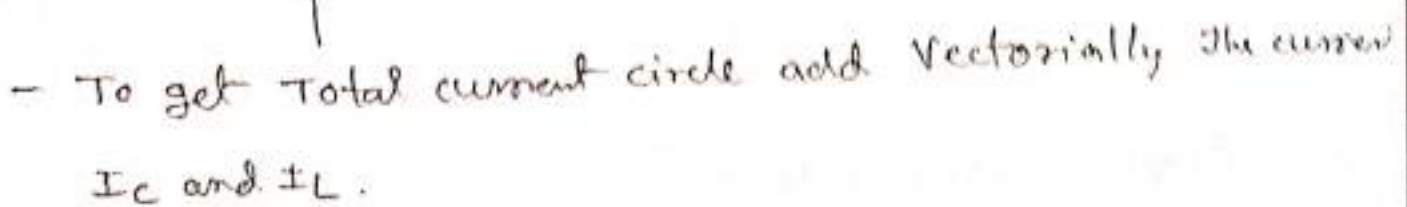
Parallel LC circuit along with internal resistances as shown in fig



In the above circuit, there are two branch currents i_L & i_C along with total current i

Case (1): Varying X_L :

- In this case, X_L is variable, X_C, R_L, R_C are fixed. and I_C is through capacitor is constant since R_C, R_L are fixed and it leads the voltage vector OV by an angle $\theta_C \left[\theta_C = \tan^{-1} \frac{X_C}{R_C} \right]$
- The current I_L through the inductance is vector OI_L and its amplitude is maximum and is equal to $\frac{V}{R_L}$ when X_L is zero and it is in phase with applied voltage V .
- $I = \frac{V}{\sqrt{R^2 + X_L^2}}$, $\sin \phi = \frac{X_L}{Z}$
- when X_L is increased from 0 to infinity and current is decreased from higher value to lower value. and its phase angle will be $\theta_L = \tan^{-1} \left(\frac{X_L}{R_L} \right)$ and it is same as series R-L circuit.



- here current I_L through inductor is constant since R_L and X_L are fixed and it lags the voltage vector OV by an angle $\theta_L = \tan^{-1} \left(\frac{X_L}{R_L} \right)$
- The current I_C through the capacitance is the vector OI_C . its amplitude is maximum and equal to V/R_C when $X_C = 0$

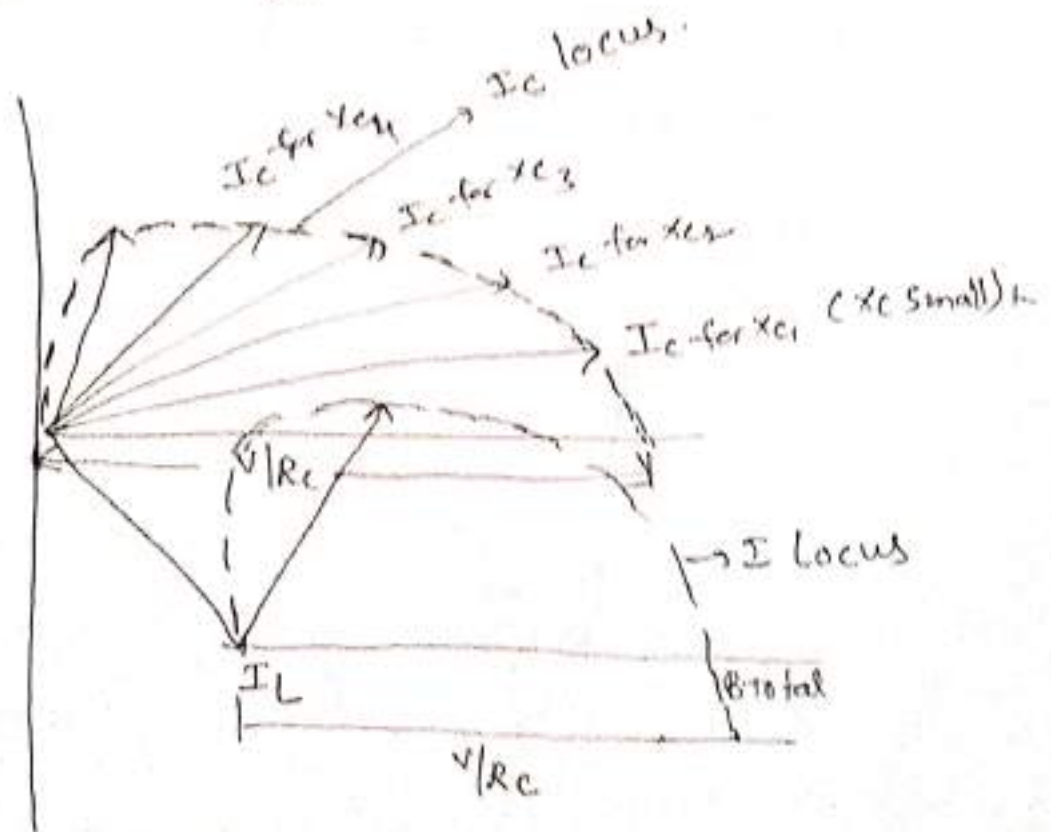
and it is in phase with applied voltage V .

$$\Rightarrow I = \frac{V}{\sqrt{R^2 + X_C^2}}$$

$$\sin \phi = \frac{X_C}{Z}$$

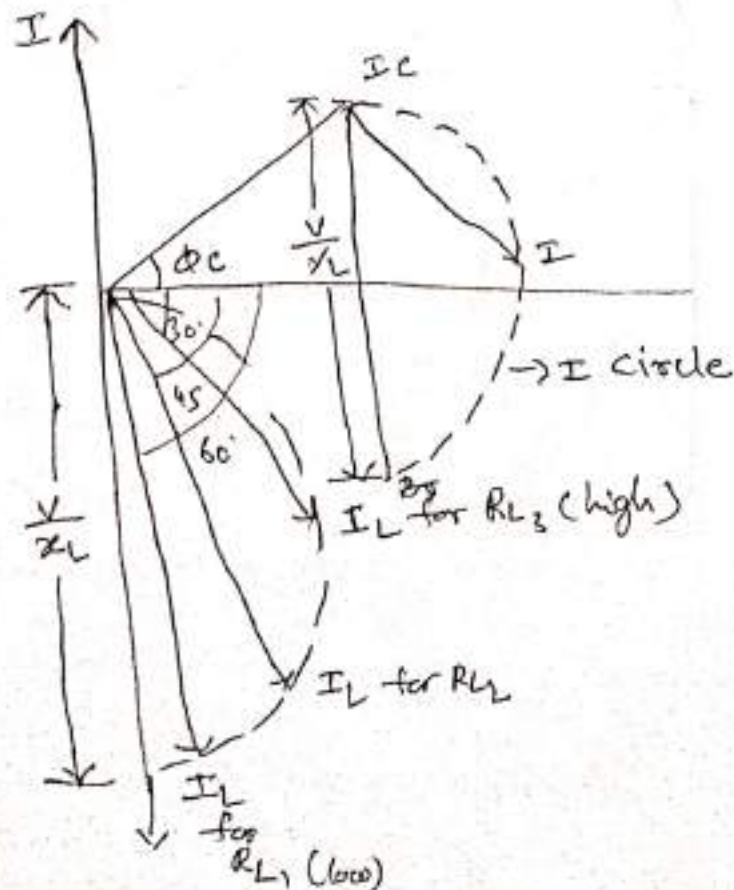
$$\begin{aligned} \sin 36^\circ &= 0.5 \\ \sin 45^\circ &= 0.707 \\ \sin 60^\circ &= 0.866 \end{aligned}$$

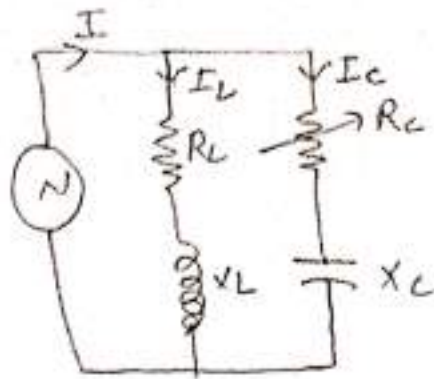
- When X_C is increased from 0 to infinity, its amplitude decreases to lower value and phase will be lead by 90°
- Phase angle $\theta_c = \tan^{-1}\left(\frac{X_C}{R_C}\right)$
- The ~~cur~~ locus of current is a semicircle with diameter of length equal to $\frac{V}{R_C}$



Case(3): Varying R_L :

- The current I_C through capacitance is constant since R_C & X_C are fixed and it leads the voltage vector OV by an angle $\theta_C = \tan^{-1}\left(\frac{X_C}{R_C}\right)$.
- The current I_L through the inductance is OI_L . its amplitude is maximum and is equal to $\frac{V}{X_L}$ where R_L is zero. It's phase will be lagging the voltage by 90° .
- when R_L is increased from '0' to infinity, its amplitude decreases to lower value $I = \frac{V}{\sqrt{R_L^2 + X_L^2}}$ $\cos \phi = \frac{R_L}{Z}$
- phase angle is lagging the voltage 'V' by an angle $\theta_L = \tan^{-1}\left(\frac{X_L}{R_L}\right)$
- Locus of current is a semi circle with diameter is equal to V/X_L

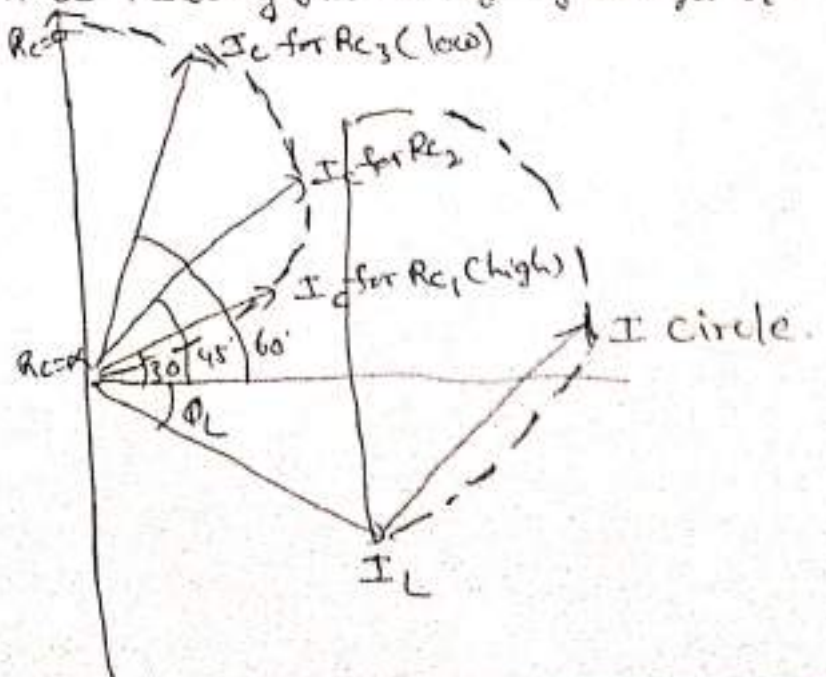




- phase angle will be leading the voltage by an angle $\theta_c = \tan^{-1} \left(\frac{X_c}{R_c} \right)$

$$- \cos \phi = \frac{R_L}{Z}$$

$$-I = \frac{V}{\sqrt{R_c^2 + X_c^2}}$$



Introduction:-

Resonance:- Resonance is the phenomenon in which applied voltage and resultant current are always in phase with each other.

[or]

An AC circuit is said to be in resonance if it exhibits unity power factor.

[or]

At resonance impedance becomes resistance only. i.e., $Z = R$.

[or]

At resonance net reactance reactances become zero.

Series Resonance:-

In Series RLC circuit, if any one of the parameter is varied such that applied voltage & resultant current are in phase with each other.

$$Z = R + jX$$

$$= R + j(X_L - X_C)$$

At resonance, $X = 0$

$$X_L - X_C = 0$$

$$X_L = X_C$$

$$Z = R$$



At resonance,

$$X_L = X_C$$

$$\omega L = \frac{1}{\omega C}$$

$$2\pi f_r L = \frac{1}{2\pi f_r C}$$

$$f_r = \frac{1}{4\pi^2 LC}$$

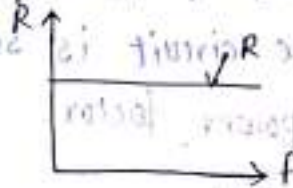
$$f_r = \frac{1}{2\pi\sqrt{LC}} \text{ Hz} \rightarrow \text{Resonant frequency.}$$

PF = 1 $\rightarrow$ power factor

Graphical representation of various elements in series resonance

(1) Resistance:-

R is independent of frequency (f).

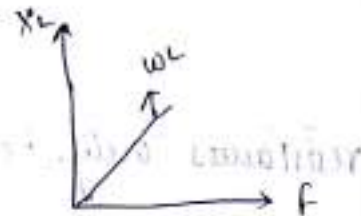


(2) Inductive reactance:-

$$jX_L = j\omega L$$

$$jX_L = j2\pi fL$$

$$X_L \propto f$$



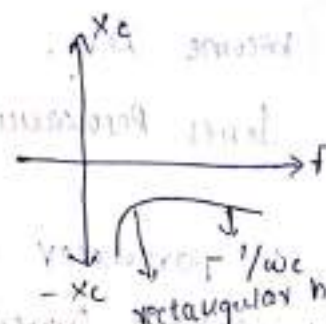
(3) Capacitance:-

$$-jX_C = -j \times \frac{1}{\omega C}$$

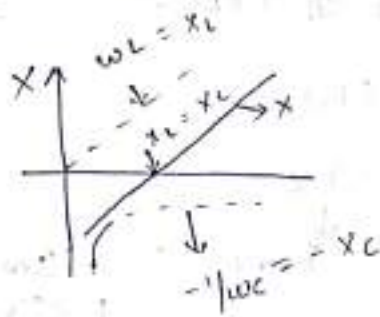
$$= j \left(-\frac{1}{\omega C} \right)$$

$$= j \left(-\frac{1}{2\pi fC} \right)$$

$$X_C \propto \frac{1}{f}$$

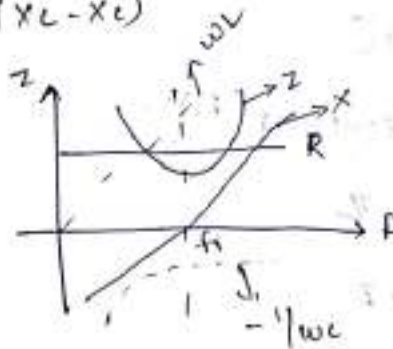


(4) Reactance (X):-



(5) Impedance:-

$$Z = R + j(X_L - X_C)$$



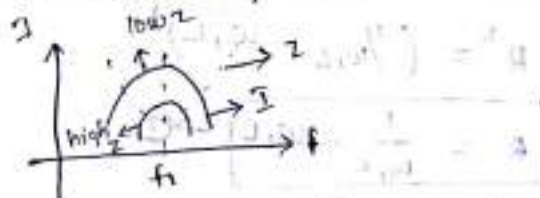
At resonant frequency, Z is minimum.

(6) Current:-

$$I = V/Z$$

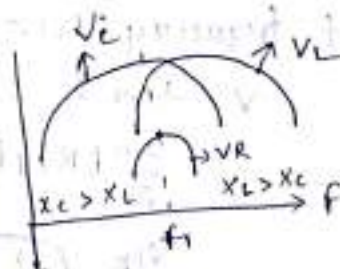
$$I \propto 1/Z$$

At resonant frequency, Z is minimum, I is maximum.



(7) Voltage:-

$$V_R = IR$$



Bandwidth :-

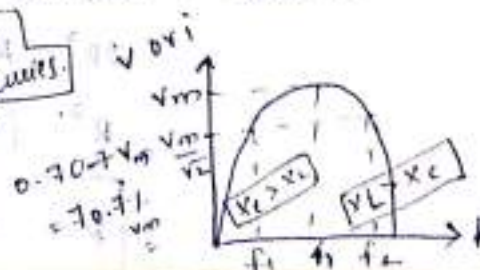
Bandwidth is the difference in frequencies at which voltage or current is equal to 70.7% of its maximum value.

$$B.W = f_2 - f_1$$

$f_1 \rightarrow$ lower cut off frequency

$f_2 \rightarrow$ Upper " " " "

Half power frequencies.



Imp
**

Relationship b/w Bandwidth & QLC :-

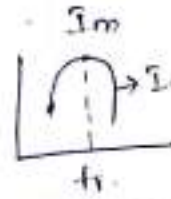
At resonant frequency (f_r),

we know, $V = I Z$
 $\downarrow \quad \downarrow$
 $V_{rms} \quad I_{rms}$

At f_r :-

$$V = I_m R$$

$$I_m = \frac{V}{R} \rightarrow (1)$$



At lower cut off frequency (f_1):-

we know, $V_{rms} = I_{rms} Z$

$$V = \frac{I_m}{\sqrt{2}} \cdot Z$$

$$V = \frac{V}{\sqrt{2}} (R + j(X_{C1} - X_{L1}))$$

$$\sqrt{2} R = \sqrt{R^2 + (X_{C1} - X_{L1})^2}$$

both sides on squaring

$$2R^2 = R^2 + \left(\frac{1}{\omega_1 C} - \omega_1 L\right)^2$$

$$R^2 = \left(\frac{1}{\omega_1 C} - \omega_1 L\right)^2$$

$$R = \frac{1}{\omega_1 C} - \omega_1 L \rightarrow (2)$$

At high cut off frequency (f_2):-

$$V = I_{rms} Z$$

$$= \frac{I_m}{\sqrt{2}} (R + j(X_{L2} - X_{C2}))$$

$$V = \frac{V}{\sqrt{2}} \sqrt{R^2 + (X_{L2} - X_{C2})^2}$$

$$\sqrt{2} R = \sqrt{R^2 + (X_{L2} - X_{C2})^2}$$

Squaring on both sides.

$$2R^2 = R^2 + (X_{L2} - X_{C2})^2$$

$$R^2 = \left(\omega_2 L - \frac{1}{\omega_2 C}\right)^2$$

$$R = \omega_2 L - \frac{1}{\omega_2 C} \rightarrow (3)$$

Equating Eq (2) & (3)

$$R = \omega_2 L - \frac{1}{\omega_2 C} = \frac{1}{\omega_1 C} - \omega_1 L$$

$$\omega_2 L + \omega_1 L = \frac{1}{\omega_1 C} + \frac{1}{\omega_2 C}$$

$$L(\omega_1 + \omega_2) = \frac{1}{C} \left(\frac{\omega_2 + \omega_1}{\omega_1 \omega_2} \right)$$

$$L = \frac{1}{C} \left(\frac{1}{\omega_1 \omega_2} \right)$$

$$\boxed{\omega_1 \omega_2 = \frac{1}{LC}} \rightarrow (4)$$

we know, $f_r = \frac{1}{2\pi\sqrt{LC}}$

$$2\pi f_r = \frac{1}{\sqrt{LC}}$$

$$\omega_r = \frac{1}{\sqrt{LC}}$$

$$\boxed{\omega_r^2 = \frac{1}{LC}} \rightarrow (5)$$

Substitute eq (5) in eq (4)

$$\omega_1 \omega_2 = \omega_r^2$$

$$\boxed{\omega_r = \sqrt{\omega_1 \omega_2}}$$

Hence $\boxed{f_r = \sqrt{f_1 f_2}}$

$\therefore$ Resonant frequency is the geometrical mean of f_1 & f_2 .

Adding Eq (2) & (3)

$$R + R = \frac{1}{\omega_1 C} - \omega_1 L + \omega_2 L - \frac{1}{\omega_2 C}$$

$$2R = \frac{1}{C} \left(\frac{1}{\omega_1} - \frac{1}{\omega_2} \right) + L(\omega_2 - \omega_1)$$

$$2R = \frac{1}{C} \left(\frac{\omega_2 - \omega_1}{\omega_1 \omega_2} \right) + L(\omega_2 - \omega_1)$$

we know $\omega_1 \omega_2 = \frac{1}{LC}$ (from eq (4)).

$$2R = \frac{1}{C} \left(\frac{\omega_2 - \omega_1}{1/LC} \right) + L(\omega_2 - \omega_1)$$

$$2R = \frac{1}{C} (\omega_2 - \omega_1) LC + L(\omega_2 - \omega_1)$$

$$2R = (\omega_2 - \omega_1)(L + \frac{R^2}{L})$$

$$2R = \frac{R}{L} (\omega_2 - \omega_1)$$

$$\omega_2 - \omega_1 = \frac{R}{L}$$

$$2\pi f_2 - 2\pi f_1 = \frac{R}{L}$$

$$2\pi (f_2 - f_1) = \frac{R}{L}$$

$$(f_2 - f_1) = \frac{R}{2\pi L}$$

$$\boxed{\text{Bandwidth} = \frac{R}{2\pi L}}$$

* **Q Quality factor:-**

Quality factor (Q) is defined as the ratio of voltage across inductor (or) capacitor to be the Sup Voltage.

It is the indication of quality of coil (R & L)

$$\therefore Q = \frac{V_L}{V}$$

At resonance,

$$V = V_R$$

$$Q = \frac{V_L}{V_R}$$

$$= \frac{I X_L}{I R}$$

$$\boxed{Q = \frac{X_L}{R}}$$

$$\boxed{Q = \frac{\omega L}{R}}$$

$$Q = \frac{V_C}{V}$$

At resonance,

$$V = V_R$$

$$Q = \frac{V_C}{V_R}$$

$$= \frac{I X_C}{I R}$$

$$\boxed{Q = \frac{X_C}{R}}$$

$$\boxed{Q = \frac{1}{\omega C R}}$$

Imp.

$$\text{Quality factor} = Q = 2\pi \times \frac{\text{Energy stored/cycle}}{\text{Energy dissipated/cycle}}$$

Relation ship b/w Bandwidth, Quality factor & Resonant frequency:-

$$\text{we know, Bandwidth} = \frac{P \times f_1}{2\pi f_1 L}$$

$$= \frac{R \times f_r}{\omega_r L}$$

$$= \frac{R \times f_r}{X_L}$$

$$B.W = \frac{f_r}{(X_L/R)} = \frac{f_r}{Q} \Rightarrow \boxed{B.W = \frac{f_r}{Q}}$$

Half power frequencies in terms of resonance frequency:-

For Symmetrical wave,

$$f_r = \frac{f_1 + f_2}{2} \rightarrow (1)$$



From eq (1),

$$f_1 = 2f_r - f_2 \rightarrow (2)$$

we know, Bandwidth = $f_2 - f_1 = \frac{R}{2\pi L} \rightarrow (3)$

Substitute (f_1) in eq (3)

$$f_2 - 2f_r + f_2 = \frac{R}{2\pi L}$$

$$2f_2 - 2f_r = \frac{R}{2\pi L}$$

$$2(f_2 - f_r) = \frac{R}{2\pi L}$$

$$\boxed{f_2 = \frac{R}{4\pi L} + f_r} \rightarrow (4)$$

From eq (1),

$$f_2 = 2f_r - f_1 \rightarrow (5)$$

Substitute eq (5) in eq (3)

$$2f_r - f_1 - f_1 = \frac{R}{2\pi L}$$

$$2f_r - 2f_1 = \frac{R}{2\pi L}$$

$$2(f_r - f_1) = \frac{R}{2\pi L}$$

$$f_r - f_1 = \frac{R}{4\pi L}$$

$$\boxed{f_1 = f_r - \frac{R}{4\pi L}} \rightarrow (6)$$

Selectivity (S): It is reciprocal of Quality factor (Q).

$$S = \frac{1}{Q}$$

$$= \frac{1}{\frac{R}{2\pi L}}$$

$$\boxed{S = \frac{2\pi L}{R}} \quad (\text{or}) \quad S = \frac{1}{Q}$$

$$= \frac{1}{f/B \cdot \omega} = \frac{B \cdot \omega}{f}$$

Voltage across L and C at resonance:-

$$V_L = I (jX_L)$$

$$\text{at resonance} \Rightarrow I = V/R$$

$$V_L = \frac{V}{R} \cdot j(\omega_r L)$$

$$= j \cdot \frac{\omega_r L}{R} \cdot V$$

$$= j \cdot \frac{X_L}{R} \cdot V$$

$$\therefore \boxed{V_L = j Q V}$$

$$V_C = I (-jX_C)$$

$$\text{At resonance} \rightarrow I = V/R$$

$$V_C = V/R (-jX_C) = -j \cdot \left(\frac{X_C}{R} \right) V$$

$$\therefore \boxed{V_C = -j Q V}$$

Frequency for voltage across Inductor is maximum:-

we know,

$$V_L = I X_L \quad (\text{in magnitude})$$

$$= \frac{V}{Z} X_L$$

$$= \frac{X_L \omega L}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$= \frac{V \omega L}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

$$= \frac{V \omega L}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

$$= \frac{V \omega L}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

$$V_L^2 = \frac{V^2 \omega^2 L^2}{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

$$\frac{V^2 \omega^2 L^2}{R^2 + (\omega L - \frac{1}{\omega C})^2} \rightarrow \frac{V^2 \omega^2 L^2}{R^2 + \omega^2 L^2 - 2 + \frac{1}{\omega^2 C^2}}$$

$$V_c^2 = \frac{V^2}{\omega^2 C^2 [R^2 + (\omega L - 1/\omega C)^2]}$$

$$V_c^2 = \frac{V^2 \omega^2 c^2}{\omega^2 c^2 (R^2 \omega^2 c^2 + (\omega^2 L(-1))^2)}$$

$$V_c^* = \frac{V^L}{R^L w^L c^L + (w^L (L-1))^L}$$

To get frequency, V_c is maximum.

$$\frac{dv_c^2}{dw} = 0$$

$$(R^2 \omega^4 c^2 + (\omega^4 L(-1))^2) (0) = V^2 (2\omega R^2 c^2 + 4\omega^5 L^2 c^2 - 4\omega L(-1) = 0$$

$$2\omega R^2 C^2 + 4\omega^2 L^2 C^2 - 4\omega L C = 0$$

$$j\omega C (R^2 C + j\omega L^2 C - jL) = 0$$

$$R^2 C + 2\omega^2 L^2 C - 2L = 0$$

$$2\omega^2 L^2 C = 2L - R^2 C$$

$$\omega^2 = \frac{\partial^2 \mathcal{L}}{\partial \psi^2} = \frac{\partial^2 \mathcal{L}}{\partial \psi^2}$$

$$\omega^2 = \frac{1}{LC} - \frac{R^2}{2L^2}$$

$$\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$f_c = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}}$$

1) A Series ckt with $R = 10\ \Omega$, $L = 0.1\text{ H}$, & $C = 50\ \mu\text{F}$ as an applied voltage $v = 50$ at an angle 0° with ω variable frequency. find (a) resonant frequency, (b) frequency at which voltage across inductor is maximum, (c) frequency at which voltage across capacitor is max. d) voltage across inductor at resonant.

8) An inductance of 0.5 H , Resistance of 5Ω & Capacitance $8 \mu\text{F}$ are in series across a 220 V AC Supply, (a) calculate the frequency at which ckt resonance, (b) find the current at resonance, bandwidth of power frequency & Voltage across inductance & capacitance.

Given data,

$$R = 5 \Omega,$$

$$L = 0.5 \text{ H},$$

$$C = 8 \mu\text{F} = 8 \times 10^{-6} \text{ F}$$

(a) frequency at resonance :- $f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.5 \times 8 \times 10^{-6}}}$

$$f_r = 79.58 \text{ Hz.}$$

(b) $V = IR$

at resonance $Z = R$,

$$V = IR$$

$$I = \frac{V}{R}$$

$$= \frac{220}{5} = 44 \text{ A.}$$

Bandwidth = $B.W = \frac{R}{2\pi L}$

$$= \frac{5}{2 \times 3.14 \times 0.5}$$

Bandwidth = 1.59 Hz

half power frequencies = $f_1 = f_r - \frac{R}{4\pi L}$
 $= 79.58 - \frac{5}{4 \times 3.14 \times 0.5}$

$$= 78.79 \text{ Hz.}$$

$f_2 = f_r + \frac{R}{4\pi L}$

$$= 79.58 + \frac{5}{4 \times 3.14 \times 0.5} = 80.40 \text{ Hz.}$$

Quality factor :- $Q = \frac{P_r}{B \cdot \omega}$

$$= \frac{79.58}{1.59}$$

$$= 50$$

Voltage across Inductor (V_L) :-

we know, $Q = \frac{V_L}{V_R} = \frac{V_L}{V}$

$$V_L = Q \times V$$

$$V_L = 50 \times 220$$

$$= 11 \text{ kV}$$

Voltage across Capacitor (V_C) :-

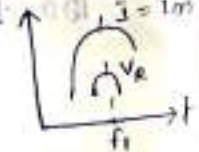
$$Q = \frac{V_C}{V_R} = \frac{V_C}{V}$$

$$V_C = Q \times V$$

$$= 11 \text{ kV}$$

Note:-

In Series resonance, Current $I = I_m$ & Voltage $V = V_R$



3) In Series RLC ckt has quality factor of 5 at 50 Radians/sec. The current flowing through ckt at resonance is 10 amperes & applied voltage 100 V. The total impedance of the ckt is 20Ω , find the ckt elements.

A) Given data,

Quality factor = 5
 $\omega = 50 \text{ rad/sec}$

At resonance $I = I_m = 10 \text{ A}$

$$V = 100V$$

$$Z = 20 \Omega$$

we know,

$$V = I_m Z$$

At resonance, $V = I_m R$

$$R = \frac{V}{I_m} = \frac{100}{10} = 10 \Omega$$

we know,

$$Q = \frac{X_L}{R} = \frac{\omega L}{R}$$

$$L = \frac{QR}{\omega}$$

$$= \frac{5 \times 10}{50}$$

$$= 1 H$$

Also

we know,

$$Q = \frac{X_C}{R} = \frac{1}{\omega CR}$$

$$Q \omega R = \frac{1}{C}$$

$$C = \frac{1}{Q \omega R}$$

$$= \frac{1}{5 \times 50 \times 10}$$

$$= 400 \mu F$$

4) In the Series RLC ckt with $L = 0.5 H$, as an instantaneous Voltage $v = 70.7 \sin(500t + 30^\circ) V$ & Current $i = 1.5 \sin(500t) A$, find the value's of R & C . At what frequency will the ckt be resonance.

A) Given data, $L = 0.5 H$

$$v = 70.7 \sin(500t + 30^\circ) V$$

$$i = 1.5 \sin(500t) A$$

$$v = 70.7 \angle 30^\circ \Rightarrow V_m = 70.7$$

$$i = 1.5 \angle 0^\circ \Rightarrow I_m = 1.5$$

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{70.7}{\sqrt{2}} = 49.9 \text{ V} \approx 50 \text{ V}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{1.5}{\sqrt{2}} = 1.06 \text{ A}$$

we know,

$$Z = \frac{V_{rms}}{I_{rms}} = \frac{49.9 \angle 30^\circ}{1.06 \angle 0^\circ}$$

$$= 47.07 \angle 30$$

$$= 47.07 (\cos 30 + j \sin 30)$$

$$= 47.07 \left(\frac{\sqrt{3}}{2} + j \cdot \frac{1}{2} \right)$$

$$= 40.76 + j 23.53$$

$$= R + jX$$

$$\text{where } R = 40.76 \Omega, X = 23.53 \Omega$$

we know,

$$X = X_L - X_C$$

$$23.53 = \omega L - \frac{1}{\omega C}$$

$$23.53 = 500(0.5) - \frac{1}{500 \times C}$$

$$23.53 = 250 - \frac{1}{500 \times C}$$

$$23.53 - 250 = - \frac{1}{500 C}$$

$$-226.47 = - \frac{1}{500 C}$$

$$C = \frac{1}{500 \times 226.47}$$

$$113235$$

$$C = 8.83 \mu\text{F}$$

$$(b) f_r = \frac{1}{2\pi \sqrt{LC}} = \frac{1}{2 \times 3.14 \sqrt{0.5 \times 8.83 \times 10^{-6}}} = 75.78 \text{ Hz}$$

5) A Series RLC ckt with $R = 25 \Omega$, $L = 0.6 H$ at frequency of $40 Hz$. Find the leading phase angle of 60° at what frequency the ckt will be resonant.

Given data,

$$R = 25 \Omega$$

$$L = 0.6 H$$

$$\text{leading phase angle } (\phi) = 60^\circ$$

$$f = 40 Hz$$

here leading phase angle, so $Z = R + j(X_C - X_L)$.

$$\text{then } \tan \phi = \frac{X_C - X_L}{R}$$

$$\tan 60^\circ = \frac{\frac{1}{\omega C} - \omega L}{25}$$

$$2 \times 3.14 \times 40$$

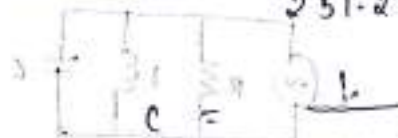
$$\sqrt{3} (25) = \frac{1}{2\pi f C} - 2\pi f L$$

$$= \frac{1}{251.2 C} - 251.2 (0.6)$$

$$43.30 = \frac{1}{251.2 C} - 150.72$$

$$43.30 + 150.72 = \frac{1}{251.2 C}$$

$$194.02 = \frac{1}{251.2 C}$$



$$251.2 (194.02)$$

$$C = 2.05 \times 10^{-5} F$$

$$\text{Resonant frequency } f_r = \frac{1}{2\pi \sqrt{LC}} = \frac{1}{2 \times 3.14 \sqrt{0.6 \times 2.05 \times 10^{-5}}}$$

$$f_r = 45.40 Hz$$

Admittance :- $[Y]$

It is the reciprocal of impedance (Z)

$$Y = \frac{1}{Z}$$

Impedance parameter

Z

R

X_L

X_C

$$Z = R + j(X_L - X_C)$$

Admittance parameter

$$Y = \frac{1}{Z}$$

$$\frac{1}{R} = G \text{ (conductance)}$$

$$\frac{1}{X_L} = B_L \text{ (susceptive Inductance)}$$

$$\frac{1}{X_C} = B_C \text{ (susceptive capacitance)}$$

$$Y = G + j(B_C - B_L)$$

Impedance :- $V = IZ$

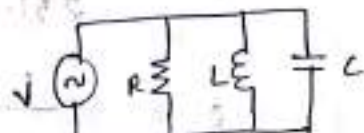
$$I = \frac{V}{Z}$$

$$I = V \cdot Y$$

$$I = V \cdot Y \rightarrow \text{Admittance.}$$

Parallel Resonance:-

Consider parallel RLC ckt excited by alternating voltage as shown in figure.



Now express all the parameters in terms of ohms & ckt is redrawn as



Take $Y_1 = \frac{1}{R}$, $Y_2 = \frac{1}{jX_L} = \frac{-j}{X_L}$, $Y_3 = \frac{1}{-jX_C} = \frac{j}{X_C}$

The admittance $Y = Y_1 + Y_2 + Y_3$.

$$= \frac{1}{R} - \frac{j}{X_L} + \frac{j}{X_C}$$

$$= \frac{1}{R} + j \left(\frac{1}{X_C} - \frac{1}{X_L} \right)$$

$$= \frac{1}{R} + j \left(\omega C - \frac{1}{\omega L} \right)$$

$$= \frac{1}{R} + j (B_C - B_L)$$

$$Y = G + j(B_C - B_L) \rightarrow (2)$$

$$Y = G + jB \rightarrow (1)$$

At resonance, $B = 0$

$$B_C - B_L = 0$$

$$B_C = B_L$$

$$\frac{1}{X_C} = \frac{1}{X_L}$$

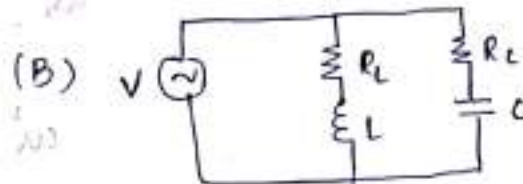
$$\omega_r C = \frac{1}{\omega_r L}$$

$$\frac{1}{\omega_r} = \frac{1}{\omega_r^2 LC} \Rightarrow \frac{1}{\omega_r^2} = \frac{1}{LC}$$

$$\omega_r = \frac{1}{\sqrt{LC}}$$

$$\text{Resonant frequency} \Rightarrow f_r = \frac{1}{2\pi\sqrt{LC}} \text{ Hz.}$$

1) Find the resonant frequency for the following ckt's.



$$Y_1 = \frac{1}{R + jX_L}$$

$$Y_2 = \frac{1}{-jX_C} = \frac{j}{X_C}$$

Total admittance: $Y = Y_1 + Y_2$

$$Y = \frac{1}{R+jX_L} + \frac{j}{X_C} \quad [\text{on Rationalization}]$$

$$= \frac{R-jX_L}{R^2+X_L^2} + \frac{j}{X_C}$$

$$Y = \frac{R}{R^2+X_L^2} + j \left(\frac{1}{X_C} - \frac{X_L}{R^2+X_L^2} \right) \rightarrow (1)$$

At resonance,

Imaginary part = 0.

$$\frac{1}{X_C} - \frac{X_L}{R^2+X_L^2} = 0$$

$$\omega C - \frac{\omega L}{R^2+(\omega L)^2} = 0$$

$$\omega C = \frac{\omega L}{R^2+(\omega L)^2}$$

$$R^2+(\omega L)^2 = \frac{L}{C}$$

$$(\omega L)^2 = \frac{L}{C} - R^2$$

$$\omega^2 L^2 = \frac{L}{C} - R^2$$

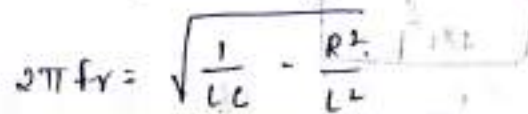
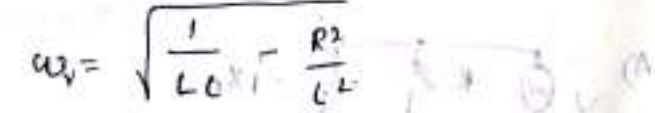
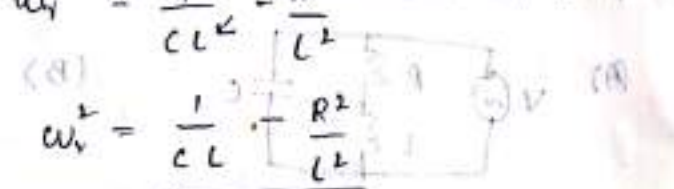
$$\omega^2 = \frac{1}{LC} - \frac{R^2}{L^2}$$

$$\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

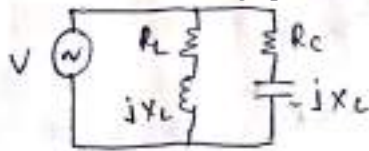
$$\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$2\pi f_r = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$\text{Resonant frequency} = f_r = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$



B)



$$Y_1 = \frac{1}{R_L + jX_L}, \quad Y_2 = \frac{1}{R_C - jX_C}$$

Total admittance :- $Y = Y_1 + Y_2$

$$Y = \frac{1}{R_L + jX_L} + \frac{1}{R_C - jX_C}$$

$$= \frac{R_L - jX_L}{R_L^2 + X_L^2} + \frac{R_C + jX_C}{R_C^2 + X_C^2}$$

$$= \frac{R_L}{R_L^2 + X_L^2} - j \frac{X_L}{R_L^2 + X_L^2} + \frac{R_C}{R_C^2 + X_C^2} + j \frac{X_C}{R_C^2 + X_C^2}$$

$$= \frac{R_L + R_C}{(R_L^2 + X_L^2)(R_C^2 + X_C^2)} + j \left(\frac{X_C}{R_C^2 + X_C^2} - \frac{X_L}{R_L^2 + X_L^2} \right)$$

At resonance,

imaginary part = 0

$$\frac{X_C}{R_C^2 + X_C^2} - \frac{X_L}{R_L^2 + X_L^2} = 0$$

$$\frac{X_C}{R_C^2 + X_C^2} = \frac{X_L}{R_L^2 + X_L^2}$$

$$\frac{1}{\omega_r C} = \omega_r L$$

$$\frac{1}{\omega_r C} (R_L^2 + (\omega_r L)^2) = \omega_r L (R_C^2 + (\frac{1}{\omega_r C})^2)$$

$$\frac{\omega_r L}{(R_C^2 (\omega_r C)^2 + 1)} = \frac{\omega_r L}{R_L^2 + (\omega_r L)^2}$$

$$C(R_L^2 + (\omega_r L)^2) = L(R_C^2 \omega_r^2 C^2 + 1)$$

$$CR_L^2 + C\omega_Y^2 L = L R_L^2 \omega_Y^2 C + L$$

$$GR_0 \omega_Y^2 (CL^2 - LR_L^2 C^2) = L - CR_L^2$$

$$\omega_Y^2 = \frac{L - CR_L^2}{CL^2 - LR_L^2 C^2}$$

$$\omega_Y^2 = \frac{L \left[\frac{1}{C} - R_L^2 \right]}{C^2 L \left[\frac{L}{C} - R_L^2 \right]}$$

$$\omega_Y^2 = \frac{R_L^2 - 1/C}{LC (R_L^2 - 1/C)}$$

$$\omega_Y = \frac{\sqrt{R_L^2 - 1/C}}{\sqrt{LC (R_L^2 - 1/C)}}$$

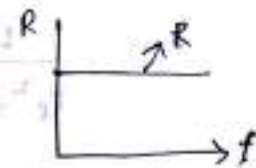
$$2\pi f_Y = \frac{\sqrt{R_L^2 - 1/C}}{\sqrt{LC (R_L^2 - 1/C)}}$$

$$f_r = \frac{1}{2\pi \sqrt{LC}} \sqrt{\frac{R_L^2 - 1/C}{R_L^2 - 1/C}} \rightarrow \text{Resonant frequency.}$$

Graphical representation of Various elements in parallel Resonance

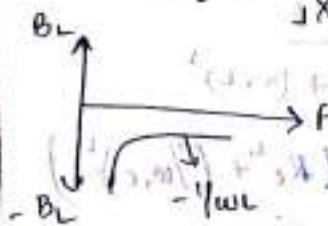
(1) Resistance (R):-

R is independent of frequency



(2) Inductive Reactance :- (B_L)

$$B_L = \frac{1}{jX_L} = \frac{-j}{X_L} = j \left(-\frac{1}{X_L} \right) = j \left(-\frac{1}{\omega L} \right)$$



$$B_L = \frac{1}{jX_L} = \frac{-j}{X_L} = j \left(-\frac{1}{\omega L} \right)$$

(3) Capacitive reactance :-

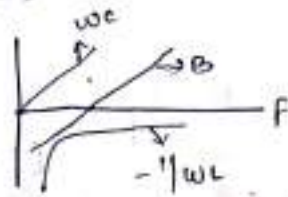
$$B_C = \frac{1}{-jX_C} = \frac{j}{X_C} = \frac{j}{\omega C}$$

$$B_C = j\omega C = j \cdot 2\pi f C$$

$$B_c \neq f$$



(4) Susceptance (B)



$$Z = R + j(X_L - X_C)$$

$$Y = G + j(B_C - B_L)$$

$$= G + j\left(\omega C - \frac{1}{\omega L}\right)$$

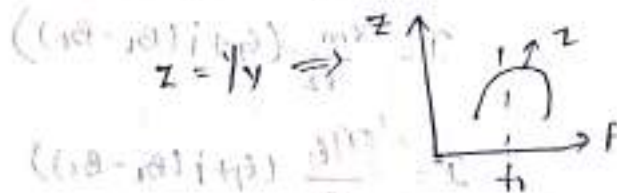
(5) Impedance & admittance:

we know,

$$Y = G + jB$$

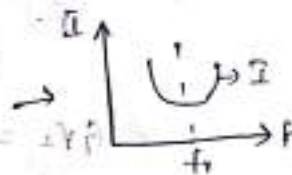
At resonance, $B = 0$

$\therefore Y = G$, i.e. Y is minimum



(6) Voltage & Current :-

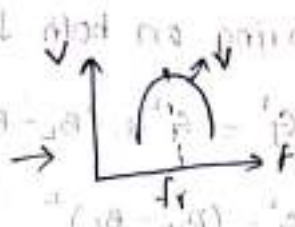
we know, $I = VY$



$$I = VY$$

$$V = I/Y$$

$$V \propto 1/Y$$

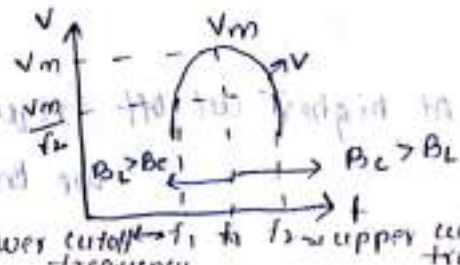


Note :- In parallel resonance, $V = V_m$
 $I = I_R$

Bandwidth (BW) :-

Like Series Resonance, bandwidth of parallel resonance also defined as the difference in frequency at which voltage is equal to 70.7% of maximum value.

$$\text{Bandwidth} = f_2 - f_1 \text{ Hz}$$



Relation b/w Bandwidth & RLC:

* At f_r :-

we know, $i = vY$

At resonance,

$$[\because Y = G + jB]$$

$$I = V_m \cdot G$$

$$Y = G$$

$$V_m = \frac{I}{G} \rightarrow (1)$$

* At lower cut off frequency (f_1):

we know,

$$I = VY$$

$$I = \frac{V_m}{\sqrt{2}} (G + j(B_L - B_C))$$

$$Y = \frac{I/G}{\sqrt{2}} (G + j(B_L - B_C))$$

$$1 = \frac{1}{\sqrt{2}G} (G + j(B_L - B_C))$$

$$G\sqrt{2} = G + j(B_L - B_C) \Rightarrow G\sqrt{2} = \sqrt{G^2 + (B_L - B_C)^2}$$

Squaring on both sides,

$$2G^2 = G^2 + (B_L - B_C)^2$$

$$G^2 = (B_L - B_C)^2$$

$$G = B_L - B_C$$

$$\frac{1}{R} = \frac{1}{X_L} - \frac{1}{X_C}$$

$$\frac{1}{R} = \frac{1}{\omega L} - \frac{1}{\omega C}$$

$$\frac{1}{R} = \frac{1}{\omega L} - \frac{1}{\omega C} \rightarrow (2)$$

* At higher cut off frequency: (f_2)

we know;

$$I = VY$$

$$Z = \frac{V_m}{I_m} \cdot (G + j(B_C - B_L))$$

$$Z = \frac{V/I}{I_m} (G + j(B_C - B_L))$$

$$G\sqrt{2} = G + j(B_C - B_L)$$

$$G\sqrt{2} = \sqrt{G^2 + (B_C - B_L)^2}$$

squaring on both sides

$$2G^2 = G^2 + (B_C - B_L)^2$$

$$G^2 = (B_C - B_L)^2$$

$$G = (B_C - B_L)$$

$$\frac{1}{R} = \left(\frac{1}{X_C} \right) - \frac{1}{X_L}$$

$$\frac{1}{R} = \omega_2 C - \frac{1}{\omega_2 L} \rightarrow (3)$$

Equating (2) & (3)

$$\frac{1}{\omega_1 L} - \omega_1 C = \omega_2 C - \frac{1}{\omega_2 L}$$

$$\frac{1}{\omega_1 L} + \frac{1}{\omega_2 L} = \omega_1 C + \omega_2 C$$

$$\frac{1}{L} \left(\frac{1}{\omega_1} + \frac{1}{\omega_2} \right) = (\omega_1 + \omega_2) C$$

$$\frac{1}{L} \left(\frac{\omega_2 + \omega_1}{\omega_1 \omega_2} \right) = (\omega_1 + \omega_2) C$$

$$\frac{1}{L(\omega_1 \omega_2)} = C$$

we know from RLC

parallel ckt $\omega_1 \omega_2 = \frac{1}{LC}$

$$\omega_1 \omega_2 = \frac{1}{LC}$$

$$\omega_1 \omega_2 = \omega_r^2$$

$$\omega_r = \sqrt{\omega_1 \omega_2}$$

$$f_r = \sqrt{f_1 f_2} \rightarrow (5)$$

$\therefore$ Resonant frequency (f_r) is the geometrical mean of f_1 & f_2 .

Adding Eq (2) & (3)

$$\frac{1}{R} + \frac{1}{R} = \frac{1}{\omega_1 L} - \omega_1 C + \omega_2 C - \frac{1}{\omega_2 L}$$

$$\frac{2}{R} = \frac{1}{L} \left[\frac{1}{\omega_1} - \frac{1}{\omega_2} \right] + C (\omega_2 - \omega_1)$$

$$\frac{2}{R} = \frac{1}{L} \left(\frac{\omega_2 - \omega_1}{\omega_1 \omega_2} \right) + C (\omega_2 - \omega_1)$$

$$= (\omega_2 - \omega_1) \left[\frac{1}{L(\omega_1 \omega_2)} + C \right]$$

$$= (\omega_2 - \omega_1) \left(\frac{1}{\cancel{L} \times \frac{1}{\cancel{L} C}} + C \right)$$

$$= (\omega_2 - \omega_1) (C + C) = \frac{2}{R}$$

$$\frac{2}{R} = (\omega_2 - \omega_1) (2C)$$

$$\omega_2 - \omega_1 = \frac{1}{RC}$$

$$f_2 - f_1 = \frac{1}{2\pi RC}$$

$$\boxed{\text{Band width} = \frac{1}{2\pi RC}}$$

Half power frequencies in terms of resonant frequency (f_r):-

for Symmetrical wave,

$$f_r = \frac{f_1 + f_2}{2}$$

$$\text{Let } f_1 = 2f_r - f_2 \rightarrow (1)$$

we know,

$$B.W = f_2 - f_1 = \frac{1}{2\pi RC} \rightarrow (2)$$

(2) in (1)

$$f_2 = 2f_r + f_2 = \frac{1}{2\pi RC}$$

$$2f_2 = \frac{1}{2\pi RC} + 2f_r$$

$$f_2 = \frac{f_1 + f_r}{2}$$

Similarly, $f_r = \frac{f_1 + f_2}{2}$

$$f_2 = 2f_r - f_1 \rightarrow (3)$$

(3) in eqn (2)

$$2f_r - f_1 - f_1 = \frac{1}{2\pi RC}$$

$$2f_r - 2f_1 = \frac{1}{2\pi RC}$$

$$f_r - f_1 = \frac{1}{4\pi RC}$$

$$-f_1 = \frac{1}{4\pi RC} - f_r$$

$$f_1 = f_r - \frac{1}{4\pi RC}$$

Quality factor:-

It is defined as "The ratio of current through the inductor (or) capacitor to the Supply current."

$$Q = \frac{I_L}{I}$$

At resonance, $I = I_R$

$$Q = \frac{I_L}{I_R} = \frac{V/X_L}{V/R}$$

$$Q = \frac{R}{X_L}$$

$$Q = \frac{R}{\omega L}$$

$$Q = \frac{I_C}{I}$$

At resonance, $I = I_R$

$$Q = \frac{I_C}{I_R} = \frac{V/X_C}{V/R}$$

$$R = \frac{R}{X_C}$$

$$R = \frac{R}{1/\omega C}$$

$$Q = R\omega C$$

Relation b/w Bandwidth, Quality factor & Resonant frequency:-

we know, Bandwidth = $\frac{1}{2\pi RC}$

Multiply numerator & denominator by f_r

$$B.W = \frac{f_r}{2\pi f_r RC}$$

$$B.W = \frac{f_r}{\omega_r RC}$$

$$B.W = \frac{f_r}{Q}$$

where $Q = \frac{\omega_r L}{R}$

Also,

$$B.W = \frac{f_r}{\omega_r RC}$$

$$= \frac{f_r \cdot \frac{X_L}{R}}{1} \quad \therefore$$

Selectivity :- (S) =

Selectivity is the reciprocal of Quality factor.

$$S = \frac{1}{Q}$$

$$= \frac{1}{R/\omega L}$$

$$S = \frac{1}{Q}$$

$$= \frac{1}{\omega RC}$$

$$S = \frac{\omega L}{R}$$

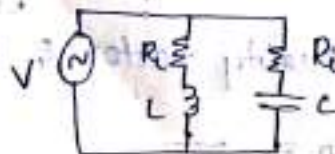
1) What is dynamic impedance?

Ans. Dynamic impedance :- Dynamic impedance or simply dynamic resistance is a pure resistance which is defined only at resonant frequency.

$$\text{Dynamic impedance :- } Z_{dy} \text{ or } R_{dy} = \frac{L}{CR}$$

2) Write the Equation for R_L & R_C which causes the ckt to resonant to at all frequencies.

A)



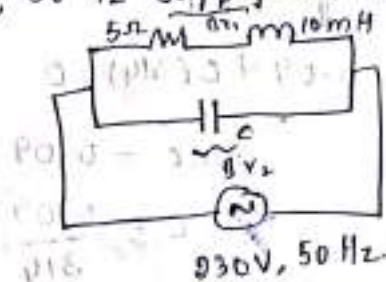
$$f_r = \frac{1}{2\pi\sqrt{LC}} \sqrt{\frac{R_L^2 - L/C}{R_C^2 - L/C}}$$

if R_L & R_C are very small,

$$\begin{aligned} R_L^2 - L/C &= 0 & R_C^2 - L/C &= 0 \\ R_L^2 &= \frac{L}{C} & R_C^2 &= \frac{L}{C} \\ R_L &= \sqrt{\frac{L}{C}} & R_C &= \sqrt{\frac{L}{C}} \end{aligned}$$

problems on Parallel Resonance:-

- 1) A parallel ckt has 2nd branches, 1st branch has a resistance of 5Ω connected in series with an inductance of 10mH . A capacitor is connected in 2nd branch. The parallel ckt is connected across a 230V , 50Hz supply. If the ckt to be in resonance,



find the value of Capacitance & also current drawn from the supply & also find currents in branches 1 & 2.

$$\begin{aligned} \text{Let } Y_1 &= \frac{1}{R + jX_L} = \frac{1}{R + j\omega L} \\ &= \frac{1}{R + j2\pi fL} \\ &= \frac{1}{5 + j(2 \times 3.14 \times 50 \times 10 \times 10^{-3})} \\ &= \frac{1}{5 + j(3.14)} \end{aligned}$$

$$\begin{aligned} Y_2 &= \frac{1}{-jX_C} = \frac{1}{-j\omega C} = j\omega C \\ &= j(2\pi f)C \\ &= j(2 \times 3.14 \times 50)(C) \end{aligned}$$

$$Y = Y_1 + Y_2 = \frac{1}{5 + j(3.14)} + j(314)C$$

$$Y_1 = \frac{1}{5 + j(314)}$$

$$= \frac{5 - j(314)}{5^2 + (314)^2}$$

$$= \frac{5 - j(314)}{25 + 98596}$$

$$= \frac{5 - j(314)}{98621}$$

$$= 0.14 - j(0.09)$$

$$Y = Y_1 + Y_2$$

$$= 0.14 - j(0.09) + j(314)C$$

$$= 0.14 + j(-0.09 + C(314))$$

At resonance, imaginary part = 0

$$-0.09 + C(314) = 0$$

$$314C = 0.09$$

$$C = \frac{0.09}{314}$$

$$= 2.86 \times 10^{-4}$$

$$= 286 \times 10^{-6}$$

$$= 286 \mu F$$

Current in branch 1,

$$I_1 = \frac{V}{Z_1} = \frac{230 \angle 0}{5 + j(314)}$$

$$= \frac{230 \angle 0}{5.9 \angle 32.12}$$

$$= 38.98 \angle -32.12$$

Current in branch 2, $I_2 = \frac{V}{Z_2} = \frac{V}{-j \times C}$

$$= \frac{230 \angle 0}{-j 11.10}$$

$$= 20.72 \angle 90$$

$$= 20.72 \angle 90$$

$$= 20.72 \angle 90$$

$$= 20.72 \angle 90$$

$$= 20.72 \angle 90$$

$$= 20.72 \angle 90$$

$$= 20.72 \angle 90$$

$$= 20.72 \angle 90$$

$$= 20.72 \angle 90$$

$$= 20.72 \angle 90$$

$$11.10 \angle 90^\circ$$

$$= 22.52 \angle 90^\circ$$

$$\text{Supply Current } (I) = I_1 + I_2$$

$$= 38.98 \angle 32.12 + 22.52 \angle 90^\circ$$

$$= 38.98 (\cos(-32.12) + j \sin(-32.12)) +$$

$$22.52 (\cos 90 + j \sin 90)$$

$$= 38.98 [0.84 + j(-0.53)] + 22.52(0 + j)$$

$$= 32.74 - j(20.65) + j(22.52)$$

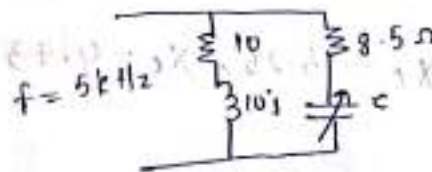
$$= 32.74 + j(22.52 - 20.65)$$

$$= 32.74 + j(1.87)$$

$$|Z_{eq}| = 32.79 \angle 3.26^\circ$$

2) An impedance of $Z_1 = 10 + 10j \Omega$ is connected with 11e1 to another impedance of resistances 8.5Ω & Variable Capacitance connected in series. find C such that the ckt is in resonance at 5 kHz .

A)



$$Y_1 = \frac{1}{10 + 10j}, \quad Y_2 = \frac{1}{8.5 - jX_C}$$

$$Y_1 = \frac{10 - j10}{10^2 + 10^2}, \quad Y_2 = \frac{1}{8.5 - jX_C}$$

$$Y = Y_1 + Y_2$$

$$= \frac{10 - j10}{200} + \frac{8.5 + jX_C}{(8.5)^2 + (X_C)^2}$$

$$= \frac{10}{200} - j \frac{10}{200} + \frac{8.5 + jX_C}{72.25 + X_C^2}$$

$$= \frac{10}{500} + \frac{8.5}{72.25 + X_c^2} + j \left(\frac{-10}{500} + \frac{X_c}{72.25 + X_c^2} \right)$$

$$= \frac{1}{50} + \frac{8.5}{72.25 + X_c^2} + j \left(\frac{-1}{50} + \frac{X_c}{72.25 + X_c^2} \right)$$

$$= 0.16 + j \left(-0.05 + \frac{X_c}{72.25 + X_c^2} \right)$$

At resonance, imaginary part = 0

$$-0.05 + \frac{X_c}{72.25 + X_c^2} = 0$$

$$\frac{X_c}{72.25 + X_c^2} = 0.05$$

$$X_c = (0.05)(72.25 + X_c^2)$$

$$X_c = 3.61 + 0.05 X_c^2$$

$$3.61 + 0.05 X_c^2 - X_c = 0$$

$$3.61 + X_c(0.05 X_c - 1) = 0$$

$$3.61 = -X_c(0.05 X_c - 1)$$

$$3.61 = X_c(1 - 0.05 X_c)$$

$$X_c = 15.26, X_c = 4.73$$

$$C_1 = X_c = \frac{1}{\omega C_1}$$

$$15.26 = \frac{1}{\omega C_1}$$

$$15.26 = \frac{1}{2\pi f C_1}$$

$$15.26 = \frac{1}{2 \times 3.14 \times 5 \times 10^3 C_1}$$

$$\Rightarrow 15.26 = \frac{3.18 \times 10^{-5}}{C_1}$$

$$15.26 = \frac{3.18 \times 10^{-5}}{C_1}$$

$$C_1 = \frac{3.18 \times 10^{-5}}{15.26}$$

$$C_1 = \frac{3.18 \times 10^{-5}}{15.26}$$

$$= 2.08 \times 10^{-6} = 2.08 \mu F$$

$$C_2 = X_c = \frac{1}{\omega C}$$

$$4.73 = \frac{1}{2\pi f C}$$

$$4.73 = \frac{1}{2 \times 3.14 \times 5 \times 10^3 \times C}$$

$$C_2 =$$

$$2 \times 3.14 \times 5000 \times 4.73$$

$$= 6.73 \times 10^{-6}$$

$$C_2 = 6.73 \mu F$$



Unit-2

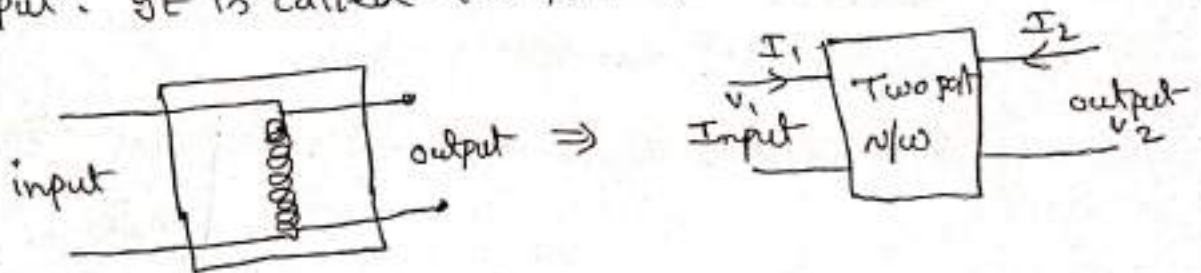
Two port Networks

Two port Parameters - Impedance - admittance - Transmission & hybrid parameters and their relation - concept of Transformed n/w - Two port n/w Parameters using transformed variables.

x

Two port n/w

It is the rectangular box represents a network, consists of two pairs of network [Four terminals] where one pair of terminals can be designated as input, the other pair being output. It is called two port n/w.



Types of two port parameters

- In Two port n/w, four variables are V_1, I_1, V_2, I_2
- out of four variables, two variables are dependent and other two variables are independent variables.
- These dependent & Independent variables are depends on the type of Parameter.

There are five types of Parameters

- ① Z-Parameters (∞) open ckt parameters
- ② Y-Parameters (∞) short ckt parameters

③ Hybrid or h-Parameters

④ Transmission or ABCD Parameters

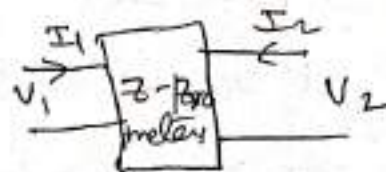
⑤ Inverse hybrid or Inverse-h parameters (or) g-Parameters

① z-Parameters (or) open ckt parameters:

z-Parameters are represented as

$$V_1 = z_{11}I_1 + z_{12}I_2 \quad (1)$$

$$V_2 = z_{21}I_1 + z_{22}I_2 \quad (2)$$



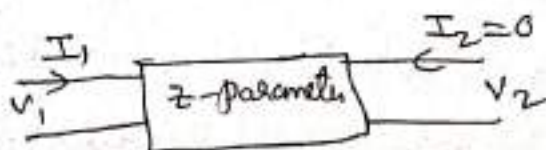
From the above equations

Dependent variables: V_1 & V_2

Independent variables: I_1 & I_2

From eq (1) & (2) z-Parameters are $\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$

Case 1: To find z_{11} & z_{21} by open ckt the output port i.e. $I_2 = 0$.



Put $I_2 = 0$ in eq (1) & (2)

From eq (1) $V_1 = z_{11}I_1$

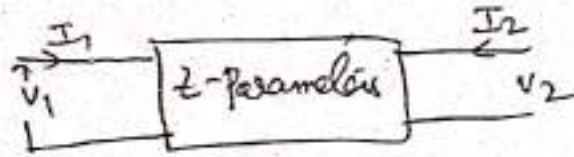
$$z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0} \rightarrow \text{Driving Point Input Impedance}$$

From eq (2)

$$V_2 = z_{21}I_1$$

$$z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0} \rightarrow \text{Forward transfer Impedance}$$

Case(2): To find z_{12} & z_{22} by open circuiting port 1
i.e $I_1 = 0$.



Put $I_1 = 0$ in eq (1) & (2)

From eq(1), $V_1 = z_{12} I_2$

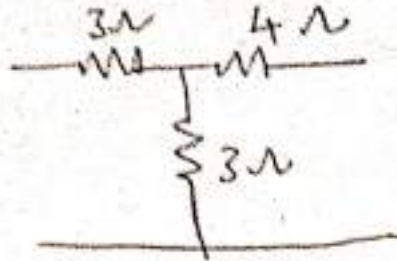
$$\boxed{z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0}} \rightarrow \text{Reverse transfer Impedance.}$$

From eq (2)

$$V_2 = z_{22} I_2$$

$$\boxed{z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0}} - \text{Driving point output Impedance.}$$

① Find z -parameters for the following ckt.



Sol

$$V_1 = z_{11} I_1 + z_{12} I_2 \quad \text{--- (1)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad - (2)$$

Case (1) To find z_{11} & z_{21} , put $I_2 = 0$ $\left[z_{11} = \frac{V_1}{I_1}, z_{21} = \frac{V_2}{I_1} \right]$

Apply KVL for loop 1

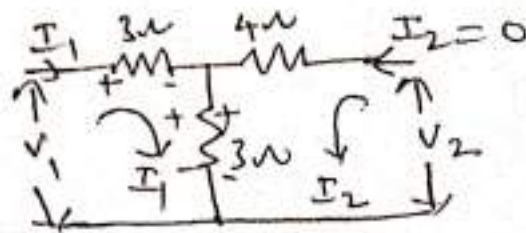
$$V_1 - 3I_1 - 3I_1 - 3I_2 = 0$$

put $I_2 = 0$

$$V_1 - 6I_1 = 0$$

$$V_1 = 6 I_1$$

$$\frac{V_1}{I_1} = 6 \Omega = Z_{11}$$



Apply KVL for loop 2

$$V_2 - 4I_2 - 3I_2 - 3I_1 = 0$$

put $I_2 = 0$

$$V_2 - 3I_1 = 0$$

$$V_2 = 3 \text{ I}_1$$

$$\boxed{\frac{v_2}{T} = z_{21} = 3} \sim$$

Case (2) : To obtain z_{12} & z_{22} , put $I_1 = 0$

$$z_{12} = \frac{V_1}{I_2}, \quad z_{22} = \frac{V_2}{I_2}$$

apply KVL for loop 1

$$V_1 - 3I_1 - 3I_1 - 3I_2 = 0$$

$$\text{put } I_1 = 0$$

$$V_1 - 3I_2 = 0$$

$$V_1 = 3I_2$$

$$\boxed{\frac{V_1}{I_2} = z_{12} = 3 \Omega}$$

apply KVL for loop 2

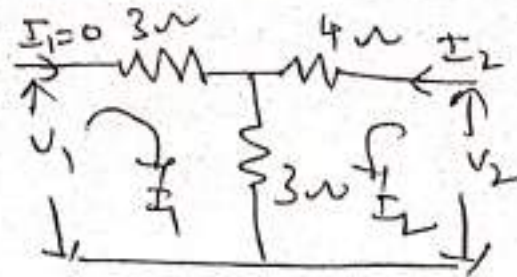
$$V_2 - 4I_2 - 3I_2 - 3I_1 = 0$$

$$\text{put } I_1 = 0$$

$$V_2 - 7I_2 = 0$$

$$V_2 = 7I_2$$

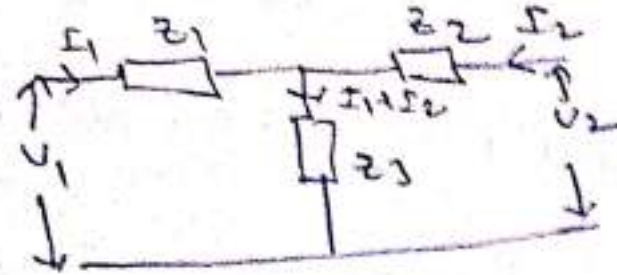
$$\boxed{\frac{V_2}{I_2} = z_{22} = 7 \Omega}$$



$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} = \begin{bmatrix} 6 & 3 \\ 3 & 7 \end{bmatrix}$$

Prob 1

① Find the z-parameter for the n/w



Sol

using KVL

$$V_1 = I_1 z_1 + (I_1 + I_2) z_3$$

$$V_1 = I_1 (z_1 + z_3) + I_2 z_3 \quad \text{--- (1)}$$

$$V_2 = I_2 z_2 + (I_1 + I_2) z_3$$

$$V_2 = I_1 z_3 + I_2 (z_2 + z_3) \quad \text{--- (2)}$$

z - parameters

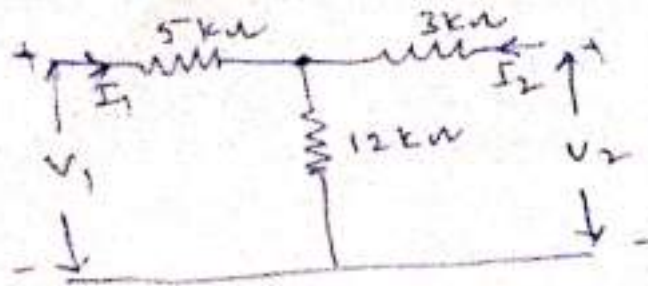
$$z_{11} = z_1 + z_3, \quad z_{12} = z_3$$

$$z_{21} = z_3, \quad z_{22} = z_2 + z_3$$

2

Find the z-parameter for the n/w shown in fig

3



Sol

Using KVL $\Rightarrow$

$$V_1 = 5 \times 10^3 I_1 + (I_1 + I_2) 12 \times 10^3$$

$$= 17 \times 10^3 I_1 + 12 \times 10^3 I_2 \quad \text{--- (1)}$$

$$V_2 = 3 \times 10^3 I_2 + 12 \times 10^3 (I_1 + I_2)$$

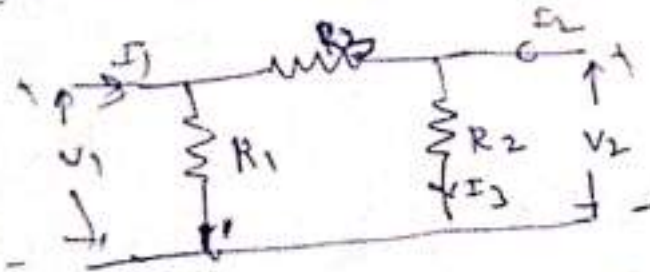
$$= 12 \times 10^3 I_1 + 15 \times 10^3 I_2 \quad \text{--- (2)}$$

Comparing two equations

$$z_{11} = 17 \times 10^3 \Omega, \quad z_{12} = 12 \times 10^3$$

$$z_{21} = 12 \times 10^3, \quad z_{22} = 15 \times 10^3$$

3 Determine the z-parameters of the π -type attenuator section.



$$V_1 = z_{11} I_1 + z_{12} I_2$$

$$V_2 = z_{21} I_1 + z_{22} I_2$$

Sol

Assume that output port is open circuited
i.e. $I_2 = 0$

$$z_{11} = \frac{V_1}{I_1} \Big|_{I_2 = 0}$$

$$= R_1 \parallel (R_2 + R_3)$$

$$= \frac{R_1 (R_2 + R_3)}{R_1 + R_2 + R_3} \quad \text{--- (1)}$$

However

$$I_3 = I_1 \times \frac{R_1}{R_1 + R_2 + R_3} \quad \text{--- (2)}$$

$$V_2 = I_3 R_2$$

$$= I_1 \frac{R_1 R_2}{R_1 + R_2 + R_3} \quad \text{--- (3)}$$

Again

$$z_{21} = \frac{V_2}{I_1} \Big|_{I_2 = 0}$$

$$= \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

now assume input open ckted, i.e. $I_1 = 0$

$$z_{22} = \frac{V_2}{I_2} \Big|_{I_1 = 0}$$

$$= R_2 \parallel (R_1 + R_3)$$

$$= \frac{R_2 (R_1 + R_3)}{R_1 + R_2 + R_3}$$

assume I_3' current through R_1 resistor

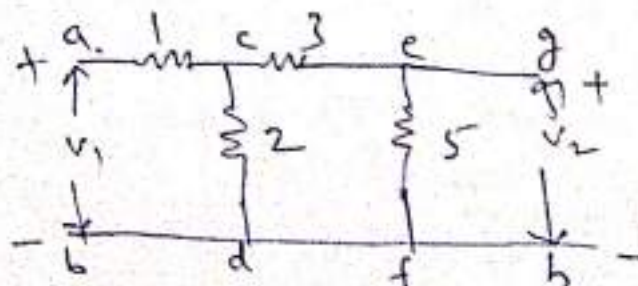
$$V_1 = I_3' R_1$$

where $I_3' = I_2 \times \frac{R_2}{R_1 + R_2 + R_3}$

$$V_1 = I_2 \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

$$z_{12} = \frac{V_1}{I_2} = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

(6) Find the z-parameters for the ckt shown in fig



Sol

(u)

Assuming o/p terminals open circuited i.e. $I_2 = 0$

$$\begin{aligned}
 R_{eq1} &= (3+5) \parallel 2 + 1 \\
 &= \frac{8 \times 2}{10} + 1 \\
 &= 1.6 + 1 \\
 &= 2.6 \Omega
 \end{aligned}$$

Let us assume input current I_1

$$V_1 = I_1 R_{eq1}$$

$$\begin{aligned}
 Z_{11} = \frac{V_1}{I_1} &= R_{eq1} \\
 &= 2.6 \Omega
 \end{aligned}$$

Let us assume that current through 5Ω resistor is I_x

$$\therefore I_x = I_1 \times \frac{2}{2+3+5} = \frac{I_1}{5} \text{ A.}$$

$$\begin{aligned}
 V_2 &= I_x \cdot 5 \\
 &= \frac{I_1}{5} \times 5 = I_1 \text{ volts.}
 \end{aligned}$$

$$\frac{V_2}{I_1} = Z_{21} = 1 \Omega.$$

Now assume i/p open circuited i.e. $I_1 = 0$, then

$$\begin{aligned}
 R_{eq2} &= (3+2) \parallel 5 \\
 &= 2.5 \Omega.
 \end{aligned}$$

$$V_2 = I_2 R_{eq2}$$

$$Z_{22} = \frac{V_2}{I_2} = R_{eq2} = 2.5 \Omega.$$

Let I_y be the current through 2Ω resistor

$$V_1 = I_y \cdot 2$$

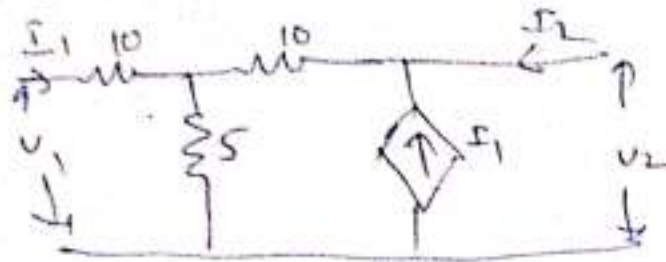
$$I_y = I_2 \times \frac{5}{2+3+5} = \frac{I_2}{2} \text{ A.}$$

$$\therefore V_1 = I_1 \times 2$$

$$= \frac{I_2}{2} \times 2$$

$$\frac{V_1}{I_2} = z_{12} = 1 \Omega$$

⑤ Determine the z-parameters



Sol First of all open circuit $I_2 = 0$, and applying vol source of V_1 at input port, the loop equations are given by

$$\text{Loop 1} = 10I_1 + 5(I_1 - I_2')$$

$$V_1 = (10 + 5)I_1 - 5I_2' \quad \text{--- (1)}$$

$$V_1 = 15I_1 - 5I_2' \quad \text{--- (1)}$$

but from I_2' loop

$$I_1 = -I_2' \quad \text{--- (2)}$$

Sub eq (2) in eq (1)

$$V_1 = 15I_1 + 5I_1$$

$$V_1 = 20I_1 \quad \text{--- (3)}$$

$$z_{11} = \frac{V_1}{I_1} = 20 \Omega$$

$$\therefore I_2' = -I_1 = -\frac{V_1}{20}$$

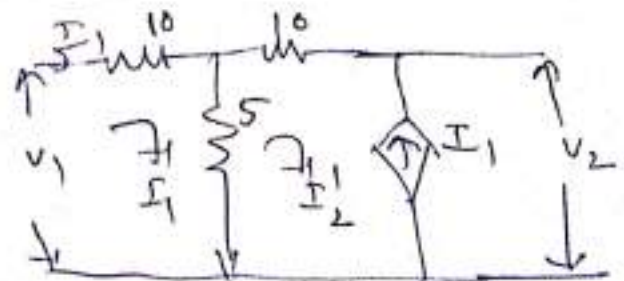
$$\text{Again } V_2 = (I_1 - I_2')5 - 10I_2'$$

$$= (I_1 + I_1)5 + 10I_1$$

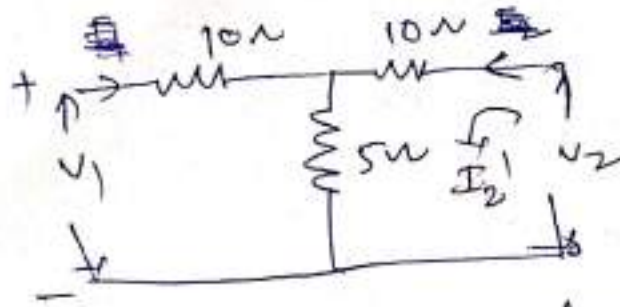
$$(I_2' = -I_1)$$

$$V_2 = 20I_1$$

$$z_{21} = \frac{V_2}{I_1} = 20 \Omega$$



next Input is opened $I_1 = 0$



here The current source becomes useless as with input open.

$$\text{here } I_2 \hat{=} I_2'$$

$$V_2 = I_2 (10 + 5) = 15 I_2$$

$$Z_{22} = \frac{V_2}{I_2} = 15 \Omega$$

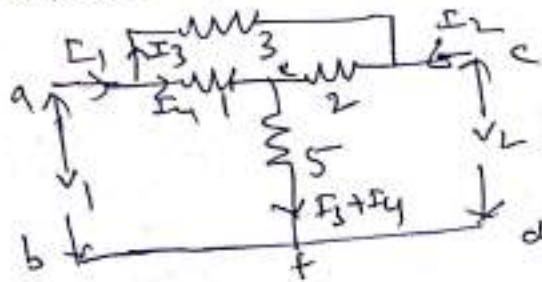
$$\text{also } V_1 = I_2' \times 5 = 5 I_2$$

$$Z_{12} = \frac{V_1}{I_2} = 5 \Omega$$

$$\therefore Z_{11} = 20 \Omega, Z_{12} = 5 \Omega$$

$$Z_{21} = 20 \Omega, Z_{22} = 15 \Omega$$

⑥ obtain the open circuit parameters and loop equations of the network



34 Let o/p open circuited i.e. $I_2 = 0$. Then

$$\begin{aligned} V_1 &= R_{eq} I_1 \\ &= ((3+2) \parallel 1 + 5) I_1 \\ &= \left(\frac{5 \times 1}{6} + 5 \right) I_1 = \frac{35}{6} \Omega (I_1) \end{aligned}$$

$$Z_{11} = \frac{V_1}{I_1} = \frac{35}{6} \Omega$$

$$V_1 = (I_3 + I_1) \times 5 \quad V_2 = I_3 \times 2 + I_1 \times 5$$

$$I_3 = I_1 \times \frac{1}{1+5}$$

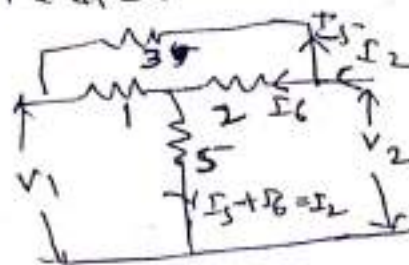
$$I_3 = I_1 \times \frac{1}{6}$$

$$\begin{aligned} V_2 &= \frac{I_1}{6} \times 2 + I_1 \times 5 \\ &= \frac{16}{3} I_1 \end{aligned}$$

$$Z_{21} = \frac{V_2}{I_1} = \frac{16}{3} \Omega$$

Similarly Input open circuited i.e. $I_1 = 0$

$$\begin{aligned} V_2 &= I_2 \cdot R_{eq} \\ &= I_2 (3 + 1) \parallel 2 + 5 \\ &= I_2 \left(\frac{4 \times 2}{6} + 5 \right) \\ &= I_2 \left(\frac{19}{3} \right) \end{aligned}$$



$$Z_{22} = \frac{V_2}{I_2} = \frac{19}{3} \Omega$$

$$\begin{aligned} V_1 &= I_5 \times 1 + I_2 \times 5 \\ &= I_2 \times \frac{2}{2+4} \times 1 + I_2 \times 5 \\ &= \frac{I_2}{3} + I_2 \times 5 \\ &= I_2 \frac{16}{3} \end{aligned}$$

$$Z_{12} = \frac{V_1}{I_2} = \frac{16}{3} \Omega$$

y-parameter:-

In this parameter the dependent variable are I_1 & I_2 and independent variable are V_1 & V_2 . then y-parameter are usually expressed as the following equation.



$$I_1 \text{ \& \& } I_2 \rightarrow \text{D}$$

$$V_1 \text{ \& \& } V_2 \rightarrow \text{I}$$

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad \text{--- (1)}$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad \text{--- (2)}$$

In matrix form

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \rightarrow (3)$$

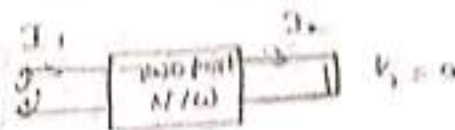
Case (i):- To find Y_{11} & Y_{12}

$$\text{Let } V_2 = 0 \quad \text{--- (4)}$$

$$\text{①} \Rightarrow I_1 = Y_{11}V_1 + Y_{12}(0)$$

$$Y_{11} = I_1 / V_1 \quad | V_2 = 0$$

→ short circuited driving input admittance.



$$\text{②} \Rightarrow I_2 = Y_{21}V_1 + 0$$

$$Y_{21} = I_2 / V_1 \quad | V_2 = 0$$

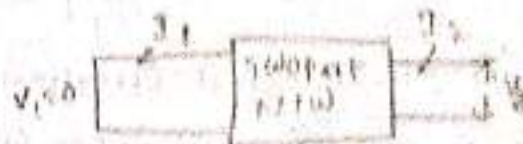
short circuited forward transfer admittance

Case (ii):- To find Y_{21} & Y_{22}

$$\text{Let } V_1 = 0 \quad \text{--- (5)}$$

$$\text{③} \Rightarrow I_1 = Y_{12}V_2$$

$$Y_{12} = I_1 / V_2 \quad | V_1 = 0$$



Short circuited reverse transform admittance.

$$\textcircled{2} \Rightarrow I_2 = Y_{21} V_1 + Y_{22} V_2$$

$$Y_{22} = \frac{I_2}{V_2}$$

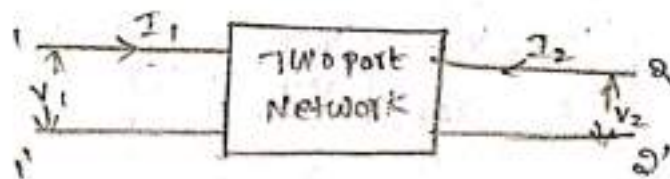
Short circuited driving output admittance.

H-parameters (Hybrid Parameters) :-

In this case the dependent variables are V_1 & I_2 and independent variables are I_1 & V_2 . Then the h-parameters are usually expressed by following two equations.

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

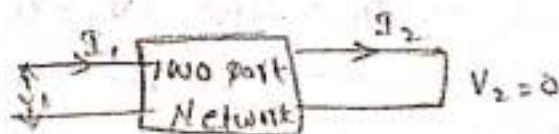


In Matrix form

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

Case (i) :-

$$V_2 = 0$$



$$\textcircled{1} \Rightarrow V_1 = h_{11} I_1 + 0$$

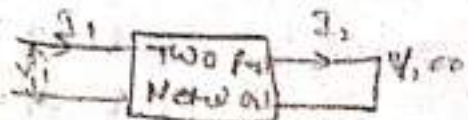
$$h_{11} = V_1 / I_1 \quad | \quad V_2 = 0$$

Short circuited driving input impedance.

$$\textcircled{2} \Rightarrow I_2 = h_{21} I_1$$

$$h_{21} = I_2 / I_1 \quad | \quad V_2 = 0$$

Short circuited forward transfer current gain.



Case (i):-

$$I_1 = 0$$

$$\textcircled{1} \Rightarrow V_1 = 0 + h_{12} V_2$$

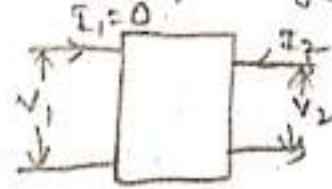
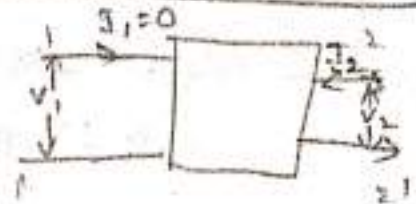
$$h_{12} = V_1 / V_2 \mid I_1 = 0$$

open circuited transfer voltage gain.

$$\textcircled{2} \Rightarrow I_2 = 0 + h_{22} V_2$$

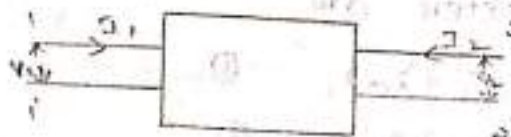
$$h_{22} = \frac{I_2}{V_2} \mid I_1 = 0 \text{ (mho)}$$

open circuited driving o/p admittance.



TRANSMISSION PARAMETERS (ABCD PARAMETERS):

In this the dependent variables are V_1 & I_1 and independent variables are V_2 & I_2 . Then ABCD Parameters are usually expressed by the following figure equations.



$$V_1 = AV_2 - BI_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}$$

Case (ii):-

$$I_2 = 0$$

$$\textcircled{1} \Rightarrow V_1 = AV_2$$

$$A = V_1 / V_2 \text{ open circuited reverse transfer voltage gain.}$$

$$\textcircled{2} \Rightarrow I_1 = CV_2$$

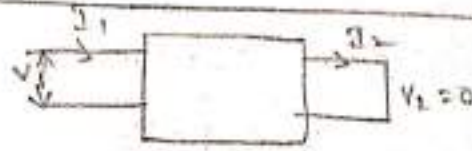
$$C = I_1 / V_2 \text{ (mho)}$$

open circuited reverse transfer admittance.



Case (ii):-

$$V_2 = 0$$



$$V_1 = -B I_2$$

$B = -V_1 / I_2$ short circuited reverse transfer impedance.

$$I_1 = -D I_2$$

$D = -I_1 / I_2$ short circuited reverse transfer current gain.

Relation b/w Parameters

Z-Parameters in terms of various Parameters:-

(A) Z-Parameters in terms of Y-Parameters:-

We know the

Z-Parameters are

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{--- (1)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{--- (2)}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

and the Y-parameters are

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \text{--- (3)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \text{--- (4)}$$

$$\begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} = \begin{pmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{pmatrix}^{-1} = \frac{1}{Y_{11}Y_{22} - Y_{12}Y_{21}} \begin{bmatrix} Y_{22} & -Y_{12} \\ -Y_{21} & Y_{11} \end{bmatrix}$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{Y_{22}}{\Delta Y} & \frac{-Y_{12}}{\Delta Y} \\ \frac{-Y_{21}}{\Delta Y} & \frac{Y_{11}}{\Delta Y} \end{bmatrix}$$

$$\text{So } Z_{11} = \frac{V_{11}}{\Delta Y}, \quad Z_{22} = \frac{V_{22}}{\Delta Y}$$

$$Z_{12} = \frac{-V_{12}}{\Delta Y}$$

$$Z_{21} = \frac{-V_{21}}{\Delta Y}$$

(b) Z-Parameters in terms of h-Parameters:-

We know the

Z-Parameters are

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \text{--- (1)}$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \text{--- (2)}$$

h-Parameters are

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \text{--- (3)}$$

$$I_2 = h_{21}I_1 + h_{22}V_2 \quad \text{--- (4)}$$

From eq (4)

$$h_{22}V_2 = I_2 - h_{21}I_1$$

$$V_2 = \frac{1}{h_{22}}I_2 - \frac{h_{21}}{h_{22}}I_1$$

$$V_2 = -\frac{h_{21}}{h_{22}}I_1 + \frac{1}{h_{22}}I_2 \quad \text{--- (5)}$$

Comparing (5) & (2)

$$Z_{21} = -\frac{h_{21}}{h_{22}}, \quad Z_{22} = \frac{1}{h_{22}}$$

Sub eq (5) in eq (3)

$$V_1 = h_{11}I_1 + h_{12} \left[-\frac{h_{21}}{h_{22}}I_1 + \frac{1}{h_{22}}I_2 \right]$$

$$V_1 = I_1 \left[h_{11} - \frac{h_{21}h_{12}}{h_{22}} \right] + \frac{h_{12}}{h_{22}}I_2$$

$$= I_1 \left[\frac{h_{11}h_{22} - h_{21}h_{12}}{h_{22}} \right] + \frac{h_{12}}{h_{22}}I_2$$

$$V_1 = I_1 \left(\frac{\Delta h}{h_{22}} \right) + \frac{h_{12}}{h_{22}}I_2 \quad \text{--- (6)}$$

compare (1) & (4)

$$Z_{11} = \frac{\Delta h}{h_{22}} \quad , \quad Z_{12} = \frac{h_{12}}{h_{22}}$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta h}{h_{22}} & \frac{h_{12}}{h_{22}} \\ \frac{-h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$$

(c) Z Parameters in terms of ABCD Parameters.

We know the

Z-Parameters are

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad - (1)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad - (2)$$

and the ABCD parameters are

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_1 = h_{21} V_2$$

$$V_1 = A V_2 - B I_2 \quad - (3)$$

$$I_1 = C V_2 - D I_2 \quad - (4)$$

from (4)

$$C V_2 = I_1 + D I_2$$

$$V_2 = \frac{1}{C} I_1 + D/C I_2 \quad - (5)$$

compare (2) & (5)

$$Z_{21} = 1/C \quad , \quad Z_{22} = D/C$$

sub eq(5) in eq(3)

$$V_1 = A \left[\frac{1}{C} I_1 + D/C I_2 \right] - B I_2$$

$$V_1 = A/C I_1 + [AD/C - B] I_2$$

$$V_1 = A/C I_1 + \left[\frac{AD - BC}{C} \right] I_2 \quad - (6)$$

compare (1) & (6)

$$Z_{11} = A/C \quad , \quad Z_{12} = \frac{AD - BC}{C} = \frac{\Delta T}{C}$$

$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} = \begin{bmatrix} \frac{V_1}{I_1} & \frac{\Delta I_1}{I_2} \\ \frac{V_2}{I_1} & \frac{\Delta I_2}{I_2} \end{bmatrix}$$

Y-parameters in terms of various parameters:-

Y-Parameters are

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \text{--- (2)}$$

Z-parameters are

$$V_1 = z_{11} I_1 + z_{12} I_2 \quad \text{--- (3)}$$

$$V_2 = z_{21} I_1 + z_{22} I_2 \quad \text{--- (4)}$$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}^{-1}$$

$$= \frac{1}{z_{11}z_{22} - z_{12}z_{21}} \begin{bmatrix} z_{22} & -z_{12} \\ -z_{21} & z_{11} \end{bmatrix}$$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{z_{22}}{\Delta Z} & \frac{-z_{12}}{\Delta Z} \\ \frac{-z_{21}}{\Delta Z} & \frac{z_{11}}{\Delta Z} \end{bmatrix}$$

Y-Parameters in terms of h-parameters:-

We know that h-parameters are

$$I_1 = h_{11} V_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} V_1 + h_{22} V_2 \quad \text{--- (2)}$$

h-parameters are

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (3)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (4)}$$

③ ⇒

$$h_{11} I_1 = V_1 - h_{12} V_2$$

$$I_1 = \frac{1}{h_{11}} V_1 - \frac{h_{12}}{h_{11}} V_2 \quad \text{--- (5)}$$

compare eq(1) & (3)

$$Y_{11} = \frac{1}{h_{11}}, \quad Y_{12} = -\frac{h_{12}}{h_{11}}$$

substitute eq(3) in eq(4)

$$I_2 = h_{21} \left[\frac{1}{h_{11}} V_1 - \frac{h_{12}}{h_{11}} V_2 \right] + h_{22} V_2$$

$$I_2 = \frac{h_{21}}{h_{11}} V_1 - \frac{h_{21}h_{12}}{h_{11}} V_2 + h_{22} V_2$$

$$I_2 = \frac{h_{21}}{h_{11}} V_1 + V_2 \left[\frac{h_{22}h_{11} - h_{21}h_{12}}{h_{11}} \right]$$

$$I_2 = \frac{h_{21}}{h_{11}} V_1 + V_2 \frac{\Delta h}{h_{11}} \quad \text{--- (5)}$$

compare (2) & (6)

$$Y_{21} = \frac{h_{21}}{h_{11}}, \quad Y_{22} = \frac{\Delta h}{h_{11}}$$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{h_{11}} & -\frac{h_{12}}{h_{11}} \\ \frac{h_{21}}{h_{11}} & \frac{\Delta h}{h_{11}} \end{bmatrix}$$

Y-parameters in terms of ABCD Parameters:-

We know that Y parameters are

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \text{--- (2)}$$

ABCD Parameters are

$$V_1 = AV_2 - BI_2 \quad \text{--- (3)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (4)}$$

$$\text{(3)} \Rightarrow BI_2 = AV_2 - V_1$$

$$I_2 = -\frac{1}{B} V_1 + \frac{A}{B} V_2 \quad \text{--- (5)}$$

complete ③ & ④

$$Y_{21} = \frac{1}{B}, \quad Y_{22} = Y_B$$

sub ③ in eq ④

$$I_1 = CV_2 - D \left[\frac{1}{B} V_1 + \frac{A}{B} V_2 \right]$$

$$I_1 = CV_2 + D/B V_1 - AD/B V_2$$

$$I_1 = D/B V_1 + \left[\frac{-AD}{B} + C \right] V_2$$

$$I_1 = D/B V_1 + \left[\frac{-AD + BC}{B} \right] V_2$$

$$I_1 = D/B V_1 - \left[\frac{AD - BC}{B} \right] V_2$$

$$I_1 = \frac{D}{B} V_1 - \frac{NT}{B} V_2 \quad \text{--- ⑥}$$

complete ⑦ & ⑧

$$Y_{11} = D/B, \quad Y_{12} = \frac{-NT}{B}$$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} D/B & \frac{-NT}{B} \\ 1/B & A/B \end{bmatrix}$$

h-Parameters to Various Parameters.

(A) h-Parameters to z-Parameters:-

We know h-Parameters are

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21}I_1 + h_{22}V_2 \quad \text{--- (2)}$$

The z-Parameters are

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \text{--- (3)}$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \text{--- (4)}$$

From (4)

$$Z_{22}I_2 = V_2 - Z_{21}I_1$$

$$I_2 = \frac{1}{Z_{22}}V_2 - \frac{Z_{21}}{Z_{22}}I_1 \quad \text{--- (5)}$$

Compare (2) & (5)

$$h_{21} = -\frac{Z_{21}}{Z_{22}}, \quad h_{22} = \frac{1}{Z_{22}}$$

Sub eqn (5) in (3)

$$V_1 = Z_{11}I_1 + Z_{12}\left[\frac{1}{Z_{22}}V_2 - \frac{Z_{21}}{Z_{22}}I_1\right]$$

$$V_1 = I_1\left[Z_{11} - \frac{Z_{21}}{Z_{22}} \cdot Z_{12}\right] + \frac{Z_{12}}{Z_{22}}V_2$$

$$V_1 = I_1\left[\frac{Z_{11}Z_{22} - Z_{21}Z_{12}}{Z_{22}}\right] + \frac{Z_{12}}{Z_{22}}V_2$$

$$V_1 = I_1 \frac{\Delta Z}{Z_{22}} + \frac{Z_{12}}{Z_{22}}V_2 \quad \text{--- (6)} \quad (\Delta Z = Z_{11}Z_{22} - Z_{21}Z_{12})$$

Compare (6) & (7)

$$h_{11} = \frac{\Delta Z}{Z_{22}}, \quad h_{12} = \frac{Z_{12}}{Z_{22}}$$

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ \frac{-Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$$

h-parameters in terms of Y-parameters:

We know that h-parameters are

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad (1)$$

$$I_2 = h_{21}I_1 + h_{22}V_2 \quad (2)$$

The Y-parameters are

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad (3)$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad (4)$$

$$(3) \Rightarrow V_1 Y_{11} = I_1 - Y_{12}V_2$$

$$V_1 = \frac{1}{Y_{11}} I_1 - \frac{Y_{12}}{Y_{11}} V_2 \quad (5)$$

compare eq (5) & eq (1)

$$h_{11} = 1/Y_{11} \quad ; \quad h_{12} = -\frac{Y_{12}}{Y_{11}}$$

eq (5) in eq (4)

$$I_2 = Y_{21} \left(\frac{1}{Y_{11}} I_1 - \frac{Y_{12}}{Y_{11}} V_2 \right) + Y_{22} V_2$$

$$= \frac{Y_{21}}{Y_{11}} I_1 + V_2 \left[Y_{22} - \frac{Y_{12}Y_{21}}{Y_{11}} \right]$$

$$= \frac{Y_{21}}{Y_{11}} I_1 + V_2 \left[\frac{Y_{22}Y_{11} - Y_{12}Y_{21}}{Y_{11}} \right]$$

$$I_2 = \frac{Y_{21}}{Y_{11}} I_1 + V_2 \left[\frac{\Delta Y}{Y_{11}} \right] \quad (6)$$

compare (2) & (6)

$$h_{21} = \frac{Y_{21}}{Y_{11}} \quad ; \quad h_{22} = \frac{\Delta Y}{Y_{11}}$$

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{V_1}{I_1} & \frac{V_1}{I_2} \\ \frac{I_2}{I_1} & \frac{I_2}{I_2} \end{bmatrix}$$

h-parameters in terms of ABCD parameters:-

We know that

h-parameters are

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

ABCD parameters are

$$V_1 = A V_2 - B I_2 \quad \text{--- (3)}$$

$$I_1 = C V_2 - D I_2 \quad \text{--- (4)}$$

$$\textcircled{4} \Rightarrow D I_2 = C V_2 - I_1$$

$$I_2 = \frac{C}{D} V_2 - \frac{1}{D} I_1 \quad \text{--- (5)}$$

compare (5) & (2)

$$h_{21} = -\frac{1}{D}, \quad h_{22} = \frac{C}{D}$$

sub. eq (5) in eq (3)

$$V_1 = A V_2 - \left[\frac{C}{D} V_2 - \frac{1}{D} I_1 \right] B$$

$$V_1 = A V_2 - \left[\frac{BC}{D} V_2 - \frac{B}{D} I_1 \right]$$

$$V_1 = \left[A - \frac{BC}{D} \right] V_2 + \frac{B}{D} I_1$$

$$V_1 = \frac{B}{D} I_1 + \left[\frac{AD - BC}{D} \right] V_2$$

$$V_1 = \frac{B}{D} I_1 + \frac{1}{D} V_2 \quad \text{--- (6)}$$

compare (1) & (6)

$$h_{11} = \frac{B}{D}, \quad h_{12} = \frac{1}{D}$$

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} B/D & D/D \\ -1/D & C/D \end{bmatrix}$$

We know that ABCD parameters are

$$V_1 = AV_2 - B I_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - D I_2 \quad \text{--- (2)}$$

Z-parameters are

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{--- (3)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{--- (4)}$$

$$\textcircled{4} \Rightarrow Z_{21} I_1 = V_2 - Z_{22} I_2$$

$$I_1 = \frac{1}{Z_{21}} V_2 - \frac{Z_{22}}{Z_{21}} I_2 \quad \text{--- (5)}$$

compare (5) & (2)

$$C = \frac{1}{Z_{21}}, \quad D = \frac{Z_{22}}{Z_{21}}$$

eq (5) in eq (3)

$$V_1 = Z_{11} \left[\frac{1}{Z_{21}} V_2 - \frac{Z_{22}}{Z_{21}} I_2 \right] + Z_{12} I_2$$

$$V_1 = \frac{Z_{11} V_2}{Z_{21}} - \frac{Z_{11} Z_{22}}{Z_{21}} I_2 + Z_{12} I_2$$

$$V_1 = \frac{Z_{11}}{Z_{21}} V_2 - I_2 \left[\frac{Z_{11} Z_{22}}{Z_{21}} - Z_{12} \right]$$

$$V_1 = \frac{Z_{11}}{Z_{21}} V_2 - I_2 \cdot \frac{\Delta Z}{Z_{21}} \quad \text{--- (6)}$$

$$\textcircled{6} \& \textcircled{7} \quad A = \frac{Z_{11}}{Z_{21}}, \quad B = \frac{\Delta Z}{Z_{21}}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$$

We know that ABCD Parameters are

$$V_1 = AV_2 - B I_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - D I_2 \quad \text{--- (2)}$$

Y-Parameters are

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \text{--- (3)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \text{--- (4)}$$

$$\text{(4)} \Rightarrow Y_{21} V_1 = I_2 - Y_{22} V_2$$

$$V_1 = \frac{1}{Y_{21}} I_2 - \frac{Y_{22}}{Y_{21}} V_2 \quad \text{--- (5)}$$

Compare (1) & (5)

$$A = \frac{-Y_{22}}{Y_{21}}, \quad B = -\frac{1}{Y_{21}}$$

Sub. eq (5) in (3)

$$I_1 = Y_{11} \left[\frac{1}{Y_{21}} I_2 - \frac{Y_{22}}{Y_{21}} V_2 \right] + Y_{12} V_2$$

$$I_1 = \left[\frac{Y_{12} - Y_{22} Y_{11}}{Y_{21}} \right] V_2 + \frac{Y_{11}}{Y_{21}} I_2$$

$$I_1 = \left[\frac{Y_{21} Y_{12} - Y_{22} Y_{11}}{Y_{21}} \right] V_2 + \frac{Y_{11}}{Y_{21}} I_2$$

$$I_1 = \frac{-\Delta}{Y_{21}} V_2 + \frac{Y_{11}}{Y_{21}} I_2 \quad \text{--- (6)}$$

$$\text{(6) \& (2)} \quad C = \frac{-\Delta}{Y_{21}}, \quad D = \frac{-Y_{11}}{Y_{21}}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{-y_{22}}{y_{21}} & -\frac{1}{y_{21}} \\ \frac{-\Delta y}{y_{21}} & \frac{-y_{11}}{y_{21}} \end{bmatrix}$$

We know that ABCD parameters are

$$V_1 = AV_2 - BI_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

h-Parameters are

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \text{--- (3)}$$

$$I_2 = h_{21}I_1 + h_{22}V_2 \quad \text{--- (4)}$$

$$\textcircled{4} \Rightarrow h_{21}I_1 = I_2 - h_{22}V_2$$

$$I_1 = -\frac{h_{22}}{h_{21}}V_2 + \frac{1}{h_{21}}I_2 \quad \text{--- (5)}$$

compare (5) & (2)

$$C = -\frac{h_{22}}{h_{21}}, \quad D = \frac{1}{h_{21}}$$

sub. eq (5) in eq (3)

$$V_1 = h_{11} \left[-\frac{h_{22}}{h_{21}}V_2 + \frac{1}{h_{21}}I_2 \right] + h_{12}V_2$$

$$V_1 = V_2 \left[-\frac{h_{11}h_{22}}{h_{21}} + h_{12} \right] + \frac{h_{11}}{h_{21}}I_2$$

$$V_1 = V_2 \left[\frac{h_{22}h_{11} - h_{12}h_{21}}{h_{21}} \right] + \frac{h_{11}}{h_{21}}I_2 \quad \text{--- (6)}$$

compare (1) & (6)

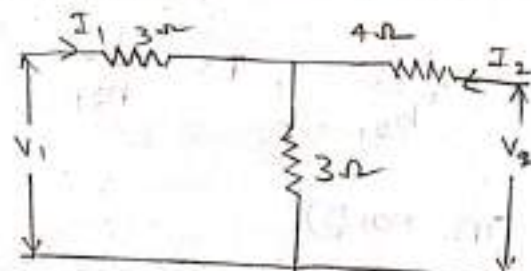
$$A = \frac{-\Delta h}{h_{21}}, \quad B = \frac{h_{11}}{h_{12}}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} -\frac{\Delta h}{h_{21}} & -\frac{h_{11}}{h_{21}} \\ -\frac{h_{22}}{h_{21}} & -\frac{1}{h_{21}} \end{bmatrix}$$

Parameters	condition for symmetry	condition for reciprocal.
Z	$z_{11} = z_{22}$	$z_{12} = z_{21}$
Y	$y_{11} = y_{22}$	$y_{12} = y_{21}$
h	$\Delta h = 1$	$h_{12} = -h_{21}$
ABCD	$A = D$	$AD - BC = 1$

PROBLEMS ON parameters:-

① Find Z & Y parameters in the following circuit.



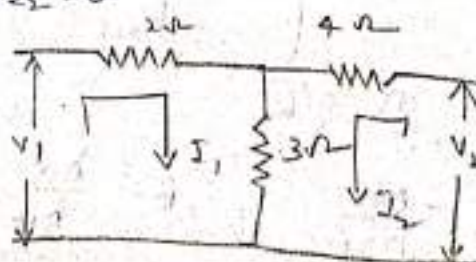
Z parameters are

$$V_1 = z_{11} I_1 + z_{12} I_2$$

$$V_2 = z_{21} I_1 + z_{22} I_2$$

Case:- To find z_{11} & z_{21} by putting $I_2 = 0$

Let $I_2 = 0$



$$\text{then } V_1 = z_{11} I_1$$

$$z_{11} = \frac{V_1}{I_1}$$

$$V_2 = z_{21} I_1$$

$$z_{21} = \frac{V_2}{I_1}$$

$$V_1 - 2I_1 - 3I_1 = 0$$

$$V_1 = 5I_1$$

$$\frac{V_1}{I_1} = 5$$

$$Z_{11} = 5$$

Apply KVL for loop 2,

$$V_2 - 4I_2 - 3I_2 - 3I_1 = 0$$

$$V_2 = 3I_1$$

$$\frac{V_2}{I_1} = 3$$

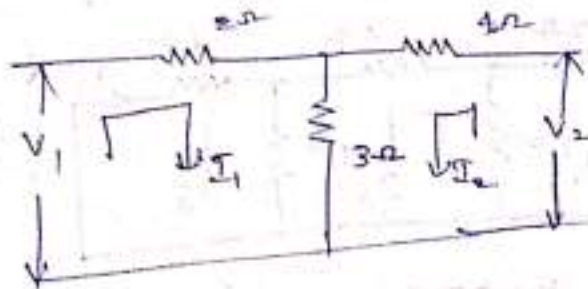
$$Z_{21} = 3$$

Case 2:- To find Z_{12} & Z_{22} , by putting $I_1 = 0$

$$I_1 = 0$$

$$V_1 = Z_{12} I_2 \quad , \quad V_2 = Z_{22} I_2$$

$$Z_{12} = \frac{V_1}{I_2} \quad , \quad Z_{22} = \frac{V_2}{I_2}$$



Apply KVL for loop 1,

$$V_1 - 2I_1 - 3I_1 - 3I_2 = 0$$

$$V_1 = 3I_2$$

$$\frac{V_1}{I_2} = Z_{12} = 3\Omega$$

Apply KVL for loop 2,

$$V_2 - 4I_2 - 3I_2 - 3I_1 = 0$$

$$V_2 = 7I_2 \Rightarrow \frac{V_2}{I_2} = Z_{22} = 7\Omega$$

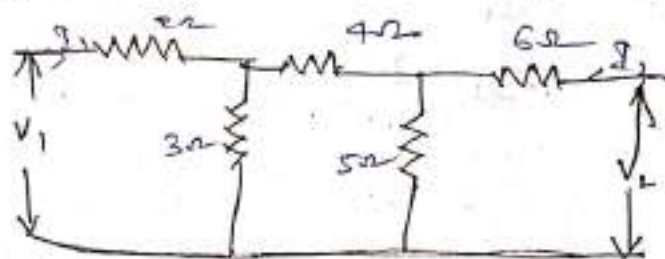
Y-Parameters

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{Z_{22}}{\Delta Z} & \frac{-Z_{12}}{\Delta Z} \\ \frac{-Z_{21}}{\Delta Z} & \frac{Z_{11}}{\Delta Z} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{7}{26} & \frac{-3}{26} \\ \frac{-3}{26} & \frac{5}{26} \end{bmatrix}$$

②

Find Y parameters for the following circ.

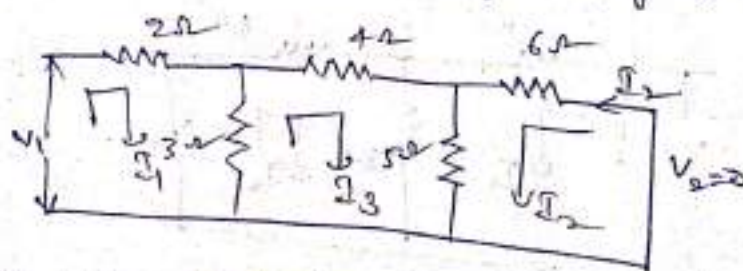


Case 1:-

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad \text{--- (1)}$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad \text{--- (2)}$$

Case 2:- To find Y_{11} & Y_{21} by putting $V_2 = 0$



Apply KVL for loop 1,

$$V_1 - 2I_1 - 3I_2 + 3I_3 = 0$$

$$V_1 = 2I_1 - 3I_3 \quad \text{--- (3)}$$

Apply KVL for loop 3

$$-4I_3 - 5I_3 - 5I_2 - 3I_3 + 3I_1 = 0$$

$$3I_1 - 5I_2 - 12I_3 = 0 \quad \text{--- (4)}$$

Apply KVL for loop 2

$$-6I_2 - 5I_2 - 5I_3 = 0$$

$$11I_2 + 5I_3 = 0$$

$$11I_2 = -5I_3 \quad \text{--- (5)}$$

$$V_2 = 0 \Rightarrow$$

$$I_1 = Y_{11} V_1$$

$$I_2 = Y_{21} V_1$$

$$Y_{11} = I_1 / V_1$$

$$Y_{21} = I_2 / V_1$$

$$\text{from (5)} \Rightarrow I_2 = -5/11 I_3 \quad \text{--- (6)}$$

$$\text{(4)} \Rightarrow 3I_1 - 5(-5/11)I_3 - 12I_3 = 0$$

$$3I_1 + 25/11 I_3 - 12I_3 = 0$$

$$3I_1 + 2.27I_3 - 12I_3 = 0$$

$$3I_1 - 9.73I_3 = 0 \quad \text{--- (7)}$$

$$3I_1 = 9.73I_3$$

$$I_3 = \frac{3I_1}{9.73}$$

$$I_3 = 0.308 I_1 \quad \text{--- (8)}$$

eq (7) in eq (3)

$$V_1 = 5I_1 - 3(0.308)I_1$$

$$V_1 = 5I_1 - 0.924 I_1$$

$$V_1 = 4.076 I_1 \Rightarrow \frac{I_1}{V_1} = 0.245 \text{ mho} = Y_{11}$$

V_1 in eq (3)

$$4.076 I_1 = 5I_1 - I_3$$

$$0.924 I_1 - I_3 = 0 \quad \text{--- (9)}$$

$$\text{(9)} \Rightarrow I_1 = \frac{1}{0.924} I_3$$

$$I_1 = 1.081 I_3 \quad \text{--- (10)}$$

eqn (1) sub in eqn (2)

$$V_1 = 5(3.26 I_3) - 3 I_3$$

$$V_1 = 13.3 I_3$$

from (1) $I_2 = -0.45 I_3$

$$I_3 = -\frac{1}{0.45} I_2$$

$$I_3 = -2.22 I_2 \quad \text{--- (7)}$$

Sub eqn (7) in $V_1 = 13.3 I_3$

$$V_1 = 13.3 (-2.22) I_2$$

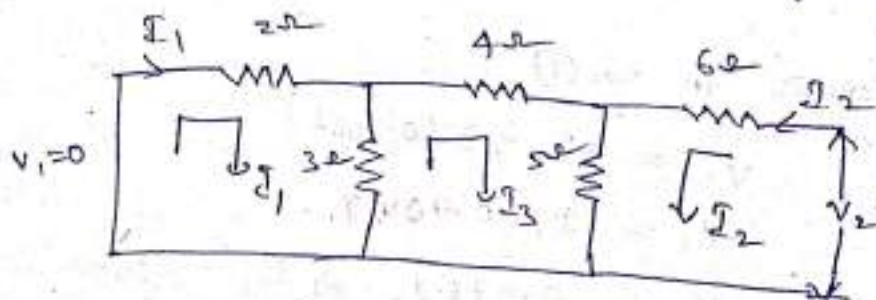
$$V_1 = -29.04 I_2$$

$$Y_{21} = \frac{V_1}{I_2} =$$

$$Y_{21} = \frac{I_2}{V_1} = \frac{-1}{29.04} = -0.03.$$

Case (iii):

To find Y_{12} & Y_{22} by putting $V_1 = 0$



Apply KVL for loop 1,

$$-2 I_1 - 3 I_1 + 3 I_3 = 0$$

$$3 I_3 - 5 I_1 = 0 \quad \text{--- (8)}$$

Apply KVL for loop 2,

$$V_2 - 6 I_2 - 5 I_2 - 5 I_3 = 0$$

$$V_2 - 11 I_2 - 5 I_3 = 0 \quad \text{--- (9)}$$

$$V_2 = 11 I_2 + 5 I_3$$

If $V_1 = 0$

$$I_1 = Y_{12} V_2$$

$$Y_{12} = \frac{I_1}{V_2}$$

$$I_2 = Y_{22} V_2$$

$$Y_{22} = \frac{I_2}{V_2}$$

Apply KVL for loop ③,

$$-4I_3 - 5I_3 - 5I_2 - 3I_3 + 3I_1 = 0$$

$$3I_1 - 5I_2 - 12I_3 = 0 \quad \text{--- (3)}$$

$$\textcircled{1} \Rightarrow 3I_2 = 5I_1$$

$$I_2 = \frac{5}{3}I_1$$

$$I_3 = \frac{5}{3}I_1 \quad \text{--- (4)}$$

$$\textcircled{3} \Rightarrow 3I_1 - 5I_2 - 12\left(\frac{5}{3}\right)I_1 = 0$$

$$3I_1 - 5I_2 - 20I_1 = 0$$

$$3I_1 - 5I_2 - 20I_1 = 0$$

$$-17I_1 = 5I_2$$

$$I_1 = -\frac{5}{17}I_2 \quad \text{--- (5)}$$

$$I_2 = -\frac{17}{5}I_1 \quad \text{--- (6)}$$

④ & ⑥ write in ②

$$V_2 = 11\left(-\frac{17}{5}\right)I_1 + 5\left(\frac{5}{3}\right)I_1$$

$$V_2 = -37.4I_1 + 8.33I_1$$

$$V_2 = -29.07I_1$$

$$\frac{I_1}{V_2} = \frac{1}{29.07}$$

$$\frac{I_1}{V_2} = 0.034 = Y_{12}$$

$$Y_{12} = 0.034$$

$$\textcircled{5} \Rightarrow I_1 = -\frac{5}{17}I_2 \quad \text{--- (6)}$$

eq ⑥ write in eq ④

$$I_3 = \frac{5}{3}\left(-\frac{5}{17}\right)I_2$$

$$I_3 = -0.49I_2$$

$$\text{eq (2)} \Rightarrow V_2 = 11I_2 + 5(-0.49)I_2$$

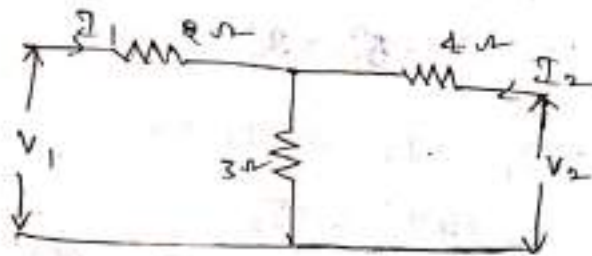
$$V_2 = 11I_2 - 2.45I_2$$

$$V_2 = 8.55I_2$$

$$\frac{I_2}{V_2} = \frac{1}{8.55}$$

$$y_{22} = \frac{I_2}{V_2} = 0.11$$

Find the Z-Parameters for the following circuit

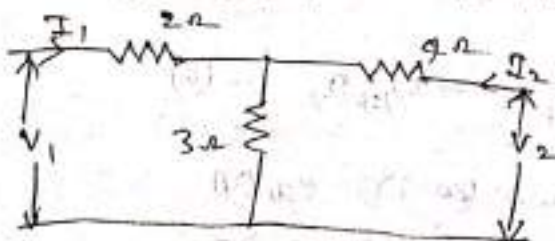


$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 2+3 & 3 \\ 3 & 4+3 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 3 \\ 3 & 7 \end{bmatrix}$$

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} 1.5 & 3 \\ 3 & 7 \end{bmatrix}^{-1}$$

Find the ABCD parameters for the following circuit.



ABCD parameters are

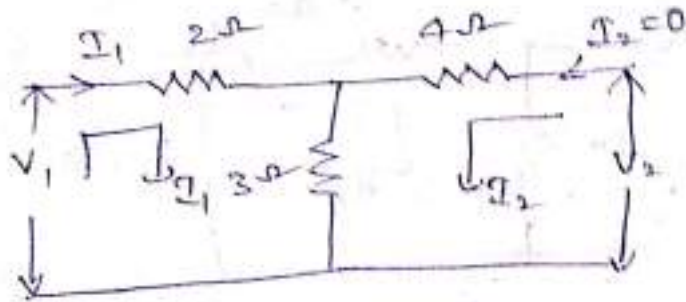
$$V_1 = AV_2 - BI_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

Case(1): To find A & C , put $I_2 = 0$

then eq ① $\Rightarrow V_1 = AV_2$
 $A = \frac{V_1}{V_2}$

eq ② $\Rightarrow I_1 = CV_2$
 $C = \frac{I_1}{V_2}$



Apply KVL for loop ①

$$V_1 - 2I_1 - 3I_1 - 3I_2 = 0$$

$$V_1 - 5I_1 = 0$$

$$V_1 = 5I_1 \quad \text{--- ③}$$

Apply KVL for loop ②

$$V_2 - 4I_2 - 3I_2 - 3I_1 = 0$$

$$V_2 + 0 - 3I_1 = 0$$

$$V_2 = 3I_1 \quad \text{--- ④}$$

$$\frac{③}{④} \Rightarrow \frac{V_1}{V_2} = \frac{5I_1}{3I_1}$$

$$A = \frac{V_1}{V_2} = 1.67$$

from ④ $\Rightarrow \frac{I_1}{V_2} = \frac{1}{3} = 0.33$

$$C = 0.33$$

1 F

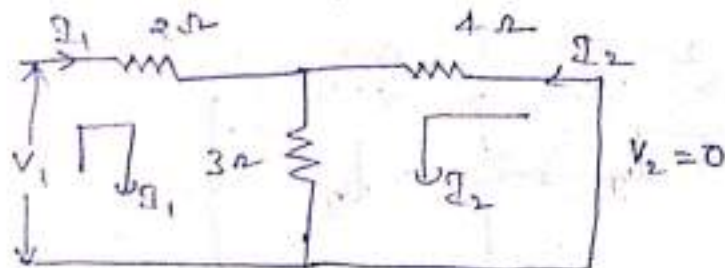
Case (iii) - To find B & D, by putting $V_2 = 0$.

$$\textcircled{1} \Rightarrow V_1 = -B I_2$$

$$B = -\frac{V_1}{I_2}$$

$$\textcircled{2} \Rightarrow I_1 = -D I_2$$

$$D = -\frac{I_1}{I_2}$$



Apply KVL for loop 1,

$$V_1 - 2I_1 - 3I_1 - 3I_2 = 0$$

$$V_1 - 5I_1 - 3I_2 = 0$$

$$V_1 = 5I_1 + 3I_2 \quad \text{--- (3)}$$

Apply KVL for loop 2,

$$-4I_2 - 3I_2 - 3I_1 = 0$$

$$-7I_2 - 3I_1 = 0$$

$$3I_1 + 7I_2 = 0 \quad \text{--- (4)}$$

$$3I_1 = -7I_2$$

$$\frac{-I_1}{I_2} = \frac{7}{3}$$

$$\frac{-I_1}{I_2} = 2.33$$

$$D = 2.33$$

$$I_1 = -\frac{7}{3}I_2$$

$$\textcircled{2} \Rightarrow I_2 = -\frac{3}{7}I_1$$

$$\textcircled{3} \Rightarrow V_1 = 5I_1 + 3\left(-\frac{7}{3}\right)I_2$$

$$V_1 = 5I_1 - 2I_2$$

$$= 5I_1 - 3I_2$$

$$V_1 = 3.72I_1$$

$$V_1 = 2I_2$$

$$-\frac{V_1}{I_2} = -2$$

$$B = -2$$

$$\textcircled{3} \Rightarrow V_1 = 5(-7/3)I_2 + 3I_2$$

$$= -\frac{35}{3}I_2 + 3I_2$$

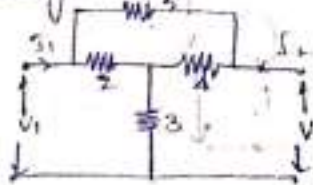
$$= -11.66I_2 + 3I_2$$

$$V_1 = -8.6I_2$$

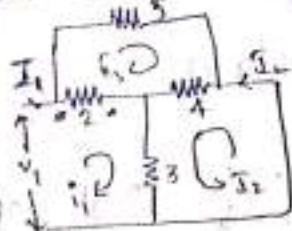
$$-\frac{V_1}{I_2} = +8.6$$

$$B = 8.6$$

1. Calculate V_1 parameter for the following circuit



Case:- Short circuit port ② i.e. $V_2 = 0$



Loop ①

$$V_1 = 5(I_1 - I_3) + 3(I_1 + I_2)$$

$$5I_1 + 3I_2 - 2I_3 = V_1 \rightarrow (1)$$

Loop ②

$$4(I_2 - I_3) + 3(I_3 + I_1) = 0$$

$$3I_1 + 4I_2 + I_3 = 0 \rightarrow (2)$$

$$3I_1 - 4I_2 - I_3 = 0 \rightarrow (3)$$

Loop ③

$$2(I_3 - I_1) + 5I_3 + 4(I_3 + I_2) = 0$$

$$2I_1 - 4I_2 - 11I_3 = 0 \rightarrow (4)$$

from eq (2)

$$I_2 = \frac{3}{4}I_1 - \frac{1}{4}I_3$$

$$I_2 = 0.75I_1 - 0.25I_3 \rightarrow (5)$$

from eq (3)

$$I_3 = -3/4I_1 - 7/4I_2$$

$$I_3 = -0.75I_1 - 1.75I_2 \rightarrow (6)$$

put eq (5) in (4)

$$I_2 = 0.75I_1 - 0.25(-0.75I_1 - 1.75I_2)$$

$$I_2 = 0.75I_1 + 0.0625I_1 + 0.4375I_2$$

$$-3.8125I_2 = 0.8125I_1$$

$$I_2 = -0.672I_1 \rightarrow (7)$$

put eq (6) & (7) in eq (1) we get

$$5I_1 + 3(-0.672I_1) - 2(-0.75I_1 - 1.75I_2) = V_1$$

$$5I_1 - 0.016I_1 + 1.5I_1 + 3.5I_2 = V_1$$

$$4.484I_1 + 3.5(-0.672I_1) = V_1$$

$$2.132I_1 = V_1$$

$$\frac{I_1}{V_1} = \frac{1}{2.132} = 0.469 \text{ V} = V_1$$

Ans

$$I_1 = -2.33I_2 - 1.33I_3 \rightarrow (8)$$

put eq (8) in eq (1) we get

$$5I_1 + 3I_2 - 2(-0.75I_1 - 1.75I_2) = V_1$$

$$5I_1 + 3I_2 + 1.5I_1 + 3.5I_2 = V_1$$

$$6.5I_1 + 7.5I_2 = V_1 \rightarrow (9)$$

from eq (8)

$$I_3 = \frac{2.33}{11}I_1 - \frac{4}{11}I_2$$

$$I_3 = 0.181I_1 - 0.363I_2 \rightarrow (10)$$

Put eq (10) in (7) we get

$$I_1 = -2.33I_2 - 1.33(0.181I_1 - 0.363I_2)$$

$$= -2.33I_2 - 0.241I_1 + 0.483I_2$$

$$I_1 = -0.241I_1 - 1.847I_2$$

$$1.241I_1 = -1.847I_2 \Rightarrow I_1 = -1.488I_2$$

eq (10) $\rightarrow$

$$6.5(-1.488I_2) + 7.5I_2 = V_1$$

$$-9.672I_2 + 7.5I_2 = V_1$$

$$-2.172I_2 = V_1$$

$$\frac{I_2}{V_1} = \frac{1}{-2.172} = -0.46$$

$$V_{21} = -0.913 \text{ V}$$

Case: 2 Short circuit loop ② i.e. $V_1 = 0$



Apply KVL to loop ①

$$2(I_1 - I_3) + 3(I_1 + I_2) = 0$$

$$5I_1 + 3I_2 - 2I_3 = 0 \rightarrow (1)$$

$$V_2 = 4(I_2 + I_3) + 3(I_1 + I_1)$$

$$8I_1 + 7I_2 + 4I_3 = V_2 \rightarrow (2)$$

$$2(I_3 - I_1) + 5I_3 + 4(I_3 + I_2) = 0$$

$$-2I_1 + 4I_2 + 11I_3 = 0 \rightarrow (3)$$

$$I_1 = 2I_2 + 5.5I_3 \rightarrow (4)$$

$$I_3 = 2.5I_1 + 1.5I_2 \rightarrow (5)$$

$$I_1 = 2I_2 + 5.5(2.5I_1 + 1.5I_2)$$

$$I_1 = 13.75I_1 + 10.25I_2 \Rightarrow I_1 = -0.804I_2$$

$$3\vec{i}_1 + 7\vec{i}_2 + 4[3.5\vec{i}_1 + 1.5\vec{i}_2] = V_2$$

$$12\vec{i}_1 + 13\vec{i}_2 = V_2$$

$$13(0.804\vec{i}_2) + 13\vec{i}_2 = V_2$$

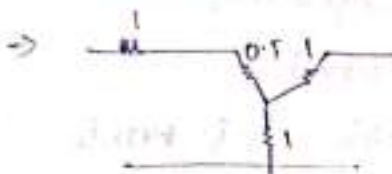
$$(0.492 + 13)\vec{i}_2 = V_2$$

$$\vec{i}_2/V_2 = \frac{1}{13.492}$$

$$\boxed{Y_{22} = 0.42 \text{ S}}$$

$$Y_{12} = -0.315 \text{ S} \quad \therefore -0.315 \text{ S}$$

2] find y-parameters



$$\begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} = \begin{pmatrix} 2.5 & 1 \\ 1 & 1 \end{pmatrix}$$

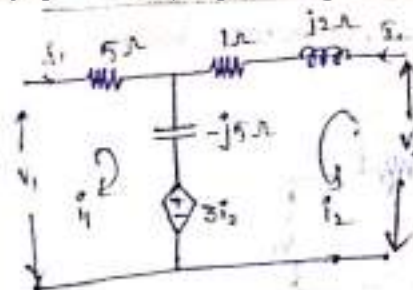
$$Y_{11} = \frac{Z_{22}}{\Delta} = \frac{1}{4} = 0.25$$

$$Y_{12} = -\frac{Z_{12}}{\Delta} = -\frac{1}{4} = -0.25$$

$$Y_{21} = -0.25$$

$$Y_{22} = \frac{Z_{11}}{\Delta} = 0.625$$

$$\begin{pmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{pmatrix} = \begin{pmatrix} 0.25 & -0.25 \\ -0.25 & 0.625 \end{pmatrix}$$



Case:1 Open circuit port 2 (i.e.) $\vec{i}_2 = 0$

Loop 1

$$V_1 = 5\vec{i}_1 - j5\vec{i}_1 - j5\vec{i}_2 + 3\vec{i}_2$$

$$V_1 = (5 - j5)\vec{i}_1$$

$$\rightarrow \frac{V_1}{\vec{i}_1} = \boxed{5 - j5 = Z_{11}}$$

$$\frac{\vec{i}_1}{V_1} = \frac{1}{5 - j5}$$

Loop 2

$$V_2 = (1 + j2)\vec{i}_2 - j5\vec{i}_2 - j5\vec{i}_1 + 3\vec{i}_2$$

$$V_2 = -j5\vec{i}_1$$

$$\frac{V_2}{\vec{i}_1} = \boxed{Z_{21} = -j5}$$

Case:2 open circuit Loop 1 (i.e.) $\vec{i}_1 = 0$

Loop 2

$$V_1 = 5\vec{i}_1 - j5\vec{i}_1 - j5\vec{i}_2 + 3\vec{i}_2$$

$$V_1 = (3 - j5)\vec{i}_2 \rightarrow \frac{V_1}{\vec{i}_2} = \boxed{3 - j5 = Z_{12}}$$

Loop 2

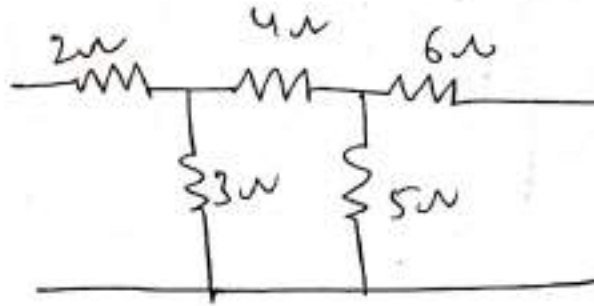
$$V_2 = (1 + j2)\vec{i}_2 - j5\vec{i}_2 - j5\vec{i}_1 + 3\vec{i}_2$$

$$V_2 = (4 - j3)\vec{i}_2$$

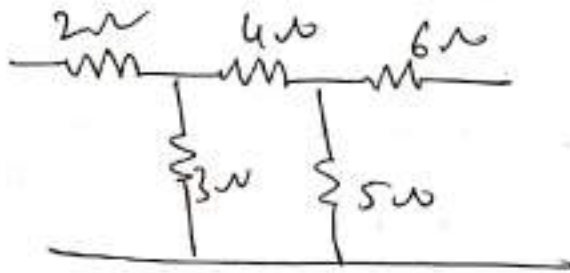
$$\rightarrow \frac{V_2}{\vec{i}_2} = \boxed{Z_{22} = 4 - j3}$$

②

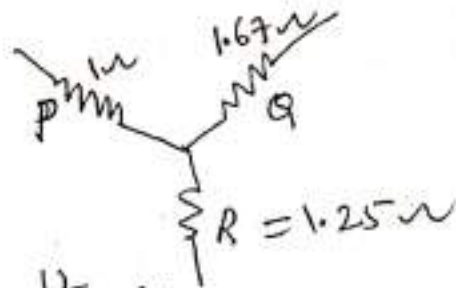
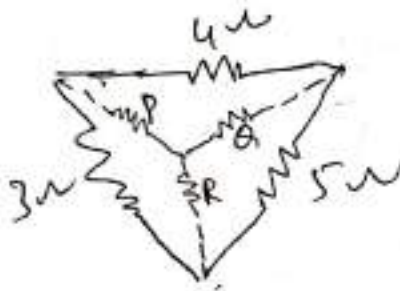
Find γ -parameters for the following ckt



Sol



First convert Δ to γ

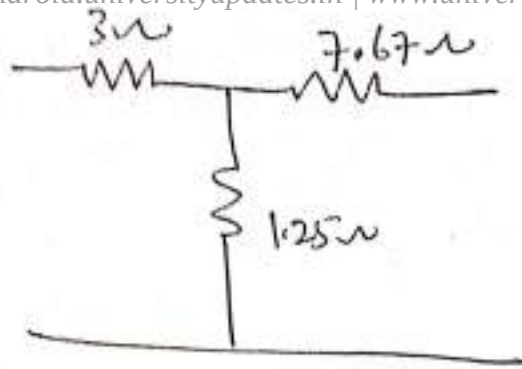


$$P = \frac{3 \times 4}{3 + 4 + 5} = \frac{12}{12} = 1 \Omega$$

$$Q = \frac{4 \times 5}{3 + 4 + 5} = \frac{20}{12} = \frac{5}{3} = 1.67 \Omega$$

$$R = \frac{3 \times 5}{3 + 4 + 5} = \frac{15}{12} = \frac{5}{4} = 1.25 \Omega$$





Y-Parameters

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \text{--- (2)}$$

case (1) To find Y_{11} & Y_{21} put $V_2 = 0$

apply KVL for loop 1

$$-V_1 + 3I_1 + 1.25I_1 + 1.25I_2 = 0$$

$$V_1 - 4.25I_1 - 1.25I_2 = 0 \quad \text{--- (1)}$$

apply KVL for loop 2

$$-V_2 + 7.67I_2 + 1.25I_2 + 1.25I_1 = 0$$

$$\text{put } V_2 = 0$$

$$8.92I_2 + 1.25I_1 = 0$$

$$I_2 = \frac{-1.25}{8.92} I_1 = -0.14I_1 \quad \text{--- (2)}$$

put eq (2) in eq (1)

$$V_1 - 4.25I_1 - (-0.14 \times 1.25)I_1 = 0$$

$$V_1 - 4.25I_1 + 0.175I_1 = 0$$

$$V_1 = 4.075I_1 \Rightarrow \frac{I_1}{V_1} = \frac{1}{4.075} = \boxed{0.246 = Y_{11}}$$

From eq (2)

$$I_1 = -7.14 I_2 \quad (3)$$

Sub eq (3) in eq (1)

$$V_1 - 4.25 [-7.14 I_2] - 1.25 I_2 = 0$$

$$V_1 + 30.34 I_2 - 1.25 I_2 = 0$$

$$V_1 + 29.09 I_2 = 0$$

$$29.09 I_2 = -V_1$$

$$\frac{I_2}{V_1} = -0.034$$

$$Y_{21} = -0.034 \text{ S}$$

Case (2): To find Y_{12} & Y_{22} , put $V_1 = 0$

$$Y_{12} = \frac{I_1}{V_2}, \quad Y_{22} = \frac{I_2}{V_2}$$

apply KVL for loop 1

$$-V_1 + 3 I_1 + 1.25 I_1 + 1.25 I_2 = 0$$

$$V_1 - 4.25 I_1 - 1.25 I_2 = 0$$

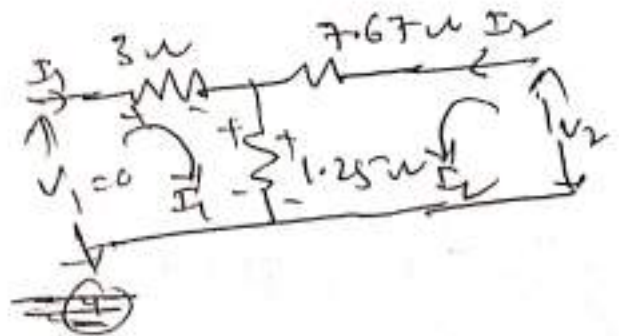
put $V_1 = 0$

$$4.25 I_1 = -1.25 I_2 \quad (4)$$

apply KVL for loop 2

$$-V_2 + 7.67 I_2 + 1.25 I_2 + 1.25 I_1 = 0$$

$$V_2 - 8.92 I_2 - 1.25 I_1 = 0 \quad (5)$$



From eq (3) $\Rightarrow I_2 = \frac{-4.25}{1.25} I_1$

put eq (5) in eq (5) $I_2 = -3.4 I_1 - (5.1)$

~~$V_2 - 8.42 I_2 - 1.25 I_1 = 0$~~

$V_2 - 8.42 (-3.4 I_1) - 1.25 I_1 = 0$

$V_2 + 28.62 I_1 - 1.25 I_1 = 0$

$V_2 + 27.37 I_1 = 0 \quad \text{--- (6)}$

$\frac{I_1}{V_2} = \frac{-1}{27.37} \quad v$

$\boxed{Y_{12} = -0.0365} \quad v \quad \text{--- (7)}$

also from eq (5)

$I_1 = \frac{-1.25}{4.25} I_2$

$I_1 = -0.29 I_2 \quad \text{--- (8)}$

sub eq (8) in eq (5)

$V_2 - 8.92 I_2 + 1.25 \times 0.29 I_2 = 0$

$V_2 - 8.92 I_2 + 0.36 I_2$

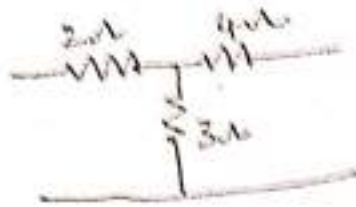
$V_2 = 8.92 I_2 - 0.36 I_2$

$V_2 = 8.56 I_2$

$\frac{I_2}{V_2} = 0.117 \Rightarrow \boxed{Y_{22} = 0.117}$

$\begin{pmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{pmatrix} = \begin{bmatrix} 0.246 & -0.036 \\ -0.034 & 0.117 \end{bmatrix}$

③ Find h-parameters for the following circuit



Sol

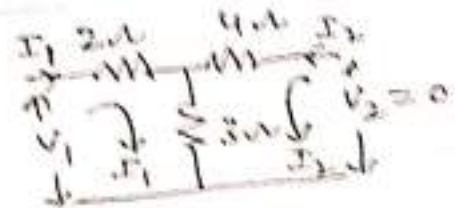
h-parameters

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21}I_1 + h_{22}V_2 \quad \text{--- (2)}$$

case (i) To find h_{11} & h_{21} , put $V_2 = 0$

$$\boxed{h_{11} = \frac{V_1}{I_1}, \quad h_{21} = \frac{I_2}{I_1}}$$



apply KVL for loop 1

$$-V_1 + 2I_1 + 3I_1 + 3I_2 = 0$$

$$V_1 - 5I_1 - 3I_2 = 0 \quad \text{--- (3)}$$

apply KVL for loop 2

$$-V_2 + 4I_2 + 3I_2 + 3I_1 = 0$$

$$\text{put } V_2 = 0$$

$$7I_2 + 3I_1 = 0$$

$$7I_2 = -3I_1 \quad \text{--- (4)}$$

$$\boxed{h_{21} = \frac{I_2}{I_1} = -\frac{3}{7} = -0.428}$$

From eq (4)

$$I_2 = -0.428 I_1 \quad \text{--- (5)}$$

put eq (5) in eq (3)

$$V_1 - 5I_1 - 3(-0.428)I_1 = 0$$

$$V_1 - 5I_1 + 1.285 I_1 = 0$$

$$\boxed{h_{11} = \frac{V_1}{I_1} = 3.715 \Omega}$$

Case (2) : To find ~~the~~ h_{12} & h_{22} , put $I_1 = 0$

loop $\boxed{h_{12} = \frac{V_1}{V_2}}, \boxed{h_{22} = \frac{I_2}{V_2}}$

Apply KVL for loop 1

$$-V_1 + 2I_1 + 3I_1 + 3I_2 = 0$$

$$V_1 - 5I_1 - 3I_2 = 0$$

Put $I_1 = 0$

$$V_1 = 3I_2 \quad \text{--- (6)}$$

apply KVL for loop 2

$$-V_2 + 4I_2 + 3I_2 + 3I_1 = 0$$

$$V_2 - 7I_2 - 3I_1 = 0$$

Put $I_1 = 0$

$$V_2 = 7I_2 \quad \text{--- (7)}$$

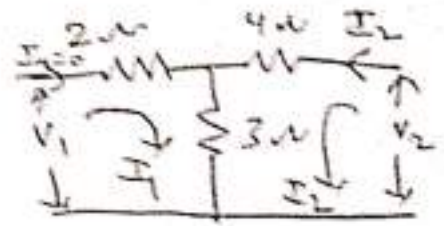
$$\boxed{h_{22} = \frac{I_2}{V_2} = \frac{1}{7}}$$

Put $I_2 = \frac{1}{7} V_2 \quad \text{--- (8)}$

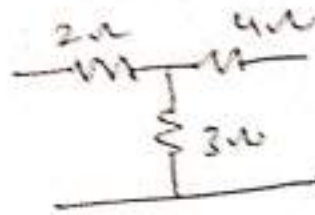
Sub eq (8) in eq (6)

$$V_1 = 3 \times \frac{1}{7} V_2$$

$$\boxed{h_{12} = \frac{V_1}{V_2} = \frac{3}{7}}$$



4 Find ABCD parameters for the following ckt



Sol

ABCD parameters

$$V_1 = AV_2 - BI_2 \quad \text{--- (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

case (i) To find A & C, put $I_2 = 0$

apply KVL for loop 1

$$-V_1 + 2I_1 + 3I_1 + 3I_2 = 0$$

$$\text{Put } I_2 = 0$$

$$V_1 - 5I_1 = 0$$

$$V_1 = 5I_1 \quad \text{--- (3)}$$

apply KVL for loop 2

$$V_2 - 4I_2 - 3I_2 - 3I_1 = 0$$

$$\text{Put } I_2 = 0$$

$$V_2 = 3I_1 \quad \text{--- (4)}$$

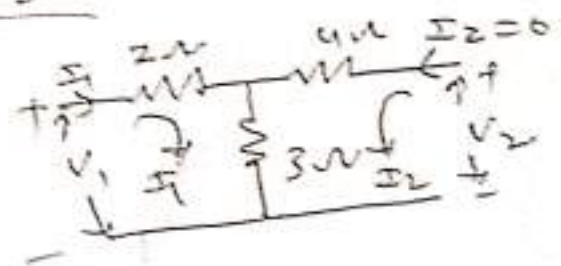
$$C = \frac{I_1}{V_2} = \frac{1}{3} \Rightarrow \boxed{C = \frac{1}{3}}$$

$$\text{From eq (4)} \quad I_1 = \frac{1}{3}V_2 \quad \text{--- (5)}$$

Put eq (5) in eq (3)

$$V_1 = 5 \times \frac{1}{3} V_2$$

$$\boxed{A = \frac{V_1}{V_2} = \frac{5}{3}}$$

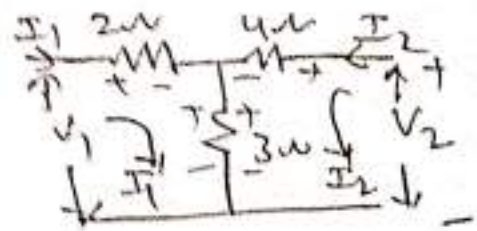


Case (2): To find B & D by putting $V_2 = 0$ $\left[B = \frac{-V_1}{I_2}, D = \frac{-I_1}{I_2} \right]$

apply KVL for loop 1

$$-V_1 + 2I_1 + 3I_1 + 2I_2 = 0$$

$$V_1 = 5I_1 + 2I_2 \quad \text{--- (6)}$$



apply KVL for loop 2

$$V_2 - 4I_2 - 3I_2 - 3I_1 = 0$$

put $V_2 = 0$

$$-7I_2 + 3I_1 = 0$$

$$-7I_2 = -3I_1 \quad \text{--- (7)}$$

$$D = \frac{-I_1}{I_2} = \frac{7}{3}$$

From eq (7) $I_1 = -\frac{7}{3}I_2 \quad \text{--- (8)}$

Sub eq (8) in eq (6)

$$V_1 = 5\left(-\frac{7}{3}I_2 + 3I_2\right)$$

$$V_1 = -8.66 I_2$$

$$B = \frac{-V_1}{I_2} = 8.66$$

TWO-PORT PARAMETER CONVERSION TABLE

DESIRED PARAMETERS	GIVEN PARAMETERS			
	[z]	[y]	[h]	[t]
[z]	$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{y_{22}}{\Delta_y} & \frac{-y_{12}}{\Delta_y} \\ \frac{-y_{21}}{\Delta_y} & \frac{y_{11}}{\Delta_y} \end{bmatrix}$	$\begin{bmatrix} \frac{\Delta_h}{h_{22}} & \frac{h_{12}}{h_{22}} \\ \frac{-h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$	$\begin{bmatrix} \frac{A}{C} & \frac{\Delta_1}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix}$
[y]	$\begin{bmatrix} \frac{z_{22}}{\Delta_z} & \frac{-z_{12}}{\Delta_z} \\ \frac{-z_{21}}{\Delta_z} & \frac{z_{11}}{\Delta_z} \end{bmatrix}$	$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{h_{11}} & \frac{-h_{12}}{h_{11}} \\ \frac{h_{21}}{h_{11}} & \frac{\Delta_h}{h_{11}} \end{bmatrix}$	$\begin{bmatrix} \frac{D}{B} & \frac{-\Delta_1}{B} \\ \frac{-1}{B} & \frac{A}{B} \end{bmatrix}$
[h]	$\begin{bmatrix} \frac{\Delta_z}{z_{22}} & \frac{z_{12}}{z_{22}} \\ \frac{z_{21}}{z_{22}} & \frac{1}{z_{22}} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{y_{11}} & \frac{-y_{12}}{y_{11}} \\ \frac{y_{21}}{y_{11}} & \frac{\Delta_y}{y_{11}} \end{bmatrix}$	$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{B}{D} & \frac{\Delta_1}{D} \\ \frac{-1}{D} & \frac{C}{D} \end{bmatrix}$
[t]	$\begin{bmatrix} \frac{z_{11}}{z_{21}} & \frac{\Delta_z}{z_{21}} \\ \frac{1}{z_{21}} & \frac{z_{22}}{z_{21}} \end{bmatrix}$	$\begin{bmatrix} \frac{-y_{22}}{y_{21}} & \frac{-1}{y_{21}} \\ \frac{y_{11}}{y_{21}} & \frac{-y_{12}}{y_{21}} \end{bmatrix}$	$\begin{bmatrix} \frac{-\Delta_h}{h_{21}} & \frac{-h_{11}}{h_{21}} \\ \frac{h_{21}}{h_{21}} & \frac{-1}{h_{21}} \end{bmatrix}$	$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$
$\Delta_z = z_{11}z_{22} - z_{12}z_{21} \quad \Delta_y = y_{11}y_{22} - y_{12}y_{21} \quad \Delta_h = h_{11}h_{22} - h_{12}h_{21} \quad \Delta_1 = AD - BC$				

Problems on interrelation between two port parameters

- ① Determine Y , h and ABCD Parameters if Z parameters are
 $z_{11}=10$, $z_{12}=5$, $z_{21}=5$, $z_{22}=9$.

Sol

Given

$$z_{11}=10, z_{12}=5, z_{21}=5, z_{22}=9.$$

$$Z = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} = \begin{bmatrix} 10 & 5 \\ 5 & 9 \end{bmatrix}$$

(i) Y -Parameters

$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} = \begin{bmatrix} \frac{y_{22}}{\Delta Y} & -\frac{y_{12}}{\Delta Y} \\ -\frac{y_{21}}{\Delta Y} & \frac{y_{11}}{\Delta Y} \end{bmatrix}$$

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} \frac{z_{22}}{\Delta Z} & -\frac{z_{12}}{\Delta Z} \\ -\frac{z_{21}}{\Delta Z} & \frac{z_{11}}{\Delta Z} \end{bmatrix}$$

$$\begin{aligned} \Delta Z &= z_{11}z_{22} - z_{12}z_{21} \\ &= 10 \times 9 - 5 \times 5 \\ &= 90 - 25 = 65 \end{aligned}$$

$$y_{11} = \frac{z_{22}}{\Delta Z} = \frac{9}{65}$$

$$y_{12} = -\frac{z_{12}}{\Delta Z} = -\frac{5}{65}$$

$$y_{21} = -\frac{z_{21}}{\Delta Z} = -\frac{5}{65}$$

$$y_{22} = \frac{z_{11}}{\Delta Z} = \frac{10}{65}$$

(2) h-Parameters

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ -\frac{Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{65}{9} & \frac{5}{9} \\ -\frac{5}{9} & \frac{1}{9} \end{bmatrix}$$

(3) ABCD Parameters

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{10}{5} & \frac{65}{5} \\ \frac{1}{5} & \frac{9}{5} \end{bmatrix}$$

II Find Z , Y , ABCD parameters if h -parameters are
 $h_{11} = 8$, $h_{12} = 5$, $h_{21} = 3$, $h_{22} = 12$

Sol ① Z-parameters

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta h}{h_{22}} & \frac{h_{12}}{h_{22}} \\ -\frac{h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{81}{12} & \frac{5}{12} \\ -\frac{3}{12} & \frac{1}{12} \end{bmatrix}$$

$$\Delta h = h_{11}h_{22} - h_{12}h_{21}$$

$$= 8 \times 12 - 5 \times 3$$

$$= 96 - 15$$

$$= 81$$

(2) Y-parameter

$$\begin{pmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{pmatrix} = \begin{pmatrix} \frac{1}{h_{11}} & -\frac{h_{12}}{h_{11}} \\ -\frac{h_{21}}{h_{11}} & \frac{\Delta h}{h_{11}} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{8} & -\frac{5}{8} \\ -\frac{3}{8} & \frac{81}{8} \end{pmatrix}$$

(3) ABCD Parameter

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} -\frac{\Delta h}{h_{21}} & -\frac{h_{11}}{h_{21}} \\ -\frac{h_{22}}{h_{21}} & -\frac{1}{h_{21}} \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{81}{3} & -\frac{8}{3} \\ -\frac{12}{3} & -\frac{1}{3} \end{pmatrix}$$

III Find $z, h, ABCD$ parameter if Y -Parameter are

$$Y_{11} = 9, Y_{12} = 6, Y_{21} = 6, Y_{22} = 15$$

$$\Delta Y = Y_{11}Y_{22} - Y_{12}Y_{21} = 15 \times 9 - 36 = 135 - 36 = 99$$

Sol

① Z Parameter

$$\begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} = \begin{pmatrix} \frac{Y_{22}}{\Delta Y} & -\frac{Y_{12}}{\Delta Y} \\ -\frac{Y_{21}}{\Delta Y} & \frac{Y_{11}}{\Delta Y} \end{pmatrix} = \begin{pmatrix} \frac{15}{99} & -\frac{6}{99} \\ -\frac{6}{99} & \frac{9}{99} \end{pmatrix}$$

② h-parameter

$$\begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} = \begin{pmatrix} \frac{1}{Y_{11}} & -\frac{Y_{12}}{Y_{11}} \\ \frac{Y_{21}}{Y_{11}} & \frac{\Delta Y}{Y_{11}} \end{pmatrix} = \begin{pmatrix} \frac{1}{9} & -\frac{6}{9} \\ \frac{6}{9} & \frac{99}{9} \end{pmatrix}$$

(3) ABCD Parameters

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} \frac{-Y_{L2}}{Y_{L1}} & \frac{-1}{Y_{L1}} \\ \frac{-\Delta Y}{Y_{L1}} & \frac{-Y_{11}}{Y_{L1}} \end{pmatrix} = \begin{pmatrix} \frac{-15}{6} & \frac{-1}{6} \\ \frac{-99}{6} & \frac{-9}{6} \end{pmatrix}$$

IV Find z, y, h parameters if $A=6, B=2, C=3, D=8$.

$$\Delta t = AD - BC = 6 \times 8 - 2 \times 3 = 48 - 6 = 42$$

Sol (1) z parameters

$$\begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} = \begin{pmatrix} \frac{A}{C} & \frac{\Delta t}{C} \\ \frac{1}{C} & \frac{D}{C} \end{pmatrix} = \begin{pmatrix} \frac{6}{3} & \frac{42}{3} \\ \frac{1}{3} & \frac{8}{3} \end{pmatrix}$$

(2) y - Parameter

$$\begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix} = \begin{pmatrix} \frac{D}{B} & \frac{-\Delta t}{B} \\ -\frac{1}{B} & \frac{A}{B} \end{pmatrix} = \begin{pmatrix} \frac{8}{2} & \frac{-42}{2} \\ -\frac{1}{2} & \frac{6}{2} \end{pmatrix} = \begin{pmatrix} 4 & -21 \\ -0.5 & 3 \end{pmatrix}$$

(3) h - Parameters

$$\begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} = \begin{pmatrix} \frac{B}{D} & \frac{\Delta t}{D} \\ -\frac{1}{D} & \frac{C}{D} \end{pmatrix} = \begin{pmatrix} \frac{2}{8} & \frac{42}{8} \\ -\frac{1}{8} & \frac{3}{8} \end{pmatrix}$$

Unit - III: Transient Analysis

D.C Transient Analysis: Transient Response of R-L, R-C, R-L-C Series Circuits for D.C Excitation - Initial Conditions in network - Initial Conditions in elements - Solution Method Using Differential Equation and Laplace Transforms - Response of R-L & R-C Networks to Pulse Excitation.

A.C Transient Analysis: Transient Response of R-L, R-C, R-L-C Series Circuits for Sinusoidal Excitations - Solution Method Using Differential Equations and Laplace Transforms.

Unit Outcomes:

The student will be able to

- Distinguish between classical method and Laplace transform approach in analysing transient phenomenon in DC excitations
- Distinguish between classical method and Laplace transform approach in analysing transient phenomenon in sinusoidal excitations

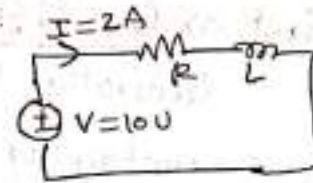
TRANSIENT RESPONSE ANALYSIS

1. Steady state :

A circuit having voltage source or current source is said to be steady state if voltage and currents doesn't change with respect to time.

Ex: $t = 2 \text{ sec}$, $V = 10 \text{ V}$, $I = 2 \text{ A}$

$t = 5 \text{ sec}$, $V = 10 \text{ V}$, $I = 2 \text{ A}$



2. Transient state :

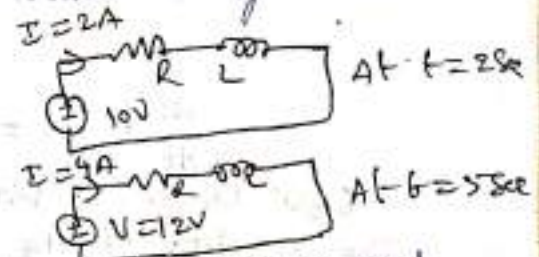
A circuit having voltage source, current source, circuit parameters like $R, L \& C$. Now either change in voltage or current source or any one of the circuit parameters, then voltage and currents are changed from one state to another state is called transient state.

(OR)

The behaviour of voltage and currents when it is changed from one steady state to another steady state is called transient state.

Ex: $t = 2 \text{ sec}$, $V = 10 \text{ V}$, $I = 2 \text{ A}$

$t = 5 \text{ sec}$, $V = 12 \text{ V}$, $I = 4 \text{ A}$



3. Transient time :

The time required for changing the voltage and currents from one state to another state is called Transient time.

4. Natural Response :

The circuit having circuit parameters like R, L, C with independent current and voltage sources, the response of output depends on the nature of the circuit, then the response is called natural response.

5. Transient Response :

The circuit having circuit parameters like R, L, C with independent voltage and current sources, then the energy storing elements ($L \& C$) which delivers energy to the resistors, the response or o/p change with time and get saturated.

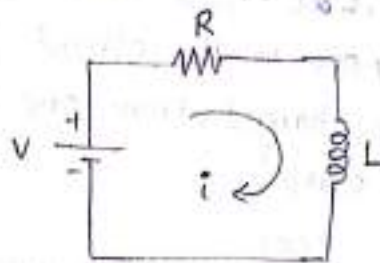
6 Forced Response :

The circuits having circuit parameters like R, L , with dependent voltage and current sources, the response or output depends on the sources, then it is called forced response.

First order differential equations

Generally kirchoff's voltage law is applied to circuit containing $R-L, R-C$ & $R-L-C$ which gives differential equation.

for example, take $R-L$ circuit



$$V = IR + L \frac{di}{dt}$$

$$\frac{di}{dt} + \frac{R}{L} i = \frac{V}{L} \longrightarrow (1)$$

Eqn (1) is called First order differential equation

The solution of the differential equation is obtained by two cases.

Case 1: Differential eqn's method

Case 2: Laplace transforms method

Differential Equation Methods :-

In the case of DE method, the eqns are of two types.

1. Non-homogeneous DE

2. Homogeneous DE

Non-Homogeneous Differential Equation:-

$$\frac{di}{dt} + Pi = Q \longrightarrow (2)$$

$$\frac{di}{dt} + \frac{R}{L} i = \frac{V}{L}$$

$$\text{where } P = \frac{R}{L}, Q = \frac{V}{L}$$

The solution is

$i(t)$ = Complimentary func. + partial integer

= Transient response + steady state

= Natural response + steady state.

$$i(t) = i_{CF} + i_{ss}$$

$$= K e^{-Pt} + e^{-Pt} \int Q e^{Pt} dt$$

$$= K e^{-Pt} + e^{-Pt} Q \frac{e^{Pt}}{P}$$

$$i(t) = K e^{-Pt} + \frac{Q}{P}$$

Homogeneous Differential Equation:-

$$\frac{di}{dt} + Pi = 0$$

The solution is

$i(t)$ = complementary func. + Partial integral

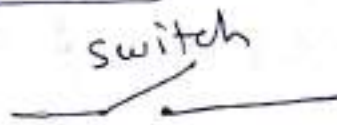
= transient response + s.s

$$i(t) = K e^{-Pt} + e^{-Pt} \int 0 \cdot e^{Pt} dt$$

$$i(t) = K e^{-Pt}$$

Initial conditions in elements

Assume that



$t=0$, The switch is ready to operate

$t=0^-$, The time just before the switch is closed.

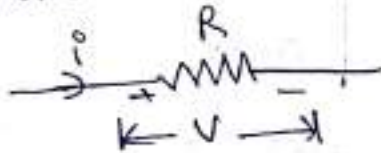
$t=0^+$, The time just immediately after closing the switch.

Now let us study the switching operation on passive elements such as R, L, C.

Resistor (R)

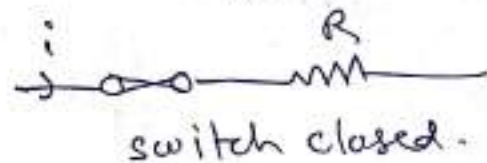
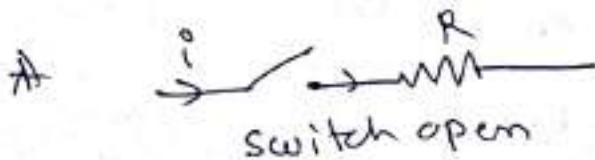
* It consumes energy.

* The property of resistor is it opposes the flow of current.



Voltage across resistor $(V) = iR$

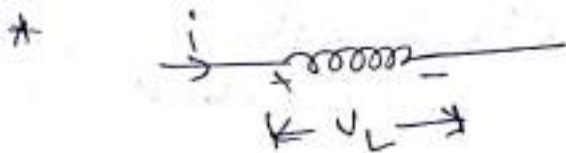
Current through resistor $(i) = \frac{V}{R}$



* If switch is closed the current through the resistor changes instantaneously when voltage changes.

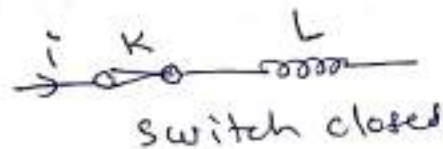
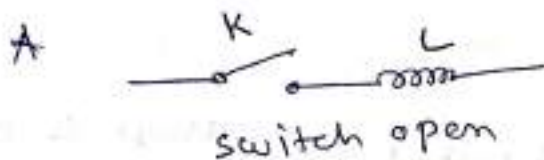
Inductor (L)

- * Inductor stores energy in magnetic field.
- * The property of Inductor is it opposes the flow of change in current. (or) does not allow sudden change in current.



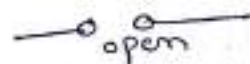
voltage across inductor (L) is $V_L = L \frac{di}{dt}$

current through inductor is $i_L = \frac{1}{L} \int_0^t V dt$

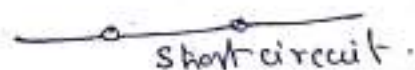


* If switch is closed ~~at t=0~~

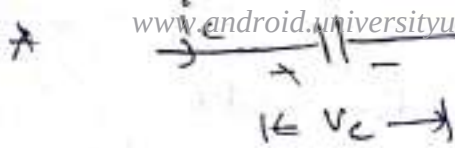
At $t=0^+$, Inductor does not allow sudden change of current. so it acts as open circuit



At $t=\infty$ (steady state); Inductor continuously takes full current. so it acts as short circuit

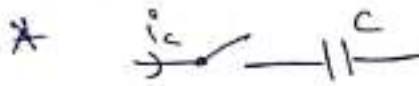
Capacitor (C)

- * It stores energy in electric field.
- * The property of capacitor is it does not allow sudden change in voltage.



current through the capacitor is $i_c = C \frac{dV_c}{dt}$

voltage across capacitor is $V_c = \frac{1}{C} \int_0^t i_c dt$.



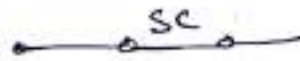
Switch is opened



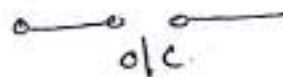
Switch is closed.

A If the switch is closed

At $t=0^+$, capacitor does not allow sudden change in voltage. so it acts as short circuit

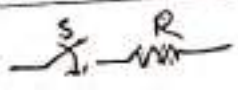
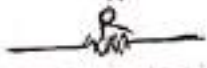
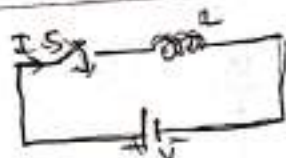
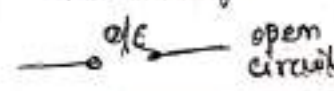
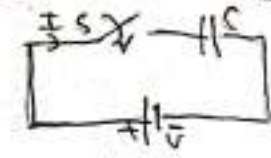
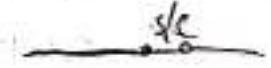
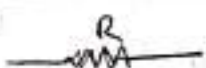
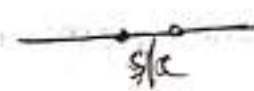
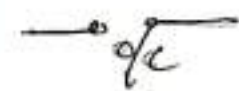


At $t=\infty$, capacitor builds up full voltage & no current so it acts as open circuit



S.No	Parameter	At $t=0^+$ (Transient state)	At $t=\infty$ (Steady state)
1.			
2.			
3.			
4.			
5.			

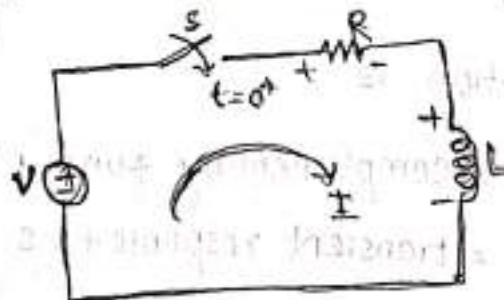
* Initial Conditions:

Resistor (R)	Inductor (L)	Capacitor (C)
 At $t = 0^+$ (transient state) $V = IR$ $I = \frac{V}{R}$ 	 At $t = 0^+$, $I = 0$ full voltage. 	 At $t = 0^+$, $V = 0$ full load current 
 At $t = \infty$ (steady state) $V = IR$ $I = \frac{V}{R}$	At $t = \infty$ (steady state)  Full load current $V = 0$	At $t = \infty$ (s.s)  full voltage $I = 0$

Note: The main purpose of initial conditions is to find out the unknown value of k .

* Transient Analysis of Series R-L circuit:

Series R-L circuit is excited by DC input which is connected through switch as shown in the figure.



At $t = 0^-$, $I = 0$; At $t = 0^+$, $I = 0$

Initial Conditions:

$t = 0$, $I = 0 \rightarrow (1)$

Now, apply KVL for the closed circuit

$$-V + IR + L \frac{di}{dt} = 0$$

$$V - IR - L \frac{di}{dt} = 0$$

$$V = IR + L \frac{di}{dt}$$

divide both sides with 'L'

$$\frac{di}{dt} + \frac{R}{L} i = \frac{V}{L} \longrightarrow (2)$$

It is a non-homogeneous equation

$$\frac{di}{dt} + P i = Q, \text{ where } P = \frac{R}{L}, Q = \frac{V}{L}$$

The solution for eqn (2) is

$$i(t) = K e^{-Pt} + \frac{Q}{P}$$

$$i(t) = K e^{-\frac{R}{L}t} + \frac{V}{R} \longrightarrow (3)$$

Subst. eqn (1) in eqn (3)

$$0 = K e^0 + \frac{V}{R}$$

$$\boxed{K = -\frac{V}{R}} \longrightarrow (4)$$

Subst. eqn (4) in eqn (3)

$$i(t) = -\frac{V}{R} e^{-\frac{R}{L}t} + \frac{V}{R}$$

$$\boxed{i(t) = \frac{V}{R} (1 - e^{-\frac{R}{L}t})} \longrightarrow (5)$$

In case of series R-L circuit, the time constant

$$\tau = \frac{L}{R}$$

subst. τ in eqn (5)

$$i(t) = \frac{V}{R} (1 - e^{-\frac{t}{\tau}}) \longrightarrow (6)$$

Now, $t = \tau$, then

$$i(t) = \frac{V}{R} (1 - e^{-1})$$

$$i(t) = \frac{V}{R} (1 - 0.367)$$

$$i(t) = 0.633 \frac{V}{R}$$

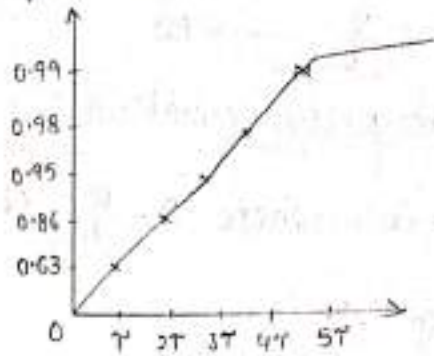
$$\text{at } t = 2\tau, i(t) = \frac{V}{R} (1 - e^{-2}) = 0.865 \frac{V}{R}$$

$$t = 3\tau, i(t) = \frac{V}{R} (1 - e^{-3}) = 0.951 \frac{V}{R}$$

$$t = 4\tau, i(t) = \frac{V}{R} (1 - e^{-4}) = 0.98 \frac{V}{R}$$

$$t = 5\tau, i(t) = \frac{V}{R} (1 - e^{-5}) = 0.99 \frac{V}{R}$$

Now, the graph b/w $i(t)$ vs t



Voltage across resistor:

$$V_R = iR$$

$$= \frac{V}{R} (1 - e^{-\frac{t}{T}}) R$$

$$V_R = V (1 - e^{-\frac{t}{T}})$$

Voltage across inductor:

$$V = V_R + V_L$$

$$V_L = V - V_R$$

$$= V - V (1 - e^{-\frac{t}{T}})$$

$$V_L = V e^{-\frac{t}{T}}$$

Q: Find the current in series R-L series circuit having $R = 2\Omega$ and $L = 10H$, while a DC voltage of $100V$ is applied. What is the value of current after 5 sec of switch ON.

Sol: Given that what is the time taken by the current to reach half of its final value.

$$R = 2\Omega$$

$$L = 10H$$

$$V = 100V$$

$$t = 5 \text{ sec}$$

$$\text{WKT } i(t) = \frac{V}{R} (1 - e^{-\frac{R}{L}t})$$

$$= \frac{100}{2} (1 - e^{-\frac{2}{10}t})$$

$$= 50 (1 - e^{-\frac{1}{5}t})$$

$$\text{at } t = 5, i = 50 (1 - e^{-\frac{5}{5}})$$

$$i = 31.606A$$

$$\text{Final value } i = \frac{V}{R} = \frac{100}{2} = 50A$$

$$\text{half of its final value} = 25A$$

$$-25 = -50(1 - e^{-t/5})$$

$$-25 = -50e^{-t/5}$$

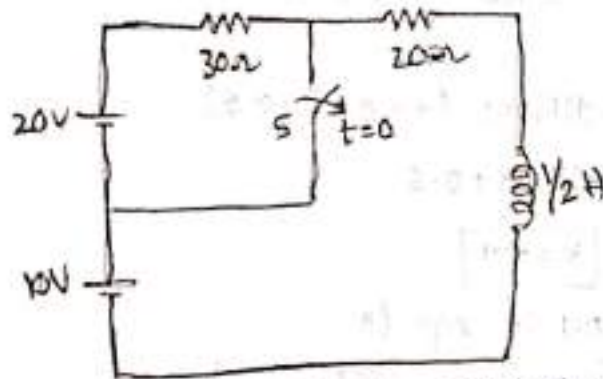
$$\frac{1}{2} = e^{-t/5}$$

$$\log\left(\frac{1}{2}\right) = \frac{-t}{5}$$

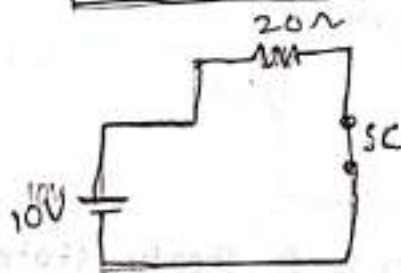
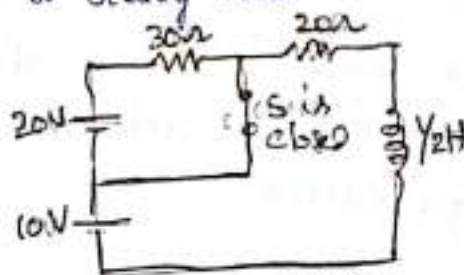
$$t = -5 \ln\left(\frac{1}{2}\right)$$

$$t = 3.46 \text{ sec}$$

2. In the figure switch 's' is closed and steady state is reached. At $t=0$, the switch 's' is opened, find $i_L(t)$ for $t > 0$

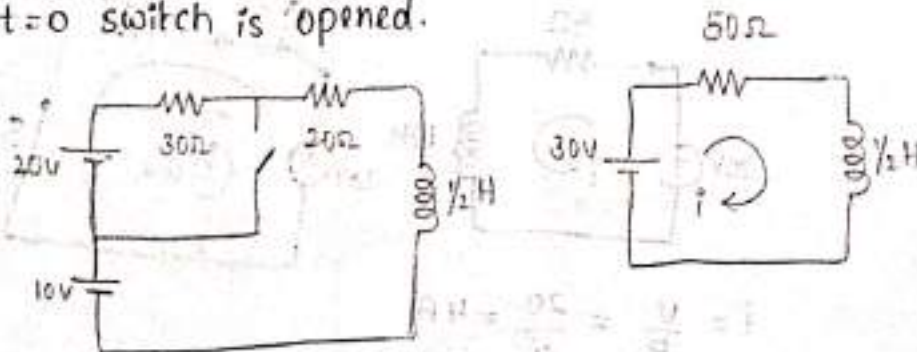


Ans: s is closed & steady state is reached:



$$i = \frac{V}{R} = \frac{10}{20} = 0.5 \text{ A}$$

At $t=0$ switch is opened.



Initial conditions are $t=0, i=0.5 \text{ A}$

Applying KVL to the closed circuit

$$30 - 50 i_L(t) - \frac{1}{2} \frac{d i_L(t)}{dt} = 0$$

$$50 i_L(t) + \frac{1}{2} \frac{d i_L(t)}{dt} = 30$$

$$\frac{d i_L(t)}{dt} + 100 i_L(t) = 60 \longrightarrow (2)$$

It is like a non-homogeneous eqn, the solution is

$$i_L(t) = k e^{-Pt} + \frac{Q}{P} \quad [P=100, Q=60]$$

$$= k e^{-100t} + 0.6 \longrightarrow (3)$$

To find k :

Use initial conditions ($t=0, i=0.5$)

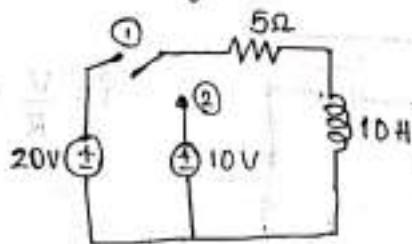
$$0.5 = k + 0.6$$

$$k = -0.1$$

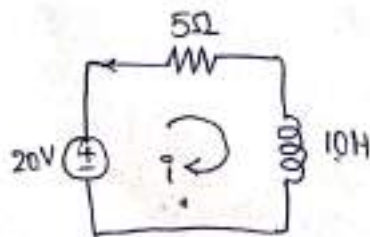
Substitute $k = -0.1$ in eqn (3)

$$i_L(t) = -0.1 e^{-100t} + 0.6$$

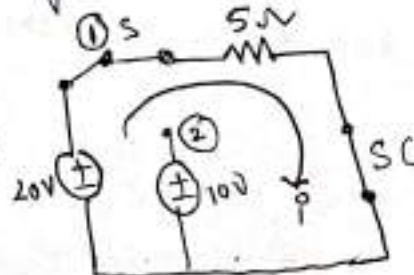
3. In the ckt shown in figure switch is at position 1 and steady state is reached. At $t=0$, switch is moved to position 2 find the expression for current.



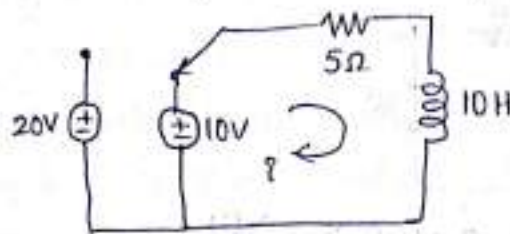
Ans: Switch is at position 1 & steady state is reached.



$$i = \frac{V}{R} = \frac{20}{5} = 4 \text{ A}$$



At $t=0$, switch is moved to position 2.



Initial conditions:

$$t=0, i=4A$$

Now, apply KVL to the circuit:

$$10 - 5i - 10 \frac{di}{dt} = 0$$

$$10 \frac{di}{dt} + 5i = 10$$

$$\frac{di}{dt} + \frac{1}{2}i = 1 \rightarrow (1)$$

It is like a non-homogeneous eqn, the solution is

$$i(t) = Ke^{-pt} + \frac{Q}{P} \quad [P = \frac{1}{2}, Q = 1]$$

$$= Ke^{-t/2} + 2 \rightarrow (2)$$

To find K:

Use initial conditions, $(t=0, i=4A)$

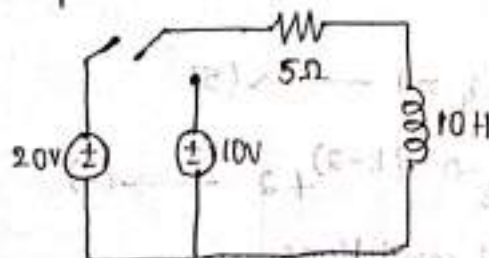
$$4 = K + 2$$

$$\boxed{K=2}$$

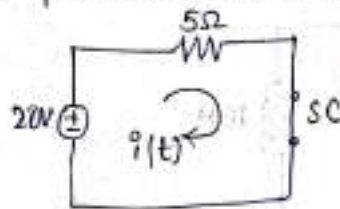
substitute $K=2$ in eqn (2)

$$\boxed{i(t) = 2e^{-t/2} + 2}$$

4. In the following circuit at $t=0$, switch is moved to position 1. At $t=2$ sec, switch is moved to position 2. Find the current through inductor.



Ans. Switch is at position 1 at $t=0$



Initial conditions, $t=0, i=0 \rightarrow (1)$

applying KVL to the circuit

$$t=0, 20 - 5i = 10 \frac{di}{dt}$$

$$5i = 20$$

$$\boxed{i = 4A}$$

Switch is at position 2 at t :

$$10 \frac{di}{dt} + 5i = 20$$

$$\frac{di}{dt} + \frac{1}{2}i = 2 \rightarrow (2)$$

It is in non-homogeneous eqn, the solution is

$$i(t) = Ke^{-Pt} + \frac{Q}{P} \quad [P = \frac{1}{2}, Q = 2]$$

$$i(t) = -4e^{-0.5t} + 4 \rightarrow (3)$$

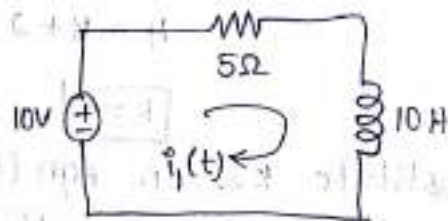
Now switch is moved to position 2 at $t=2$:

Initial condition:

$$t=2, i = 4 - 4e^{-0.5(2)}$$

$$= 4 - 4e^{-1}$$

$$t=2, i = 2.5A \rightarrow (4)$$



Now, apply KVL to the closed circuit

$$10 - 5i_1 - 10 \frac{di_1}{dt} = 0$$

$$10 \frac{di_1}{dt} + 5i_1 = 10$$

$$\frac{di_1}{dt} + \frac{1}{2}i_1 = 1 \rightarrow (5)$$

$$\text{solution is } i(t) = Ke^{-0.5(t-2)} + 2 \rightarrow (6)$$

to find k , use initial conditions,

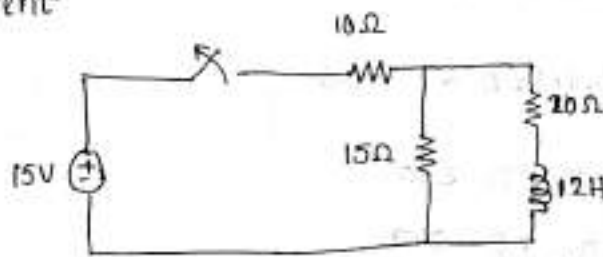
$$t=2, i = 2.5A$$

$$\boxed{k = 0.5}$$

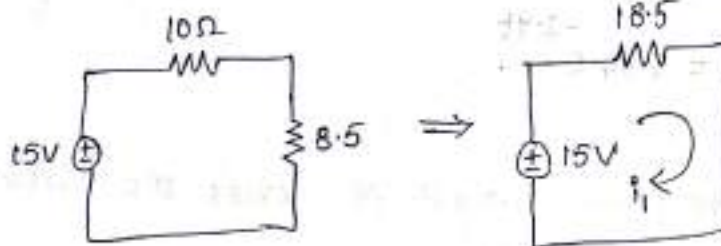
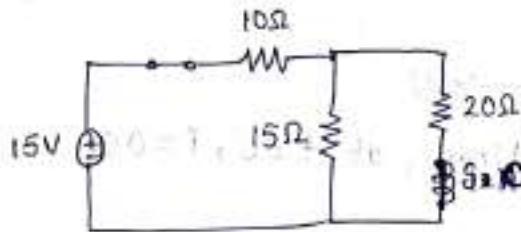
Substitute $k=0.5$ in eqn (6)

$$i_1(t) = 0.5e^{-0.5(t-2)} + 2$$

5. In the following circuit switch is closed and steady state is reached. At $t=0$, switch is suddenly opened, find the expression for current



Sol: Switch is closed and s.s is reached.



$$i_1 = \frac{15}{18.5} = 0.81 \text{ A}$$

According to current division rule

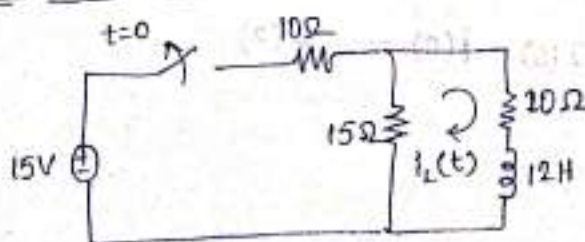
$$i_3 = i_1 \times \frac{15}{15+20}$$

$$= 0.81 \times \frac{15}{35}$$

$$i_3 = 0.34 \text{ A}$$

Initial current $i_0 = i_3 = 0.34 \text{ A} \rightarrow$ (1) at $t=0$

switch is opened suddenly at $t=0$:



apply KVL to the closed ckt

$$-15 i_L(t) - 20 i_L(t) - 12 \frac{di_L(t)}{dt} = 0$$

$$-35 i_L(t) - 12 \frac{di_L(t)}{dt} = 0$$

$$\frac{di_L(t)}{dt} + \frac{35}{12} i_L(t) = 0$$

$$\frac{di_L(t)}{dt} + 2.9 i_L(t) = 0 \rightarrow (2)$$

eqn (2) is a homogeneous eqn.

$$\frac{di}{dt} + Pi = 0, P = 2.9$$

solution is $i_L(t) = K e^{-Pt}$

$$i_L(t) = K e^{-2.9t} \rightarrow (3)$$

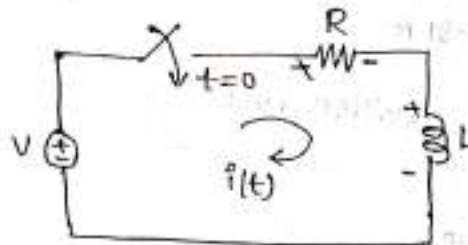
to find k, use initial conditions, at $t=0, i = 0.34$

$$0.34 = K$$

$$i_L(t) = 0.34 e^{-2.9t}$$

Method 2:

DC transient response analysis of series R-L circuit by Laplace transform method:



$$\text{KVL equation, } -V + Ri(t) + L \frac{di(t)}{dt} = 0$$

$$V = Ri(t) + L \frac{di(t)}{dt} \rightarrow (1)$$

In Laplace transform

$$L(V) = V/s$$

$$L[i(t)] = I(s)$$

$$L\left[\frac{di(t)}{dt}\right] = sI(s) - i(0) \rightarrow (2)$$

Initial condition

Initial conditions:

$$t=0, i=0$$

apply initial current $i=0$ in eqn (2)

$$L\left[\frac{di(t)}{dt}\right] = sI(s)$$

by apply Laplace transform

$$\frac{V}{s} = R I(s) + L [s I(s)]$$

$$I(s) [R + sL] = \frac{V}{s}$$

$$I(s) = \frac{V/s}{R + sL}$$

$$I(s) = \frac{V/s}{L(s + \frac{R}{L})}$$

$$I(s) = \frac{(V/L)}{s(s + \frac{R}{L})} \rightarrow (3)$$

Using partial fractions,

$$\frac{(V/L)}{s(s + R/L)} = \frac{A}{s} + \frac{B}{s + \frac{R}{L}}$$

$$\frac{A}{s} = \frac{V/L}{s(s + R/L)} \Big|_{s=0} = \frac{V/L}{R/L}$$

$$\boxed{A = \frac{V}{R}}$$

$$\frac{B}{s + R/L} = \frac{V/L}{s(s + R/L)} \Big|_{s = -\frac{R}{L}}$$

$$B = \frac{(V/L)}{-\frac{R}{L}}$$

$$\boxed{B = -\frac{V}{R}}$$

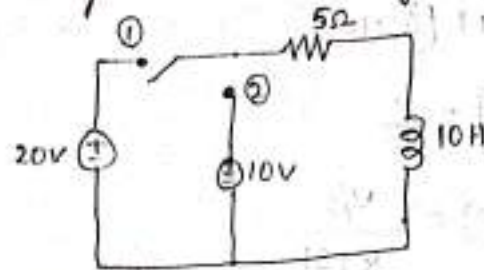
$$\begin{aligned} \text{from eqn (3)} \quad I(s) &= \frac{V/R}{s} - \frac{V/R}{s + R/L} \\ &= \frac{V}{R} \left[\frac{1}{s} - \frac{1}{s + R/L} \right] \rightarrow (4) \end{aligned}$$

apply inverse Laplace

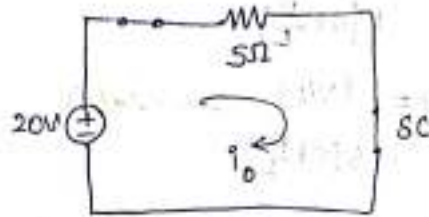
$$i(t) = \frac{V}{R} (1 - e^{-\frac{R}{L}t})$$

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1. In the circuit shown in fig. switch is at position 1 & steady state is reached. At $t=0$ switch is moved to position 2. Find current through inductor using Laplace transform method.

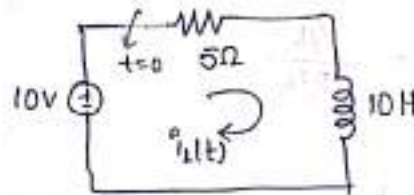


Ans: Switch is at position 1 & steady state is reached.



$$i_0 = \frac{20}{5} = 4A$$

Switch is moved to position 2 at $t=0$



Initial condition,

$$t=0, i_0 = 4A \rightarrow (1)$$

Apply KVL,

$$10 - 5i_L(t) - 10 \frac{di_L(t)}{dt} = 0$$

$$10 \frac{di_L(t)}{dt} + 5i_L(t) = 10$$

$$\frac{di_L(t)}{dt} + \frac{1}{2} i_L(t) = 1 \rightarrow (2)$$

apply Laplace transform,

$$L\left[\frac{di_L(t)}{dt} + 0.5 i_L(t)\right] = L[1]$$

$$[sI(s) - i(0)] + 0.5 I(s) = \frac{1}{s}$$

$$I(s) [s + 0.5] = \frac{1}{s} + 4$$

$$I(s) = \frac{1 + 4s}{s(s + 0.5)} \rightarrow (3)$$

Ans

$$\frac{1+4s}{s(s+0.5)} = \frac{A}{s} + \frac{B}{s+0.5}$$

$$\frac{A}{s} = \frac{1+4s}{s(s+0.5)} \Big|_{s=0}$$

$$A = \frac{1+4(0)}{0+0.5}$$

$$\boxed{A = 2}$$

$$\frac{B}{s+0.5} = \frac{1+4s}{s(s+0.5)} \Big|_{s=-0.5}$$

$$B = \frac{1+4(-0.5)}{-0.5}$$

$$= \frac{1-2}{-0.5}$$

$$\boxed{B = 2}$$

$$\therefore I(s) = \frac{2}{s} + \frac{2}{s+0.5}$$

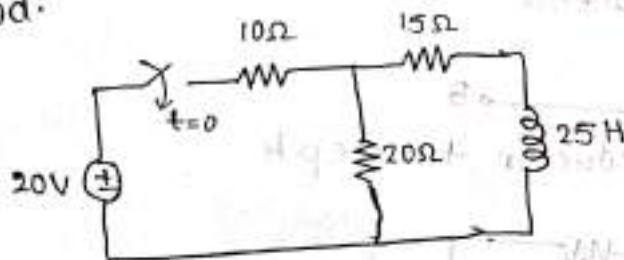
By using inverse Laplace

$$L^{-1} I(s) = L^{-1} \left\{ \frac{2}{s} \right\} + L^{-1} \left\{ \frac{2}{s+0.5} \right\}$$

$$i(t) = 2 + 2e^{-0.5t}$$

$$i(t) = 2(1 + e^{-0.5t})$$

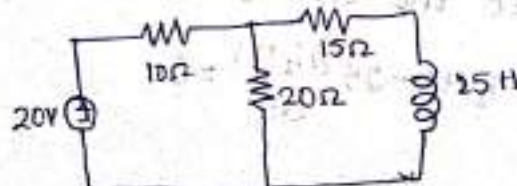
2. In the following figure, the switch is closed at $t=0$, find current through inductor using Laplace transform method.



Ans. Initial condition,

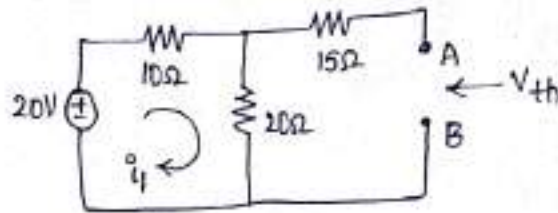
$$\text{at } t=0, i=0 \rightarrow (1)$$

Switch is closed at $t=0$



Apply Thevenin theorem to the above circuit

Step 1: remove 25 H inductor from the circuit



Step 2: find V_{th}

apply KVL to the closed circuit

$$20 - 10i_1 - 20i_1 = 0$$

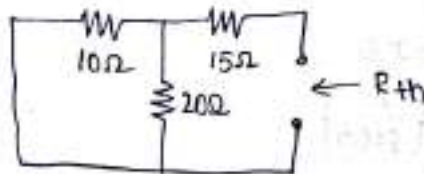
$$30i_1 = 20$$

$$i_1 = 0.66$$

$$V_{th} - 20 \times 0.66 = 0$$

$$V_{th} = 13.2 \text{ V}$$

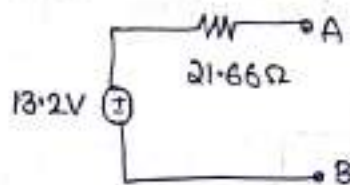
Step 3: To find R_{th} , short circuit the voltage source



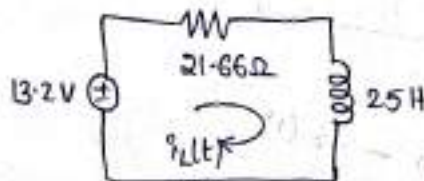
$$R_{th} = 15 + \frac{10 \times 20}{30}$$

$$R_{th} = 21.66 \Omega$$

Step 4: Draw thevenin's equivalent circuit



Step 5: Add 25 H inductor to step 4.



Now apply KVL to the circuit

$$13.2 - 21.66 i_L(t) - 25 \frac{di_L(t)}{dt} = 0$$

$$21.66 i_L(t) + 25 \frac{di_L(t)}{dt} = 13.2 \rightarrow (2)$$

Apply Laplace transform,

$$21.66 i_L(s) + 25 [s i_L(s) - i_L(0)] = \frac{13.2}{s}$$

$$21.66 i_L(s) + 25 s i_L(s) = \frac{13.2}{s}$$

$$i_L(s) [21.66 + 25s] = \frac{13.2}{s}$$

$$i_L(s) = \frac{13.2}{s(21.66 + 25s)}$$

$$i_L(s) = \frac{\frac{13.2}{25}}{s(s + \frac{21.66}{25})}$$

$$i_L(s) = \frac{13.2/25}{s(s + 0.8664)}$$

$$i_L(s) = \frac{0.528}{s(s + 0.8664)} \rightarrow (3)$$

$$\text{Let } \frac{0.528}{s(s + 0.8664)} = \frac{A}{s} + \frac{B}{s + 0.8664}$$

$$0.528 = A(s + 0.8664) + Bs$$

$$0.528 = (A + B)s + 0.8664A$$

$$A + B = 0, A = \frac{0.528}{0.8664}$$

$$B = -0.613, A = 0.613$$

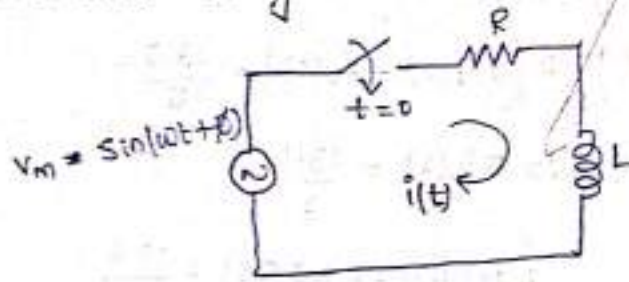
$$i_L(s) = \frac{0.613}{s} - \frac{0.613}{s + 0.8664}$$

apply inverse Laplace transform,

$$i_L(t) = 0.613 [1 - e^{-0.8664t}]$$

Method 3: AC transient response analysis Series

R-L circuit using DE method.



Initial condition $t=0, i=0 \rightarrow (1)$

Apply KVL to the circuit

$$\frac{iR}{L} + L \frac{di}{dt} = v_m \sin(\omega t + \phi) \rightarrow (2)$$

It is non-homogeneous eqn.

$$\frac{di}{dt} + Pi = Q$$

The solution is $i = i_{CF} + i_{PI}$

$$i_{CF} = ke^{-Pt} = ke^{-\frac{R}{L}t} \rightarrow (3)$$

Consider, $i_{PI} = A \cos(\omega t + \phi) + B \sin(\omega t + \phi) \rightarrow (4)$

$$\frac{di}{dt} = -A\omega \sin(\omega t + \phi) + B\omega \cos(\omega t + \phi) \rightarrow (5)$$

Substitute eqn (4) & eqn (5) in eqn (2)

$$\begin{aligned} -A\omega \sin(\omega t + \phi) + B\omega \cos(\omega t + \phi) + \frac{R}{L}[A \cos(\omega t + \phi) + B \sin(\omega t + \phi)] \\ = \frac{V_m}{L} \sin(\omega t + \phi) \rightarrow (6) \end{aligned}$$

Equating like terms, we get

$$(-A\omega + B \frac{R}{L}) \sin(\omega t + \phi) = \frac{V_m}{L} \sin(\omega t + \phi)$$

$$-A\omega + B \frac{R}{L} = \frac{V_m}{L} \rightarrow (7)$$

$$\text{Ily, } B\omega + A \frac{R}{L} = 0 \rightarrow (8)$$

$$\text{from (8) } B = -\frac{AR}{\omega L} \rightarrow (9)$$

substitute eqn (9) in eqn (7)

$$-A\omega + \left[-\frac{AR}{\omega L} \cdot \frac{R}{L}\right] = \frac{V_m}{L}$$

$$-A \left[\omega + \frac{R^2}{\omega L^2} \right] = \frac{V_m}{L}$$

$$-A \left[\frac{(\omega L)^2 + R^2}{\omega L} \right] = V_m$$

$$A = \frac{V_m \omega L}{(\omega L)^2 + R^2} \rightarrow (10)$$

Substitute eqn (10) in eqn (9)

$$B = \frac{-(-V_m \omega L)}{R^2 + (\omega L)^2} \times \frac{R}{\omega L}$$

$$B = \frac{V_m R}{R^2 + (\omega L)^2} \rightarrow (11)$$

from eqn (4)

$$i_{PI} = \frac{V_m \omega L}{(\omega L)^2 + R^2} \cos(\omega t + \phi) + \frac{V_m R}{R^2 + (\omega L)^2} \sin(\omega t + \phi) \rightarrow (12)$$

$$\text{take, } \frac{V_m R}{R^2 + (\omega L)^2} = K_1 \cos \theta \rightarrow (13)$$

$$\tan \theta = \frac{-\omega L}{R}$$

$$\frac{-V_m \omega L}{R^2 + (\omega L)^2} = K_1 \sin \theta \rightarrow (14)$$

$$\theta = \tan^{-1} \left(\frac{-\omega L}{R} \right)$$

Solve eqn (13) & (14) & add

$$K_1^2 (\cos^2 \theta + \sin^2 \theta) = \frac{V_m^2 (R^2 + (\omega L)^2)}{(R^2 + (\omega L)^2)^2}$$

$$K_1 = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \rightarrow (15)$$

eqn (12) becomes

$$i_{PI} = K_1 \sin \theta \cos(\omega t + \phi) + K_1 \cos \theta \sin(\omega t + \phi)$$

$$i_{PI} = K_1 [\sin(\omega t + \phi + \theta)] \rightarrow (16)$$

Substitute K_1 in eqn (16)

$$i_{PI} = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \left[\sin(\omega t + \phi - \tan^{-1} \left(\frac{\omega L}{R} \right)) \right] \rightarrow (17)$$

The complete solution is

$$i(t) = i_{CF} + i_{PI}$$

$$i(t) = K e^{-\frac{R}{L}t} + \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \left[\sin(\omega t + \theta - \tan^{-1} \frac{\omega L}{R}) \right]$$

To find K , use initial condition, i.e., $t=0$, $i=0$

$$K = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \left[\sin(\omega t + \theta - \tan^{-1} \left(\frac{\omega L}{R} \right)) \right]_{t=0}$$

1. Frequency 50 Hz, 400 V (peak value) sinusoidal voltage is applied at $t=0$ to a series R-L circuit having resistance 5Ω and inductance 0.2H . Obtain an expression of current at any instant 't'. Calculate the value of the transient current after 0.01 sec switching ON.

sol. Given data,

$$\text{frequency} = 50 \text{ Hz}$$

$$\text{voltage peak value} = 400 \text{ V}$$

$$R = 5\Omega$$

$$L = 0.2 \text{ H}$$

$$\text{Sinusoidal voltage } (V) = V \sin \omega t$$

$$= 400 \sin 2\pi f t$$

$$= 400 \sin (2 \times 3.14 \times 50) t$$

$$(V) = 400 \sin 314 t$$

for series RL circuit in AC

$$Ri + L \frac{di}{dt} = 400 \sin 314 t$$

$$\frac{di}{dt} + \frac{R}{L} i = \frac{400}{0.2} \sin 314 t$$

$$\frac{di}{dt} + \frac{5}{0.2} i = 2000 \sin 314 t$$

$$\frac{di}{dt} + 25 i = 2000 \sin 314 t \rightarrow (1)$$

The solution is

$$i(t) = i_{CF} + i_{PF}$$

$$i(t) = K e^{-25t} + \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \sin \left(314t - \tan^{-1} \frac{\omega L}{R} \right)$$

$$i(t) = K e^{-25t} + \frac{400}{\sqrt{5^2 + (2\pi \times 0.2 \times 50)^2}} \sin (314t - \tan^{-1} 12.56)$$

$$i(t) = K e^{-25t} + 6.34 \sin (314t - 85.4) \rightarrow (2)$$

To find k , use initial conditions

i.e., $t=0, i=0$

$$0 = ke^0 + 6.34 \sin(314(0) - 85.4)$$

$$= k + 6.34 \sin(-85.4)$$

$$= k + 6.34(-0.997)$$

$$= k - 6.320$$

$$\boxed{k = 6.320}$$

Substitute in eqn (2)

$$i(t) = 6.32 e^{-25t} + 6.34 \sin(314t - 85.4)$$

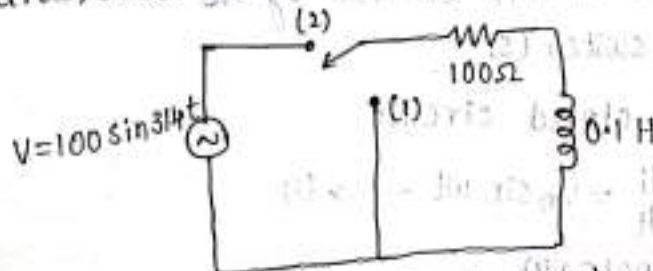
Transient current:

$$i(t) = 6.32 e^{-25 \times 0.01}$$

$$i(t) = 4.92 \text{ amp}$$

$$\boxed{i_t = 4.92 \text{ A}}$$

2. Obtain the current and t_{70} , if AC voltage be is applied when the switch 's' is moved to position 2 from position 1, at $t=0$. Assume steady state current of 1A in the RL circuit, when the switch was at position 1.



Sol: Given initial conditions $i = 1 \text{ A}, t = 0$

sinusoidal voltage $(V) = 100 \sin 314t$

for R-L series circuit

$$Ri + L \frac{di}{dt} = 100 \sin 314t \rightarrow (1)$$

$$\frac{di}{dt} + \frac{R}{L} i = \frac{100}{L} \sin 314t$$

$$\frac{di}{dt} + 1000i = 1000 \sin 314t$$

The solution is $i(t) = i_{CF} + PI$

$$i(t) = K e^{-Pt} + \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \sin(314t - \tan^{-1} \frac{\omega L}{R})$$

$$i(t) = K e^{-1000t} + 0.9541 \sin(314t - 17.432) \rightarrow (2)$$

To find K:

use initial conditions $t=0, i=1A$

Substitute these values in eqn (2)

$$1 = K e^{-1000(0)} + 0.9541 \sin(-17.432)$$

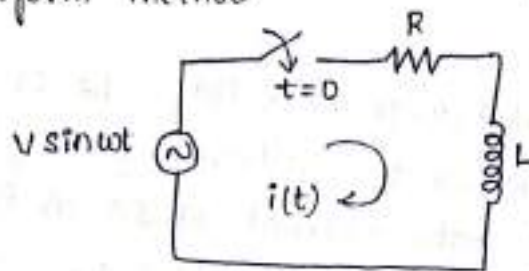
$$1 = K - 0.286$$

$$K = 1.286$$

$$\therefore i(t) = 1.3 e^{-1000t} + 0.9541 \sin(314t - 17.432)$$

Method 4:

AC transient analysis of sound RL circuit using Laplace transform method.



Consider series RL circuit excited by AC source $V_m \sin \omega t$ through switch (S)

Apply KVL to the closed circuit

$$iR + L \frac{di}{dt} = V_m \sin \omega t \rightarrow (1)$$

Apply Laplace transform

$$I(s)R + L[sI(s) - I(0)] = V_m \frac{\omega}{s^2 + \omega^2} \rightarrow (2)$$

$$I(s)[R + sL] = \frac{V_m \omega}{s^2 + \omega^2}$$

$$I(s) = \frac{V_m \omega}{L(s + \frac{R}{L})(s^2 + \omega^2)}$$

$$I(s) = \frac{(V_m \omega / L)}{(s + \frac{R}{L})(s^2 + \omega^2)} \rightarrow (3)$$

Using Partial Fractions

$$\frac{V_m \omega / L}{(s + \frac{R}{L})(s^2 + \omega^2)} = \frac{A}{s + \frac{R}{L}} + \frac{B}{s^2 + \omega^2} \rightarrow (4)$$

$$A = \frac{\frac{V_m \omega}{L}}{s^2 + \omega^2} \Big|_{s = -\frac{R}{L}}$$

$$= \frac{(\frac{V_m \omega}{L})}{(\frac{R}{L})^2 + \omega^2}$$

$$= \frac{V_m \omega L}{R^2 + \omega^2 L^2}$$

$$A = \frac{V_m \omega L}{R^2 + (\omega L)^2}$$

Cross multiplying eqn (4)

$$\frac{V_m \omega}{L} = A(s^2 + \omega^2) + (Bs + C)(s + \frac{R}{L})$$

$$\frac{V_m \omega}{L} = As^2 + A\omega^2 + Bs^2 + Bs\frac{R}{L} + Cs + \frac{CR}{L}$$

s^2 terms

$$A + B = 0 \rightarrow (6)$$

$$B = -A = \frac{-V_m \omega L}{R^2 + (\omega L)^2}$$

Comparing 's' terms

$$\frac{BR}{L} + C = 0 \rightarrow (7)$$

$$C = -\frac{BR}{L}$$

$$= \frac{V_m \omega R L}{L [R^2 + (\omega L)^2]}$$

$$C = \frac{V_m \omega R}{R^2 + (\omega L)^2}$$

Comparing constants

$$\frac{CR}{L} + A\omega^2 = \frac{V_m \omega}{L}$$

$$\frac{CR}{L} + \frac{V_m \omega L}{R^2 + (\omega L)^2} = \frac{V_m \omega}{L}$$

$$\frac{CR}{L} = \frac{V_m \omega [R^2 + (\omega L)^2 - L]}{L [R^2 + (\omega L)^2]}$$

Substitute A, B & C values in eqn (4)

$$I(s) = \frac{\left(\frac{V_m \omega L}{R^2 + (\omega L)^2}\right)}{s + \frac{R}{L}} + \frac{\frac{-V_m \omega L}{R^2 + (\omega L)^2} s + \frac{V_m \omega R}{R^2 + (\omega L)^2}}{s^2 + \omega^2}$$

$$= \frac{\left(\frac{V_m \omega L}{R^2 + (\omega L)^2}\right)}{s + \frac{R}{L}} - \frac{V_m \omega L}{R^2 + (\omega L)^2} \times \frac{s}{s^2 + \omega^2} + \frac{V_m R}{R^2 + (\omega L)^2} \times \frac{\omega}{s^2 + \omega^2} \rightarrow (5)$$

Apply inverse Laplace

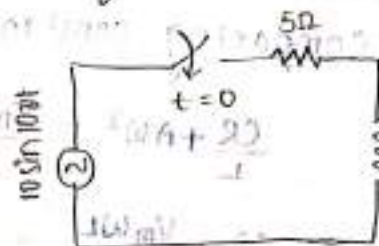
$$i(t) = \frac{V_m \omega L}{R^2 + (\omega L)^2} e^{-\frac{R}{L}t} - \frac{V_m \omega L}{R^2 + (\omega L)^2} \cos \omega t + \frac{V_m R}{R^2 + (\omega L)^2} \sin \omega t \rightarrow (6)$$

$$= \frac{V_m \omega L}{R^2 + (\omega L)^2} e^{-\frac{R}{L}t} - \frac{V_m}{R^2 + (\omega L)^2} \left[\omega L \cos \omega t - R \sin \omega t \right] \rightarrow (7)$$

$$= \frac{V_m \omega L}{R^2 + (\omega L)^2} e^{-\frac{R}{L}t} - \frac{V_m}{R^2 + (\omega L)^2} \left[\sqrt{R^2 + (\omega L)^2} \sin \phi \cos \omega t - \sqrt{R^2 + (\omega L)^2} \cos \phi \sin \omega t \right]$$

$$= \frac{V_m \omega L}{R^2 + (\omega L)^2} e^{-\frac{R}{L}t} - \frac{V_m}{R^2 + (\omega L)^2} \left[\sin(\phi - \omega t) \right]$$

1. A sinusoidal voltage $10 \sin 100t$ is applied to a series R-L circuit having $R = 5\Omega$ & $L = 10H$. Then find current flowing through the conductor after switch is closed at $t = 0$. Using Laplace transform method.



Initial condition, at $t = 0$, $i = 0 \rightarrow (1)$

Apply KVL to the closed circuit

$$5i + 10 \frac{di}{dt} = 10 \sin 100t \rightarrow (2)$$

Apply Laplace transform,

$$5I(s) + 10[sI(s) - I(0)] = 10 \sin 100t \rightarrow (3)$$

$$5I(s) + 10[sI(s) - 0] = 10 \frac{\omega}{s^2 + \omega^2}$$

$$I(s) [5 + 10s] = 10 \frac{\omega}{s^2 + \omega^2}$$

$$I(s) = \frac{10\omega}{(10s+5)(s^2+\omega^2)}$$

$$= \frac{10\omega}{L(s+\frac{R}{L})(s^2+\omega^2)}$$

$$= \frac{10\omega/L}{(s+R/L)(s^2+\omega^2)}$$

Using partial fractions

$$\frac{10\omega/L}{(s+\frac{R}{L})(s^2+\omega^2)} = \frac{A}{s+\frac{R}{L}} + \frac{B}{s^2+\omega^2}$$

$$A = \frac{10\omega/L}{s^2+\omega^2} \Big|_{s=-\frac{R}{L}}$$

$$= \frac{10\omega}{10}$$

$$\left(-\frac{R}{L}\right)^2 + \omega^2$$

$$= \frac{\omega}{\frac{5^2}{10^2} + 100}$$

$$\frac{0.01}{0.01} = \frac{5^2}{10^2} + 100$$

$$= \frac{\omega \times 100}{25 + 100^2 \times 100}$$

$$\frac{0.01}{0.01} = \frac{25 + 100^2 \times 100}{100 \times 100}$$

$$0.01 \times 2 = \frac{100 \times 100}{25 + (100)^2 \times 100}$$

$$0.01 \times 2 = \frac{100 \times 100}{25 + (100)^2 \times 100}$$

$$0.01 \times 2 = \frac{100 \times 100}{25 + (100)^2 \times 100}$$

$$A = 0.009$$

Now, $\frac{Bs+C}{s^2+\omega^2} = \frac{10\omega/L}{(s+R/L)(s^2+\omega^2)}$

$$\frac{(10\omega/L)}{(s+R/L)(s^2+\omega^2)} = \frac{A(s^2+\omega^2)}{(s+R/L)} + \frac{(Bs+C)(s+R/L)}{(s^2+\omega^2)}$$

$$\frac{(10 \times 100)}{(s+\frac{R}{L})(s^2+\omega^2)} = \frac{A(s^2+\omega^2) + (Bs+C)(s+\frac{R}{L})}{(s+\frac{R}{L})(s^2+\omega^2)}$$

$$100 = As^2 + Aw^2 + Bs^2 + Bs \frac{R}{L} + Cs + \frac{CR}{L}$$

Comparing "s" terms

$$A+B=0 \rightarrow (5)$$

$$B = -A$$

$$B = -0.009$$

Comparing 's' terms

$$\frac{BR}{L} + C = 0$$

$$C = -\frac{BR}{L}$$

$$C = -\frac{(0.009) 5}{10}$$

$$\boxed{C = 0.0045}$$

Substitute A, B & C values in eqn (4)

$$I(s) = \frac{0.009}{s+1/2} + \frac{-0.009s + 0.0045}{s^2 + 10000}$$

$$I(s) = \frac{0.009}{s+2.5} - \frac{0.009s}{s^2+100^2} + \frac{0.0045}{s^2+100^2}$$

Apply inverse Laplace

$$i(t) = 0.009 e^{-1/2 t} - 0.009 \cos 100t + \frac{0.0045}{s^2+100^2} \times \frac{100}{100}$$

$$= 0.009 e^{-1/2 t} - 0.009 \cos \omega t + \frac{0.0045}{100} \times \frac{100}{s^2+100^2}$$

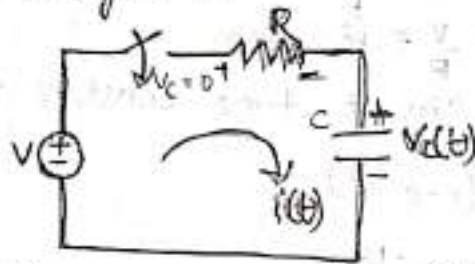
$$= 0.009 e^{-1/2 t} - 0.009 \cos \omega t + 0.0045 \sin 100t$$

$$i(t) = 0.009 e^{-1/2 t} - 0.009 \cos \omega t + 0.0045 \sin \omega t$$

* Series R-C circuit

Method 1:

DC transient analysis of series R-C circuit using DE method:



Initial conditions, $t=0, V_c=0 \rightarrow (1)$

Now apply KVL, $-V + i(t)R + V_c(t) = 0$

$$V - RI(t) - V_c(t) = 0$$

$$RI(t) + V_c(t) = V \rightarrow (2)$$

$$I(t) = \frac{V - V_c(t)}{R}$$

$$\text{Now, } i(t) = C \frac{dV_c(t)}{dt}$$

$$RC \frac{dV_c(t)}{dt} + V_c(t) = V \rightarrow (3)$$

$$\frac{dV_c(t)}{dt} + \frac{1}{RC} V_c(t) = \frac{V}{RC} \rightarrow (4)$$

It is non-homogeneous eqn

$$\frac{dV_c(t)}{dt} + PV_c(t) = Q$$

The solution is

$$V_c(t) = ke^{-\frac{t}{RC}} + \frac{V/RC}{1/RC}$$

$$V_c(t) = ke^{-\frac{t}{RC}} + V \rightarrow (5)$$

To find k, use initial conditions

$$\text{i.e., } t=0, V_c=0$$

$$\boxed{k = -V}$$

substitute $k = -V$ in eqn (5)

$$V_c(t) = -Ve^{-\frac{t}{RC}} + V$$

$$V_c(t) = -V \left[1 - e^{-\frac{t}{RC}} \right] \text{ Volts}$$

$$i(t) = C \frac{dv}{dt}$$

$$= C(-V e^{-\frac{t}{RC}}) \times \frac{-1}{RC}$$

$$i(t) = \frac{V}{R} e^{-\frac{t}{RC}} \text{ amp.}$$

In case of series RC circuit time constant $\tau = RC$

Now, $V_c(t) = V(1 - e^{-t/\tau})$

$$i(t) = \frac{V}{R} (e^{-t/\tau})$$

The graph for $V_c(t)$ is

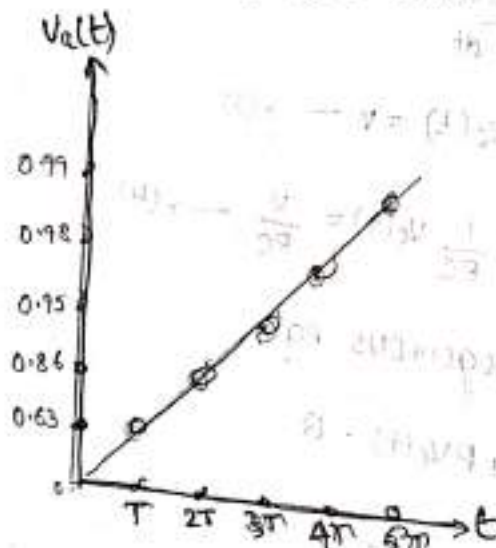
$$t = \tau, V_c(t) = 0.63V$$

$$t = 2\tau, V_c(t) = 0.86V$$

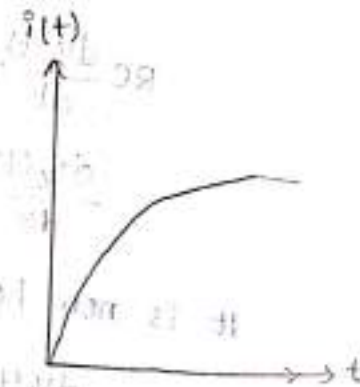
$$t = 3\tau, V_c(t) = 0.95V$$

$$t = 4\tau, V_c(t) = 0.98V$$

$$t = 5\tau, V_c(t) = 0.99V$$

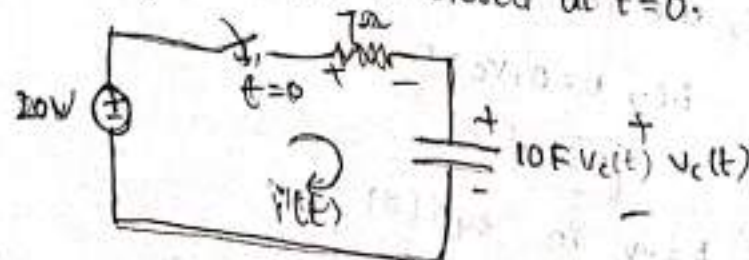


voltage wave form



current wave form

1. A 20V dc supply is applied to a series RC circuit with $R=7\Omega$ and $C=10F$. Find voltage and current through the capacitor after switch is closed at $t=0$.



Now apply KVL to the circuit is, ~~$-20 + 7i(t) + V_c(t) = 0$~~

$$-20 + 7i(t) + V_c(t) = 0$$

$$7i(t) + V_c(t) = 20 \rightarrow (2)$$

$$7C \frac{dV_c(t)}{dt} + V_c(t) = 20 \rightarrow (3)$$

$$70 \frac{dV_c(t)}{dt} + V_c(t) = 20$$

$$[C = 10F]$$

$$\frac{dV_c(t)}{dt} + \frac{1}{70} V_c(t) = \frac{20}{70} \rightarrow (4)$$

It is a non-homogeneous eqn

$$P = \frac{1}{70}, Q = \frac{20}{70}$$

Solution is $V_c(t) = k e^{-Pt} + \frac{Q}{P}$

$$= k e^{-\frac{1}{70}t} + \frac{20/70}{1/70}$$

$$V_c(t) = k e^{-\frac{1}{70}t} + 20 \rightarrow (5)$$

To find k:-

Use initial conditions $t=0, V_c=0$

$$k e^0 + 20 = 0$$

$$k = -20$$

Substitute 'k' in eqn (5)

$$V_c(t) = -20 e^{-\frac{1}{70}t} + 20$$

$$= 20 [1 - e^{-\frac{1}{70}t}]$$

differentiate the eqn

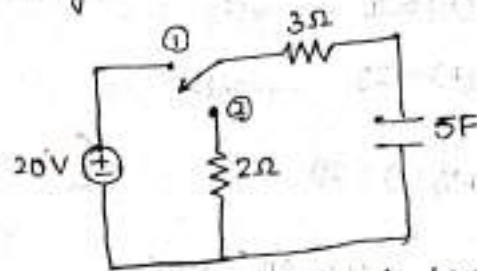
$$i(t) = C \cdot \frac{dV_c}{dt}$$

$$= C \left[-ve^{-\frac{t}{RC}} \right] \times \frac{-1}{RC}$$

$$= \frac{V}{R} \cdot e^{-\frac{t}{RC}}$$

$$i(t) = \frac{20}{7} e^{-\frac{1}{70}t} A$$

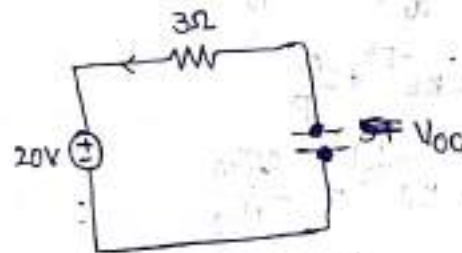
The circuit shown in fig: switch is at position 1 for long time. At $t=0$, switch is moved to position 2, find voltage and current through the capacitor.



Initially switch at position 1 for long time:

long time = steady state

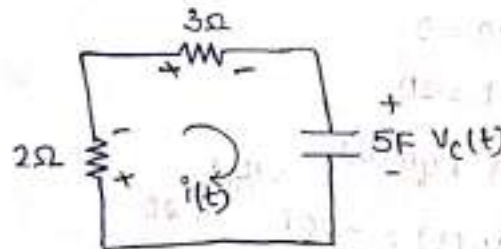
At steady state, capacitor acts as open circuit



$$V_{oc} - 20 = 0$$

$$V_{oc} = 20$$

Switch is moved to position 2 at $t=0$



Initial condition $t=0, V_0 = 20V \rightarrow (1)$

Apply KVL to the circuit

$$(+2+3)i(t) + V_c(t) = 0$$

$$+5 \left[C \frac{dV_c(t)}{dt} \right] + V_c(t) = 0$$

$$\frac{dV_c(t)}{dt} + \frac{1}{25} V_c(t) = 0 \rightarrow (2)$$

$$P = \frac{1}{25}$$

$$\text{Solution is } V_c(t) = k e^{-\frac{1}{25}t} \rightarrow (3)$$

To find k , use initial condition

$$20 = k e^0$$

$$k = 20$$

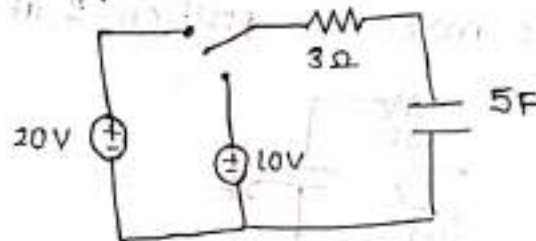
$$V_c(t) = 20 e^{-\frac{t}{25}} \text{ volts}$$

$$\text{current}(i) \Rightarrow i(t) = C \frac{dV}{dt}$$

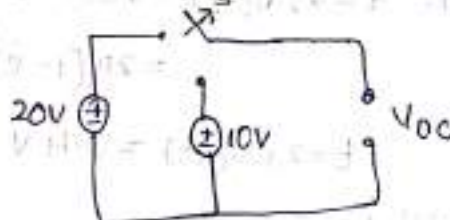
$$= 5 \times 20 \times \frac{-1}{25} e^{-t/25}$$

$$= -4 e^{-t/25} \text{ amps}$$

In the circuit shown in fig. switch is moved to position 1 at $t=0$ switch is moved to position 2. Then find the voltage and current through the capacitor.

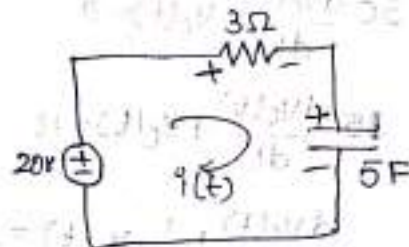


Initially switch is open for long time



$$V_{OC} = 0$$

Now, switch is moved to position 1 at $t=0$



Initial condition,

$$t=0, V_c = 0 \rightarrow (1)$$

Apply KVL for the circuit.

$$-20 + 3i(t) + V_c(t) = 0$$

$$20 - 3i(t) - V_c(t) = 0$$

$$20 - 3C \frac{dV_c(t)}{dt} - V_c(t) = 0$$

$$15 \frac{dV_c(t)}{dt} + V_c(t) = 20$$

$$\frac{dV_c(t)}{dt} + \frac{1}{15} V_c(t) = \frac{20}{15} \rightarrow (2)$$

Non-homogeneous eqn

$$P = \frac{1}{15}, Q = \frac{20}{15}$$

$$\text{Solution is } V_c(t) = K e^{-Pt} + \frac{Q}{P}$$

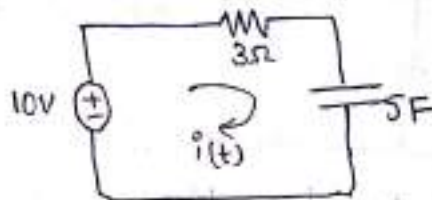
$$V_c(t) = K e^{-\frac{t}{15}} + 20 \rightarrow (3)$$

To find k use initial condition

$$K = -20$$

$$V_c(t) = -20 e^{-\frac{t}{15}} + 20 \rightarrow (4)$$

Now, switch is moved to position 2 at $t=2$ sec



$$\begin{aligned} \text{Initial conditions } t=2, V_c(t) &= 20(1 - e^{-\frac{t}{15}}) \\ &= 20(1 - e^{-\frac{2}{15}}) \end{aligned}$$

$$t=2, V_c(t) = 2.4 \text{ V} \rightarrow (5)$$

Apply KVL,

$$10 - 3i_c(t) - V_c(t) = 0$$

$$10 - 3C \frac{dV_c(t)}{dt} - V_c(t) = 0$$

$$15 \frac{dV_c(t)}{dt} + V_c(t) = 10$$

$$\frac{dV_c(t)}{dt} + \frac{1}{15} V_c(t) = \frac{10}{15}$$

Non-homogeneous eqn, $P = \frac{1}{15}, Q = \frac{10}{15}$

$$\text{Solution is } P = \frac{1}{15}, Q = \frac{10}{15}$$

$$V_c(t) = K e^{-\frac{1}{15}(t-2)} + 10$$

To find K, use initial conditions

$$2.4 = K e^{-\frac{1}{15}(0)} + 10$$

$$2.4 = K e^0 + 10$$

$$K = -7.6$$

$$\boxed{K = -7.6}$$

$$V_c(t) = -7.6 e^{-\frac{1}{15}(t-2)} + 10$$

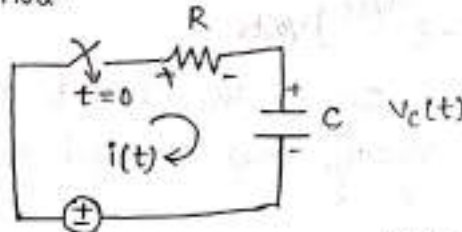
$$i_c(t) = C \frac{dV_c(t)}{dt}$$

$$= 5(-7.6) e^{-\frac{1}{15}(t-2)} \times \left(-\frac{1}{15}\right)$$

$$= 2.53 e^{-\frac{1}{15}(t-2)} \text{ amp.}$$

Method 2:

De transient analysis of series RC circuit using Laplace transform method.



Apply KVL to the circuit, $-V + i(t)R + V_c(t) = 0$

$$V = I(t)R + V_c(t)$$

$$I(t) = C \frac{dV_c(t)}{dt}$$

$$C \frac{dV_c(t)}{dt} R + V_c(t) = V \rightarrow (1)$$

Apply Laplace transform:

$$CR[sV_c(s) - V_c(0)] + V_c(s) = \frac{V}{s} \quad V_c(0) = 0$$

$$CRsV_c(s) + V_c(s) = \frac{V}{s}$$

$$V_c(s) = \frac{V/s}{CR(s + \frac{1}{CR})}$$

$$V_c(s) = \frac{V/CR}{s(s + \frac{1}{CR})} \rightarrow (2)$$

To find K, use initial conditions

$$2.4 = K e^{-\frac{1}{15}(0)} + 10$$

$$2.4 = K e^0 + 10$$

$$K = -7.6$$

$$K = -7.6$$

$$V_c(t) = -7.6 e^{-\frac{1}{15}(t-2)} + 10$$

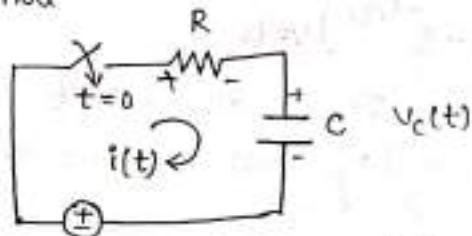
$$i_c(t) = C \frac{dV_c(t)}{dt}$$

$$= 5(-7.6) e^{-\frac{1}{15}(t-2)} \times \left(-\frac{1}{15}\right)$$

$$= 2.53 e^{-\frac{1}{15}(t-2)} \text{ amp.}$$

Method 2:

DC transient analysis of series RC circuit using Laplace transform method.



Apply KVL to the circuit, $-V + i(t)R + V_c(t) = 0$

$$V = I(t)R + V_c(t)$$

$$I(t) = C \frac{dV_c(t)}{dt}$$

$$C \frac{dV_c(t)}{dt} R + V_c(t) = V \rightarrow (1)$$

Apply Laplace transform.

$$CR[s V_c(s) - V_c(0)] + V_c(s) = \frac{V}{s} \quad V_c(0) = 0$$

$$CRs V_c(s) + V_c(s) = \frac{V}{s}$$

$$V_c(s) = \frac{V/s}{CR(s + \frac{1}{CR})}$$

$$V_c(s) = \frac{V/CR}{s(s + \frac{1}{CR})} \rightarrow (2)$$

Apply Laplace transform by partial fraction

$$\frac{V/CR}{s(s + \frac{1}{CR})} = \frac{A}{s} + \frac{B}{s + \frac{1}{CR}}$$

at $s=0$ $A = \frac{V/CR}{\frac{1}{CR}}$

$$A = V$$

By, $B = -V$

$$V_C(s) = \frac{V}{s} - \frac{V}{s + \frac{1}{RC}}$$

$$V_C(s) = V \left(\frac{1}{s} - \frac{1}{s + \frac{1}{RC}} \right) \rightarrow (3)$$

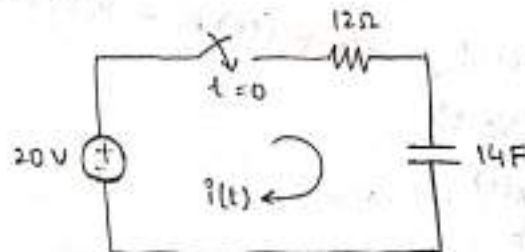
$$B = -V$$

Apply inverse Laplace transform

$$V_C(t) = V - Ve^{-t/RC}$$

$$V_C(t) = V [1 - e^{-t/RC}] \text{ volts}$$

1. A 20V is applied to series RC circuit having $R=12\Omega$ and $C=14F$. Find the voltage and current after switch is closed at $t=0$.



Initial conditions $t=0, V_C=0$

Apply KVL to the circuit

$$20 - i(t)12 - V_C(t) = 0$$

$$12 i_C(t) + V_C(t) = 20$$

$$i_C(t) = C \frac{dV_C(t)}{dt}$$

$$12C \frac{dV_C(t)}{dt} + V_C(t) = 20$$

$$168 \frac{dV_C(t)}{dt} + V_C(t) = 20 \rightarrow (1)$$

Apply Laplace transform

$$168 [sV_C(s) - V_C(0)] + V_C(s) = \frac{20}{s}$$

$$168 sV_c(s) + V_c(s) = \frac{20}{s}$$

$$V_c(s) = \frac{20}{s(168s+1)}$$

$$V_c(s) = \frac{20}{168s(s+\frac{1}{168})}$$

$$V_c(s) = \frac{20/168}{s(s+\frac{1}{168})}$$

Apply partial fractions

$$\frac{20/168}{s(s+\frac{1}{168})} = \frac{A}{s} + \frac{B}{s+\frac{1}{168}}$$

$$A = 20$$

$$\text{Hly, } B = -20$$

$$V_c(s) = \frac{20}{s} - \frac{20}{s+\frac{1}{168}}$$

$$\frac{A}{s} = \frac{\frac{20}{168}}{s(s+\frac{1}{168})} \Big|_{s=0}$$

$$A = 20$$

$$\frac{B}{s+\frac{1}{168}} = \frac{20/168}{s(s+\frac{1}{168})} \Big|_{s=-\frac{1}{168}}$$

$$B = -20$$

$$V_c(s) = 20 \left[\frac{1}{s} - \frac{1}{s+\frac{1}{168}} \right]$$

Apply inverse Laplace transform

$$V_c(t) = 20 \left[1 - e^{-\frac{t}{168}} \right] \text{ volts}$$

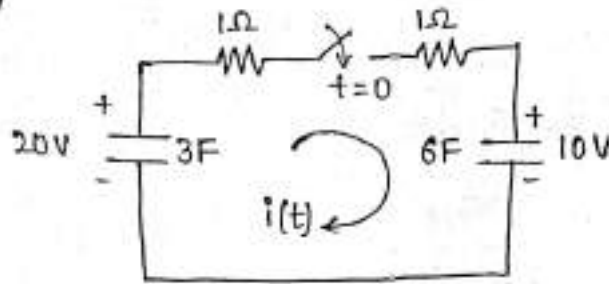
$$i(t) = C \frac{dV_c(t)}{dt}$$

$$= 14 \times 20 \frac{d}{dt} \left[1 - e^{-\frac{t}{168}} \right]$$

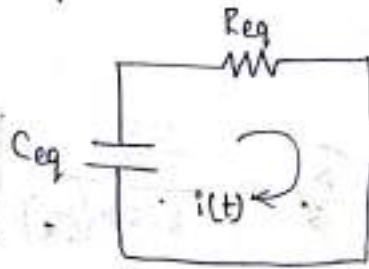
$$= 280 e^{-\frac{t}{168}} \left[\frac{-1}{168} \right]$$

$$i(t) = -1.666 e^{-\frac{t}{168}} \text{ amp}$$

2. Find $i(t)$ and $V(t)$ for the following figure in which capacitor initially charged to 20V and 6F capacitor is charged to 6V. Assume initial capacitor voltage is 9V.



Sol. By closing switch



$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}, \quad R_{eq} = R_1 + R_2$$

$$= \frac{3 \times 6}{3 + 6} = 2 \text{ F}$$

$$= 1 + 1 = 2 \Omega$$

$$V_{eq} = 20 - 10 = 10 \text{ V}$$

Initial conditions $t=0$, $V_c(0) = 9 \text{ V}$

Apply KVL to the circuit

$$i_R + V_c(t) = 0$$

$$C \frac{dV_c(t)}{dt} + R + V_c(t) = 0$$

$$2 \frac{dV_c(t)}{dt} + R + V_c(t) = 0$$

Apply Laplace transform.

$$4[sV_c(s) - V_c(0)] + V_c(s) = 0$$

$$4sV_c(s) - 4 \times 9 + V_c(s) = 0$$

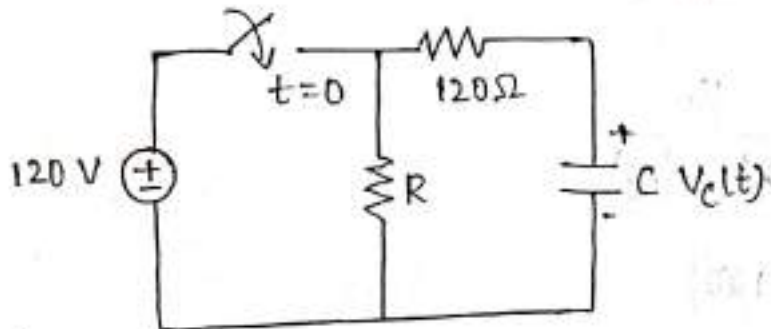
$$V_c(s) = \frac{36}{4s + 1}$$

$$V_c(s) = \frac{9}{s + \frac{1}{4}}$$

inverse Laplace Transform

$$V_c(t) = 9e^{-\frac{t}{4}} \text{ volt}$$

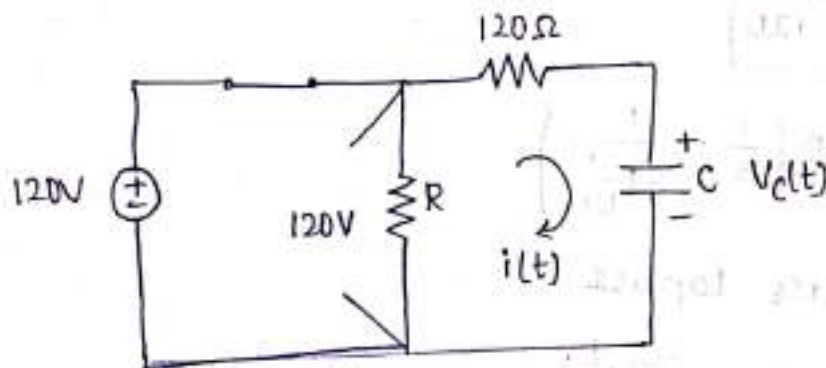
3. In the circuit shown in the fig. the switch is open for long time and closing the switch at $t=0$, the capacitor voltage rises to 70V in 10ms. After steady state is reached and the switch is opened again and capacitor voltage is 90V in 0.5s. Find the values of R & C.



sol: Initial conditions

$$t=0, V_C(0) = 0 \rightarrow (1)$$

Now, switch is closed at $t=0$



Apply KVL to the circuit

$$120 - i(t) \cdot 120 - V_C(t) = 0$$

$$i(t) \cdot 120 + V_C(t) = 120$$

$$C \frac{dV_C(t)}{dt} \cdot 120 + V_C(t) = 120 \rightarrow (2)$$

Apply Laplace transform

$$120C [s V_C(s) - V_C(0)] + V_C(s) = \frac{120}{s}$$

$$V_C(s) [Cs(120) + 1] = \frac{120}{s}$$

$$V_c(s) = \frac{120/s}{cs(120)+1}$$

$$= \frac{120}{120cs(s + \frac{1}{120c})}$$

$$= \frac{1/c}{s(s + \frac{1}{120c})} \rightarrow (3)$$

Partial fractions,

$$\frac{1/c}{s(s + \frac{1}{120c})} = \frac{A}{s} + \frac{B}{s + \frac{1}{120c}}$$

$$\text{at } s=0, A = \frac{1/c}{\frac{1}{120c}}$$

$$\boxed{A = 120}$$

$$B = \frac{1/c}{s} \Big|_{s = -\frac{1}{120c}}$$

$$\boxed{B = -120}$$

$$V_c(s) = 120 \left(\frac{1}{s} - \frac{1}{s + \frac{1}{120c}} \right)$$

Apply inverse Laplace

$$V_c(t) = 120 (1 - e^{-\frac{1}{120c}t}) \rightarrow (4)$$

from problem, $t = 10 \text{ msec} = 10^{-2} \text{ sec}$

$$V_c = 10 \text{ V}$$

from eqn (4)

$$10 = 120 (1 - e^{-\frac{10^{-2}}{120c}})$$

$$-50 = -120 e^{-\frac{10^{-2}}{120c}}$$

$$0.416 = e^{-\frac{10^{-2}}{120c}}$$

Apply ln on both sides

$$\ln(0.416) = \ln\left(e^{-\frac{10^3}{120C}}\right)$$

$$-0.87 = -\frac{10^3}{120C}$$

$$120C \times 0.87 = 10^3$$

$$104.4C = 10^3$$

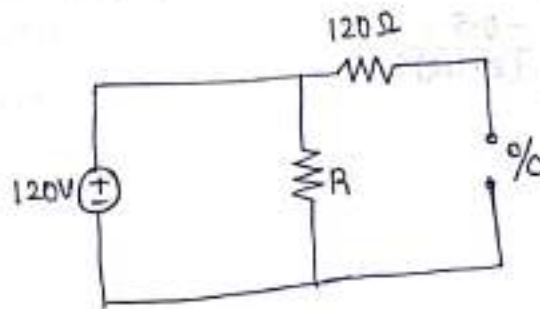
$$C = \frac{10^3}{104.4}$$

$$= 9.57 \times 10^{-3} \times 10^{-3}$$

$$= 95.7 \times 10^{-6} \text{ F}$$

$$C = 95.7 \mu\text{F}$$

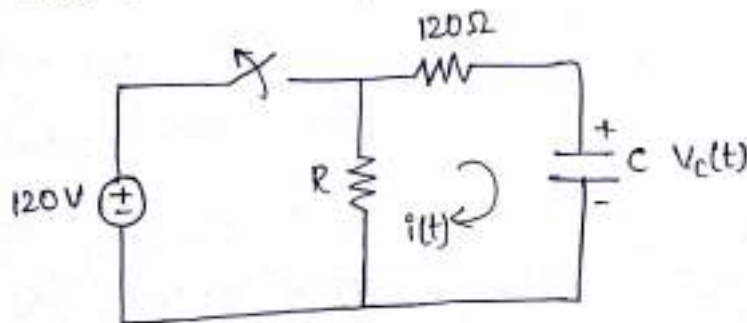
After steady state is reached the switch is again opened then the circuit is



Initial conditions,

$$t = 0, V_C(0) = 120 \text{ V} \longrightarrow (5)$$

Now switch is open



Apply KVL to the circuit

$$(R+120) i(t) + V_C(t) = 0$$

$$(R+120) \left[C \frac{dV_C(t)}{dt} \right] + V_C(t) = 0 \longrightarrow (6)$$

Apply Laplace transform

$$(R+120) \times C [sV_c(s) - V_c(0)] + V_c(s) = 0$$

$$V_c(s) [(R+120)Cs + 1] = 120 \times (R+120)C$$

$$V_c(s) = \frac{120 \times (R+120)C}{(120+R)C \left[s + \frac{1}{(R+120)C} \right]}$$

$$V_c(s) = \frac{120}{s + \frac{1}{(R+120)C}} \rightarrow (7)$$

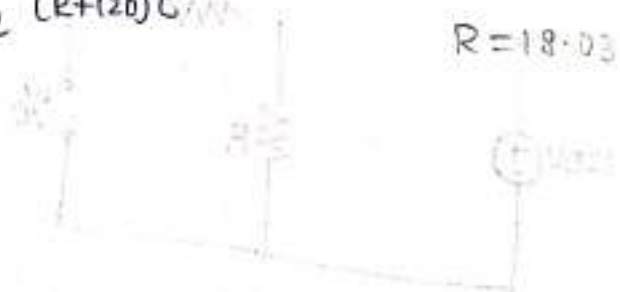
Apply inverse Laplace transform

$$V_c(t) = 120 e^{-\frac{t}{(R+120)C}} \rightarrow (8)$$

from problem $V_c = 90V$, $t = 0.5 \text{ sec}$

$$90 = 120 e^{-\frac{0.5}{(R+120)C}}$$

$$R = 18.03 \text{ k}\Omega$$



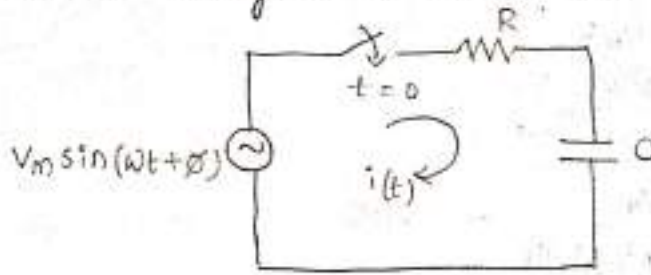
When switch is open $V_{AB} = 120V$, $C = 1 \mu F$

When switch is closed



Method 3:

AC transient analysis of series R-C circuit by DE method.



Apply KVL to the circuit,

$$V_m \sin(\omega t + \phi) - i(t)R - \frac{1}{C} \int i dt = 0$$

$$V_m \sin(\omega t + \phi) = i(t)R + \frac{1}{C} \int i dt \rightarrow (1)$$

differentiate eqn (1) wrt 't'

$$V_m \omega \cos(\omega t + \phi) = \frac{di}{dt} R + \frac{1}{C} i$$

$$\frac{di}{dt} + \frac{1}{RC} i = \frac{V_m \omega}{R} \cos(\omega t + \phi) \rightarrow (2)$$

It is like non-homogeneous eqn.

$$\text{Solution is } i(t) = i_{CF} + i_{PI} \rightarrow (3)$$

$$i_{CF} = k e^{-Pt} = k e^{-\frac{1}{RC}t} \rightarrow (4)$$

consider,

$$i_{PI} = A \cos(\omega t + \phi) + B \sin(\omega t + \phi) \rightarrow (5)$$

differentiate wrt 't'

$$\frac{di}{dt} = -A\omega \sin(\omega t + \phi) + B\omega \cos(\omega t + \phi) \rightarrow (6)$$

Substitute eqn (5) & (6) in eqn (2)

$$-A\omega \sin(\omega t + \phi) + B\omega \cos(\omega t + \phi) + \frac{1}{RC} [A \cos(\omega t + \phi) + B \sin(\omega t + \phi)] = \frac{V_m \omega}{R} \cos(\omega t + \phi) \rightarrow (7)$$

Compare, $\sin(\omega t + \phi)$ terms

$$-A\omega + \frac{B}{RC} = 0 \rightarrow (8)$$

Compare $\cos(\omega t + \phi)$ terms

$$B\omega + \frac{A}{RC} = \frac{V_m \omega}{R} \rightarrow (9)$$

$$\text{from eqn (8)} A = \frac{B}{\omega RC} \rightarrow (10)$$

substitute eqn (10) in eqn (9)

$$B\omega + \frac{B}{\omega R^2 C^2} = \frac{V_m \omega}{R}$$

$$B = \frac{V_m \omega^2 L^2 R}{\omega^2 R^2 C^2 + 1}$$

$$B = \frac{V_m R}{R^2 + \left(\frac{1}{\omega C}\right)^2} \quad \text{--- (11)}$$

$$A = \frac{V_m}{\omega C \left(R^2 + \left(\frac{1}{\omega C}\right)^2\right)} \quad \text{--- (12)}$$

Substitute eqn(11) & eqn(12) in eqn(5)

$$i_{PI} = \frac{\left(\frac{V_m}{\omega C}\right)}{R^2 + \left(\frac{1}{\omega C}\right)^2} \cos(\omega t + \phi) + \frac{V_m R}{R^2 + \left(\frac{1}{\omega C}\right)^2} \sin(\omega t + \phi) \quad \text{--- (13)}$$

$$\frac{k_1 \sin \theta}{k_1 \cos \theta}$$

$$= k_1 \sin \theta \cos(\omega t + \phi) + k_1 \cos \theta \sin(\omega t + \phi)$$

$$k_1^2 (\sin^2 \theta + \cos^2 \theta) = \frac{\left(\left(\frac{1}{\omega C}\right)^2 + R^2\right) V_m^2}{\left(R^2 + \left(\frac{1}{\omega C}\right)^2\right)^2}$$

$$k_1 = \frac{V_m}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$$

$$\tan \theta = \frac{1}{\omega C R}$$

$$\theta = \tan^{-1}\left(\frac{1}{\omega C R}\right)$$

$$i_{PI} = k_1 \sin(\omega t + \phi + \theta)$$

$$= \frac{V_m}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} \sin\left(\omega t + \phi + \tan^{-1}\left(\frac{1}{\omega C R}\right)\right) \quad \text{--- (14)}$$

Substitute eqn(14) & eqn(4) in eqn(3)

$$i(t) = k e^{-\frac{t}{RC}} + \frac{V_m}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} \sin\left(\omega t + \phi + \tan^{-1}\left(\frac{1}{\omega C R}\right)\right) \quad \text{--- (15)}$$

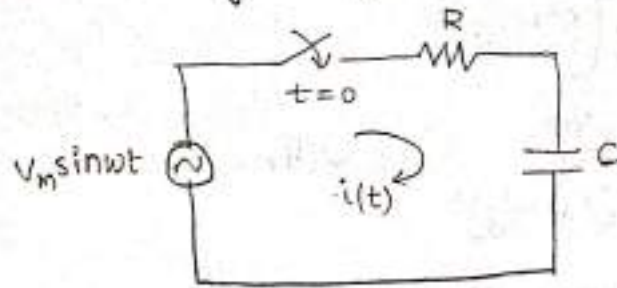
to find K, use initial conditions $t=0, i(t)=0$

$$0 = k e^0 + \frac{V_m}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} \sin\left[\omega t + \phi + \tan^{-1}\left(\frac{1}{\omega C R}\right)\right]$$

$$K = \frac{-V_m}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} \sin\left[\omega t + \phi - \tan^{-1}\left(\frac{1}{\omega C R}\right)\right]$$

Method 4:

AC Transient analysis of series RC circuit by LT method:



Initial conditions.

$$t=0, i=0 \rightarrow (1)$$

Apply KVL to the circuit

$$v_m \sin wt = i(t)R + \frac{1}{C} \int i dt \rightarrow (2)$$

Apply Laplace transform

$$\frac{v_m \omega}{s^2 + \omega^2} = I(s)R + \frac{1}{C} \left(\frac{I(s)}{s} - \frac{I(0)}{0} \right)$$

$$I(s) \left[R + \frac{1}{Cs} \right] = \frac{v_m \omega}{s^2 + \omega^2}$$

$$I(s) = \frac{v_m \omega}{\left(R + \frac{1}{Cs} \right) (s^2 + \omega^2)}$$

$$= \frac{v_m \omega}{\frac{R}{s} \left(s + \frac{1}{RC} \right) (s^2 + \omega^2)}$$

$$= \frac{\left(\frac{v_m \omega s}{R} \right)}{\left(s + \frac{1}{RC} \right) (s^2 + \omega^2)} \rightarrow (3)$$

Apply partial fractions

$$\frac{\left(\frac{v_m \omega s}{R} \right)}{\left(s + \frac{1}{RC} \right) (s^2 + \omega^2)} = \frac{A}{s + \frac{1}{RC}} + \frac{Bs + D}{s^2 + \omega^2} \rightarrow (4)$$

$$A = \frac{v_m \omega s}{\left(s^2 + \omega^2 \right)} \bigg|_{s = -\frac{1}{RC}} = \frac{\frac{v_m \omega}{R} \times -\frac{1}{RC}}{\left(\frac{1}{RC} \right)^2 + \omega^2}$$

$$= \frac{\left(-\frac{v_m \omega}{R^2 C} \right)}{\omega^2 \left[1 + \frac{1}{R^2 C^2 \omega^2} \right]}$$

$$A = \frac{\left(-\frac{V_m \omega}{R^2 C}\right)}{\frac{\omega^2}{R^2} \left[R^2 + \left(\frac{1}{\omega C}\right)^2\right]}$$

$$A = \frac{-\frac{V_m}{\omega C}}{R^2 + \left(\frac{1}{\omega C}\right)^2} \rightarrow (5)$$

To find B & D, cross multiply eqn (4)

$$\frac{V_m \omega S}{R} = A(s^2 + \omega^2) + (Bs + D)\left(s + \frac{1}{RC}\right)$$

$$\frac{V_m \omega S}{R} = As^2 + A\omega^2 + Bs^2 + \frac{BS}{RC} + Ds + \frac{D}{RC} \rightarrow (6)$$

Compare s^2 terms,

$$A + B = 0$$

$$B = -A$$

$$B = \frac{\left(\frac{V_m}{\omega C}\right)}{R^2 + \left(\frac{1}{\omega C}\right)^2} \rightarrow (7)$$

Compare constant terms

$$A\omega^2 + \frac{D}{RC} = 0$$

$$\frac{D}{RC} = -A\omega^2$$

$$D = \frac{V_m \omega R}{R^2 + \left(\frac{1}{\omega C}\right)^2} \rightarrow (8)$$

Substitute A, B & D values in eqn (3)

$$I(s) = \frac{\left(-\frac{V_m}{\omega C}\right)}{\left[R^2 + \left(\frac{1}{\omega C}\right)^2\right]\left(s + \frac{1}{RC}\right)} + \frac{\frac{V_m s}{\omega C} + \frac{V_m \omega R}{R^2 + \left(\frac{1}{\omega C}\right)^2}}{\left[R^2 + \left(\frac{1}{\omega C}\right)^2\right](s^2 + \omega^2)} \rightarrow (9)$$

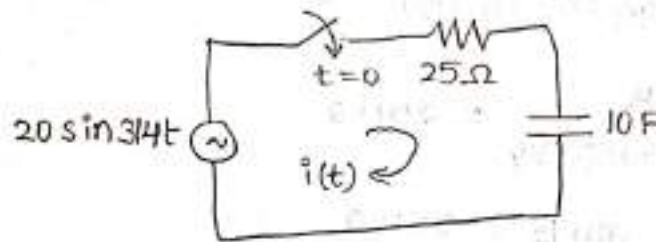
Apply Inverse Laplace transform.

$$i(t) = \frac{\left(-\frac{V_m}{\omega C}\right)}{R^2 + \left(\frac{1}{\omega C}\right)^2} e^{-\frac{t}{RC}} + \frac{\left(\frac{V_m}{\omega C}\right)}{R^2 + \left(\frac{1}{\omega C}\right)^2} \cos \omega t + \frac{\frac{V_m R}{R^2 + \left(\frac{1}{\omega C}\right)^2}}{R^2 + \left(\frac{1}{\omega C}\right)^2} \sin \omega t$$

$$\left(\frac{1}{\omega^2 R^2 + 1}\right)^{1/2} \omega$$

1. A sinusoidal voltage $20 \sin 314t$ having $R = 25 \Omega$ & $C = 10F$. Calculate the current flowing through the capacitor after switch is closed at $t = 0$ using
(a) DE method.
(b) Laplace transform method.

Sol:



(a) DE method:

Using initial conditions

$$\text{at } t=0, i=0 \rightarrow (1)$$

Apply KVL to the circuit

$$20 \sin 314t = 25 i(t) + \frac{1}{10} \int i dt \rightarrow (2)$$

differentiate wrt 't'

$$6280 \cos wt = \frac{di}{dt} 25 + \frac{1}{10} i$$

$$\frac{di}{dt} + \frac{1}{250} i = \frac{6280}{25} \cos wt \rightarrow (3)$$

It is like non-homogeneous eqn

$$\text{solution is } i(t) = i_{CF} + i_{PI} \rightarrow (4)$$

$$i_{CF} = K e^{-\frac{t}{250}} \rightarrow (5)$$

Consider, $i_{PI} = A \cos wt + B \sin wt \rightarrow (6)$

differentiate wrt 't'

$$\frac{di}{dt} = -A w \sin wt + B w \cos wt \rightarrow (7)$$

$$\frac{di}{dt} = -A 314 \sin 314t + B 314 \cos 314t \rightarrow (8)$$

Substitute eqn (6) & (8) in eqn (3)

$$-A 314 \sin 314t + B 314 \cos 314t + \frac{1}{250} (A \cos wt + B \sin wt) = \frac{6280}{25} \cos wt \rightarrow (9)$$

Compare $\sin 314t$ terms.

$$-314 A + \frac{B}{250} = 0 \rightarrow (10)$$

Compare $\cos 314t$ terms

$$314B + \frac{A}{250} = \frac{6280}{25} \rightarrow (10)$$

from eqn (9) $A = \frac{B}{250 \times 314} \rightarrow (11)$

Substitute eqn (11) in (10)

$$314B + \frac{B}{250 \times 250 \times 314} = 251.2$$

$$314B = 251.2$$

$$\boxed{B = 0.8} \rightarrow (12)$$

$$A = \frac{0.8}{250 \times 314}$$

$$\boxed{A = 1.019 \times 10^{-5}} \rightarrow (13)$$

Substitute eqn (12) & (13) in eqn (6)

$$i_{PI} = (1.019 \times 10^{-5}) \cos 314t + 0.8 \sin 314t \rightarrow (14)$$

Substitute eqn (5) & eqn (14) in eqn (4)

$$i(t) = K e^{-\frac{t}{250}} + 0.00001 \cos 314t + 0.8 \sin 314t$$

To find K, use initial conditions.

$$\text{at } t=0, i=0$$

where,

$$K = -0.00001 \cos 314t + 0.8 \sin 314t$$

(b) Laplace Transform Method:

$$\text{Initial condition } t=0, i=0 \rightarrow (1)$$

Apply KVL to the circuit.

$$20 \sin 314t = 25 i(t) + \frac{1}{10} \int i dt \rightarrow (2)$$

Apply Laplace transform

$$\frac{20 \times 314}{s^2 + 314^2} = 25 I(s) + \frac{1}{10} \frac{I(s)}{s}$$

$$\frac{20 \times 314}{s^2 + 314^2} = I(s) \left[25 + \frac{1}{10s} \right]$$

$$I(s) = \frac{20 \times 314}{(s^2 + 314^2)(25 + \frac{1}{10s})}$$

$$= \frac{20 \times 314}{\frac{25}{s}(s + \frac{1}{250})(s^2 + 314^2)}$$

$$I(s) = \frac{\left(\frac{20 \times 314 \times s}{25}\right)}{\left(s + \frac{1}{250}\right)(s^2 + 314^2)} \rightarrow (3)$$

Apply partial fractions

$$\frac{\left(\frac{20 \times 314 \times s}{25}\right)}{\left(s + \frac{1}{250}\right)(s^2 + 314^2)} = \frac{A}{s + \frac{1}{250}} + \frac{Bs + C}{s^2 + 314^2}$$

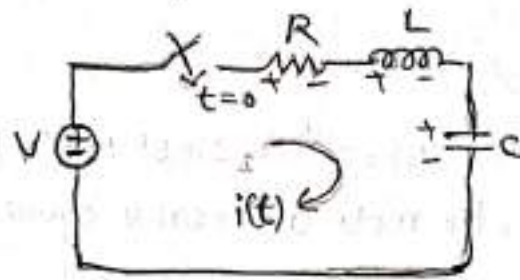
$$A = \left. \frac{\left(\frac{20 \times 314 \times s}{25}\right)}{s^2 + 314^2} \right|_{s = -\frac{1}{250}} = \frac{\frac{20 \times 314 \times -\frac{1}{250}}{25}}{\left(\frac{-1}{250}\right)^2 + 314^2}$$

$$= \frac{-\frac{20 \times 314}{25 \times 25 \times 10}}{314^2 \left[1 + \frac{1}{250^2 \times 314^2}\right]}$$

$$=$$

Method 1:

DC transient analysis of series R-L-C circuit using DE method



Apply KVL to the circuit, $-V + Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i dt = 0$

$$V = iR + L \frac{di}{dt} + \frac{1}{C} \int i dt \rightarrow (1)$$

differentiate eqn (1) wrt "t"

$$R \frac{di}{dt} + \frac{d^2i}{dt^2} L + \frac{1}{C} i = 0 \rightarrow (2)$$

$$\frac{d^2i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0$$

$$\text{Let } \frac{d}{dt} = s$$

$$s^2 i + \frac{R}{L} s i + \frac{1}{LC} i = 0 \rightarrow (3)$$

from eqn (3)

$$s^2 + \frac{R}{L} s + \frac{1}{LC} = 0 \rightarrow (4)$$

It is a second order DE. So, the solution is

$$s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-\frac{R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}}}{2}$$

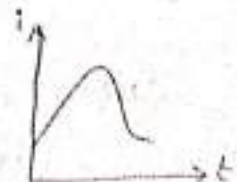
$$= \frac{1}{2} \left[\frac{-R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}} \right]$$

$$s_{1,2} = \underbrace{\frac{-R}{2L}}_{\alpha} \pm \underbrace{\sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}}_{\beta} \rightarrow (5)$$

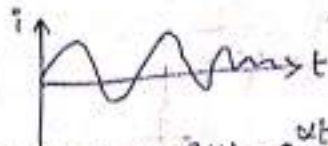
$$s_{1,2} = \alpha \pm \beta$$

Case (i): If $\left(\frac{R}{2L}\right)^2 > \frac{1}{LC}$. The roots are real & unequal. The response is over damped.

The solution is $i(t) = e^{\alpha t} [k_1 e^{\beta t} + k_2 e^{-\beta t}]$.



Case 2: Similarly, If $\left(\frac{R}{2L}\right)^2 < \frac{1}{LC}$, the complex & conjugate response is under damped.



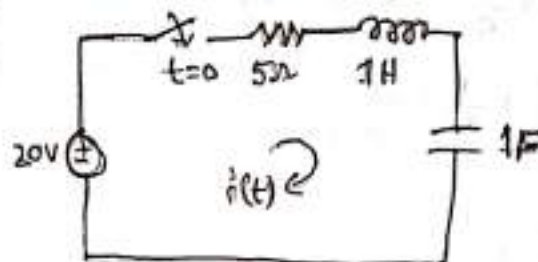
The solution is $i(t) = e^{\alpha t} [k_1 \sin \beta t + k_2 \cos \beta t]$

Case 3: If $\left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$, the roots are real & equal. The response is critically damped.

The solution is $i(t) = e^{\alpha t} [k_1 + k_2 t]$



1. In a series R-L-C circuit, $R = 5\Omega$, $L = 1\text{ H}$ & $C = 1\text{ F}$. A DC voltage of 20V is applied to a circuit at $t = 0$. Obtain $i(t)$.



Sol: Apply KVL to the circuit, $-20 + 5i(t) + 1 \cdot \frac{di}{dt} + \frac{1}{1} \int ci dt$

$$20 = 5i + \frac{di}{dt} + \int i dt \rightarrow (1)$$

differentiate eqn (1)

$$5 \frac{di}{dt} + \frac{d^2 i}{dt^2} + i = 0$$

$$\frac{d^2 i}{dt^2} + 5 \frac{di}{dt} + i = 0$$

$$\text{Let } \frac{d}{dt} = s$$

$$s^2 + 5s + 1 = 0$$

$$s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-5 \pm \sqrt{25 - 4}}{2}$$

$$= \frac{-5 \pm \sqrt{21}}{2} = -0.21, -4.79 \rightarrow (2)$$

Case (i): $25 > 4$. The roots are real and unequal. The response is over damped.

The solution is

$$i(t) = e^{-\frac{5}{2}t} \left[K_1 e^{\frac{\sqrt{21}}{2}t} + K_2 e^{-\frac{\sqrt{21}}{2}t} \right]$$

$$i(t) = e^{-2.5t} \left[K_1 e^{2.29t} + K_2 e^{-2.29t} \right] \rightarrow (3)$$

To find K:

Use initial condition $t=0, i=0$.

$$i(0) = e^0 [K_1 + K_2]$$

$$0 = K_1 + K_2$$

$$K_1 = -K_2 \rightarrow (4)$$

again use $\frac{di}{dt}$ for eqn (3)

$$\frac{di}{dt} = -0.21 K_1 e^{-0.21t} - 4.79 K_2 e^{-4.79t} \rightarrow (5)$$

again put initial conditions in eqn (5).

$$\frac{di}{dt} = -0.21 K_1 - 4.79 K_2 \rightarrow (6)$$

Substitute initial conditions in eqn (1)

$$20 = 5(0) + 1 \frac{di}{dt} + 1 \cdot 0$$

$$\frac{di}{dt} = 20 \rightarrow (7)$$

put eqn (7) in eqn (6)

$$20 = -0.21 K_1 - 4.79 K_2 \rightarrow (8)$$

Solve eqn (4) & eqn (8)

$$20 = 0.21 K_2 - 4.79 K_2$$

$$20 = -4.58 K_2$$

$$K_2 = -4.366$$

$$K_1 = 4.366$$

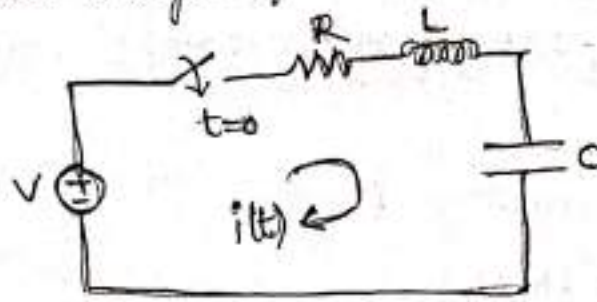
$$\therefore i(t) = e^{-2.5t} [4.366 e^{2.29t} - 4.366 e^{-2.29t}]$$

$$\begin{cases} e^{-2.5+2.29} \\ = e^{-0.21} \end{cases}$$

$$\begin{cases} e^{2.5-2.29} \\ = e^{0.21} \end{cases}$$

Method 2:

DC transient analysis of series RLC circuit by Laplace method



$$L \left(\frac{di}{dt} \right) = sI(s)$$

Apply KVL to the circuit

$$V = Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i dt \rightarrow (1)$$

Apply Laplace

$$\frac{V}{s} = I(s)R + L sI(s) + \frac{1}{C} \frac{I(s)}{s} \rightarrow (2)$$

$$\frac{V}{s} = I(s)R \left[R + sL + \frac{1}{Cs} \right]$$

$$I(s) = \frac{(V/s)}{R + sL + \frac{1}{Cs}}$$

$$= \frac{(V/s)}{L \left(\frac{R}{L} + s + \frac{1}{LCs} \right)} = \frac{\frac{V}{s}}{\frac{L}{s} \left(\frac{RS}{L} + s^2 + \frac{1}{LC} \right)}$$

$$I(s) = \frac{V/L}{s^2 + \frac{RS}{L} + \frac{1}{LC}} \rightarrow (3)$$

take critical resistance $R_{cr} = 2\sqrt{\frac{L}{C}}$

$$(\text{zeta}) \frac{R}{R_{cr}} = \frac{R}{2\sqrt{\frac{L}{C}}}$$

(natural frequency) $\omega_n^2 = \frac{1}{LC}$

$$\omega_n = \frac{1}{\sqrt{LC}}$$

$$2\zeta\omega_n = 2 \times \frac{R}{2} \sqrt{\frac{C}{L}} \times \frac{1}{\sqrt{LC}}$$

$$2\zeta\omega_n = \frac{R}{L}$$

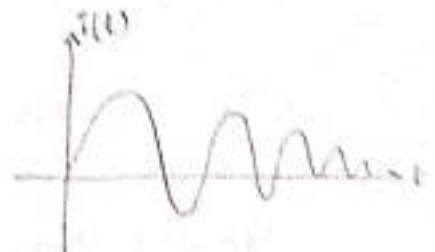
Now, eqn (3) becomes

$$\begin{aligned} I(s) &= \frac{V/L}{s^2 + 2\zeta\omega_n s + \omega_n^2} \\ &= \frac{V/L}{s^2 - \zeta\omega_n s + \omega_n^2} \rightarrow (4) \end{aligned}$$

Case (i): If $\xi < 1$, roots of the eqn are complex conjugate the roots are underdamped.

$$i(t) = \frac{V}{L\omega_d} e^{-\xi\omega_n t} \sin\omega_d t$$

$$\text{where } \omega_d = \omega_n \sqrt{1 - \xi^2}$$



Case (ii): If $\xi = 1$, then the $I(s)$ is given by

$$I(s) = \frac{(V/L)}{s^2 + 2\omega_n s + \omega_n^2} = \frac{(V/L)}{(s + \omega_n)^2}$$

By applying inverse Laplace,

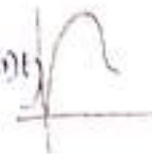
$$i(t) = \frac{V}{L} t e^{-\omega_n t}$$

$$L[t e^{-at}] = \frac{1}{(s+a)^2}$$

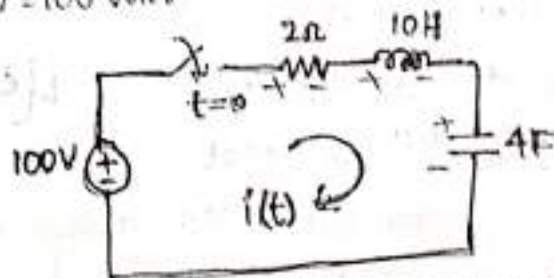
Case (iii): If $\xi > 1$, then damped is over damped.

The solution of the eqn is

$$i(t) = \frac{V}{2L\omega_n \sqrt{\xi^2 - 1}} e^{-\xi\omega_n t} \left(e^{(\omega_n \sqrt{\xi^2 - 1})t} - e^{-(\omega_n \sqrt{\xi^2 - 1})t} \right)$$



1. In RLC Series circuit $R = 2\Omega$, $L = 10H$, $C = 4F$ and the switch is closed at $t = 0$, find the expression for current by using LT. where $V = 100$ Volts.



Apply KVL to the circuit, $-100 + 2i(t) + 10 \frac{di(t)}{dt} + \frac{1}{4} \int i dt = 0$

$$100 = 2i(t) + 10 \frac{di(t)}{dt} + \frac{1}{4} \int i dt \rightarrow (1)$$

Apply Laplace to the eqn (1)

$$\frac{100}{s} = 2I(s) + 10[sI(s) - I(0)] + \frac{1}{4} \frac{I(s)}{s}$$

$$I(s) \left[2 + 10s + \frac{1}{4s} \right] = \frac{100}{s}$$

$$I(s) = \frac{100}{s \left[2 + 10s + \frac{1}{4s} \right]}$$

$$= \frac{100}{2s + 10s^2 + \frac{1}{4}}$$

$$I(s) = \frac{100}{10 \left(s^2 + \frac{3}{10}s + \frac{10}{40} \right)}$$

$$I(s) = \frac{10}{s^2 + 0.3s + 0.25} \rightarrow (2)$$

$$s^2 + 0.3s + 0.25 \Rightarrow s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-0.3 \pm \sqrt{0.3^2 - 4(0.25)}}{2}$$

$$= -0.1 \pm 0.122i$$

$$I(s) = \frac{10}{s^2 + 2 \times 0.1 \times s + (0.1)^2 - (0.1)^2 + 0.025}$$

$$= \frac{10}{(s + 0.1)^2 + 0.015}$$

$$I(s) = \frac{10}{(s + 0.1)^2 + (0.122)^2} \rightarrow (3)$$

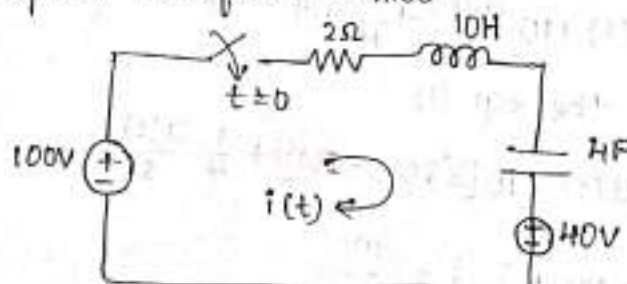
$$I(s) = \frac{10}{(s + 0.1)^2 + (0.122)^2} \times \frac{0.122}{0.122} \rightarrow (4)$$

Apply inverse Laplace to the eqn (4).

$$i(t) = 81.96 e^{-0.1t} \sin 0.122t$$

$$L[e^{-at} \sin \omega t] = \frac{\omega}{(s+a)^2 + \omega^2}$$

2. In RLC circuit, the capacitor has initial voltage of 40V & the switch is closed at $t=0$. Find the expression for current by using Laplace transform method.



Apply KVL to the circuit,

$$100 = 2i(t) + 10 \frac{di}{dt} + \frac{1}{4} \int i dt + 40$$

$$60 = 2i(t) + 10 \frac{di}{dt} + \frac{1}{4} \int i dt \rightarrow (1)$$

Apply Laplace transform,

$$\frac{60}{s} = 2I(s) + 10sI(s) + \frac{1}{4} \frac{I(s)}{s}$$

$$I(s) \left[s^2 + 10s + \frac{1}{4} \right] = \frac{60}{s}$$

$$I(s) = \frac{60}{s \left[s^2 + 10s + \frac{1}{4} \right]}$$

$$= \frac{60}{10s^2 + 2s + \frac{1}{4}}$$

$$= \frac{6}{s^2 + \frac{1}{5}s + \frac{1}{40}}$$

$$I(s) = \frac{6}{s^2 + 0.2s + 0.025} \rightarrow (2)$$

$$s^2 + 0.2s + 0.025 \Rightarrow s_{1,2} = \frac{-0.2 \pm \sqrt{(0.2)^2 - 4(0.025)}}{2}$$

$$= -0.1 \pm 0.122i$$

$$I(s) = \frac{6}{s^2 + 2 \times 0.1 \times s + (0.1)^2 + 0.1^2 + 0.025}$$

$$= \frac{6}{(s+0.1)^2 + 0.015}$$

$$= \frac{6}{(s+0.1)^2 + (0.122)^2} \rightarrow (3)$$

$$= \frac{6}{(s+0.1)^2 + (0.122)^2} \times \frac{0.122}{0.122}$$

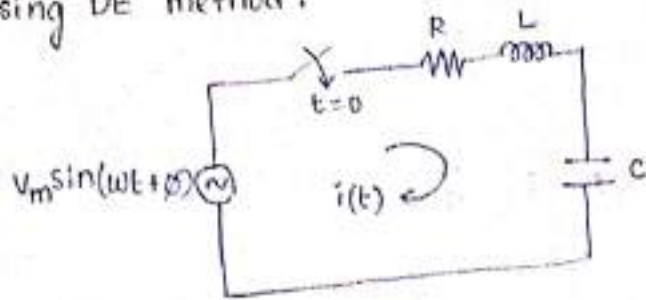
$$= \frac{6}{0.122} \left[\frac{0.122}{(s+0.1)^2 + (0.122)^2} \right]$$

apply
inverse laplace

$$i(t) = \frac{6}{0.122} \left[e^{-0.1t} \sin 0.122t \right] \text{ amp}$$

Method 3:

AC transient analysis of series RLC series circuit by using DE method:



Apply KVL to the circuit.

$$v_m \sin(\omega t + \phi) = i(t)R + L \frac{di}{dt} + \frac{1}{C} \int i dt \rightarrow (1)$$

differentiate wrt 't'.

$$v_m \omega \cos(\omega t + \phi) = \frac{di}{dt} R + L \frac{d^2 i}{dt^2} + \frac{1}{C} i \rightarrow (2)$$

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = \frac{v_m \omega}{L} \cos(\omega t + \phi) \rightarrow (3)$$

It is a second order differential equation.

Let us consider, the solution is

$$i(t) = i_{CF} + i_{PI} \rightarrow (4)$$

and take

$$i_{PI} = A \cos(\omega t + \phi) + B \sin(\omega t + \phi) \rightarrow (5)$$

$$\frac{di}{dt} = -A\omega \sin(\omega t + \phi) + B\omega \cos(\omega t + \phi) \rightarrow (6)$$

$$\frac{d^2 i}{dt^2} = -A\omega^2 \cos(\omega t + \phi) - B\omega^2 \sin(\omega t + \phi) \rightarrow (7)$$

substitute eqn (5), (6), (7) in eqn (3)

$$[-A\omega^2 \cos(\omega t + \phi) - B\omega^2 \sin(\omega t + \phi)] + \frac{R}{L} [-A\omega \sin(\omega t + \phi) + B\omega \cos(\omega t + \phi)] + \frac{1}{LC} [A \cos(\omega t + \phi) + B \sin(\omega t + \phi)] = \frac{v_m \omega}{L} \cos(\omega t + \phi) \rightarrow (8)$$

Compare $\cos(\omega t + \phi)$ terms

$$-A\omega^2 + \frac{B\omega R}{L} + \frac{A}{LC} = \frac{v_m \omega}{L} \rightarrow (9)$$

Compare $\sin(\omega t + \phi)$ terms

$$-B\omega^2 - \frac{AR\omega}{L} + \frac{B}{LC} = 0 \rightarrow (10)$$

from eqn (10) $B \left(\frac{1}{LC} - \omega^2 \right) - \frac{AR\omega}{L} = 0$

$$B = \frac{V_m R / L}{\omega^2 - \frac{1}{LC}} \rightarrow (11)$$

from eqn (9)

$$A \left(\frac{1}{LC} - \omega^2 \right) + \frac{B \omega R}{L} = \frac{V_m \omega}{L} \rightarrow (12)$$

$$A \left(\frac{1}{LC} - \omega^2 \right) + \left(\frac{A \omega R / L}{\frac{1}{LC} - \omega^2} \right) \left(\frac{\omega R}{L} \right) = \frac{V_m \omega}{L}$$

$$A \left(\frac{1}{LC} - \omega^2 + \frac{R^2 \omega^2}{\frac{1}{LC} - \omega^2} \right) = \frac{V_m \omega}{L}$$

$$A \left(\frac{1}{LC} - \omega^2 + \frac{LC R^2 \omega^2}{1 - LC \omega^2} \right) = \frac{V_m \omega}{L}$$

$$A \left(\frac{1 - LC \omega^2}{LC} + \frac{LC R^2 \omega^2}{1 - LC \omega^2} \right) = \frac{V_m \omega}{L}$$

substitute eqn (11) in eqn (12)

$$A \left(\omega^2 - \frac{1}{LC} \right) + A \left(\frac{\omega R}{L} \right)^2 \times \frac{1}{\omega^2 - \frac{1}{LC}} = -\frac{V_m \omega}{L}$$

$$A \left[\left(\omega^2 - \frac{1}{LC} \right)^2 + \left(\frac{\omega R}{L} \right)^2 \right] = -\frac{V_m \omega}{L} \left(\omega^2 - \frac{1}{LC} \right)$$

$$A = \frac{-\frac{V_m \omega}{L} \left(\omega^2 - \frac{1}{LC} \right)}{\left(\omega^2 - \frac{1}{LC} \right)^2 + \left(\frac{\omega R}{L} \right)^2}$$

$$A = \frac{-\frac{V_m \omega}{L} \times \frac{\omega}{L} \left[\omega L - \frac{1}{\omega C} \right]}{\frac{\omega^2}{L^2} \left[R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2 \right]}$$

$$A = \frac{-V_m \left[\omega L - \frac{1}{\omega C} \right]}{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2} \rightarrow (13)$$

Substitute eqn (13) in eqn (11)

$$B = \frac{V_m R}{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

Substitute $A \sin \theta$ in eqn (14)

$$i_{PI} = \frac{-V_m(\omega L - \frac{1}{\omega C})}{R^2 + (\omega L - \frac{1}{\omega C})^2} \cos(\omega t + \phi) + \frac{V_m R}{R^2 + (\omega L - \frac{1}{\omega C})^2} \sin(\omega t + \phi) \rightarrow (14)$$

$\underbrace{\hspace{10em}}_{K_I \sin \theta} \qquad \underbrace{\hspace{10em}}_{K_I \cos \theta}$

$$K_I^2 (\sin^2 \theta + \cos^2 \theta) = \frac{V_m^2 (\omega L - \frac{1}{\omega C})^2}{[R^2 + (\omega L - \frac{1}{\omega C})^2]^2} \times \frac{R^2 V_m^2}{[R^2 + (\omega L - \frac{1}{\omega C})^2]^2}$$

$$K_I^2 = \frac{V_m^2 [R^2 + (\omega L - \frac{1}{\omega C})^2]}{[R^2 + (\omega L - \frac{1}{\omega C})^2]^2}$$

$$K_I^2 = \frac{V_m^2}{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

$$K_I = \frac{V_m}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{-V_m (\omega L - \frac{1}{\omega C})}{V_m R}$$

$$\boxed{\theta = \tan^{-1} \left(\frac{\frac{1}{\omega C} - \omega L}{R} \right)}$$

then eqn (14) becomes

$$i_{PI} = \frac{V_m}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} [-\sin \theta \cdot \cos(\omega t + \phi) + \cos \theta \cdot \sin(\omega t + \phi)]$$

$$i_{PI} = \frac{V_m}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} \sin(\omega t + \phi - \theta) \rightarrow (15)$$

To find i_{CF}:

Consider $\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0$

take $\frac{di}{dt} = s$

$$\left(s^2 + \frac{R}{L} s + \frac{1}{LC} \right) i = 0$$

$$s_{1,2} = \underbrace{-\frac{R}{2L}}_{\alpha} \pm \underbrace{\sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}}_{\beta}$$

Case 1: $\left(\frac{R}{2L}\right)^2 > \frac{1}{LC}$ (over damped)

The roots are real and unequal.

Then the solution is

$$i_{CF} = e^{\alpha t} [k_2 e^{\beta t} + k_3 e^{-\beta t}]$$

Now the complete solution is

$$i(t) = e^{-\frac{R}{2L}t} \left[k_2 e^{\sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}t} + k_3 e^{-\sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}t} \right] + \frac{V_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \sin(\omega t + \phi - \theta)$$

Case 2: $\left(\frac{R}{2L}\right)^2 < \frac{1}{LC}$ (under damped)

The roots are complex conjugate

$$i_{CF} = e^{\alpha t} [k_2 \cos \beta t + k_3 \sin \beta t]$$

$$i(t) = e^{-\frac{R}{2L}t} \left[k_2 \cos \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}t + k_3 \sin \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}t \right] + \frac{V_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \sin(\omega t + \phi - \theta)$$

Case 3: $\left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$ (critically damped)

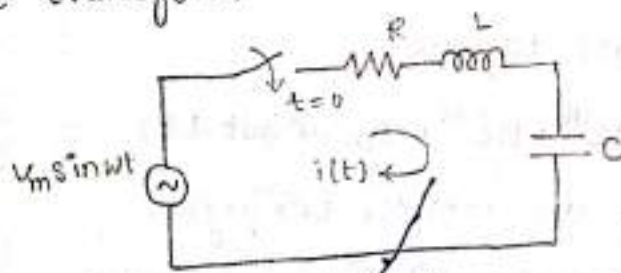
The roots are real and equal.

$$i_{CF} = e^{\alpha t} [k_2 + k_3 t]$$

$$i(t) = e^{-\frac{R}{2L}t} [k_2 + k_3 t] + \frac{V_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \sin(\omega t + \phi - \theta)$$

Method 4:

AC transient analysis of series RLC circuit by Laplace transform method.



Apply KVL to the circuit

$$V_m \sin \omega t = iR + L \frac{di}{dt} + \frac{1}{C} \int i dt \rightarrow (1)$$

Apply Laplace transform

$$\frac{V_m \omega}{s^2 + \omega^2} = R I(s) + sL I(s) + \frac{I(s)}{Cs} \rightarrow (2)$$

$$I(s) = \frac{V_m \omega}{(s^2 + \omega^2) \left(R + sL + \frac{1}{Cs} \right)}$$

$$I(s) = \frac{V_m \omega s}{(s^2 + \omega^2) \left(s^2 + \frac{R}{L}s + \frac{1}{LC} \right)}$$

$$I(s) = \frac{V_m \omega s / L}{(s^2 + \omega^2) \left(s^2 + \frac{R}{L}s + \frac{1}{LC} \right)} \rightarrow (3)$$

Case 1: Roots are real and equal, then

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = (s+a)^2$$

$$I(s) = \frac{\left(\frac{V_m \omega s}{L} \right)}{(s^2 + \omega^2)(s+a)^2} \rightarrow (4)$$

Apply partial fractions

$$I(s) = \frac{A}{s+a} + \frac{B}{(s+a)^2} + \frac{Cs+D}{s^2+\omega^2}$$

By inverse Laplace transform

$$i(t) = A e^{-at} + B t e^{-at} + I_m \sin(\omega t + \theta)$$

Case 2:

The roots are real and unequal,

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = (s+a)(s+b)$$

$$I(s) = \frac{\left(\frac{V_m \omega s}{L} \right)}{(s^2 + \omega^2)(s+a)(s+b)}$$

$$I(s) = \frac{A}{s+a} + \frac{B}{s+b} + \frac{Cs+D}{s^2+\omega^2}$$

Apply inverse Laplace

$$i(t) = A e^{-at} + B e^{-bt} + I_m \sin(\omega t \pm \theta)$$

Case 3: The roots are complex conjugate.

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = \left[s^2 + \frac{R}{L}s + \left(\frac{R}{2L} \right)^2 \right] + \left(\frac{1}{LC} - \frac{R^2}{4L^2} \right)$$

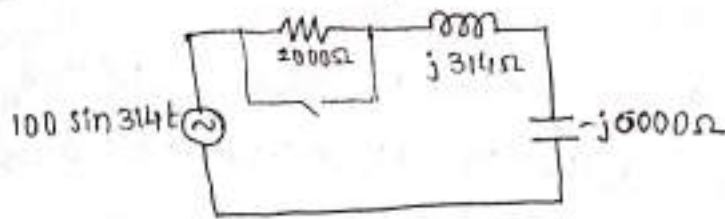
$$= (s+a)^2 + \omega_d^2$$

$$I(s) = \frac{(V_m \omega s / L)}{(s^2 + \omega^2) [(s+a)^2 + \omega_d^2]} = \frac{As+B}{s^2+\omega^2} + \frac{Cs+D}{(s+a)^2 + \omega_d^2}$$

Inverse Laplace,

$$i(t) = I_m \sin(\omega t \pm \theta) + I_0 e^{-at} \sin(\omega_d t \pm \theta)$$

1. In the figure the switch is open, steady state is reached with $v = 100 \sin 314t$. The switch is closed at $t=0$, the circuit is allowed to come to steady state again. Determine the steady state current and complete transient current.



sol: Case 1: Switch is open

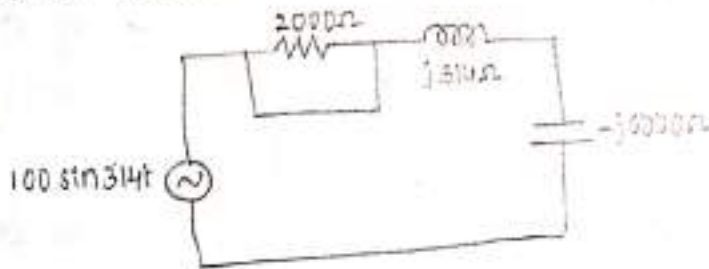
$$Z = 2000 + j(314 - 6000)$$

$$Z = 6027.48 \angle -70.62$$

$$I = \frac{V}{Z}$$

$$I = 0.017 \angle 70.62 \rightarrow (1)$$

Case 2: Switch is closed.



$$Z = j(314 - 6000)$$

$$Z = 5686 \angle -90$$

$$|I| = \frac{V}{Z}$$

$$I = 0.018 \angle 90$$

Steady state current (I_{ss})

$$I_{ss} = 0.018 \sin(314t - 90) \rightarrow PI$$

To find transient current (complementary function)

$$(s^2 + \frac{R}{L}s + \frac{1}{LC})i = 0$$

$$s_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$\text{put } R = 0$$

$$s_{1,2} = \pm \sqrt{-\frac{1}{LC}}$$

$$= \pm j0.001$$

roots are complex conjugate.

$$i_{CF} = e^{at} (k_1 \cos \beta t + k_2 \sin \beta t)$$

$$= k_1 \cos \beta t + k_2 \sin \beta t$$

$$= k_1 \cos 0.001t + k_2 \sin 0.001t$$

complete current $i(t) = i_{CF} + i_{PI}$

$$i(t) = k_1 \cos 0.001t + k_2 \sin 0.001t + 0.018 \sin(314t - 90^\circ)$$

$$= k_1 \cos 0.001t + k_2 \sin 0.001t - 0.018 \cos 314t \rightarrow (2)$$

Use initial conditions

$$t=0, i=0$$

$$0.017 = k_1 - 0.0175$$

$$k_1 = 0.0365$$

UNIT-III

FOURIER TRANSFORM

Fourier theorem — Trigonometric form and exponential form of fourier series — Conditions of symmetry — line spectra and phase angle spectra — Analysis of electrical circuit excited by non-sinusoidal sources of periodic-wave forms — fourier integrals and fourier transforms — properties of fourier transform and applications of electrical circuit.

Fourier Series :

→ Fourier series means a non sinusoidal wave form like triangular ~~plus~~, rectangular plus (or) square plus etc are applied to the inputs to the electrical circuit.

→ It is represented by $f(t)$.

→ In general fourier series expansion is given below

$$f(t) = a_0 + a_1 \cos \omega t + a_2 \cos 2\omega t + \dots + a_n \cos n\omega t + b_1 \sin \omega t + \dots$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos n\omega t + b_n \sin n\omega t] \rightarrow (1)$$

Exponential form of fourier series :

We know,

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos n\omega t + b_n \sin n\omega t] \rightarrow (1)$$

$$= a_0 + \sum_{n=1}^{\infty} (\sqrt{a_n^2 + b_n^2}) \left[\frac{a_n}{\sqrt{a_n^2 + b_n^2}} \cos n\omega t + \frac{b_n}{\sqrt{a_n^2 + b_n^2}} \sin n\omega t \right]$$

$$= a_0 + \sum_{n=1}^{\infty} A_n \cos (n\omega t - \theta_n) \rightarrow (2)$$

Using Euler's Identity Expansion

$$f(t) = a_0 + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} C_n e^{jn\omega t} \rightarrow f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega t} \rightarrow (3)$$

Multiplying both sides by $e^{-jk\omega t}$ & integrating with t_1 to t_1+T

$$\int_{t_1}^{t_1+T} f(t) e^{-jk\omega t} dt = \int_{t_1}^{t_1+T} \sum_{n=-\infty}^{\infty} C_n e^{jn\omega t} e^{-jk\omega t} dt$$

$$(\text{for } n=k) = C_n T_0 \rightarrow (4) \quad \text{where } \int_{t_1}^{t_1+T} \sum_{n=-\infty}^{\infty} e^{j(n-k)\omega t} dt = T_0$$

$$= (t)_{t_1}^{t_1+T}$$

$$= t_1 + T - t_1$$

$$= T$$

$$C_n = \frac{1}{T_0} \int_{t_1}^{t_1+T} f(t) e^{-jn\omega t} dt$$

Trigonometric Form:

We know,

$$C_n = \frac{1}{T_0} \int_{t_1}^{t_1+T} f(t) e^{-jn\omega t} dt \rightarrow (1)$$

multiplied by '2' on both sides

$$2C_n = \frac{2}{T_0} \int_{t_1}^{t_1+T} f(t) e^{-jn\omega t} dt \rightarrow (2)$$

$$a_n - jb_n = \frac{2}{T_0} \int_{t_1}^{t_1+T} f(t) [\cos n\omega t - j \sin n\omega t] dt$$

$$a_n - jb_n = \frac{2}{T_0} \int_{t_1}^{t_1+T} f(t) \cos n\omega t dt - j \frac{2}{T_0} \int_{t_1}^{t_1+T} f(t) \sin n\omega t dt$$

on comparing,

$$a_n = \frac{2}{T_0} \int_{t_1}^{t_1+T} f(t) \cos n\omega t dt \rightarrow (3)$$

$$b_n = \frac{2}{T_0} \int_{t_1}^{t_1+T} f(t) \sin n\omega t dt \rightarrow (4)$$

$$a_0 = \frac{1}{T_0} \int_{t_1}^{t_1+T} f(t) dt \rightarrow (5)$$

Symmetry in Fourier Series:

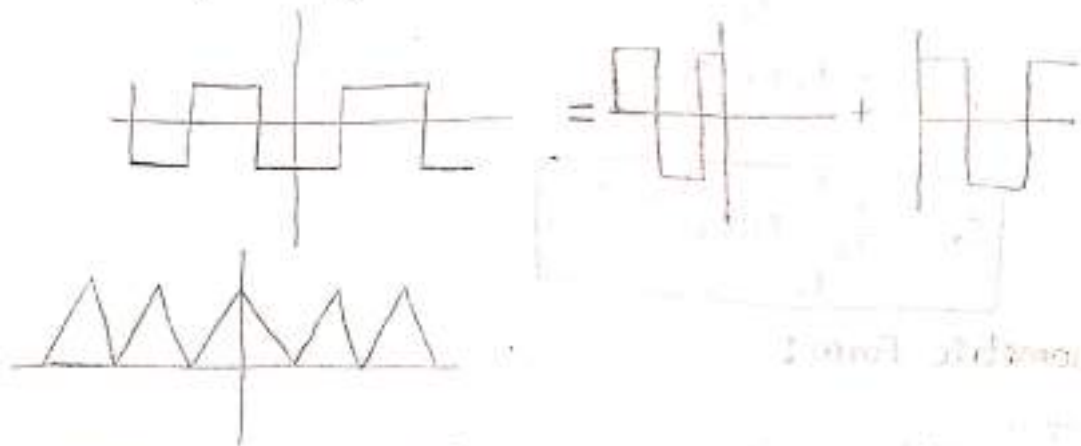
There are three types of symmetry. They are:

1. Even

2. Odd

3. Half wave

1. Even Symmetry :



condition $f(-t) = f(t)$

Taking $a_0 = \frac{1}{T_0} \int_{t_1}^{t_1+T} f(t) dt$, $t_1 = -\frac{T_0}{2}$

$$a_0 = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{-\frac{T_0}{2}+T_0} f(t) dt$$

$$= \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f(t) dt$$

$$f(-t) = f(t) \text{ , } d(-t) = -dt$$

$$= \frac{1}{T_0} \int_{-\frac{T_0}{2}}^0 f(t) dt + \frac{1}{T_0} \int_0^{\frac{T_0}{2}} f(t) dt$$

$$= \frac{1}{T_0} \int_{-\frac{T_0}{2}}^0 f(-t)(-dt) + \frac{1}{T_0} \int_0^{\frac{T_0}{2}} f(t) dt$$

$$a_0 = \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) dt$$

$$a_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \cos n\omega t dt$$

put $t_1 = -\frac{T_0}{2}$

$$= \frac{2}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f(t) \cos n\omega t dt$$

$$= \frac{2}{T_0} \left[\int_{-\frac{T_0}{2}}^0 f(t) \cos n\omega t dt + \int_0^{\frac{T_0}{2}} f(t) \cos n\omega t dt \right]$$

put $f(-t) = f(t)$

$$a_n = \frac{2}{T_0} \left[\int_{-T_0/2}^0 f(-t) \cos(n\omega(-t)) (-dt) + \int_0^{T_0/2} f(t) \cos n\omega t dt \right]$$

$$= \frac{2}{T_0} \left[\int_{-T_0/2}^0 f(t) \cos n\omega t (-dt) + \int_0^{T_0/2} f(t) \cos n\omega t dt \right]$$

$$a_n = \frac{4}{T_0} \int_0^{T_0/2} f(t) \cos n\omega t dt$$

$$b_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \sin \omega t dt$$

put $t_1 = -\frac{T_0}{2}$

$$= \frac{2}{T_0} \int_{-T_0/2}^{T_0/2} f(t) \sin \omega t dt$$

$$= \frac{2}{T_0} \left[\int_{-T_0/2}^0 f(t) \sin \omega t dt + \int_0^{T_0/2} f(t) \sin \omega t dt \right]$$

$$= \frac{2}{T_0} \left\{ \int_0^{T_0/2} f(t) \sin \omega t dt + \int_0^{T_0/2} f(t) \sin \omega t dt \right\}$$

$$b_n = \frac{4}{T_0} \int_0^{T_0/2} f(t) \sin \omega t dt$$

2. Odd Symmetry:

$$f(-t) = -f(t)$$

$$(i) a_0 = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} f(t) dt$$

$$\text{putting } t_1 = -\frac{T_0}{2}$$

$$a_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) dt$$

$$= \frac{1}{T_0} \left[\int_{-T_0/2}^0 f(t) dt + \int_0^{T_0/2} f(t) dt \right]$$

put $f(-t) = -f(t)$ in 1st term

$$= \frac{1}{T_0} \int_{-T_0/2}^0 f(-t) d(-t) + \frac{1}{T_0} \int_0^{T_0/2} f(t) dt$$

$$= -\frac{1}{T_0} \int_{-T_0/2}^0 f(t) (-d(t)) + \frac{1}{T_0} \int_0^{T_0/2} f(t) dt$$

$$= -\frac{1}{T_0} \int_0^{T_0/2} f(t) dt + \frac{1}{T_0} \int_0^{T_0/2} f(t) dt$$

$$\boxed{a_0 = 0}$$

$$a_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \cos n\omega t dt$$

$$\text{put } t_1 = -\frac{T_0}{2}$$

$$a_n = \frac{2}{T_0} \int_{-T_0/2}^{T_0/2} f(t) \cos n\omega t dt$$

$$= \frac{2}{T_0} \int_{-T_0/2}^0 f(t) \cos n\omega t dt + \frac{2}{T_0} \int_0^{T_0/2} f(t) \cos n\omega t dt$$

$$\text{put } f(-t) = -f(t)$$

$$a_n = \frac{2}{T_0} \int_{-T_0/2}^0 f(-t) \cos n\omega(-t) dt + \frac{2}{T_0} \int_0^{T_0/2} f(t) \cos n\omega t dt$$

$$= -\frac{2}{T_0} \int_0^{T_0/2} f(t) \cos n\omega t dt + \frac{2}{T_0} \int_0^{T_0/2} f(t) \cos n\omega t dt$$

$$a_n = 0$$

$$b_n = \frac{2}{T_0} \int_t^{t+T_0} f(t) \sin n\omega t dt$$

$$\text{put } t_1 = -T_0/2$$

$$b_n = \frac{2}{T_0} \int_{-T_0/2}^{T_0/2} f(t) \sin n\omega t dt$$

$$= \frac{2}{T_0} \int_{-T_0/2}^0 f(t) \sin n\omega t dt + \frac{2}{T_0} \int_0^{T_0/2} f(t) \sin n\omega t dt$$

$$\text{put } f(-t) = -f(t), d(-t) = -dt$$

$$= \frac{2}{T_0} \int_{-T_0/2}^0 f(-t) \sin n\omega(-t) \cdot d(-t) + \frac{2}{T_0} \int_0^{T_0/2} f(t) \sin n\omega t dt$$

$$= -\frac{2}{T_0} \int_{-T_0/2}^0 f(t) \sin n\omega t dt + \frac{2}{T_0} \int_0^{T_0/2} f(t) \sin n\omega t dt$$

$$= \frac{2}{T_0} \int_0^{T_0/2} f(t) \sin n\omega t dt + \frac{2}{T_0} \int_0^{T_0/2} f(t) \sin n\omega t dt$$

$$b_n = \frac{4}{T_0} \int_0^{T_0/2} f(t) \sin n\omega t dt$$

$$\left[\frac{\pi}{n} (f) \sin - \frac{\pi}{n} (f) \cos \right] \frac{1}{n\omega}$$

$$\left[(\pi - \pi \cos) - (0 - \pi) \right] \frac{1}{n\omega}$$

$$[0 = \pi]$$

3. Half Wave Symmetry:

$$f(t) = -f(t - T_0/2)$$

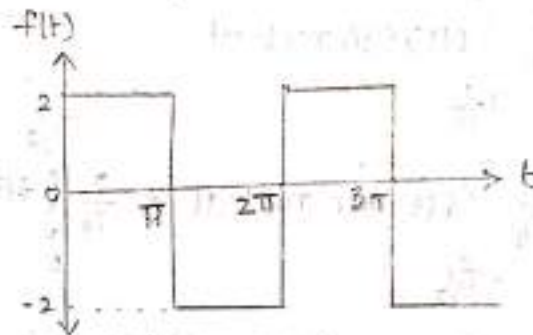
If n is even, $a_0 = 0$, $a_n = 0$, $b_n = 0$

If n is odd, $a_0 = 0$

$$a_n = \frac{4}{T_0} \int_0^{T_0/2} f(t) \cos n\omega t dt$$

$$b_n = \frac{4}{T_0} \int_0^{T_0/2} f(t) \sin n\omega t dt$$

1. Obtain Fourier series for the following wave form



Ans: We know that Fourier Series Expansion is

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos n\omega t + b_n \sin n\omega t]$$

$$= a_0 + a_1 \cos \omega t + a_2 \cos 2\omega t + \dots + b_1 \sin \omega t + b_2 \sin 2\omega t + \dots$$

$$\text{One cycle } T = 2\pi = T_0$$

$$(1) a_0 = \frac{1}{T_0} \int_0^{2\pi} f(t) dt$$

$$= \frac{1}{T_0} \left[\int_0^{\pi} f(t) dt + \int_{\pi}^{2\pi} f(t) dt \right]$$

$$= \frac{1}{2\pi} \left[\int_0^{\pi} 2 dt + \int_{\pi}^{2\pi} (-2) dt \right]$$

$$= \frac{1}{2\pi} \left[2(t)_0^{\pi} - 2(t)_{\pi}^{2\pi} \right]$$

$$= \frac{2}{2\pi} \left[(\pi - 0) - (2\pi - \pi) \right]$$

$$\boxed{a_0 = 0}$$

$$a_n = \frac{2}{T_0} \int_0^{2\pi} f(t) \cos n\omega t \, dt \quad \text{Here } \omega = 2\pi f = 2\pi \frac{1}{T}$$

$$\omega = \frac{2\pi}{2\pi} = 1$$

$$= \frac{2}{T_0} \left[\int_0^{\pi} f(t) \cos nt \, dt + \int_{\pi}^{2\pi} f(t) \cos nt \, dt \right]$$

$$= \frac{2}{2\pi} \left[\int_0^{\pi} 2 \cos nt \, dt + \int_{\pi}^{2\pi} (-2) \cos nt \, dt \right]$$

$$= \frac{2}{\pi} \left[\left(\frac{\sin nt}{n} \right)_0^{\pi} - \left(\frac{\sin nt}{n} \right)_{\pi}^{2\pi} \right]$$

$$= \frac{2}{\pi} \left[\frac{\sin n\pi}{n} - \frac{\sin 0}{n} - \frac{\sin 2n\pi}{n} + \frac{\sin n\pi}{n} \right]$$

for $n=1, 2, 3, \dots$

$$\boxed{a_n = 0}$$

$$b_n = \frac{2}{T_0} \int_0^{2\pi} f(t) \sin n\omega t \, dt$$

$$= \frac{2}{2\pi} \left[\int_0^{\pi} f(t) \sin n\omega t \, dt + \int_{\pi}^{2\pi} f(t) \sin n\omega t \, dt \right]$$

$$= \frac{1}{\pi} \left[\int_0^{\pi} 2 \sin n\omega t \, dt + \int_{\pi}^{2\pi} -2 \sin n\omega t \, dt \right]$$

$$= \frac{2}{\pi} \left[\int_0^{\pi} \sin nt \, dt + \int_{\pi}^{2\pi} -\sin nt \, dt \right]$$

$$= \frac{2}{\pi} \left[\left(-\frac{\cos nt}{n} \right)_0^{\pi} + \left(\frac{\cos nt}{n} \right)_{\pi}^{2\pi} \right]$$

$$b_n = \frac{2}{\pi} \left[\frac{-\cos n\pi}{n} + \frac{\cos 0}{n} + \frac{\cos 2n\pi}{n} - \frac{\cos n\pi}{n} \right]$$

For $n=1$,

$$b_1 = \frac{2}{\pi} [-(-1) + 1 + 1 + 1]$$

$$\boxed{b_1 = \frac{8}{\pi}}$$

$$b_2 = \frac{8}{\pi} \left[\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \right]$$

$$b_2 = 0$$

$$b_3 = \frac{8}{\pi} \left[\frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} \right]$$

$$b_3 = \frac{8}{3\pi}$$

Fourier Series:

$$f(t) = b_1 \sin \omega t + b_3 \sin 3\omega t + \dots$$

$$= \frac{8}{\pi} \sin \omega t + \frac{8}{3\pi} \sin 3\omega t + \dots$$

2. Obtain Fourier series



Ans: Fourier Series is

$$f(t) = a_0 + a_1 \cos \omega t + a_2 \cos 2\omega t + \dots + b_1 \sin \omega t + b_3 \sin 3\omega t + \dots$$

$$(i) a_0 = \frac{1}{T_0} \int_0^{T_0} f(t) dt$$

$$= \frac{1}{2} \int_0^{1/2} f(t) dt + \frac{1}{2} \int_{3/2}^2 f(t) dt$$

$$= \frac{1}{2} \int_0^{1/2} 2 dt + \frac{1}{2} \int_{3/2}^2 2 dt$$

$$= (t)_0^{1/2} + (t)_{3/2}^2$$

$$= \frac{1}{2} - 0 + 2 - \frac{3}{2}$$

$$= \frac{5}{2} - \frac{3}{2}$$

$$a_0 = 1$$

$$\frac{8}{\pi} = 1$$

$$(ii) a_n = \frac{2}{T_0} \int_0^{T_0} f(t) \cos n\omega t dt$$

$$= \frac{2}{2} \left[\int_0^{1/2} 2 \cos n\omega t dt + \int_{3/2}^2 2 \cos n\omega t dt \right]$$

$$= 2 \int_0^{1/2} \cos n\omega t dt + \int_{3/2}^2 \cos n\omega t dt$$

$$= 2 \left[\frac{\sin n\pi t}{n\pi} \right]_0^{1/2} + \left[\frac{\sin n\pi t}{n\pi} \right]_{3/2}^2$$

$$= 2 \left[\frac{\sin \frac{n\pi}{2}}{n\pi} - \sin 0 + \frac{\sin 2n\pi}{n\pi} - \frac{\sin \frac{3n\pi}{2}}{n\pi} \right]$$

$$= 2 \left[\frac{\sin \frac{n\pi}{2}}{n\pi} - \sin \frac{3n\pi}{2} \right]$$

For $n=1$, $a_1 = \left[\frac{1}{2\pi} + \frac{1}{\pi} \right] 2$

$$\boxed{a_1 = \frac{4}{\pi}}$$

For $n=2$, $a_2 = 0$

For $n=3$, $a_3 = 2 \left[\frac{-1}{3\pi} - \frac{-1}{3\pi} \right]$

$$\boxed{a_3 = -\frac{4}{3\pi}}$$

$$(iii) b_n = \frac{2}{T_0} \int_0^{T_0} f(t) \sin n\omega t dt$$

$$= \frac{2}{2} \left[\int_0^{1/2} 2 \sin n\omega t dt + \int_{3/2}^2 2 \sin n\omega t dt \right]$$

$$= 2 \left[\left[-\frac{\cos n\pi t}{n\pi} \right]_0^{1/2} + \left[-\frac{\cos n\pi t}{n\pi} \right]_{3/2}^2 \right]$$

$$= 2 \left[-\frac{\cos \frac{n\pi}{2}}{n\pi} + \frac{\cos 0}{n\pi} - \frac{\cos 2n\pi}{n\pi} + \frac{\cos \frac{3n\pi}{2}}{n\pi} \right]$$

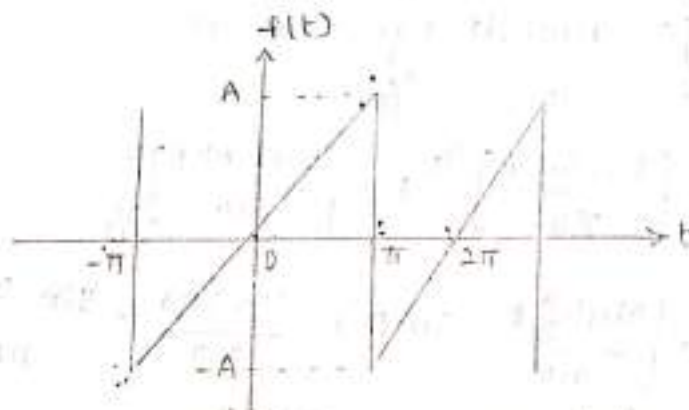
$$b_1 = 2 \left[0 + \frac{1}{\pi} + \frac{1}{2\pi} \right] \frac{1}{\pi} = \frac{0}{\pi} = 0$$

$$b_2 = 2 [0] = 0$$

Fourier series is

$$f(t) = 1 + \frac{4}{\pi} \cos \omega t - \frac{4}{3\pi} \cos 3\omega t + \dots$$

3. Obtain Fourier series for the following wave form.



* For the above problem, we can take function in 2 ways. 1. splitting $0 - 2\pi$ into $(0 - \pi) + (\pi - 2\pi)$.

$$\int_0^{2\pi} f(t) dt = \int_0^{\pi} f(t) dt + \int_{\pi}^{2\pi} f(t) dt$$

2. Take only one function with limits $(-\pi \text{ to } \pi)$

$$\int_{-\pi}^{\pi} f(t) dt$$

(i) $T = \text{one cycle} = \pi - (-\pi) = 2\pi = T_0$

$$a_0 = \frac{1}{T_0} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) t dt$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{A(-A)}{\pi - (-\pi)} t dt$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{2A}{2\pi} t dt$$

$$= \frac{A}{2\pi^2} \left[\frac{t^2}{2} \right]_{-\pi}^{\pi}$$

$$a_0 = 0$$

$$a_n = \frac{2}{2\pi} \int_{-\pi}^{\pi} \left(\frac{A}{\pi}\right) t \cos n\omega t dt$$

$$= \frac{A}{\pi^2} \int_{-\pi}^{\pi} t \cos n\omega t dt$$

$$= \frac{A}{\pi^2} \int_{-\pi}^{\pi} t \cos nt dt$$

$$= \frac{A}{\pi^2} \left[\left(\frac{t \sin nt}{n} \right)_{-\pi}^{\pi} - \frac{1}{n} \int_{-\pi}^{\pi} \sin nt dt \right]$$

$$= \frac{A}{\pi^2} \left[-\frac{1}{n} \left(-\frac{\cos nt}{n} \right)_{-\pi}^{\pi} \right]$$

$$= \frac{-A}{n^2 \pi^2} [-\cos n\pi + \cos n\pi]$$

$$= \frac{2A}{n^2 \pi^2} [\cos n\pi - \cos n\pi]$$

$$\boxed{a_n = 0}$$

$$b_n = \frac{2}{2\pi} \int_{-\pi}^{\pi} \left(\frac{A}{\pi}\right) t \sin n\omega t dt$$

$$= \frac{A}{\pi^2} \int_{-\pi}^{\pi} t \sin nt dt$$

$$= \frac{A}{\pi^2} \left[\left(-\frac{t \cos nt}{n} \right)_{-\pi}^{\pi} + \frac{1}{n} \int_{-\pi}^{\pi} \cos nt dt \right]$$

$$= \frac{A}{\pi^2} \left[\left(-\frac{t \cos nt}{n} \right)_{-\pi}^{\pi} + \frac{1}{n^2} (\sin nt)_{-\pi}^{\pi} \right]$$

$$= \frac{A}{\pi^2} \left[\frac{-\pi \cos n\pi}{n} + \frac{\pi \cos n\pi}{n} + \frac{1}{n^2} [\sin n\pi + \sin n\pi] \right]$$

$$= \frac{-2A\pi \cos n\pi}{\pi^2 n^2}$$

$$\boxed{b_n = \frac{-2A}{n\pi} \cos n\pi}$$

$$b_1 = \frac{2A}{\pi}, \quad b_2 = \frac{-A}{\pi}, \quad b_3 = \frac{2A}{3\pi}$$

Fourier expansion is

$$f(t) = \frac{2A}{\pi} \sin \omega t - \frac{A}{\pi} \sin 2\omega t + \frac{2A}{3\pi} \sin 3\omega t - \dots$$

Fourier Transform:

→ Fourier transform also used to find out the solution for electrical circuit by using non-sinusoidal, non-periodic and periodic wave forms.

→ It is represented by symbol $f(j\omega)$. Then,

$$f(j\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

properties:

1. Linearity: If $f(j\omega)$ is the Fourier transform of time function $f(t)$. Then it satisfies the following relation $a_1 f_1(t) + a_2 f_2(t) = a_1 F_1(j\omega) + a_2 F_2(j\omega)$.

2. Time Scaling property: If $f(t)$ is the time function and $f(j\omega)$ is the Fourier transform function, Then, $F[f(at)] = \frac{1}{|a|} \left(\frac{F(j\omega)}{a} \right)$

3. Time Shifting property: If $f(t)$ is the time function and it is shifted to t_0 and $F(j\omega)$ is the Fourier transform, Then $F[f(t-t_0)] = e^{-j\omega t_0} F(j\omega)$.

4. Frequency shifting property: $F(j\omega)$ is the Fourier transform function and $f(t)$ is the time function, and ω_0 is the frequency shifting. Then

$$F[f(t) e^{j\omega_0 t}] = F[j(\omega - \omega_0)]$$

5. Modulation property: $F(j\omega)$ is the Fourier transform function and $f(t)$ is the time function, Then

$$\text{modulation property } F[f(t) \cos \omega_0 t] = \frac{1}{2} [F(j(\omega - \omega_0)) + F(j(\omega + \omega_0))]$$

$$\text{Ily } F[f(t) \sin \omega_0 t] = \frac{1}{2j} [F(j(\omega - \omega_0)) - F(j(\omega + \omega_0))]$$

6. Time differentiation property: If $F(j\omega)$ is the Fourier transform function and $f(t)$ is the time function, Then $F\left[\frac{d}{dt} f(t)\right] = (j\omega) F(j\omega)$.

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7. Frequency differentiation property: If $F(j\omega)$ is the Fourier transform function and $f(t)$ is the time function. Then,

$$\frac{d}{dt} f(t) \rightarrow F(j\omega) = F(j\omega) \cdot (-jt)$$

8. Complex Translation property: If $F(j\omega)$ is the Fourier transform function and $f(t)$ is the time function. Then,

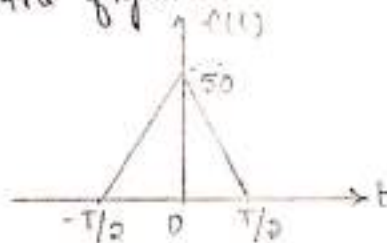
$$F[f(t)e^{at}] = F(j\omega - a)$$

$$\text{or } F[f(t)e^{-at}] = F(j\omega + a)$$

9. Convolution property: If $F_1(j\omega)$ is the Fourier transform function and $f_1(t)$ is the time function. Then,

$$F[f_1(t) * f_2(t)] = F_1(j\omega) \times F_2(j\omega).$$

1. Find the Fourier transform of a single triangular pulse as shown in the figure.

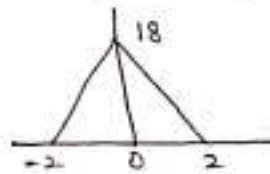


$$F(j\omega) = \int_{-T/2}^{T/2} f(t) e^{-j\omega t} dt$$

$$= \int_{-T/2}^0 f(t) e^{-j\omega t} dt + \int_0^{T/2} f(t) e^{-j\omega t} dt$$

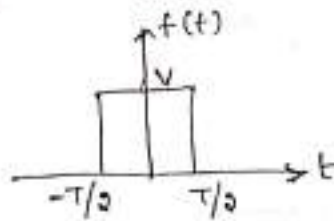
$$= \int_{-T/2}^0 \frac{y_2 - y_1}{x_2 - x_1} t e^{-j\omega t} dt + \int_0^{T/2} \frac{y_2 - y_1}{x_2 - x_1} t e^{-j\omega t} dt$$

Find the Fourier transform of a single triangular pulse as shown in the figure.



$$\begin{aligned}
 \text{Sol: } F(j\omega) &= \int_{-2}^2 f(t) e^{-j\omega t} dt \\
 &= \int_{-2}^0 f(t) e^{-j\omega t} dt + \int_0^2 f(t) e^{-j\omega t} dt \\
 &= \int_{-2}^0 \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (t+2) e^{-j\omega t} dt + \int_0^2 \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (t-2) e^{-j\omega t} dt \\
 &= \int_{-2}^0 \frac{18}{2} (t+2) e^{-j\omega t} dt + \int_0^2 \frac{-18}{2} (t-2) e^{-j\omega t} dt \\
 &= 9 \left\{ \int_{-2}^0 (t+2) e^{-j\omega t} dt + \int_0^2 -(t-2) e^{-j\omega t} dt \right\} \\
 &= 9 \left\{ \left[(t+2) \frac{e^{-j\omega t}}{-j\omega} - \int \frac{e^{-j\omega t}}{-j\omega} dt \right]_{-2}^0 - \left[(t-2) \frac{e^{-j\omega t}}{-j\omega} - \int \frac{e^{-j\omega t}}{-j\omega} dt \right]_0^2 \right\} \\
 &= 9 \left\{ \left[(t+2) \frac{e^{-j\omega t}}{-j\omega} - \frac{e^{-j\omega t}}{(-j\omega)^2} \right]_{-2}^0 - \left[(t-2) \frac{e^{-j\omega t}}{-j\omega} - \frac{e^{-j\omega t}}{(-j\omega)^2} \right]_0^2 \right\} \\
 &= 9 \left\{ \left(\frac{-2}{j\omega} + \frac{1}{(j\omega)^2} + \frac{e^{j2\omega}}{(j\omega)^2} \right) - \left(-\frac{e^{-j2\omega}}{(j\omega)^2} - \frac{2}{j\omega} + \frac{1}{(j\omega)^2} \right) \right\} \\
 &= 9 \left\{ \frac{e^{j2\omega}}{(j\omega)^2} + \frac{e^{-j2\omega}}{(j\omega)^2} - \frac{2}{(j\omega)^2} \right\} \\
 &= 9 \left\{ \frac{e^{j2\omega} + e^{-j2\omega}}{(j\omega)^2} - \frac{2}{(j\omega)^2} \right\} \\
 &= 9 \left\{ \frac{(e^{j2\omega} + e^{-j2\omega}) - 2}{(j\omega)^2 \times \frac{4}{4}} \right\} \\
 &= 36 \left\{ \frac{e^{-j2\omega} + e^{j2\omega}}{(j2\omega)^2} - \frac{2}{(j2\omega)^2} \right\}
 \end{aligned}$$

3. Find the fourier transform for the following wave form.



$$\text{Sol: } F(j\omega) = \int_{-T/2}^{T/2} f(t) e^{-j\omega t} dt$$

$$\sin \omega = \frac{e^{j\omega} - e^{-j\omega}}{2j}$$

$$\cos \omega = \frac{e^{j\omega} + e^{-j\omega}}{2}$$

$$= \int_{-T/2}^0 f(t) e^{-j\omega t} dt + \int_0^{T/2} f(t) e^{-j\omega t} dt$$

$$= \int_{-T/2}^0 V e^{-j\omega t} dt + \int_0^{T/2} V e^{-j\omega t} dt$$

$$= V \left\{ \left(\frac{e^{-j\omega t}}{-j\omega} \right)_{-T/2}^0 + \left(\frac{e^{-j\omega t}}{-j\omega} \right)_0^{T/2} \right\}$$

$$= V \left\{ \frac{-1}{j\omega} - \frac{e^{j\frac{T}{2}\omega}}{-j\omega} + \frac{e^{-j\frac{T}{2}\omega}}{-j\omega} + \frac{1}{j\omega} \right\}$$

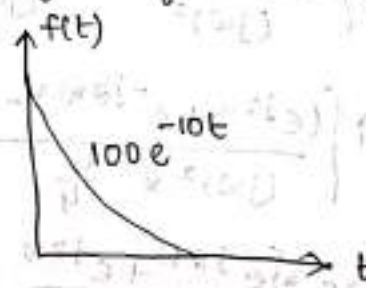
$$= V \left\{ \frac{j\omega \frac{T}{2}}{j\omega} + \frac{-j\omega \frac{T}{2}}{j\omega} \right\}$$

$$= V \left\{ \frac{e^{j\omega T/2} - e^{-j\omega T/2}}{j\omega \times \frac{2}{2}} \right\}$$

$$= 2V \left\{ \frac{e^{j\omega T/2} - e^{-j\omega T/2}}{2j\omega} \right\}$$

$$= 2V \left(\frac{\sin \omega T/2}{\omega} \right)$$

4. Obtain Fourier transform of following figure.



Sol: $F(j\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$

$$= \int_{-\infty}^0 f(t) e^{-j\omega t} dt + \int_0^{\infty} f(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 (0) e^{-j\omega t} dt + \int_0^{\infty} 100 e^{-10t} e^{-j\omega t} dt$$

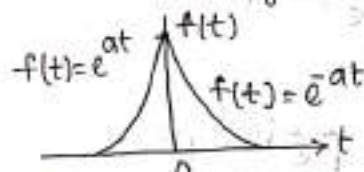
$$= 100 \int_0^{\infty} e^{-(10+j\omega)t} dt$$

$$= 100 \left[\frac{e^{-(10+j\omega)t}}{-(10+j\omega)} \right]_0^{\infty}$$

$$= 100 \times \left(0 - \left(\frac{-1}{10+j\omega} \right) \right)$$

$$F(j\omega) = \frac{100}{10+j\omega}$$

4. Find the Fourier transform of the following signal.



Sol: From the figure,

$$f(t) = e^{-at} \quad , \quad 0 < t < \infty$$

$$= e^{at} \quad , \quad -\infty < t < 0$$

$$F(j\omega) = \int_{-\infty}^0 f(t) e^{-j\omega t} dt + \int_0^{\infty} f(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 e^{at} e^{-j\omega t} dt + \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 e^{(a-j\omega)t} dt + \int_0^{\infty} e^{(-a-j\omega)t} dt$$

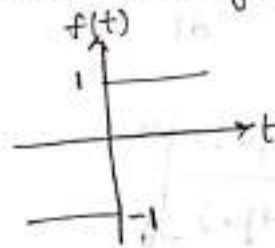
$$= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0 + \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$f(j\omega) = \frac{1}{a-j\omega} + \frac{1}{a+j\omega}$$

$$f(j\omega) = \frac{a+j\omega+a-j\omega}{a^2+\omega^2}$$

$$f(j\omega) = \frac{2a}{a^2+\omega^2}$$

5. Determine the Fourier transform of the following signal.



Sol: Here, $f(t) = 1, 0 < t < \alpha$
 $f(t) = -1, -\alpha < t < 0$

$$F(j\omega) = \int_{-\infty}^0 f(t) e^{-j\omega t} dt + \int_0^{\infty} f(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 e^{-j\omega t} dt - \int_0^{\infty} e^{-j\omega t} dt$$

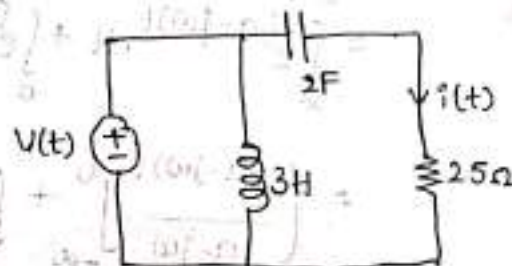
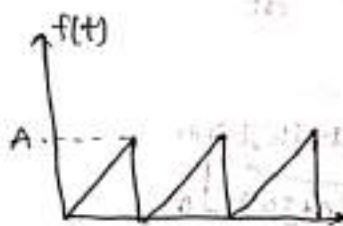
$$= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-\infty}^0 - \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^{\infty}$$

$$= \left(\frac{1}{-j\omega} + \frac{1}{-j\omega} \right)$$

$$= \frac{2}{j\omega}$$

Fourier transform application to Electrical Circuits:

6. Voltage having the waveform shown in the figure is applied to a electrical circuit which is also shown in the figure. Then determine the current flowing through the resistor.



sol:

$$V(t) = \frac{V(t)}{25 + \frac{1}{j\omega C}}$$

$$= \frac{V(t)}{25 + \frac{1}{2j\omega}}$$

$$V(t) = a_0 + a_1 \cos \omega t + a_2 \cos 2\omega t + \dots + b_1 \sin \omega t + b_2 \sin 2\omega t + \dots$$

$$a_0 = \frac{1}{T_0} \int_0^T f(t) dt$$

$$= \frac{1}{T} \int_0^T \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (t - 0) dt$$

$$= \frac{1}{T} \int_0^T \left(\frac{A - 0}{T - 0} \right) t dt$$

$$= \frac{A}{T^2} \int_0^T t dt$$

$$= \frac{A}{T^2} \left(\frac{t^2}{2} \right)_0^T$$

$$= \frac{A}{T^2} \left(\frac{T^2}{2} \right)$$

$$\boxed{a_0 = \frac{A}{2}}$$

$$a_n = \frac{2}{T_0} \int_0^T f(t) \cos n\omega t dt$$

$$= \frac{2}{T} \int_0^T \frac{A}{T} t \cos \frac{2n\pi}{T} t dt$$

$$= \frac{2A}{T^2} \int_0^T t \cos \frac{2n\pi}{T} t dt$$

$$= \frac{2A}{T^2} \left[\frac{t \cdot T}{2n\pi} \sin \frac{2n\pi}{T} t - \frac{T}{2n\pi} \int \sin \frac{2n\pi}{T} t dt \right]_0^T$$

$$= \frac{2A}{T^2} \left[\frac{t \cdot T}{2n\pi} \sin \frac{2n\pi}{T} t + \left(\frac{T}{2n\pi} \right)^2 \cos \frac{2n\pi}{T} t \right]_0^T$$

$$= \frac{2A}{T^2} \left[0 + \left(\frac{T}{2n\pi} \right)^2 \cos 2n\pi - \left(\frac{T}{2n\pi} \right)^2 \cos 0 \right]$$

$$\therefore \text{For } n=1, a_1 = \frac{2A}{T^2} \left[\left(\frac{T}{2\pi} \right)^2 - \left(\frac{T}{2\pi} \right)^2 \right]$$

$$= 0$$

$$\text{For } n=2, a_2 = 0.$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t \, dt$$

$$= \frac{2}{T} \int_0^T \frac{A}{T} t \sin \frac{2n\pi}{T} t \, dt$$

$$= \frac{2A}{T^2} \int_0^T t \sin \frac{2n\pi}{T} t \, dt$$

$$= \frac{2A}{T^2} \left[t \left(\frac{-\cos \frac{2n\pi}{T} t}{\left(\frac{2n\pi}{T} \right)} \right) + \int \frac{\cos \frac{2n\pi}{T} t}{\frac{2n\pi}{T}} dt \right]_0^T$$

$$= \frac{2A}{T^2} \left[\frac{tT}{2n\pi} \left(-\cos \frac{2n\pi}{T} t \right) + \left(\frac{T}{2n\pi} \right)^2 \sin \frac{2n\pi}{T} t \right]_0^T$$

$$= \frac{2A}{T^2} \left[\frac{T^2}{2n\pi} (-1) \right]$$

$$= -\frac{A}{n\pi}$$

$$\text{For } n=1, b_1 = -\frac{A}{\pi} \quad \therefore \text{For } n=2, b_2 = -\frac{A}{2\pi}$$

$$v(t) = \frac{A}{2} - \frac{A}{\pi} \sin \omega t - \frac{A}{2\pi} \sin 2\omega t$$

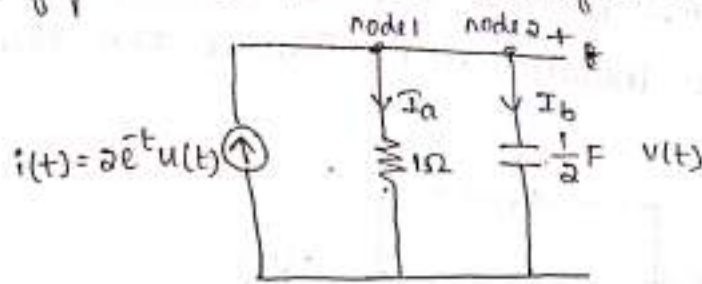
$$i(t) = \frac{v(t)}{25 + \frac{1}{2j\omega}}$$

$$= \frac{\frac{A}{2} - \frac{A}{\pi} \sin \omega t - \frac{A}{2\pi} \sin 2\omega t}{25 + \frac{1}{2j\omega}}$$

$$\left[\frac{A}{2} - \frac{A}{\pi} \sin \omega t - \frac{A}{2\pi} \sin 2\omega t \right] \frac{45}{57}$$

$$\left[\frac{A}{2} \cos^2 \left(\frac{T}{2\pi} \right) - \frac{A}{\pi} \cos^2 \left(\frac{T}{\pi} \right) + 0 \right] \frac{45}{57}$$

7. Find the response voltage in the network shown in the figure. Use Fourier transform method.



Using KCL,

$$i(t) = i_a + i_b$$

$$2e^t u(t) = \frac{v(t)}{1} + \frac{v(t)}{X_c}$$

$$2e^t u(t) = v(t) + \frac{v(t)}{\left(\frac{1}{j\omega C}\right)}$$

$$2e^t u(t) = v(t) + j\omega C v(t)$$

$$2e^t u(t) = v(t) [1 + j0.5\omega]$$

Apply Fourier transform

$$2 \times \frac{1}{j\omega + 1} = v(j\omega) [1 + j0.5\omega]$$

$$v(j\omega) = \frac{2}{(j\omega + 1)(1 + j0.5\omega)}$$

$$v(j\omega) = \frac{2}{(1 + j\omega)\left(\frac{1}{2}\right)(2 + j\omega)}$$

$$v(j\omega) = \frac{4}{(1 + j\omega)(2 + j\omega)}$$

Using partial fractions.

$$v(j\omega) = \frac{A}{1 + j\omega} + \frac{B}{2 + j\omega}$$

$$A = 4, B = -4$$

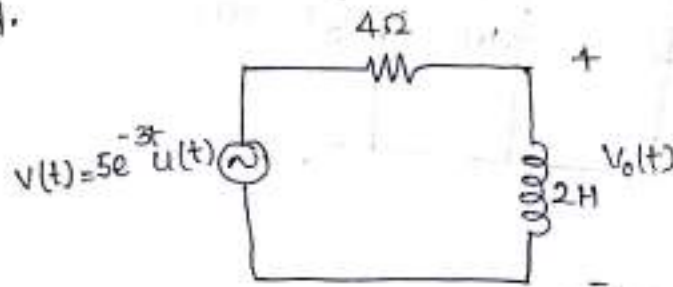
$$v(j\omega) = \frac{4}{1 + j\omega} - \frac{4}{2 + j\omega}$$

Inverse Laplace Fourier transform

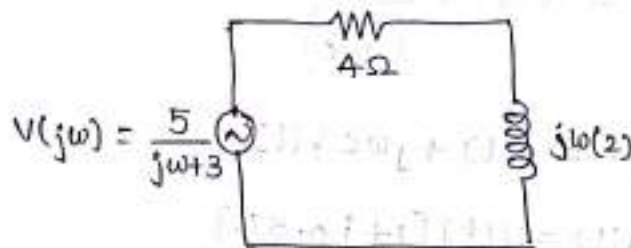
$$v(t) = 4e^t u(t) - 4e^{2t} u(t)$$

8. For input voltage $v(t) = 5e^{-3t}u(t)$ is applied to a series R-L circuit with 4Ω resistor & $L = 2H$. Find the output voltage $V_o(t)$ across inductor using Fourier transform method.

Sol:



For the above circuit first convert into Fourier transform circuit, then apply voltage division method.



Using voltage division rule,

$$\text{voltage across inductor } [V_o(t)] = \frac{j\omega(2)}{4 + j\omega(2)} \times \frac{5}{j\omega + 3}$$

$$= \frac{j10\omega}{[4 + j(2\omega)](j\omega + 3)}$$

Using partial fractions,

$$V_o(j\omega) = \frac{A}{4 + j2\omega} + \frac{B}{j\omega + 3}$$

$$A = -20, B = 20$$

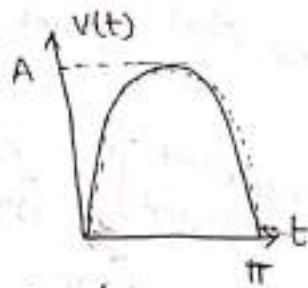
$$V_o(j\omega) = \frac{-20}{4 + j2\omega} + \frac{20}{j\omega + 3}$$

$$V_o(j\omega) = \frac{-10}{2 + j\omega} + \frac{20}{j\omega + 3}$$

Using inverse Fourier transform

$$V_o(t) = -10e^{-2t}u(t) + 20e^{-3t}u(t).$$

Find the Fourier transform of the sine pulse shown in the figure and sketch the magnitude & phase spectra.



sd: Magnitude spectra:

We know that,

$$\text{exponential form } C_n = \frac{1}{T_0} \int_{t_1}^{t_1+T} f(t) e^{-jn\omega t} dt$$

$$C_n = \frac{1}{T_0} \int_{t_1}^{t_1+T} f(t) [\cos n\omega t - j \sin n\omega t] dt$$

$$\text{Re}[C_n] + j \text{Im}[C_n] = \frac{1}{T_0} \int_{t_1}^{t_1+T} f(t) \cos n\omega t dt - \frac{j}{T_0} \int_{t_1}^{t_1+T} f(t) \sin n\omega t dt$$

$$\text{Re}[C_n] = \frac{1}{T_0} \int_{t_1}^{t_1+T} f(t) \cos n\omega t dt$$

$$\text{Im}[C_n] = -\frac{1}{T_0} \int_{t_1}^{t_1+T} f(t) \sin n\omega t dt$$

$$\text{Magnitude spectra } |C_n| = \sqrt{\text{Re}[C_n]^2 + \text{Im}[C_n]^2}$$

Phase spectra:

$$\phi = \tan^{-1} \left(\frac{\text{Im}[C_n]}{\text{Re}[C_n]} \right)$$

$$= \tan^{-1} \left(\frac{-\int_{t_1}^{t_1+T} f(t) \sin n\omega t dt}{\int_{t_1}^{t_1+T} f(t) \cos n\omega t dt} \right)$$

Now,

$$\begin{aligned} v(t) &= \frac{1}{T} \int_0^T f(t) e^{-j\omega t} dt = I \quad \text{if } \omega = 1 \\ &= \frac{1}{\pi} \int_0^\pi f(t) e^{-j\omega t} dt \quad \text{if } \omega = 1 \\ &= \frac{A}{\pi} \int_0^\pi \sin \omega t e^{-j\omega t} dt \end{aligned}$$

$$v(t) = \frac{A}{\pi} \left[\sin t \frac{e^{-j\omega t}}{-j\omega} - \int \cos t \frac{e^{-j\omega t}}{-j\omega} dt \right]_0^{\pi}$$

$$I = \frac{A}{\pi} \left[\sin t \frac{e^{-j\omega t}}{-j\omega} - \cos t \frac{e^{-j\omega t}}{(-j\omega)^2} + \int -\sin t \frac{e^{-j\omega t}}{(-j\omega)^2} dt \right]_0^{\pi}$$

$$I = \frac{A}{\pi} \left[\sin t \frac{e^{-j\omega t}}{-j\omega} - \cos t \frac{e^{-j\omega t}}{(-j\omega)^2} \right]_0^{\pi} - \frac{A}{(j\omega)^2 \pi} \int_0^{\pi} \sin t e^{-j\omega t} dt$$

$$I = \frac{A}{\pi} \left[\sin t \frac{e^{-j\omega t}}{-j\omega} - \cos t \frac{e^{-j\omega t}}{(j\omega)^2} \right]_0^{\pi} - \frac{1}{(j\omega)^2} I$$

$$I = \frac{A}{\pi} \left[\frac{e^{-j\omega \pi}}{(j\omega)^2} + \frac{1}{(j\omega)^2} \right] + \frac{I}{\omega^2}$$

$$I \left(1 - \frac{1}{\omega^2} \right) = \frac{A}{\pi} \left(\frac{e^{-j\omega \pi} + 1}{(j\omega)^2} \right)$$

$$I = \frac{A}{\pi} \left(\frac{1 + e^{-j\omega \pi}}{-\omega^2 \left(\frac{\omega^2 - 1}{\omega^2} \right)} \right)$$

$$I = \frac{A}{\pi} \left(\frac{1 + e^{-j\omega \pi}}{(1 - \omega^2)} \right)$$

Magnitude spectra:

$$\{I\} = \frac{A}{\pi(1-\omega^2)} (1 + \cos \omega \pi - j \sin \omega \pi)$$

$$|I| = \sqrt{\frac{A}{\pi(1-\omega^2)} [(1 + \cos \omega \pi)^2 + (\sin \omega \pi)^2]}$$

Phase spectra:

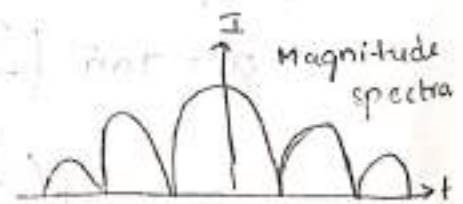
$$\phi = \tan^{-1} \left(\frac{-\sin \omega \pi}{1 + \cos \omega \pi} \right)$$

$$= \tan^{-1} \left(\frac{-2 \sin \frac{\omega \pi}{2} \cos \frac{\omega \pi}{2}}{\cos^2 \frac{\omega \pi}{2} \cdot 2} \right)$$

$$= \tan^{-1} \left(\frac{-\sin \frac{\omega \pi}{2}}{\cos \frac{\omega \pi}{2}} \right)$$

$$= \tan^{-1} \left(-\tan \frac{\omega \pi}{2} \right)$$

$$\boxed{\phi = -\frac{\omega \pi}{2}}$$



Fourier Transforming

Fourier Theorem - Trigonometric form and exponential form of four Series - conditions of symmetry - line spectra and phase angle spectra - Analysis of electrical circuits excited by non-sinusoidal sources of periodic waveforms - Fourier integrals and Fourier transforms - Properties of Fourier transforms and application to electrical circuits.

Jean Baptiste Joseph Fourier (1768-1830) developed a technique of analyzing non sinusoidal waveforms applicable to a wide range of Engineering problems. Later it was applied to analyze electric circuits excited by non sinusoidal waveforms.

Fourier Theorem

Fourier Theorem which applies to all the functions that normally arise in physical and engineering application to which Fourier series can be obtained as follows.

A function $f(t)$ can be written as

$$f(t) = \frac{a_0}{2} + a_1 \cos \omega t + a_2 \cos 2\omega t + \dots + b_1 \sin \omega t + b_2 \sin 2\omega t + \dots$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t) \quad \text{--- (1)}$$

and the coefficient a_n and b_n are given by

$$a_0 = \frac{2}{T} \int_0^T f(t) dt \quad \text{--- (2)}$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega t dt \quad \text{--- (3)}$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t dt \quad \text{--- (4)}$$

It is to be noted that the Fourier Series can be written in a more compact form either in sin or cos term by combining the sin and cos term as follows.

Let us consider the n th harmonic term

$$f_n(t) = a_n \cos n\omega t + b_n \sin n\omega t$$

$$= \sqrt{a_n^2 + b_n^2} \left[\frac{a_n}{\sqrt{a_n^2 + b_n^2}} \cos n\omega t + \frac{b_n}{\sqrt{a_n^2 + b_n^2}} \sin n\omega t \right]$$

$$= A_n \left[\cos \theta_n \cos n\omega t + \sin \theta_n \sin n\omega t \right]$$

$$= A_n \left[\cos (n\omega t - \theta_n) \right]$$

$\therefore$ Fourier Series can be written as [in terms of cos form]

$$f(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos (n\omega t - \theta_n)$$

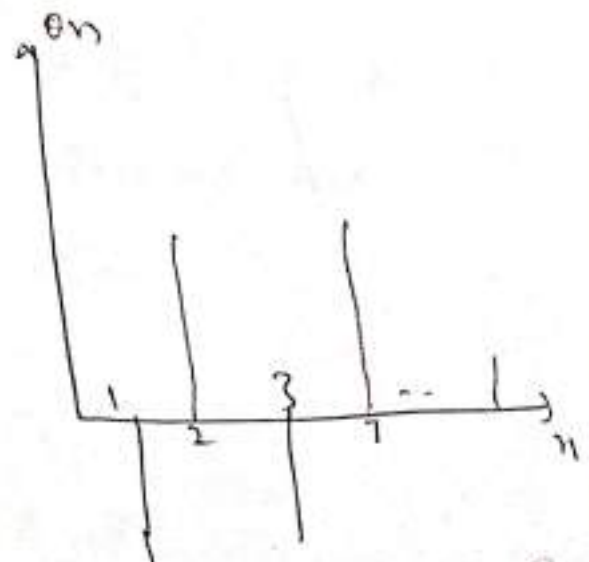
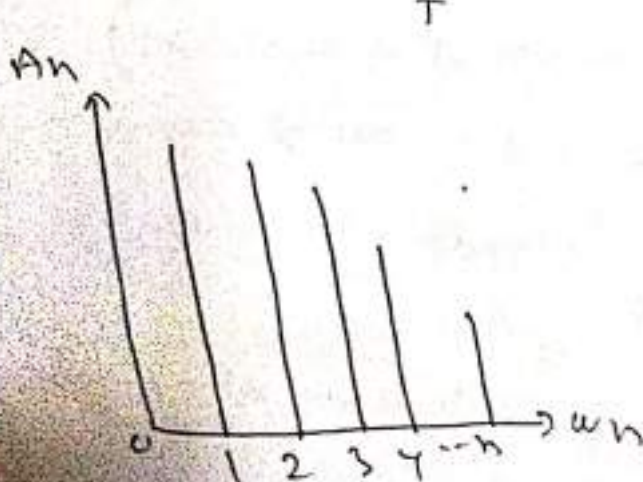
where
 $\theta_n = \tan^{-1} \frac{b_n}{a_n}$

$\therefore$ Fourier Series in terms of sin form

$$f(t) = A_0 + \sum A_n \sin (n\omega t - \theta_n)$$

where
 $\theta_n = \tan^{-1} \frac{a_n}{b_n}$

where $\omega = \frac{2\pi}{T}$



Exponential form of Fourier series

Fourier series can be written as

$$f(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t - \theta_n)$$

$$\text{where } \omega = \frac{2\pi}{T_0}$$

$n=1$, one cycle covers T_0 seconds while $A_1 \cos(\omega_0 t - \theta_1)$ is termed as fundamental harmonic

$n=2$, Two cycle covers T_0 seconds while $A_2 \cos(2\omega_0 t - \theta_2)$ is termed as 2nd harmonic

In the continued fashion

$n=k$, k cycle covers within T_0 seconds while $A_k \cos(k\omega_0 t - \theta_k)$ is termed as k^{th} harmonic.

Using Euler's identity

$$f(t) = a_0 + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} C_n e^{jn\omega_0 t}$$

where C_n = complex Fourier coefficients

$$= a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + b_n \sin n\omega_0 t$$

Fourier coefficient C_n can be evaluated as follows

$$\therefore f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

multiplying both sides by $e^{-jk\omega_0 t}$ and integrating

over the interval t_1 to $t_1 + T_0$, we get

$$\int_{t_1}^{t_1+T_0} f(t) e^{-jk\omega_0 t} dt = \int_{t_1}^{t_1+T_0} \left(\sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \right) e^{-jk\omega_0 t} dt$$

$$= C_k T_0$$

$$\text{Since } \int_{t_1}^{t_1+T_0} e^{j(n-k)\omega t} dt = \begin{cases} 0 & \text{for } n \neq k \\ = T_0 & \text{for } n = k \end{cases}$$

$\therefore$ The fourier coefficients are defined by the expression

$$C_n = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} f(t) e^{-jn\omega t} dt \quad \text{--- (1)}$$

The above expression is the exponential form of fourier series.

Trigonometric form of fourier series

From eq (1), we can write

$$\begin{aligned} 2C_n &= \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) e^{-jn\omega t} dt \\ &= \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) [\cos n\omega t - j \sin n\omega t] dt \\ &= \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \cos n\omega t dt - j \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \sin n\omega t dt \end{aligned} \quad \text{--- (2)}$$

However C_n being a complex coefficient

$$2C_n = a_n - j b_n \quad \text{--- (2)}$$

Comparing eq (1) & (2)

$$a_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \cos n\omega t dt, \quad b_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} f(t) \sin n\omega t dt \quad \text{--- (4)}$$

Also from eq (1), a_0 being written here as a_0 then

$$a_0 = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} f(t) dt \quad (5)$$

Symmetry in Fourier Series

There are three types of Symmetry

1. Even function Symmetry:

A function is said to be even if

$$f(t) = f(-t)$$

An even function is symmetrical about the vertical axis.

In order to determine the Fourier coefficients, the condition of symmetry being applied

$$\text{Let } t_1 = -\frac{T_0}{2}$$

$$a_0 = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} f(t) dt = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f(t) dt = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f(t) dt$$

$$= \frac{1}{T_0} \int_{-\frac{T_0}{2}}^0 f(t) dt + \frac{1}{T_0} \int_0^{\frac{T_0}{2}} f(t) dt$$

Now change the variable of the first integral so that $t = -x$ in the first integral. Then

$$f(-x) = f(x), \quad dt = d(-x) = -dx \quad \text{and } x = \frac{T_0}{2} \text{ to } 0$$

$$a_0 = \frac{1}{T_0} \int_{\frac{T_0}{2}}^0 f(x) (-dx) + \frac{1}{T_0} \int_0^{\frac{T_0}{2}} f(t) dt$$

$$= \frac{1}{T_0} \int_0^{\frac{T_0}{2}} f(x) dx + \frac{1}{T_0} \int_0^{\frac{T_0}{2}} f(t) dt$$

$$a_0 = \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) dt$$

Similarly

$$\begin{aligned} a_n &= \frac{2}{T_0} \int_{-\frac{T_0}{2}}^0 f(t) \cos n\omega_0 t dt + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \cos n\omega_0 t dt \\ &= \frac{2}{T_0} \int_{\frac{T_0}{2}}^0 f(x) \cos(-n\omega_0 x) (-dx) + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \cos n\omega_0 t dt \\ &= \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(x) \cos(n\omega_0 x) dx + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \cos n\omega_0 t dt \end{aligned}$$

$$a_n = \frac{4}{T_0} \int_0^{\frac{T_0}{2}} f(t) \cos n\omega_0 t dt$$

$$\begin{aligned} \text{Also } b_n &= \frac{2}{T_0} \int_{-\frac{T_0}{2}}^0 f(t) \sin n\omega_0 t dt + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \sin n\omega_0 t dt \\ &= \frac{2}{T_0} \int_{\frac{T_0}{2}}^0 f(x) \sin(-n\omega_0 x) (-dx) + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \sin n\omega_0 t dt \\ &= -\frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(x) \sin n\omega_0 x dx + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \sin n\omega_0 t dt \end{aligned}$$

$$b_n = 0$$

(2) Odd function Symmetry

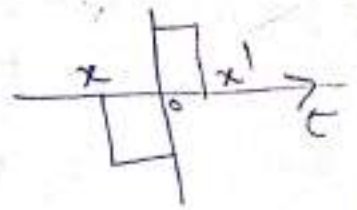
A function is said to be odd if

$$f(t) = -f(-t)$$

$$a_0 = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^0 f(t) dt + \frac{1}{T_0} \int_0^{\frac{T_0}{2}} f(t) dt$$

$$a_0 = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} -f(x) (-dx) + \frac{1}{T_0} \int_0^{\frac{T_0}{2}} f(t) dt$$

$$= -\frac{1}{T_0} \int_0^{\frac{T_0}{2}} f(x) dx + \frac{1}{T_0} \int_0^{\frac{T_0}{2}} f(t) dt$$



$$a_0 = 0$$

$$a_n = \frac{2}{T_0} \int_{-\frac{T_0}{2}}^0 -f(x) \cos(-n\omega_0 x) (-dx) + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \cos n\omega_0 t dt$$

$$= -\frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(x) \cos n\omega_0 x dx + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \cos n\omega_0 t dt$$

$$a_n = 0$$

$$b_n = \frac{2}{T_0} \int_{-\frac{T_0}{2}}^0 -f(x) \sin(-n\omega_0 x) (-dx) + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \sin n\omega_0 t dt$$

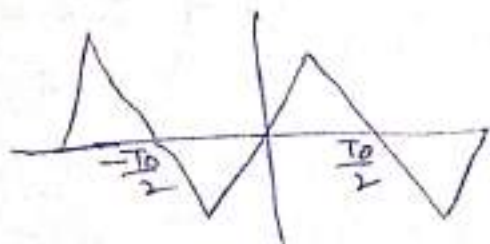
$$= \frac{2}{T_0} \int_0^{\frac{T_0}{2}} + f(x) \sin(n\omega_0 x) dx + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \sin n\omega_0 t dt$$

$$b_n = \frac{4}{T_0} \int_0^{\frac{T_0}{2}} f(t) \sin n\omega_0 t dt$$

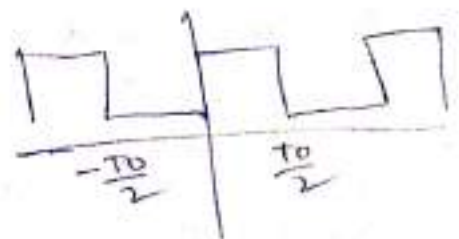
(3) Half wave Symmetry

A function is said to be half wave symmetry if

$$f(t) = -f\left(t - \frac{T_0}{2}\right)$$



half symmetry



here .

$$t - \frac{T_0}{2} = x$$

$$t = \frac{T_0}{2} + x$$

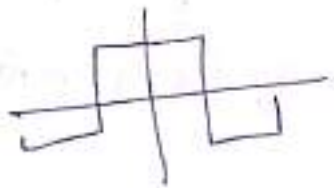
$$a_0 = 0$$

$$b_n = a_n = 0 \quad \text{when } n \text{ is even}$$

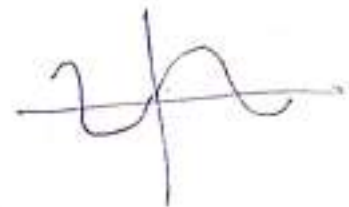
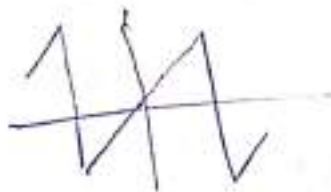
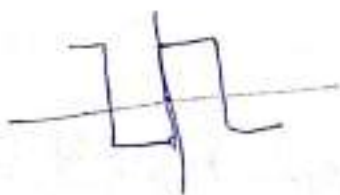
$$\text{and } a_n = \frac{4}{T_0} \int_0^{\frac{T_0}{2}} f(t) \cos n\omega t \, dt \quad \text{when } n \text{ is odd}$$

$$b_n = \frac{4}{T_0} \int_0^{\frac{T_0}{2}} f(t) \sin n\omega t \, dt \quad \text{when } n \text{ is odd}$$

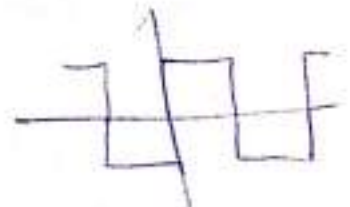
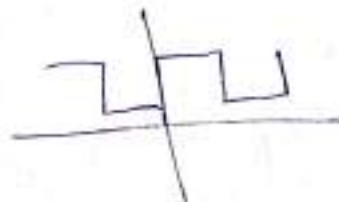
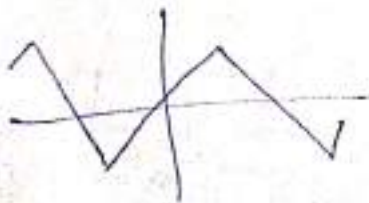
Even Symmetry



odd Symmetry



Half wave Symmetry



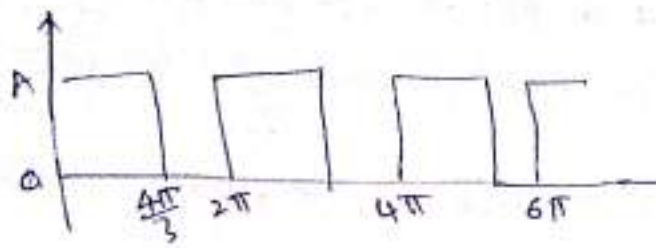
Frequency spectrum

- The frequency spectrum of the function $f(t)$ expressed as a Fourier series consists of a plot of the amplitude of the harmonics vs frequency (commonly known as amplitude spectrum).
- The plot of phase of harmonics vs frequency which we call the phase spectrum.
- However, the frequency components being discrete, the spectra are called line spectra and it illustrates the frequency content of the signal. Hence $c_0 = a_0$ and $c_n = \sqrt{a_n^2 + b_n^2}$

Properties of Fourier analysis

- ① For an even waveform, all the terms of the Fourier series are cosine terms i.e. ~~no~~ terms with a_0 coefficients do exist and no sine terms are present since $b_n = 0$.
- ② For an odd waveform, the Fourier series contains only sine terms and there is no avg value $a_0 = 0$ and no cosine terms since $a_n = 0$.
- ③ If the given waveform is of half wave symmetry, only odd harmonics are present in the series. The series would contain both sine and cosine terms when n is odd. The average value is zero.

- ① Obtain the coefficients of the exponential Fourier Series for the waveform shown in fig.



$$C_n = \frac{1}{T_0} \int_{t_1}^{t_1 + T_0} f(t) e^{-jn\omega t} dt$$

Sol

$$C_n = \frac{1}{T_0} \int_0^{T_0} f(t) e^{-jn\omega t} dt$$

$$= \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-jn\omega t} dt$$

$$= \frac{1}{2\pi} \int_0^{4\pi/3} A \cdot e^{-jn(\frac{2\pi}{T_0})t} dt$$

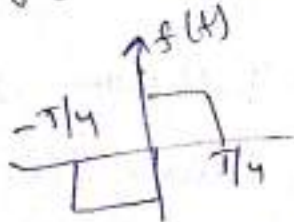
$$= \frac{1}{2\pi} \int_0^{4\pi/3} A e^{-jnt} dt$$

$$= \frac{jA}{2\pi n} \left[e^{-jn \cdot \frac{4\pi}{3}} - 1 \right]$$

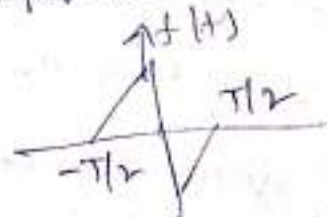
$$= \frac{jA}{2\pi n} \left[e^{-j(\frac{4n\pi}{3})} - 1 \right]$$

②

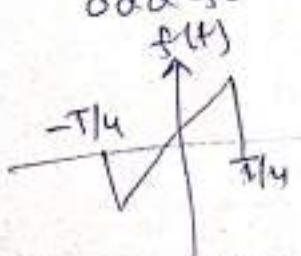
Find the waveform symmetry of the following fig.



odd function



odd function

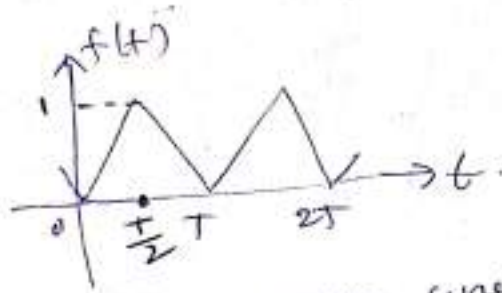


odd function



Even function

③ Find the Amplitude of the 5th harmonic of the given waveform



$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0}{\frac{T}{2} - 0} = \frac{2}{T}$$

Sol Given waveform is a even waveform, so only cosine terms would be present $a_n = \frac{4}{T} \int_0^{T/2} f(t) \cos\left(\frac{2n\pi}{T} t\right) dt$

$$a_n = \frac{4}{T} \int_0^{T/2} \frac{2t}{T} \cos\left(\frac{2n\pi}{T} t\right) dt$$

$$= \frac{4}{T} \times \frac{2}{T} \int_0^{T/2} t \cos \frac{10\pi t}{T} dt$$

$$= \frac{8}{T^2} \left[t \frac{\sin \frac{10\pi t}{T}}{\left(\frac{10\pi}{T}\right)} \right]_0^{T/2} - \int_0^{T/2} 1 \cdot \frac{\sin \frac{10\pi t}{T}}{\frac{10\pi}{T}} dt$$

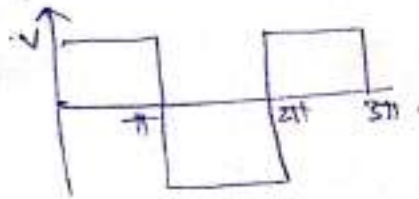
$$= \frac{8}{T^2} \left[\left(\frac{T}{2} \frac{\sin \frac{10\pi \times T}{2} \times \frac{T}{2}}{\frac{10\pi}{T}} - 0 \right) + \left(\frac{\cos \frac{10\pi t}{T}}{\frac{100\pi^2}{T^2}} \right) \right]_0^{T/2}$$

$$= \frac{8}{T^2} \left[\frac{T^2}{20\pi} (0) + \left(\frac{T^2}{100\pi^2} \times \cos \frac{5\pi T}{2T} - \frac{\cos 0}{\frac{100\pi^2}{T^2}} \right) \right]$$

$$= \frac{8}{T^2} \left[\frac{T^2}{100\pi^2} (\cos 5\pi - 1) \right]$$

$$= \frac{4}{25\pi^2} (\cos 5\pi - 1) = \frac{-4}{25\pi^2}$$

4) A square wave form is shown in fig. obtain the fourier series



So here $0 < wt < \pi$, $f(t) = V$, $\pi < wt < 2\pi$, $f(t) = -V$

Hence given waveform possesses symmetry i.e odd symmetry.

The fourier series does not contain any cosine term and the average value would be zero.

$$\therefore a_n = 0, a_0 = 0$$

$$a_n = \frac{2}{T_0} \int_{-\frac{T_0}{2}}^0 f(t) \cos n\omega_0 t dt + \frac{2}{T_0} \int_0^{\frac{T_0}{2}} f(t) \cos(n\omega_0 t) dt$$

$$= \frac{2}{2\pi} \int_0^{\pi} V \cos n\omega_0 t d(\omega t) + \frac{2}{2\pi} \int_{\pi}^{2\pi} -V \cos n\omega_0 t d(\omega t)$$

$$= \frac{V}{\pi} \left[\left(\frac{1}{n} \sin n\omega t \right)_0^{\pi} - \left(\frac{1}{n} \sin n\omega t \right)_{\pi}^{2\pi} \right]$$

$$= \frac{V}{\pi} \left[\frac{1}{n} \sin n\pi - \frac{1}{n} \sin n(2\pi) + \frac{1}{n} \sin n\pi \right]$$

$$= 0$$

$$b_n = \frac{1}{\pi} \left[\int_0^{\pi} V \sin n\omega t d(\omega t) + \int_{\pi}^{2\pi} -V \sin n\omega t d(\omega t) \right]$$

$$= \frac{V}{\pi} \left[\left(-\frac{\cos n\omega t}{n} \right)_0^{\pi} + \left(\frac{1}{n} \cos n\omega t \right)_{\pi}^{2\pi} \right]$$

$$= \frac{V}{\pi n} \times (-\cos n\pi + \cos 0 + \cos n 2\pi - \cos n\pi)$$

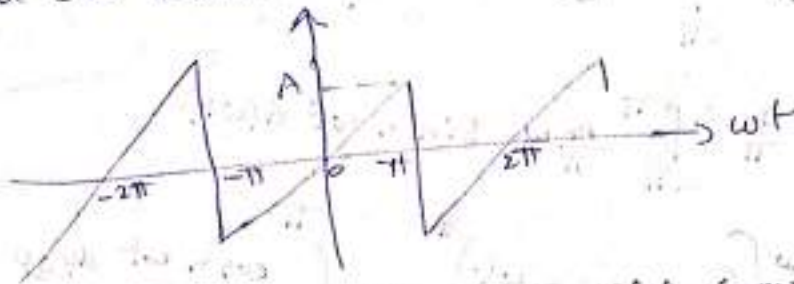
$$= \frac{2V}{\pi n} (1 - \cos n\pi) \quad \left| \begin{array}{l} n=1 \Rightarrow b_1 = \frac{4V}{\pi} \\ n=2 \Rightarrow b_2 = 0 \\ n=3 \Rightarrow b_3 = \frac{4V}{\pi} \end{array} \right.$$

$$\cos n\pi = -1 \text{ for } n = \text{odd}$$

$$= 1 \text{ for } n = \text{even}$$

$$f(t) = \frac{4V}{\pi} \sin \omega t + \frac{4V}{3\pi} \sin 3\pi t + \frac{4V}{5\pi} \sin 5\pi t + \dots$$

(5) Find the Fourier expansion of the given wave form



Sol By inspection, it is a odd symmetry.

For odd symmetry - If the period 0 to 2π is considered.

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{4}{T_0} \int_0^{T_0} f(t) \sin n(\omega t) d(\omega t)$$

$$= \frac{4}{2\pi} \int_0^{\pi} \frac{A}{\pi} t \sin n(\omega t) d(\omega t)$$

$$= \frac{4A}{2\pi} \times \frac{A}{\pi} \left[-\left(t \cdot \frac{\cos n(\omega t)}{n} \right) + \int_0^{\pi} \frac{\cos n\omega t}{n} d(\omega t) \right]$$

$$= -\frac{2A}{\pi^2} \times \frac{\pi}{n} \cos n\pi$$

$$b_n = -\frac{2A}{n\pi} \cos n\pi$$

$$n=1, b_1 = \frac{2A}{\pi}, \quad n=2, b_2 = -\frac{2A}{2\pi}, \quad n=3, b_3 = \frac{2A}{3\pi}$$

Fourier series becomes

$$f(t) = \frac{2A}{\pi} \sin \omega t - \frac{2A}{2\pi} \sin 2\omega t + \frac{2A}{3\pi} \sin 3\omega t - \frac{2A}{4\pi} \sin 4\omega t + \dots$$

(08)

$$b_n = \frac{2}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f(t) \sin n\omega_0 t \, dt$$

$$= \frac{2}{2\pi} \int_{-\pi}^{\pi} \frac{2A}{2\pi} t \sin n\omega t \, d(\omega t)$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{A}{\pi} \omega t \sin n\omega t \, d(\omega t)$$

$$= \frac{A\omega}{\pi^2} \left[-t \cdot \frac{\cos n\omega t}{n} - \int_{-\pi}^{\pi} \frac{\cos n\omega t}{n} d(\omega t) \right]$$

$$= \frac{A\omega}{\pi^2} \left[\frac{2A}{n} + \left(\frac{\sin n\omega t}{n^2} \right) \right]$$

$$= -\frac{2A}{\pi^2 n} \cos n\pi$$

$$\left(\omega = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1 \right)$$

$$= -\frac{2A}{\pi n} (\cos n\pi)$$

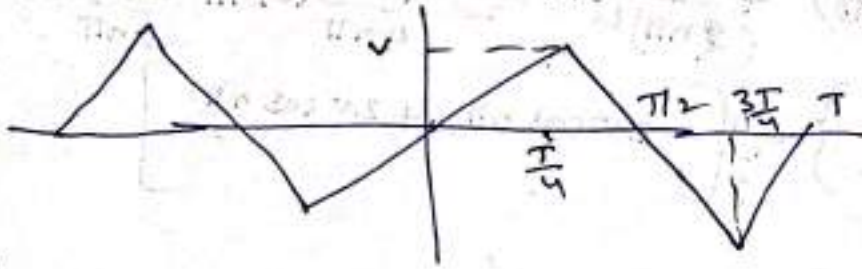
for $n=1$, $b_1 = \frac{2A}{\pi}$

$n=2$, $b_2 = -\frac{2A}{2\pi}$

$n=3$, $b_3 = +\frac{2A}{3\pi}$...

$$f(t) = \frac{2A}{\pi} \left(\sin \omega t - \frac{1}{2} \sin 2\omega t + \frac{1}{3} \sin 3\omega t - \dots \right)$$

→ obtain fourier series



Given function is odd function
So only bn apply

$$f(t) = \frac{4V}{T}t, \quad 0 < t < \frac{T}{4}$$

$$= -\frac{4V}{T}t + 2V, \quad \frac{T}{4} < t < \frac{3T}{4}$$

$$b_n = \frac{4}{T} \left[\int_0^{T/4} \frac{4V}{T}t \sin n\omega t dt + \int_{T/4}^{T/2} -\frac{4V}{T}t \sin n\omega t dt + \int_{T/2}^{3T/4} 2V \sin n\omega t dt \right]$$

but $\omega = \frac{2\pi}{T}$

$$= \frac{16V}{T^2} \left[\left(t \cdot \frac{-\cos n\omega t}{n\omega} \right)_0^{T/4} + \left(\frac{\sin n \frac{2\pi}{T} t}{(n \times \frac{2\pi}{T})^2} \right)_0^{T/4} \right] + \left(t \frac{\cos n\omega t}{n\omega} \right)_{T/4}^{T/2} - \left(\frac{\sin n \frac{2\pi}{T} t}{(n \times \frac{2\pi}{T})^2} \right)_{T/4}^{T/2} + \left[2V \frac{\cos n\omega t}{n\omega} \right]_{T/2}^{3T/4}$$

$$= \frac{16V}{T^2} \left(-\frac{T}{4} \left(\cos n \cdot \frac{2\pi}{T} \cdot \frac{T}{4} \right) - \frac{\cos 0}{n \times \frac{2\pi}{T}} \right) + \frac{\sin \frac{2\pi}{T} \cdot \frac{T}{4} \cdot 2}{(n \times \frac{2\pi}{T})^2}$$

$$+ \frac{T}{2} \frac{\cos n \frac{2\pi}{T} \cdot \frac{T}{2}}{n \times \frac{2\pi}{T}} - \frac{T}{4} \frac{\cos n \frac{2\pi}{T} \cdot \frac{T}{4}}{n \cdot \frac{2\pi}{T}}$$

$$+ \left[\frac{\sin n \frac{2\pi}{T} \cdot \frac{T}{2}}{(n \times \frac{2\pi}{T})^2} - \frac{\sin n \frac{2\pi}{T} \cdot \frac{T}{4}}{n \times \frac{2\pi}{T}} \right]$$

$$- \left[2V \frac{\cos n \frac{2\pi}{T} \cdot \frac{T}{2}}{n \frac{2\pi}{T}} + 2V \frac{\cos n \frac{2\pi}{T} \cdot \frac{T}{4}}{n \times \frac{2\pi}{T}} \right]$$

$$= \frac{16V}{T^2} \left[\frac{-T^2}{8n\pi} (\cos n\pi - 0) + \frac{T^2}{(2n\pi)^2} \sin \frac{n\pi}{2} + \frac{T^2}{4n\pi} \cos n\pi - \frac{T^2}{8n\pi} \cos \frac{n\pi}{2} \right]$$

$$+ \frac{T^2}{(2n\pi)^2} \sin \frac{n\pi}{2} \left] + \frac{4}{T} \left[-2V \cos n\pi + 2V \cos \frac{n\pi}{2} \right]$$

$$= \frac{4}{T^2} \times \frac{16V}{T^2} \times \frac{T^2}{4n\pi} \left[-\frac{1}{2} (\cos n\pi - 0) + \frac{1}{n\pi} \sin \frac{n\pi}{2} + \cos n\pi - \frac{1}{2} \cos \frac{n\pi}{2} + \frac{1}{n\pi} \sin \frac{n\pi}{2} \right] - \frac{T}{2n\pi} \times \frac{4}{T} \times 2V \left[\cos n\pi - \cos \frac{n\pi}{2} \right]$$

$$= \frac{4V}{n\pi} \left[-\cos \frac{n\pi}{2} + \frac{2}{n\pi} \sin \frac{n\pi}{2} + \cos n\pi \right] - \frac{4V}{n\pi} \left[\cos n\pi - \cos \frac{n\pi}{2} \right]$$

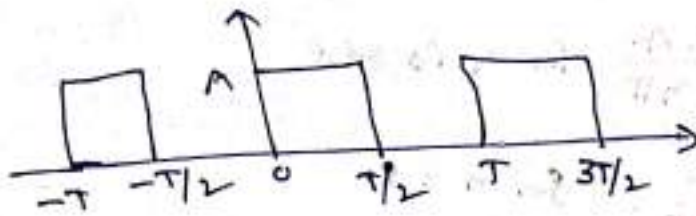
$$= \frac{4V}{n\pi} \left[\frac{1}{2} \right] = \frac{2V}{n\pi} \quad \frac{4V}{n\pi} \cdot \frac{2}{n\pi} \sin \frac{n\pi}{2} = \frac{8V}{(n\pi)^2} \sin \frac{n\pi}{2}$$

Fourier series is

$$f(t) = \frac{8V}{\pi^2} \sin \omega t + \frac{2V}{2\pi} \sin 2\omega t + \frac{2V}{3\pi} \sin 3\omega t + \dots$$

$$f(t) = \frac{8V}{\pi^2} \sin \omega t - \frac{8V}{9\pi^2} \sin 3\omega t + \frac{8V}{25\pi^2} \sin 5\omega t + \dots$$

→ obtain the fourier series



Sol

By inspection

$$f(t) = A, \quad 0 < t < T/2$$

$$= 0, \quad T/2 < t < T$$

$$a_0 = \frac{1}{T} \int_0^T f(t) dt = \frac{1}{T} \int_0^{T/2} f(t) dt + \frac{1}{T} \int_{T/2}^T f(t) dt$$

$$= \frac{1}{T} \cdot A + \frac{1}{T} \cdot 0$$

$$= \frac{A}{2}$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega t dt$$

$$= \frac{2}{T} \left[\int_0^{T/2} A \cos n\omega t dt + \int_{T/2}^T 0 \cdot \cos n\omega t dt \right]$$

$$= \frac{2A}{T} \left(\frac{\sin n\omega t}{n\omega} \right)_0^{T/2}$$

$$= \frac{2A}{nT\omega} \sin \frac{n\omega T}{2}$$

$$= \frac{1/2 A}{n \cdot T \left(\frac{2\pi}{T} \right)} \sin n \frac{2\pi}{2\pi}$$

$$= \frac{A}{n\pi} \sin n\pi$$

$$= 0 \text{ for any value of } n$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t dt = -\frac{A}{n\pi} (\cos n\pi - 1)$$

for any value (even) $b_n = 0$

$$b_n = \frac{2A}{n\pi} \text{ for } n \text{ is odd}$$

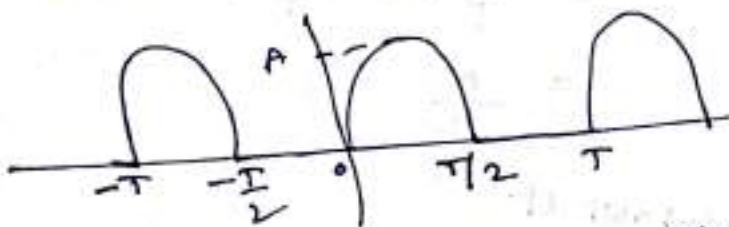
$$= \frac{2A}{\pi} \text{ for } n=1$$

$$= \frac{2A}{3\pi} \text{ for } n=3$$

Fourier Series

$$f(t) = \frac{A}{2} + \frac{2A}{\pi} \sin \omega t + \frac{2A}{3\pi} \sin 3\omega t + \dots$$

→ obtain Fourier Series



By inspection $f(t) = A \sin \omega t$ $0 \leq t < \frac{T}{2}$
 $= 0$ $\frac{T}{2} < t < T$

$$a_0 = \frac{1}{T} \int_0^{T/2} A \sin \omega t \, dt$$

$$= \frac{1}{T} \int_0^{T/2} A \sin \frac{2\pi t}{T} \, dt$$

$$= -\frac{A}{T} \left(\frac{\cos \frac{2\pi t}{T}}{\frac{2\pi}{T}} - 1 \right)$$

$$= -\frac{A}{2\pi} (\cos 2\pi - 1)$$

$$= \frac{A}{\pi}$$

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^{T/2} A \sin \frac{2\pi t}{T} \cos n \frac{2\pi}{T} t dt \\
 &= \frac{2A}{T} \int_0^{T/2} \frac{1}{2} \left[\sin \left(\frac{2\pi}{T} (1+n) t \right) + \sin \left(\frac{2\pi}{T} (1-n) t \right) \right] dt \\
 &= -\frac{2A}{2T} \left[\frac{T}{2\pi(1+n)} \cos \frac{2\pi}{T} (1+n) t + \frac{T}{2\pi(1-n)} \cos \left(\frac{2\pi}{T} (1-n) t \right) \right]_0^{T/2} \\
 &= -\frac{A}{2\pi} \left[\frac{1}{1+n} \cos(\pi(1+n)) + \frac{1}{1-n} \cos(\pi(1-n)) \right]
 \end{aligned}$$

For n odd

$$\cos(\pi(1+n)) = \cos(\pi(1-n)) = 1$$

For n even

$$\cos(\pi(1+n)) = \cos(\pi(1-n)) = -1$$

$a_n = 0$ for n is odd

$$\begin{aligned}
 &= \frac{2A}{\pi(1-n^2)} \quad n \text{ is even} \\
 b_n &= \frac{2}{T} \int_0^{T/2} A \sin \frac{2\pi t}{T} \sin \frac{2\pi}{T} n t dt
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{2A}{T} \int_0^{T/2} \frac{1}{2} \left[\cos \left(\frac{2\pi}{T} (1-n) t \right) - \cos \left(\frac{2\pi}{T} (1+n) t \right) \right] dt \\
 &= \frac{A}{n\pi} \left[\frac{1}{1-n} \sin \pi(1-n) - \frac{1}{1+n} \sin \pi(1+n) \right]
 \end{aligned}$$

For all n , Except $n=1$

$$\sin \pi (1-n) = \sin \pi (1+n) = 0$$

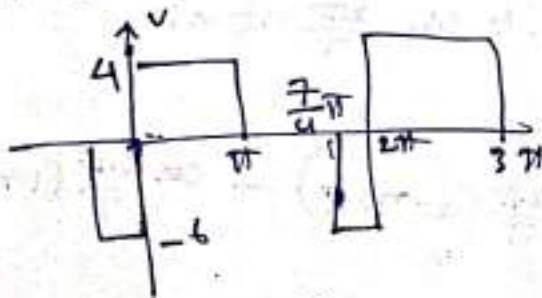
$$b_n = 0$$

$$\begin{aligned} \text{for } n=1, \quad b_1 &= \frac{2A}{T} \int_0^{T/2} \sin^2 \frac{2\pi t}{T} dt \\ &= \frac{2A}{T} \int_0^{T/2} \left(1 - \cos \frac{4\pi t}{T}\right) dt \\ &= \frac{A}{2} - \frac{A}{4\pi} \left(\sin \frac{4\pi t}{T}\right)_0^{T/2} \\ &= \frac{A}{2} \end{aligned}$$

Fourier series

$$\begin{aligned} f(t) &= \frac{A}{\pi} + \frac{A}{2} \sin \frac{2\pi t}{T} - \frac{2A}{3\pi} \cos \frac{4\pi t}{T} \\ &\quad - \frac{2A}{15\pi} \cos \frac{8\pi t}{T} + \dots \\ &= \frac{A}{\pi} + \frac{A}{2} \sin \omega t - \frac{2A}{3\pi} \cos 2\omega t - \frac{2A}{15\pi} \cos 4\omega t + \dots \end{aligned}$$

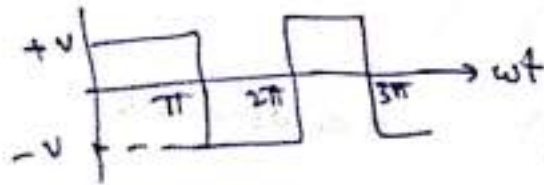
→ Find Fourier coefficients



$$\begin{aligned} a_0 &= \frac{1}{T} \int_0^T f(t) dt \\ &= \frac{1}{2\pi} \int_0^{\pi} f(t) dt + \frac{1}{2\pi} \int_{\pi}^{2\pi} f(t) dt \end{aligned}$$

$$= \frac{1}{2\pi} \int_0^{\pi} 4 \cos t \, dt$$

→ obtain Exponential Fourier Series



3) By inspection

$$f(t) = +V, \quad 0 < \omega t < \pi$$

$$= -V, \quad \pi < \omega t < 2\pi$$

Given is odd symmetry $a_0 = 0$
(avg value of the wave is zero)

$$C_n = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} f(t) e^{-jn\omega t} dt$$

$$= \frac{1}{2\pi} \int_0^{\pi} V e^{-jn\omega t} d\omega t - \frac{1}{2\pi} \int_{\pi}^{2\pi} V e^{-jn\omega t} d\omega t$$

$$= \frac{1}{2\pi} \left[\frac{jV}{n} \left(e^{-jn\omega t} \right)_0^{\pi} - \frac{jV}{n} \left(e^{-jn\omega t} \right)_{\pi}^{2\pi} \right]$$

$$= \frac{jV}{2\pi n} \left[\left(e^{-jn\pi} - e^0 \right) - \left(e^0 - e^{jn\pi} \right) \right]$$

$$n = -3, -1, 1, 3$$

$$= \frac{jV}{n\pi} (e^{jn\pi} - 1)$$

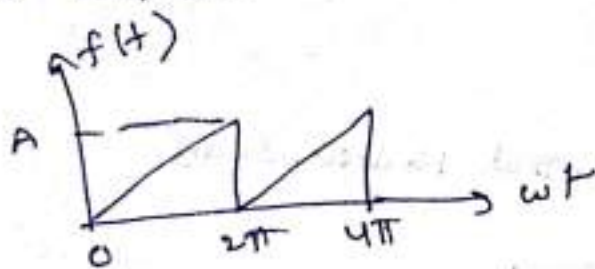
For even n , $e^{jn\pi} = 1$ & $C_n = 0$

$n = \text{odd}$ $e^{jn\pi} = -1$ & $C_n = -j \frac{2V}{n\pi}$

Fourier series

$$f(t) = \dots + j \frac{2V}{3\pi} e^{-j3\omega t} + j \frac{2V}{\pi} e^{-j\omega t} - j \frac{2V}{\pi} e^{j\omega t} - j \frac{2V}{3\pi} e^{j3\omega t} \dots$$

→ Find exponential Fourier series



So By inspection
 $f(t) = \frac{A}{2\pi} (\omega t) \quad 0 < \omega t < 2\pi$

However $c_n = \frac{1}{T_0} \int_0^{T_0} f(t) e^{-jn\omega t} dt$

In this problem $T_0 = 2\pi$

$$c_n = \frac{1}{2\pi} \int_0^{2\pi} \frac{A}{2\pi} (\omega t) e^{-jn\omega t} dt$$

$$= \frac{A}{(2\pi)^2} \left[\omega t \cdot \left(\frac{e^{-jn\omega t}}{-jn} \right) - \int_0^{2\pi} 1 \cdot \frac{e^{-jn\omega t}}{-jn} d\omega t \right]$$

$$= \frac{A}{(2\pi)^2} \left[\omega t \cdot \frac{e^{-jn\omega t}}{-jn} \times \frac{(-jn)}{(-jn)} - \frac{e^{-jn\omega t}}{(-jn)^2} \right]_0^{2\pi}$$

$$= \frac{A}{(2\pi)^2} \left[\frac{e^{-jn\omega t}}{(-jn)^2} \left[-jn\omega t - 1 \right] \right]_0^{2\pi}$$

$$= \frac{A}{(2\pi)^2} \left[\frac{e^{-j2\pi n}}{(-jn)^2} (-jn(2\pi) - 1) - \frac{e^{-0}}{(-jn)^2} \right]$$

$$= -\frac{jA}{2\pi n} \left[\frac{A}{(2\pi)^2} \left(\frac{-jn2\pi(-1)}{(-jn)^2} + \frac{1}{(-jn)^2} \frac{1}{(jn)^2} \right) \right]$$

$$= -\frac{jA}{2\pi n} \quad \left(e^{-j2\pi n} = -1 \text{ for all } n \right)$$

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega t}$$

$$= \sum_{n=-\infty}^{\infty} \left(\frac{-jA}{2\pi n} \right) e^{jn\omega t}$$

$$= \dots + j \frac{A}{4\pi} e^{-j2\omega t} + j \frac{A}{2\pi} e^{-j\omega t}$$

$$+ \frac{A}{2} + j \frac{A}{2\pi} e^{j\omega t} + j \frac{A}{4\pi} e^{j2\omega t} + \dots$$

Here avg value is $\frac{A}{2}$ which can easily be derived.

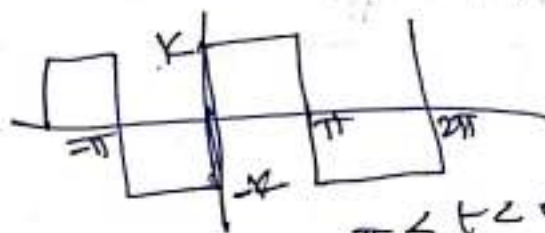
$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} \frac{A}{2\pi} t dt$$

$$= \frac{1}{2\pi} \frac{A}{2\pi} \left(\frac{t^2}{2} \right)_0^{2\pi}$$

$$= \frac{A}{(4\pi)} \frac{(2\pi)^2}{2}$$

$$= \frac{A}{2}$$

→



$$f(t) = k \quad -\pi < t < 0$$

$$= -k \quad 0 < t < \pi$$

This is odd symmetry
 $a_n = a_0 = 0$.

$$b_n = \frac{2}{T_0} \int_{-\pi}^{\pi} f(t) \sin n\omega t dt$$

$$= \frac{2}{2\pi} \int_{-\pi}^0 -k \sin n\omega t dt + \frac{2}{2\pi} \int_0^{\pi} k \sin n\omega t dt$$

obtain Fourier series



$$= -\frac{K}{\pi} \times \left(-\frac{\cos n\omega t}{n\omega} \right)_{-\pi}^0 + \frac{K}{\pi} \left(-\frac{\cos n\omega t}{n\omega} \right)_{\pi}^0$$

$$= \frac{K}{n\omega\pi} \left[\cos 0 - \cos(-n\omega\pi) - \cos n\omega\pi + \cos 0 \right]$$

$$= \frac{K}{n\pi \times 1} \left[\cos 0 - \cos(-n\pi) - \cos n\pi + \cos 0 \right]$$

$$= \frac{K}{n\pi} \left[2 - 2\cos n\pi \right]$$

$$b_n = \frac{2K}{n\pi} (1 - \cos n\pi)$$

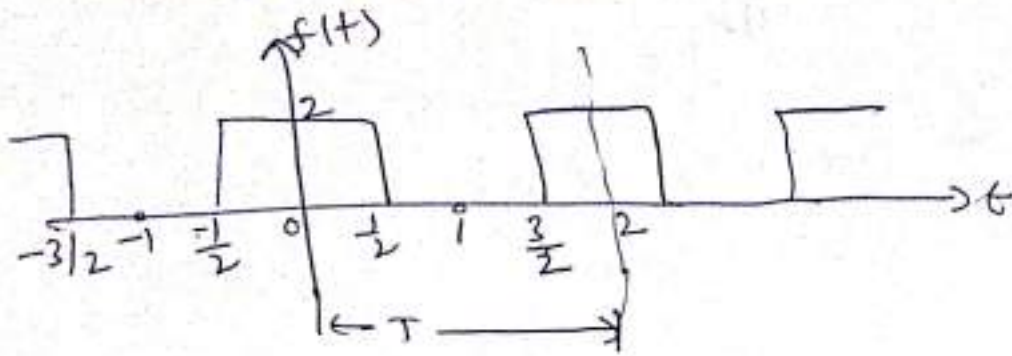
$$= \frac{4K}{n\pi} \quad \text{for } n \text{ is odd}$$

Fourier series

$$f(t) = b_1 \sin \omega t + b_2 \sin 2\omega t + b_3 \sin 3\omega t + \dots$$

$$= \frac{4K}{\pi} \sin \omega t + \frac{4K}{3\pi} \sin 3\omega t + \dots$$

→ Obtain the Fourier series



Sol

In the given problem, it is not an even or odd -

$$T = 2 \text{ sec}$$

$$a_0 = \frac{1}{T} \left\{ \int_0^{1/2} 2 dt + \int_{3/2}^2 2 dt \right\} \quad [\because T=2]$$

$$= \frac{1}{2} \left\{ 2 \cdot \frac{1}{2} + 2 \cdot \left(2 - \frac{3}{2} \right) \right\}$$

$$= \frac{1}{2} \times (1+1) = 1$$

$$\boxed{a_0 = 1}$$

$$a_n = \frac{2}{T} \left\{ \int_0^{1/2} 2 \cos n\omega t dt + \int_{3/2}^2 2 \cos n\omega t dt \right\}$$

$$= 2 \left\{ \left[\frac{+\sin n\omega t}{n\omega} \right]_0^{1/2} + \left[\frac{+\sin n\omega t}{n\omega} \right]_{3/2}^2 \right\}$$

$$= \frac{2}{n\pi} \left\{ \sin \frac{n\pi}{2} + \underbrace{\sin \frac{2n\pi}{1}}_{\rightarrow 0} - \sin \frac{3n\pi}{2} \right\}$$

$$= \frac{2}{n\pi} \left\{ \sin \frac{n\pi}{2} - \sin \frac{3n\pi}{2} \right\}$$

$$n=1, a_1 = 4/\pi$$

$$n=2, a_2 = 0$$

$$n=3, a_3 = -4/3\pi$$

$$\begin{aligned} \omega &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2} \\ \omega &= \pi \end{aligned}$$

Fourier Integral' (or) Fourier Transform

- For the signal $f(t)$, $F(j\omega)$ is called a Fourier integral or Fourier transform or continuous spectrum.
- with the help of Fourier integral, the behaviour of the linear circuits can be analyzed.
- Basically, a Fourier integral is one which transforms a function in time domain (t) into function in frequency domain (ω)
- A line or discrete spectrum becomes continuous spectrum if frequency tends to zero as time period T tends to infinity
- Fourier integral is a limiting case of a Fourier series.
- Fourier integral does not exist for a periodic function because it does not satisfy the condition of absolute integrability.
- Fourier series in time domain, there exists only one Fourier integral in frequency domain corresponding to it.
- As $T \rightarrow \infty$, $\omega \rightarrow 0$ as $\omega = \frac{2\pi}{T}$. Due to this a single pulse of $f(t)$ is generated. This pulse will not be periodic. Hence it is necessary to modify the Fourier series.

We know the Fourier series is given by

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega t}$$

$$\text{where } C_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega t} dt$$

where $f(t)$ is a periodic function

Suppose if $f(t)$ is a non periodic function

Then a few changes will have to be incorporated i.e

$$As \ T \rightarrow \infty$$

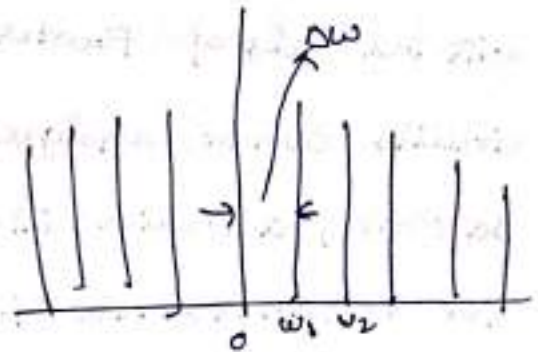
$$\omega \rightarrow 0 \quad \left(\omega = \frac{2\pi}{T} \right)$$

and n becomes irrelevant

$$n\omega \rightarrow \omega$$

$$T = \frac{2\pi}{\Delta\omega}$$

where $\Delta\omega$ is the spacing b/w adjacent component of angular frequency



Then above equation can be reduced to

$$f(t) = \sum_{n=-\infty}^{\infty} c' e^{j\omega t}$$

$$\omega = 0, \Delta\omega, 2\Delta\omega, \dots$$

③

$$\text{where } c' = \frac{\Delta\omega}{2\pi} \int_{-T/2}^{T/2} f(t) e^{-j\omega t} dt \quad \text{--- ④}$$

Sub eq ④ in eq ③

$$f(t) = \sum_{n=-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-T/2}^{T/2} f(t) e^{-j\omega t} dt \right] \cdot e^{j\omega t} \Delta\omega$$

changing summation form to integral form

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \right] e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} d\omega$$

where $F(j\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$ is a Fourier integral of $f(t)$.

Properties

The properties of Fourier transform are as follows

① Symmetry property

if $f(t)$ is function in time domain and $F(j\omega)$ is function in frequency domain i.e. Fourier transform. Then

$$\text{if } f(t) = F(j\omega)$$

$$\text{Then } F(t) = 2\pi f(-j\omega)$$

- This property states that the Fourier transform of a gate function is a sampling function and vice versa.

② Linearity property

if $f_1(t)$ & $f_2(t)$ are the two time domain functions and $F_1(j\omega)$ and $F_2(j\omega)$ are the two Fourier transforms. Then a is a arbitrary constant then

$$a f_1(t) + a_2 f_2(t) = a_1 F_1(j\omega) + a_2 F_2(j\omega)$$

③ Scaling property :

Let $f(t)$ is the time domain function and $F(j\omega)$ is the Fourier transform then

$$f(at) = \frac{1}{|a|} F\left(\frac{j\omega}{a}\right)$$

where

$f(at)$ is the function $f(t)$ compressed in time scale by factor 'a',
 $F\left(\frac{j\omega}{a}\right)$ is the function expanded by frequency scale by the same factor 'a'.

4. Frequency Shifting property

Let $f(t)$ is the time domain function
 $F(j\omega)$ is the Fourier transform.

Then if

$$f(t) = F(j\omega) \text{ then}$$

$$f(t) e^{j\omega_0 t} = F(j(\omega - \omega_0))$$

Proof:

$$F[f(t) e^{j\omega_0 t}] = \int_{-\infty}^{\infty} f(t) e^{j\omega_0 t} e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} f(t) e^{-j(\omega - \omega_0)t} dt$$

$$= F[j(\omega - \omega_0)]$$

(5) Time shifting property

Let $f(t)$ is the time domain function
 $F(j\omega)$ is the Fourier transform

$$f(t - T) = F(j\omega) e^{-j\omega T}$$

Proof

$$F[f(t - T)] = \int_{-\infty}^{\infty} f(t - T) e^{-j\omega t} dt$$

$$t - T = x, \quad dt = dx \quad \text{and} \quad t = x + T$$

Hence

$$F[f(t - T)] = \int_{-\infty}^{\infty} f(x) e^{-j\omega(x+T)} dx$$

$$= e^{-j\omega T} \int_{-\infty}^{\infty} f(x) e^{-j\omega x} dx$$

$$F[f(t - T)] = F(j\omega) e^{-j\omega T}$$

6. Differentiation property

if $f(t) \leftrightarrow F(j\omega)$ then

$$\frac{d}{dt}(f(t)) \leftrightarrow j\omega F(j\omega)$$

Proof

By the definition of the inverse fourier transform, we have

$$F^{-1}(F(j\omega)) = f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} d\omega \quad L \odot$$

$$\frac{d}{dt} f(t) = \frac{d}{dt} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} d\omega \right]$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{d}{dt} (F(j\omega) e^{j\omega t}) d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} (j\omega) d\omega$$

$$= \frac{j}{2\pi} \int_{-\infty}^{\infty} \omega F(j\omega) e^{j\omega t} d\omega$$

$$= j F^{-1}[\omega \cdot F(j\omega)]$$

$$\boxed{\frac{d}{dt} f(t) = F^{-1}(j\omega \cdot F(j\omega))}$$

$$F\left[\frac{d}{dt} f(t)\right] = j\omega \cdot F(j\omega)$$

(7) Integration property

$$F\left[\int_{-\infty}^{\infty} f(t) dt\right] = \frac{1}{j\omega} F(j\omega)$$

(8) Folding property

Let $f(t)$ & $F(j\omega)$ then

$$F(f(-t)) = F(-j\omega)$$

Prove time scaling property

Given is $f(at) = \frac{1}{|a|} F\left(\frac{j\omega}{a}\right)$

Proof: $F(f(t)) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$

case (i) when a is +ve

$$F(f(at)) = \int_{-\infty}^{\infty} f(at) e^{-j\omega t} dt \quad \text{L①}$$

$$\text{Let } x = at \Rightarrow t = \frac{x}{a}$$
$$dt = \frac{dx}{a}$$

eq (i) becomes

$$F(f(at)) = \int_{-\infty}^{\infty} f(x) e^{-j\omega \frac{x}{a}} \frac{dx}{a}$$

$$= \frac{1}{a} \int_{-\infty}^{\infty} f(x) e^{-\frac{j\omega}{a} x} dx$$

$$F(f(at)) = \frac{1}{a} F\left(\frac{j\omega}{a}\right)$$

Problem

① ~~Fourier sine & cosine integral~~ discuss the F.T of Sinusoidal function

Sol $f(t) = \sin \omega_0 t$ $\sin \omega_0 t$ is absolute integrability & hence

$$F(j\omega) = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} f(t) e^{-j\omega t} dt$$

$$= \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} \sin \omega_0 t e^{-j\omega t} dt$$

$$= \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} \left[\frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} \right] e^{-j\omega t} dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2j} \left[\int_{-T/2}^{T/2} e^{-j(\omega - \omega_0)t} dt \right] - \left[\int_{-T/2}^{T/2} e^{-j(\omega + \omega_0)t} dt \right]$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2j} \left[\frac{e^{-j(\omega - \omega_0)t}}{-j(\omega - \omega_0)} \right]_{-T/2}^{T/2} - \left[\frac{e^{-j(\omega + \omega_0)t}}{-j(\omega + \omega_0)} \right]_{-T/2}^{T/2}$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2j} \left[\frac{e^{j(\omega - \omega_0)T/2} - e^{-j(\omega - \omega_0)T/2}}{j(\omega - \omega_0)} \right] - \left[\frac{e^{j(\omega + \omega_0)T/2} - e^{-j(\omega + \omega_0)T/2}}{j(\omega + \omega_0)} \right]$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2j} \left[\frac{2j \sin(\omega - \omega_0)T/2}{j(\omega - \omega_0)} - \frac{2j \sin(\omega + \omega_0)T/2}{j(\omega + \omega_0)} \right]$$

$$= \lim_{T \rightarrow \infty} (-j) \left[\frac{\sin(\omega - \omega_0)T/2}{\omega - \omega_0} - \frac{\sin(\omega + \omega_0)T/2}{\omega + \omega_0} \right] \times \frac{T}{2}$$

From Sampling function

$$\frac{\sin \omega t}{\omega t} = \text{sinc}(\omega t)$$

$$= \lim_{T \rightarrow \infty} (-j) \left[\frac{T}{2} \sin c(\omega - \omega_0) \frac{T}{2} - \frac{T}{2} \sin c(\omega + \omega_0) \frac{T}{2} \right] \times \frac{1}{T}$$

$$= \lim_{T \rightarrow \infty} (-j\pi) \left[\frac{T}{2\pi} \sin c(\omega - \omega_0) \frac{T}{2} - \frac{T}{2\pi} \sin c(\omega + \omega_0) \frac{T}{2} \right]$$

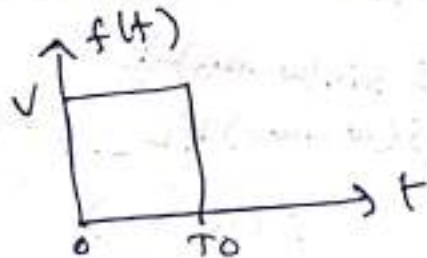
$$= (-j\pi) [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

$$F(\omega) = j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)] \quad \left(\lim_{T \rightarrow \infty} \frac{T}{2\pi} \sin c(\pi\omega) = \delta(\omega) \right)$$

$$\rightarrow f(t) = \cos \omega_0 t$$

$$F(j\omega) = \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$\rightarrow$ ~~f(t) = e~~ Determine the F.T of a pulse of duration T_0 and magnitude V shown in fig.



$$\text{Sol: } f(t) = V, \quad 0 < t < T_0$$

$$= 0, \quad t > T_0$$

Fourier transform $F(j\omega)$ is

$$F(j\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$= \int_0^{T_0} V e^{-j\omega t} dt$$

$$= V \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^{T_0} = \frac{jV}{\omega} (e^{-j\omega T_0} - 1)$$

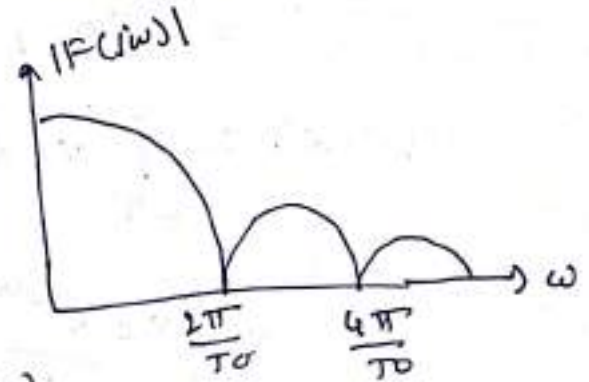
$$= -\frac{2V}{\omega} \left[\frac{e^{-j\omega T_0/2} - e^{j\omega T_0/2}}{2j} \right] e^{-j\omega T_0/2}$$

$$= \frac{2V}{\omega} \sin \frac{\omega T_0}{2} e^{-j\omega T_0/2}$$

$$F(j\omega) = V T_0 \frac{\sin \frac{\omega T_0}{2}}{\frac{\omega T_0}{2}} e^{-j\omega T_0/2}$$

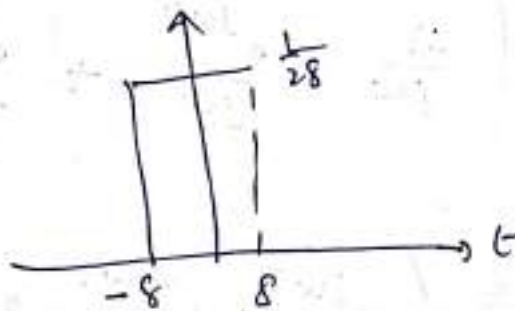
The magnitude of $|F(j\omega)|$ is of the form $\frac{\sin x}{x}$ and plot is shown in fig.

The magnitude is zero when $\frac{\omega T_0}{2} = \pi, 2\pi, 3\pi$ etc.



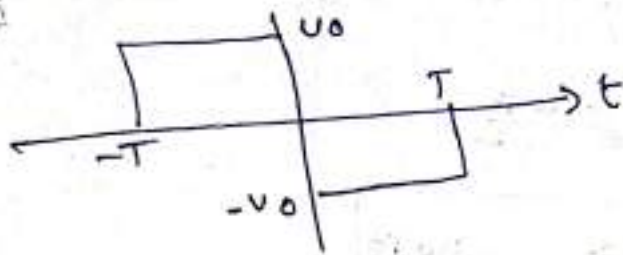
→ F.T of impulse function $\delta(t)$

$\int_{-\infty}^{\infty} \delta(t) dt = 1$



$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

→ Determine the F-T of following fig



Sol By inspection

$$f(t) = +v_0 \quad -T < t < 0$$

$$= -v_0, \quad 0 < t < T$$

$$F(j\omega) = \int_{-T}^0 v_0 e^{-j\omega t} dt + \int_0^T -v_0 e^{-j\omega t} dt$$

$$= \left(\frac{v_0 e^{-j\omega t}}{-j\omega} \right)_{-T}^0 + \left(\frac{-v_0 e^{-j\omega t}}{j\omega} \right)_0^T$$

$$= v_0 \left(\frac{-1}{j\omega} + \frac{e^{j\omega T}}{j\omega} + \frac{e^{-j\omega T}}{j\omega} - \frac{1}{j\omega} \right)$$

$$= \frac{v_0}{j\omega} (e^{j\omega T} + e^{-j\omega T} - 2)$$

$$= \frac{2jv_0}{\omega} (1 - \cos \omega T)$$

$$F(j\omega) = j\omega + 2v_0 \frac{\sin^2 \frac{\omega T}{2}}{\left(\frac{\omega T}{2}\right)^2}$$

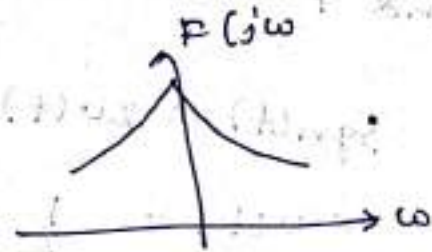
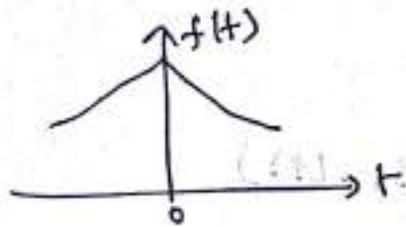
$$\therefore \left(\frac{e^{j\omega T} + e^{-j\omega T}}{2} \right)$$

$$1 - \cos \omega T = 2 \sin^2 \frac{\omega T}{2}$$

$$= \frac{2jv_0}{\omega} \times \frac{\omega^2}{4} \times \frac{2 \sin^2 \frac{\omega T}{2}}{\left(\frac{\omega T}{2}\right)^2}$$

$$= j\omega + 2v_0 \frac{\sin^2 \frac{\omega T}{2}}{\left(\frac{\omega T}{2}\right)^2}$$

→ Find the F.T of the function



Sol here $f(t) = A e^{-a|t|}$, $a > 0$

$$F(jw) = \int_{-\infty}^{\infty} f(t) e^{-jw t} dt$$

$$= \int_{-\infty}^{\infty} A e^{-a|t|} e^{-jw t} dt$$

$$= \int_{-\infty}^0 A e^{at} e^{-jw t} dt + \int_0^{\infty} A e^{-at} e^{-jw t} dt$$

$$= \int_{-\infty}^0 A e^{(a-jw)t} dt + \int_0^{\infty} A e^{(-a-jw)t} dt$$

$$= \frac{A}{a-jw} + \frac{A}{a+jw} = \frac{2aA}{a^2 + w^2}$$

→ Determine the F.T of Signum function

Sol Signum function is

$$f(t) = -1, -\infty < t < 0$$

$$= 1, 0 < t < \infty$$

$$\text{sgn}(t) = \lim_{a \rightarrow 0} e^{-a|t|} \text{sgn}(t)$$

$$F(jw) = \lim_{a \rightarrow 0} \int_{-\infty}^{\infty} e^{-a|t|} \text{sgn}(t) e^{-jw t} dt$$

$$= \lim_{a \rightarrow 0} \left[\int_{-\infty}^0 -e^{-(a-jw)t} dt + \int_0^{\infty} e^{-(a+jw)t} dt \right]$$

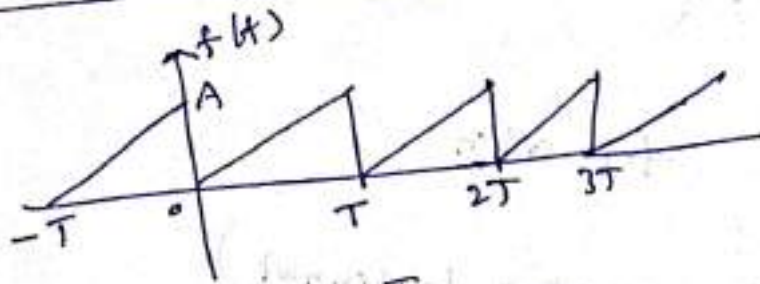
$$= \lim_{a \rightarrow 0} \left(\frac{1}{a+jw} + \frac{1}{a-jw} \right) = \frac{2}{jw}$$

Given function is shown in modelles so fig is like that

$-\infty$ to $0 \rightarrow t = -t$
 0 to $\infty \rightarrow t = t$

Analysis of electrical ckt

(1)



it is a odd function

$$a_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$= \frac{1}{T} \int_0^T \frac{A}{T} t dt$$

$$= \frac{A}{T^2} \left(\frac{t^2}{2} \right)_0^T = \frac{A}{T^2} \cdot \frac{T^2}{2} = \frac{A}{2}$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega t dt$$

$$= \frac{2}{T} \int_0^T \frac{A}{T} t \cos n\omega t dt$$

$$= \frac{2A}{T^2} \left[\left(t \cdot \frac{\sin n\omega t}{n\omega} \right)_0^T + \left(\frac{\cos n\omega t}{(n\omega)^2} \right)_0^T \right]$$

$$= \frac{2A}{T^2} \times \left(\frac{\cos n \frac{2\pi}{T} \cdot T}{(n \frac{2\pi}{T})^2} \right)_0^T$$

$$= \frac{2A}{T^2} \times \frac{T^2}{(2\pi n)^2} (\cos 2\pi n - 1)$$

$$= 0$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t dt$$

$$= \frac{2A}{T^2} \left[\left(t \cdot \frac{-\cos n\omega t}{n \cdot \frac{2\pi}{T}} \right)_0^T + \left(\frac{\sin n\omega t}{(n \cdot \frac{2\pi}{T})^2} \right)_0^T \right]$$

$$= \frac{2A}{2\pi} \times \frac{2\pi}{2\pi n} \times (-\cos 2\pi n)$$

$$b_n = \frac{-A}{\pi n} \quad \text{for all } n$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

$$= \frac{A}{2} - \frac{A}{\pi} \left(\sin \omega t + \frac{1}{2} \sin 2\omega t + \frac{\sin 3\omega t}{3} + \dots \right)$$

→ The voltage having the waveform shown in fig(1) is applied to ckt shown in fig(2). determine $i(t)$

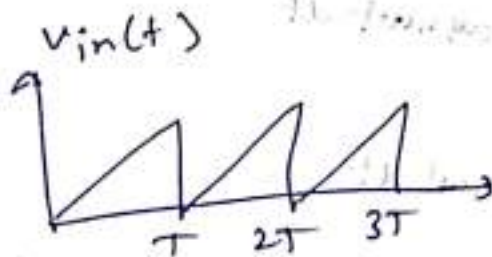


fig (1)

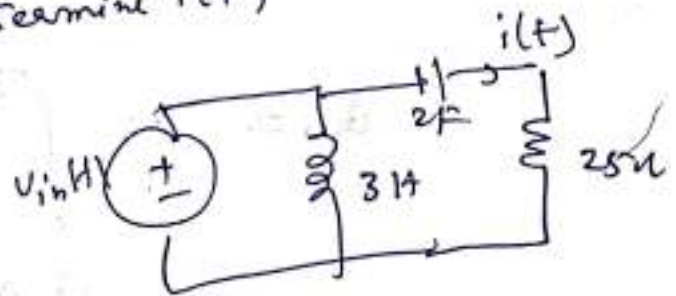


Fig (2)

Sol For fig (1)

$$v_{in}(t) = \frac{A}{2} + \sum_{n=1}^{\infty} \left(\frac{-A}{\pi n} \right) \sin n\omega t$$

$$v_{in}(t) = \frac{A}{2} - \frac{A}{\pi} \sum_{n=1}^{\infty} \frac{\sin n\omega t}{n}$$

From Fig (b)

$$i(t) = \frac{v_{in}(t)}{25 + \frac{1}{2j\omega}}$$

$$R + jX_C = R + \frac{1}{j\omega C}$$

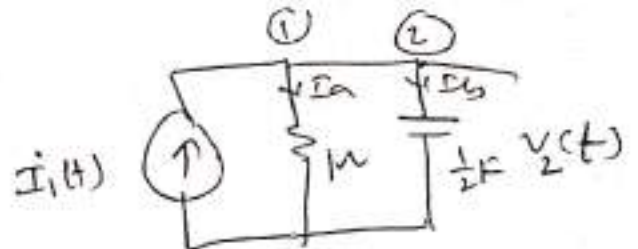
$$\frac{1}{\omega C} = \frac{1}{2\pi f C}$$

→ Find the response of the voltage in the vzw shown in fig. use F-T method



apply KCL at node 1

$$i_1(t) = i_a + i_b$$



$$2e^{-t}u(t) = \frac{v_2(t)}{R} + j\omega C v_2(t)$$

$$= v_2(t) \left(\frac{1}{2} + j\omega C \right)$$

$$= v_2(t) \left(1 + j\omega \left(\frac{1}{2} \right) \right)$$

$$2e^{-t}u(t) = v_2(t) (1 + j0.5\omega) \quad \text{--- (1)}$$

~~Let~~
 $F(v_2(t)) = v_2(j\omega)$

apply F.T on both sides of equation

$$F(2e^{-t}u(t)) = F(v_2(t) (1 + j0.5\omega))$$

$$\frac{2}{1+j\omega} = v_2(j\omega) (1 + j0.5\omega)$$

$$v_2(j\omega) = \frac{2}{(1+j\omega)(1+j0.5\omega)}$$

$$= \frac{2}{(1+j\omega) \cdot 0.5 \left(\frac{1}{0.5} + j\omega \right)}$$

$$= \frac{4}{(1+j\omega)(2+j\omega)}$$

$$= \frac{A}{1+j\omega} + \frac{B}{2+j\omega}$$

$$F(A e^{-at}u(t)) = \frac{A}{a+j\omega}$$

(Partial fraction)

$$A = 4$$

$$B = -4$$

$$V_2(j\omega) = \frac{4}{1+j\omega} - \frac{4}{2+j\omega}$$

taking ~~the~~ inverse F.T, we get

$$\mathcal{F}^{-1}(V_2(j\omega)) = \mathcal{F}^{-1}\left(\frac{4}{1+j\omega} - \frac{4}{2+j\omega}\right)$$

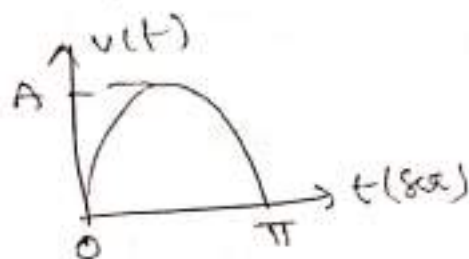
$$V_2(t) = 4 e^{-t} u(t) - 4 e^{-2t} u(t)$$

$$= 4u(t) [e^{-t} - e^{-2t}]$$

The response is

$$V_2(t) = 4u(t) [e^{-t} - e^{-2t}]$$

→ Find the F.T of sine pulse shown in fig and sketch amplitude and phase spectra. If this vol is applied to a series RL ckt with $R=1\Omega$ & $L=1H$. determine Amplitude & phase spectra for the resultant current $i(t)$.



Sol

From the above wave form

$$v(t) = A \sin t$$

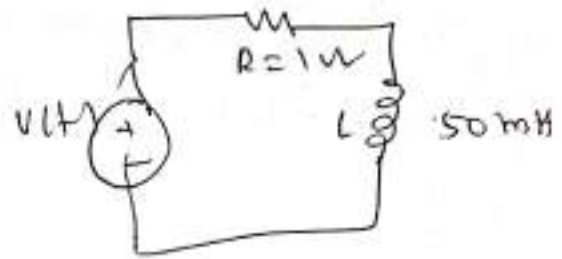
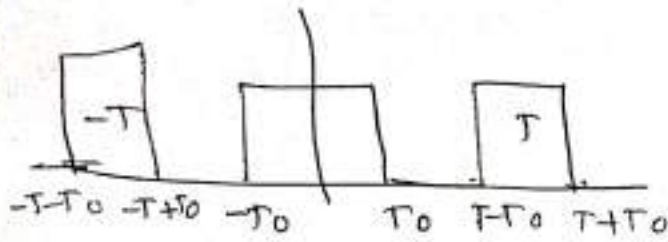
By the definition of F.T, we have

$$V(j\omega) = \int_{-\infty}^{\infty} v(t) e^{-j\omega t} dt$$

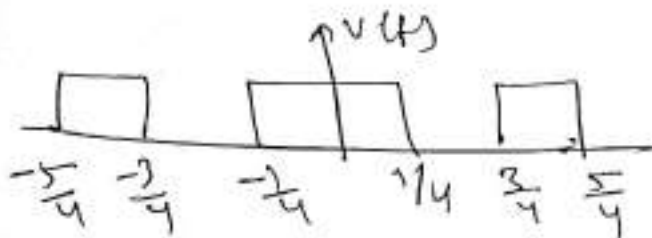
$$= \int_0^{\pi} v(t) e^{-j\omega t} dt$$

$$\text{Let } V(j\omega) = A \int_0^{\pi} \sin(t) \cdot e^{-j\omega t} dt = \pm (\text{say})$$

→ Determine & plot the vol across the resistor in the ckt shown in fig (1) using ckt analysis techniques, if the input is a square wave with $T=1$, $T_0 = \frac{1}{4}$



Sol Given $T=1$ sec
 $T_0 = \frac{1}{4}$ sec



$$v(t) = 0, -\frac{3}{4} < t < -\frac{1}{4}$$

$$= 1, -\frac{1}{4} < t < \frac{1}{4}$$

$$= 0, \text{ for } \frac{1}{4} < t < \frac{3}{4}$$

$$a_n = \frac{2}{T} \int_0^T v(t) \cos n\omega t dt$$

$$= 2 \left[\int_{-1/4}^{1/4} 1 \cdot \cos n\omega t dt \right]$$

$$= 2 \left(\frac{\sin n\omega t}{n\omega} \right)_{-1/4}^{1/4}$$

$$= \frac{2}{n \cdot \frac{2\pi}{T}} \left(\sin n \cdot \frac{2\pi}{T} \left(\frac{1}{4} \right) + \sin n \cdot \frac{2\pi}{T} \left(-\frac{1}{4} \right) \right)$$

$$= \frac{T}{n\pi} \left[\sin \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right]$$

$$= \frac{2T}{n\pi} \sin \frac{n\pi}{2} = \frac{2}{n\pi} \sin \frac{n\pi}{2}$$

$$a_n = \frac{2}{T}, -\frac{2}{3\pi}, \frac{2}{5\pi}, n = 1, 3, 5$$

$$b_n = \frac{2}{T} \int_0^T v(t) \sin n\omega t dt$$

$$= \frac{2}{1} \int_{-1/4}^{1/4} 1 \cdot \sin n\omega t dt$$

$$= 0$$

$$a_0 = \frac{1}{T} \int_{-1/4}^{1/4} v(t) dt$$

$$= (t)_{-1/4}^{1/4}$$

$$= \frac{1}{2}$$

$$v(t) = \frac{1}{2} + \sum_{n=1}^{\infty} a_n \cos n\omega t + b_n \sin n\omega t$$

$$= \frac{1}{2} + \frac{2}{\pi} \cos \omega t - \frac{2}{3\pi} \cos 3\omega t + \frac{2}{5\pi} \cos 5\omega t + \dots$$

voltage across resistor

$$v_R(t) = i(t) R$$

$$\text{where } i(t) = \frac{v(t)}{1 + j 2\pi f L} = \frac{v(t)}{1 + j \frac{2\pi}{T} L}$$

$$= \frac{v(t)}{1 + j 2\pi L}$$

$T = 25\pi \text{ ms}$, $R = 15\Omega$
 $v(t) = ?$ and $I_f = ?$
 fundamental
 frequency
 current

$$n \cdot \frac{2\pi}{T}$$

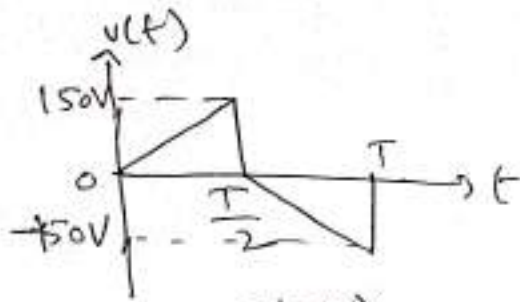
$$\frac{n\pi}{2}$$

$$\frac{n \cdot 2\pi}{T}$$

$$(n \cdot \frac{2\pi}{T})$$

$$n \cdot \frac{2\pi}{T}$$

→ Determine the Fourier Series $T = 25 \pi \text{ ms}$, $R = 15 \Omega$, $L = 0.0438 \text{ H}$. Find upto 7th harmonic $v(t) = ?$ and $I_f = ?$



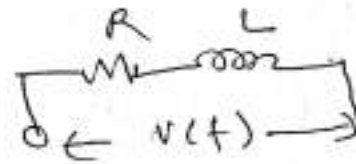
Fig(a)

given

$$T = 25 \pi \text{ ms}$$

$$R = 15 \Omega$$

$$L = 0.0438 \text{ H}$$



fundamental frequency current

Fig(a) is a half wave symmetry

$$v(t) = -v(t - \frac{T}{2})$$

$$v(t) = \frac{150t}{(\frac{T}{2})}, \quad \omega = \frac{2\pi}{T}$$

$$= \frac{300t}{T}$$

for n even $a_0 = a_n = b_n = 0$

$$\text{for n odd} \quad a_n = \frac{4}{T} \int_0^{T/2} v(t) \cos n\omega t dt$$

$$= \frac{4}{T} \int_0^{T/2} \frac{300t}{T} \cos n\omega t dt$$

$$= \frac{1200}{T^2} \int_0^{T/2} t \cos n\omega t dt$$

$$= \frac{1200}{T^2} \left[\left(t \frac{\sin n\omega t}{n \cdot \frac{2\pi}{T}} \right) \Big|_0^{T/2} + \left(\frac{\cos n \frac{2\pi}{T} t}{(n \cdot \frac{2\pi}{T})^2} \right) \Big|_0^{T/2} \right]$$

$$= \frac{600}{T^2} \times \frac{T^2}{2n\pi} \left[0 - 0 + \frac{(\cos n\pi - 1)}{2n\pi} \right]$$

$$= \frac{300}{(n\pi)^2} (\cos n\pi - 1)$$

$$= -\frac{600}{n^2\pi^2} \quad n \text{ is odd.}$$

$$b_n = \frac{4}{T} \int_0^{T/2} v(t) \sin n\omega t dt$$

$$= \frac{300}{n\pi}$$

$$v(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

$$= \frac{-600}{\pi^2} \cos \omega t + \frac{-600}{9\pi^2} \cos 3\omega t + \frac{-600}{25\pi^2} \cos 5\omega t + \frac{-600}{49\pi^2} \cos 7\omega t + \dots$$

$$+ \frac{300}{\pi} \sin \omega t + \dots + \frac{300}{3\pi} \sin 3\omega t + \dots$$

Fundamental frequency voltage

$$V_f(t) = \frac{-600}{\pi^2} \cos \omega t + \frac{300}{\pi} \sin \omega t$$

$$= \frac{600}{\pi^2} \left(-\cos \omega t + \frac{\pi}{2} \sin \omega t \right)$$

$$= \frac{600}{\pi^2} \sqrt{\left(\frac{\pi}{2}\right)^2 + 1} \cdot \sin(\omega t - \theta)$$

$$\theta = \tan^{-1} \left(\frac{2}{\pi} \right)$$

$$\left(\theta = \tan^{-1} \frac{a_n}{b_n} \right)$$

$$= \tan^{-1} \left(\frac{1}{\frac{\pi}{2}} \right)$$

From Fig (b)

$$Z = R + jX_L = R + j\omega L$$

$$= 15 + j 0.0438 \times \omega$$

$$\text{where } \omega = 2\pi f = \frac{2\pi}{T} = \frac{2\pi}{25\pi \times 10^{-3}} = \frac{2000}{25} = 80 \text{ rad/s}$$

$$\therefore Z = 15 + j 3.504 \Omega$$

$$= 15.404 \angle 13.15^\circ \Omega$$

Fundamental frequency current

$$I_f = \frac{V_f}{Z} = \frac{600}{\pi^2} \sqrt{\left(\frac{\pi}{2}\right)^2 + 1}$$

$$= 7.35 \angle 15.404^\circ$$

Rms value of fundamental current

$$I_{rms} = \frac{7.35}{\sqrt{2}} = 5.19 \text{ A}$$

→ The current source in the fig is $i(t) = 4e^{-t}$ for $t \geq 0$.
Find the voltage v_o using Fourier transform method?



→ An input voltage $v_1(t) = 5e^{-3t}$ volt is applied to a series RL circuit with $R = 4\Omega$ and $L = 2H$. Find the o/p voltage $v_o(t)$ across the inductor using frequency domain analysis (Fourier transform method)



Sol

Using Fourier transform method.

$$F(v_1(t)) = v_1(j\omega) = FT[5e^{-3t} \text{ volt}]$$

$$v_1(j\omega) = \frac{5}{j\omega + 3}$$

Using voltage division rule

$$v_o(j\omega) = \frac{j2\omega}{4 + j2\omega} \times \frac{5}{j\omega + 3}$$

$$= \frac{j\omega}{2 + j\omega} \times \frac{5}{j\omega + 3}$$

$$v_o(j\omega) = \frac{5(j\omega)}{(2 + j\omega)(j\omega + 3)}$$

UNIT-V

FILTER DESIGN AND CIRCUIT SIMULATION

Filters - low pass - High pass and band pass - RL, RC filters - derived filters and composite design filters.

Circuit simulation - Description of circuit elements - Nodes and sources, input and output variables - modelling of the above element - DC analysis.

* Filters Introduction;

→ Filters consisting of purely reactive n/w [L-C n/w] that passes desired band of frequencies and attenuates all other frequencies.

Ex: Tuning of radio, selection of channels in TV, communication system in Telecommunication engineering.

→ Filters are broadly classified into two types.

1. Active filters

2. Passive filters

1. Active Filters: The filters which consists of active elements such as operation amplifiers, transistors in addition to R, L, C, then it is called Active filters.

Advantages:

1. No resonance issue
2. It is used to reduce harmonic contents
3. It is used for voltage regulation.
4. It is used for reactive power compensation.
5. Reliable operation.

Disadvantages:

1. It requires addition power supply to operate operation amplifier so, it is expensive.

2. Passive filters: The filters which consisting of passive elements like R, L, C, then it is called passive filters.

Advantages:

1. cheap
2. Designing is easy
3. High efficiency
4. High reliable.

Disadvantages:

1. It is restricted to particular harmonics
2. It is fixed reactive power compensation
3. It is tuning for fixed frequency.
4. In this filter, L is used. These are larger in size and bulky. It needs expensive.

* Basic definitions:

1. Pass band: The range of frequencies over which attenuation is zero. Then it is called pass band.
2. Stop band: The range of frequencies over which attenuation is infinity, then it is called stop band.
3. Cut off frequency: It is the frequency which separates pass band from stop band.

* Filter Fundamentals:

→ The complete study of any filter section needs the calculation of characteristic impedance (Z_0 or Z_c).

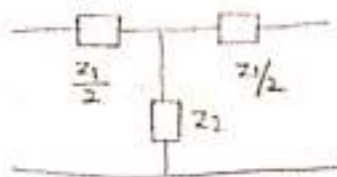
(b) Propagation constant (γ)

(c) Attenuation constant (α)

(d) Phase constant (β).

(a) Characteristic Impedance (Z_0 or Z_c):

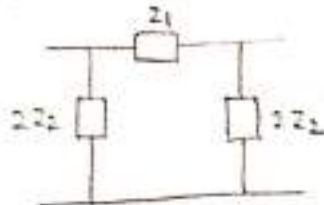
(i) For T-network:



$$Z_{0T} = \sqrt{\left(\frac{Z_1}{2}\right)^2 + Z_1 Z_2}$$

$$e^{\gamma} = 1 + \frac{Z_1}{2Z_2} + \frac{Z_{0T}}{Z_2}$$

(ii) For π -network:



$$Z_{0\pi} = \frac{Z_1 Z_2}{Z_{0T}}$$

$$e^{\gamma} = 1 + \frac{Z_1}{2Z_2} + \frac{Z_1}{Z_{0\pi}}$$

* Classification of filters:

Filters are broadly classified into 3 types.

1. constant K-filters
2. m-derived filters
3. composite filters

1. Constant K-filters:

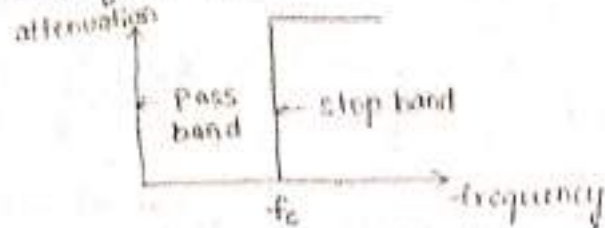
→ In this type of filters, it satisfies the following relation.

$$Z_1 Z_2 = K^2 \text{ or } R_0^2 \rightarrow \text{Designed Impedance.}$$

→ Constant K-filters are classified into 4 types:

1. Constant K low pass filters.
2. Constant K high pass filters.
3. Constant K band pass filters.
4. Constant K band elimination filters.

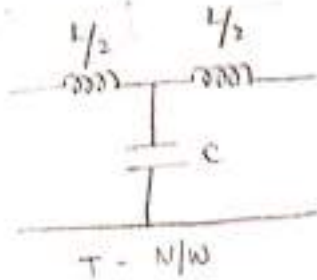
constant n wave pass filter
 → It is the filter which passes the signal upto cut off frequency and attenuates higher than the cut off frequency.
 → The wave form is



In this case, let $z_1 = j\omega L$; $z_2 = \frac{1}{j\omega C}$

$$\frac{z_1}{2} = j\omega \left(\frac{L}{2}\right) ; z_2 = \frac{1}{j\omega C}$$

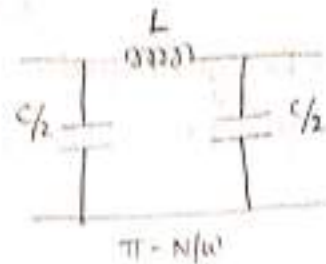
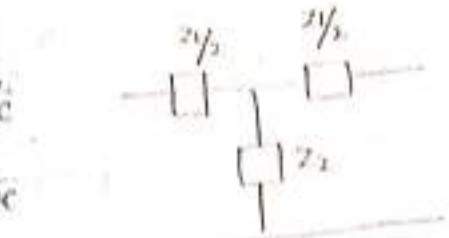
$$\frac{z_1}{2} \propto \frac{L}{2} ; z_2 \propto \frac{1}{C}$$



For π N/w

$$z_1 \propto L ; 2z_2 = \frac{2}{j\omega C}$$

$$2z_2 \propto \frac{1}{(C/2)}$$



(i) Design Impedance (R_0):

We know that, $z_1 z_2 = R_0^2$

$$R_0 = \sqrt{z_1 z_2}$$

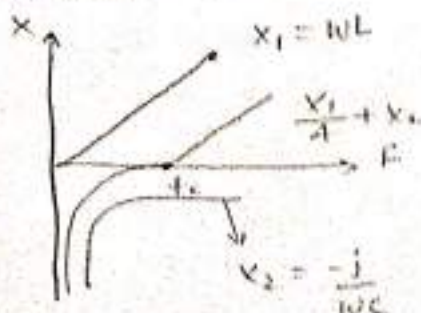
$$= \sqrt{(j\omega L) \times \frac{1}{j\omega C}}$$

$$R_0 = \sqrt{\frac{L}{C}}$$

(ii) Cut off frequency:

$$\left. \begin{array}{l} z_1 = j\omega L \\ x_1 = j\omega L \\ x_1 \propto \omega L \end{array} \right\} \begin{array}{l} z_2 = \frac{1}{j\omega C} \\ x_2 = \frac{-j}{\omega C} \\ x_2 \propto \frac{-1}{\omega C} \end{array}$$

Reactance curves:



from the graph,

$$\text{At } f_c: \frac{x_1}{2} + \frac{x_2}{2} = 0$$

$$\frac{\omega L}{2} + \frac{1}{2\omega C} = 0$$

$$\frac{\omega L}{2} = \frac{1}{2\omega C}$$

$$(2\pi f_c)^2 LC = 4$$

$$f_c^2 = \frac{1}{\pi^2 LC}$$

$$f_c \text{ (cut off frequency)} = \frac{1}{\pi\sqrt{LC}}$$

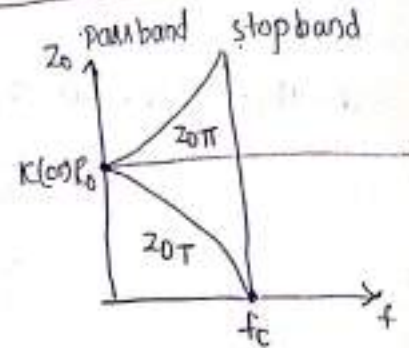
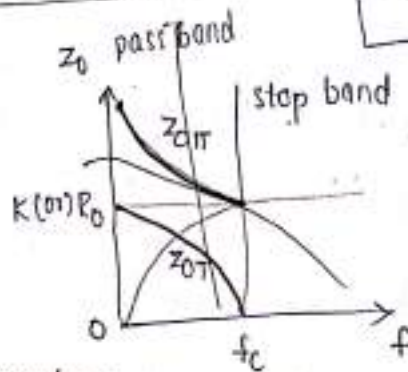
(iii) Z_{OT} & Z_{OP}

$$Z_{OT} = \sqrt{\left(\frac{Z_1}{2}\right)^2 + Z_1 Z_2}$$

$$Z_{OT} = R_0 \sqrt{1 - \left(\frac{f}{f_c}\right)^2}$$

$$Z_{OP} = \frac{Z_1 Z_2}{Z_{OT}} = \frac{R_0^2}{R_0 \sqrt{1 - \left(\frac{f}{f_c}\right)^2}}$$

$$Z_{OP} = \frac{R_0}{\sqrt{1 - \left(\frac{f}{f_c}\right)^2}}$$



(iv) Design Equations:

We know that, $R_0 = \sqrt{\frac{L}{C}}$

$$f_c = \frac{1}{\pi\sqrt{LC}}$$

case (i): $\frac{R_0}{f_c} = \sqrt{\frac{L}{C}} \times \pi\sqrt{LC}$

$$= \pi L \Rightarrow \boxed{L = \frac{R_0}{\pi f_c}} \text{ Henry (H)}$$

case (ii) $R_0 f_c = \sqrt{\frac{L}{C}} \times \frac{1}{\pi\sqrt{LC}}$

$$= \frac{1}{\pi C} \Rightarrow \boxed{C = \frac{1}{\pi R_0 f_c}} \text{ farad (F)}$$

1. Design a constant K low pass filter of T and π N/w with a cut off frequency of 2KHz with a load resistance termination of 500Ω .

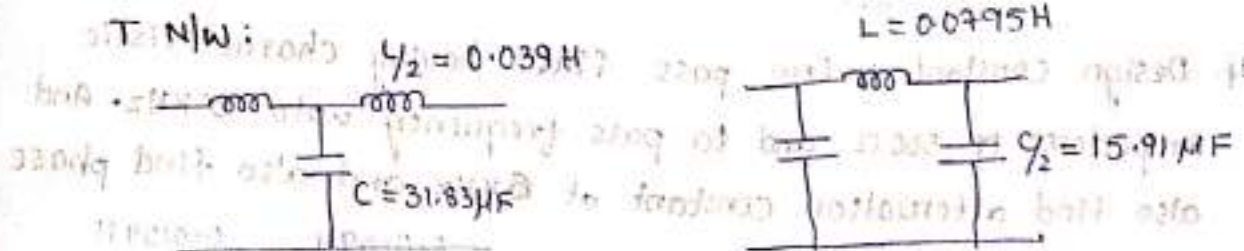
Sol: Given that,

$$f = 2000 \text{ Hz}$$

$$R_0 = 500\Omega$$

$$L = \frac{R_0}{\pi f_c} = 0.0795 \text{ H}$$

$$C = \frac{1}{R_0 \pi f_c} = 31.83 \mu\text{F}$$



2. A low pass filter of T-section has series inductance of 80 mH and shunt capacitance of $0.022 \mu\text{F}$. Determine the cut-off frequency nominal designed impedance R_0 and also design equivalent T and π networks.

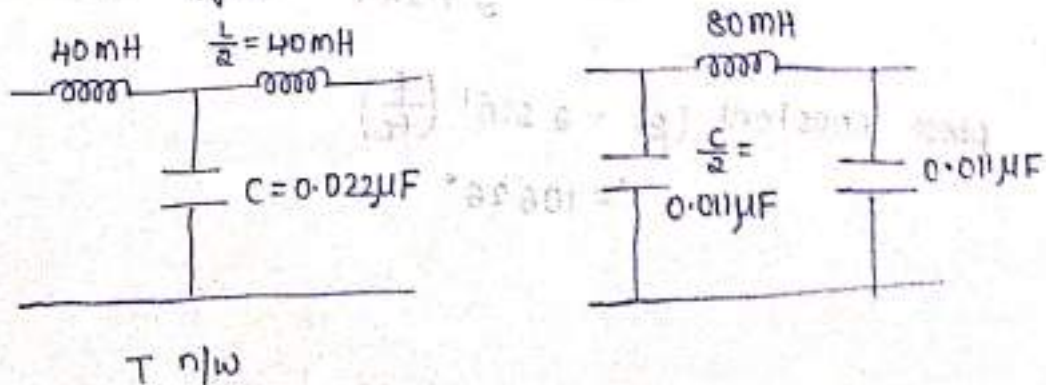
Sol: Given,

$$L = 80 \text{ mH}$$

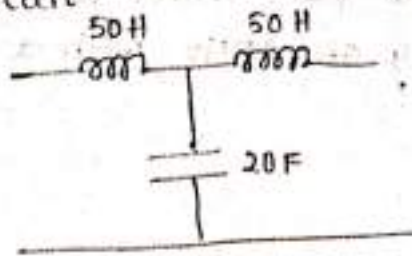
$$C = 0.022 \mu\text{F}$$

$$R_0 = \sqrt{\frac{L}{C}} = 1906.93 \Omega$$

$$f_c = \frac{1}{\pi \sqrt{LC}} = 7587.41 \text{ Hz}$$



3. Find the designed impedance and cut off frequency for the following circuit.



sol: Given,

$$L = 100 \text{ H}$$

$$C = 20 \text{ F}$$

$$R_0 = \sqrt{\frac{L}{C}} = 2.236 \Omega$$

$$f_c = \frac{1}{\pi \sqrt{LC}} = 0.001117 \text{ Hz}$$

4. Design constant K Low pass filter having characteristic impedance of 500Ω and to pass frequency upto 5 KHz . And also find attenuation constant at 6 KHz and also find phase constant at 0.4 KHz .

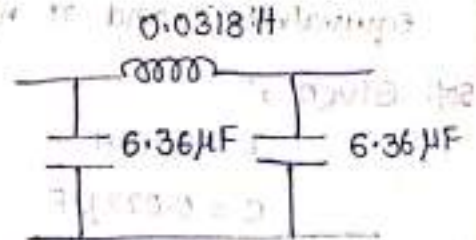
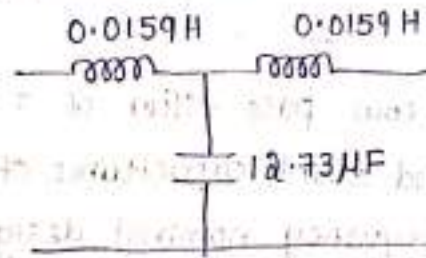
sol: Given,

$$R_0 = 500 \Omega$$

$$f_c = 5 \text{ KHz}$$

$$L = \frac{R_0}{\pi f_c} = 0.0318 \text{ H}$$

$$C = \frac{1}{\pi R_0 f_c} = 12.73 \mu\text{F}$$



$$\text{Attenuation constant } (\alpha) = 2 \cosh^{-1} \left(\frac{f}{f_c} \right)$$

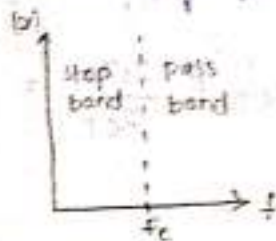
$$= 1.244$$

$$\text{phase constant } (\beta) = 2 \sinh^{-1} \left(\frac{f}{f_c} \right)$$

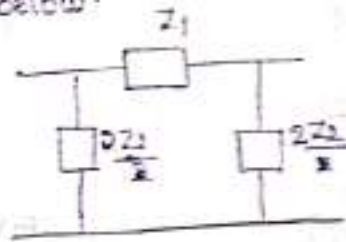
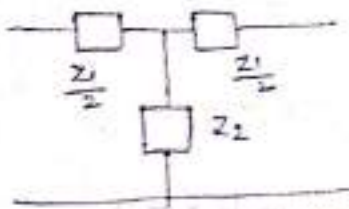
$$= 106.26^\circ$$

Constant 'K' High pass filter:

→ The filter which attenuates the signal upto cut off frequency and passes the all other frequencies above cut off frequency is called as constant 'K' high pass filter.



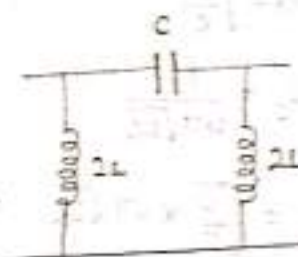
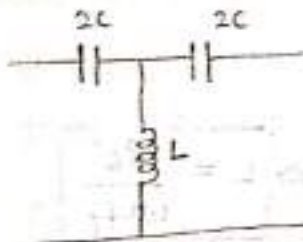
→ In this high pass filters $z_1 = \frac{1}{j\omega C}$, $z_2 = j\omega L$; then the nominal T and π networks are shown below:



$$z_1 = \frac{1}{j\omega C}, \quad \frac{z_1}{2} = \frac{1}{j\omega(2C)}, \quad \frac{z_1}{2} \propto \frac{1}{2C}$$

$$z_2 \propto L$$

$$2z_2 = j\omega(2L)$$



1. Design Impedance: $z_1 z_2 = R_0^2$ 2. Cut off frequency:

$$z_1 z_2 = R_0^2$$

$$z_1 = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$

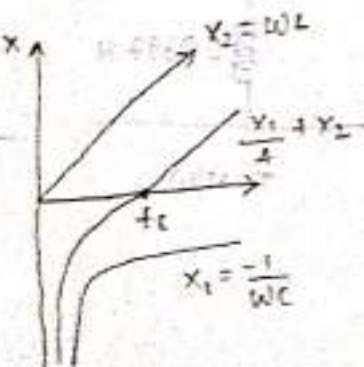
$$R_0 = \sqrt{z_1 z_2} = \sqrt{\frac{-j}{\omega C} \times j\omega L} = \sqrt{\frac{L}{C}}$$

$$R_0 = \sqrt{\frac{L}{C}}$$

$$z_2 = j\omega L$$

$$X_2 = j\omega L \Rightarrow X_2 \propto \omega L$$

$$X_1 = \frac{1}{\omega C}$$



from graph,

at f_c ,

$$\frac{x_1}{4} + x_2 = 0$$

$$\frac{-1}{4\omega L} + \omega L = 0$$

$$\frac{1}{4\omega L} = \omega L$$

$$4\omega^2 LC = 1$$

$$\omega^2 = \frac{1}{4LC}$$

$$\omega = \frac{1}{2\sqrt{LC}}$$

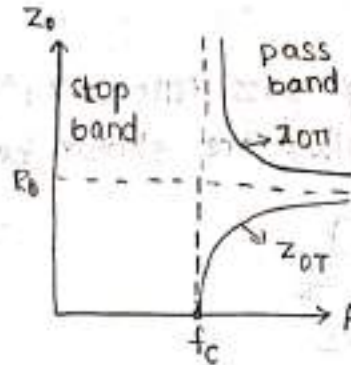
$$2\pi f_c = \frac{1}{2\sqrt{LC}}$$

$$f_c = \frac{1}{4\pi\sqrt{LC}}$$

$$Z_{OT} = \sqrt{\left(\frac{Z_1}{2}\right)^2 + Z_1 Z_2}$$

$$= R_0 \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$Z_{OT} = \frac{Z_1 Z_2}{Z_{OT}} = \frac{R_0}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$



For Z_{OT} :

put $f = f_c$, $Z_{OT} = \sqrt{1-1} = 0$

$f = \infty$, $Z_{OT} = R_0$

4 Design Equation:

We know, $R_0 = \sqrt{\frac{L}{C}}$

$$f_c = \frac{1}{4\pi\sqrt{LC}}$$

case (i): $\frac{R_0}{f_c} = \sqrt{\frac{L}{C}} \times 4\pi\sqrt{LC} = 4\pi L \Rightarrow L = \frac{R_0}{4\pi f_c}$

case (ii): $R_0 \cdot f_c = \sqrt{\frac{L}{C}} \times \frac{1}{4\pi\sqrt{LC}} = \frac{1}{4\pi C} \Rightarrow C = \frac{1}{4\pi R_0 f_c}$

1. Design constant K high pass filter if design impedance is 500Ω and cut off frequency is 1000 Hz .

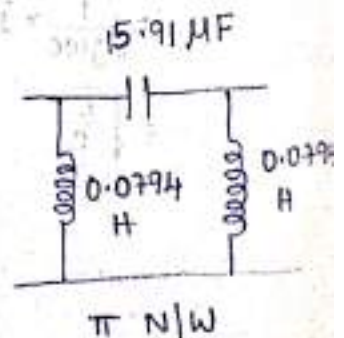
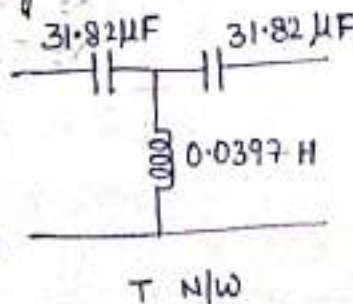
sol: Given,

$$R_0 = 500\Omega$$

$$f_c = 1000\text{ Hz}$$

$$L = \frac{R_0}{4\pi f_c} = 0.0397\text{ H}$$

$$C = \frac{1}{4\pi R_0 f_c} = 15.91\mu\text{F}$$



2. Design a T section constant K high pass filter having $f_c = 10 \text{ kHz}$ and $R_0 = 600 \Omega$. Find its characteristic impedance (Z_{OT}) and phase constant ($Z_{O\pi}$) at 25 kHz

Sol: Given,

$$f_c = 10 \text{ kHz}$$

$$R_0 = 600 \Omega$$

$$L = \frac{R_0}{4\pi f_c} = 0.0047 \text{ H} \approx 4.77 \text{ mH}$$

$$C = \frac{1}{4\pi R_0 f_c} = 0.132 \text{ nF}$$

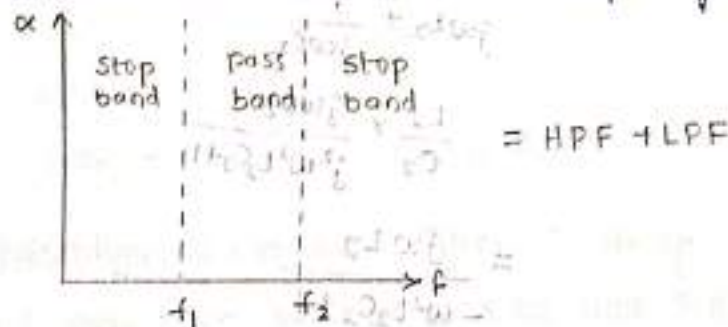
$$Z_{OT} = R_0 \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = 600 \sqrt{1 - \left(\frac{10 \times 10^3}{25 \times 10^3}\right)^2} = 549.9$$

$$Z_{O\pi} = \frac{R_0}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = 654.65$$

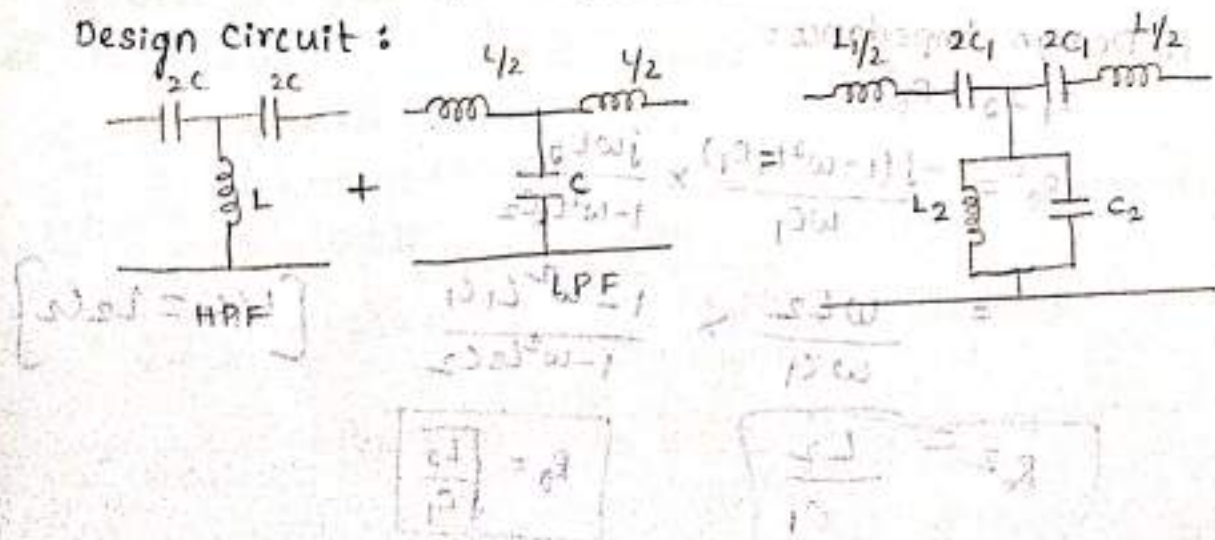
$$\beta = 2 \sin^{-1} \left(\frac{f_c}{f} \right) = 47.15^\circ$$

Constant K Band pass Filters:

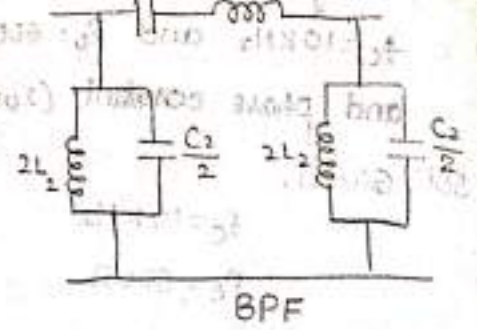
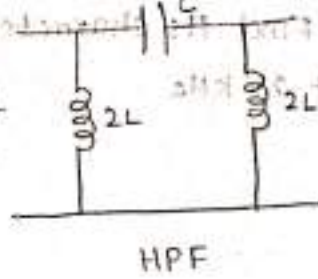
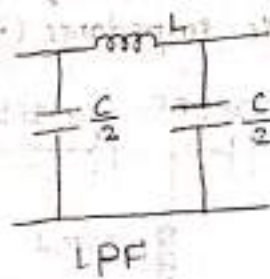
→ Band pass Filters passes certain range of frequencies i.e., f_1 to f_2 and attenuates below & above frequency band.



Design circuit:



II network:



Let us take $z_1 = j\omega L_1 + \frac{1}{j\omega C_1}$

$z_2 = j\omega L_2$ parallel with $\frac{1}{j\omega C_2}$

$$z_1 = j\omega L_1 - \frac{j}{\omega C_1}$$

$$= \frac{j\omega L_1 \times \omega C_1 - j}{\omega C_1}$$

$$= \frac{j\omega^2 L_1 C_1 - j}{\omega C_1}$$

$$= \frac{-j(1 - \omega^2 L_1 C_1)}{\omega C_1}$$

$$z_2 = \frac{j\omega L_2 \times \frac{1}{j\omega C_2}}{j\omega L_2 + \frac{1}{j\omega C_2}}$$

$$= \frac{L_2 \times \frac{j\omega C_2}{j^2 \omega^2 L_2 C_2 + 1}}{C_2}$$

$$= \frac{j\omega L_2}{- \omega^2 L_2 C_2 + 1}$$

(i) Design Impedance:

$$z_1 z_2 = R_0^2$$

$$R_0^2 = \frac{-j(1 - \omega^2 L_1 C_1)}{\omega C_1} \times \frac{j\omega L_2}{1 - \omega^2 L_2 C_2}$$

$$= \frac{\omega L_2}{\omega C_1} \times \frac{1 - \omega^2 L_1 C_1}{1 - \omega^2 L_2 C_2}$$

$$[L_1 C_1 = L_2 C_2]$$

$$R_0^2 = \frac{L_2}{C_1}$$

$$R_0 = \sqrt{\frac{L_2}{C_1}}$$

(ii) Cut-off frequency:

$$\text{At } f_c, \quad \frac{Z_1}{A} + Z_2 = 0 \rightarrow (1)$$

$$\text{where } Z_1 = \text{HPF} + \text{LPF} \\ = \frac{1}{j\omega C_1} + j\omega L_1$$

$$Z_2 = \text{LPF} + \text{HPF} \\ = j\omega L_2 + \frac{1}{j\omega C_2}$$

By solving Z_1 & Z_2 and substitute in eqn (1),

$$\text{we get, lower cut off frequency } f_1 = \frac{R_0 + \sqrt{R_0^2 + \frac{L_1}{C_1}}}{2\pi L_1}$$

$$\text{Higher cut off frequency } f_2 = -\left(\frac{R_0 + \sqrt{R_0^2 + \frac{L_1}{C_1}}}{2\pi L_1} \right)$$

$$\text{Resonant frequency } f_0 = \sqrt{f_1 f_2}$$

(iii) Design Equations:

$$Z_{OT}(\text{BPF}) = Z_{OT}(\text{HPF}) + Z_{OT}(\text{LPF})$$

$$L_1 = \frac{R_0}{\pi(f_2 - f_1)} \quad C_1 = \frac{f_2 - f_1}{4\pi R_0 f_1 f_2}$$

$$L_2 = \frac{R_0(f_2 - f_1)}{4\pi f_1 f_2} \quad C_2 = \frac{1}{\pi R_0(f_2 - f_1)}$$

1. Design a prototype band pass filter if design Impedance

$R_0 = 4 \text{ k}\Omega$ and pass band between $1.25 \text{ k}\Omega$ and $2 \text{ k}\Omega$

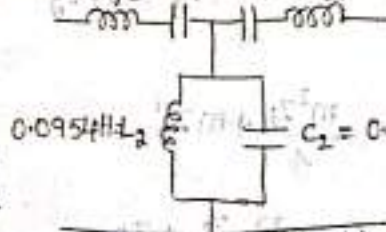
sol: Given that

$$R_0 = 4 \text{ k}\Omega$$

$$f_1 = 1250 \text{ Hz}$$

$$f_2 = 2000 \text{ Hz}$$

$$0.848 \text{ H}, L_1/2, 2C_1, 2C_1, L_2/2 = 0.848 \text{ H}$$



$$10^{-4} \times 10^{-1}$$

$$2C_1 = 11.92 \text{ nF}$$

$$L_1 = \frac{R_0}{\pi(f_2 - f_1)} = 1.697 \text{ H}$$

$$L_2 = \frac{R_0(f_2 - f_1)}{4\pi f_1 f_2} = 0.095 \text{ H}$$

$$C_2 = \frac{1}{\pi R_0(f_2 - f_1)} = 0.1061 \text{ }\mu\text{F}$$

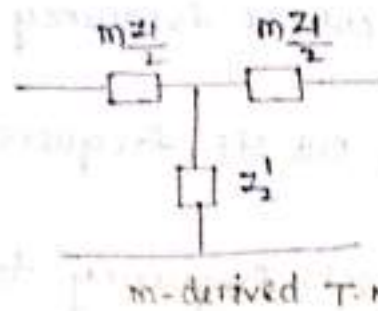
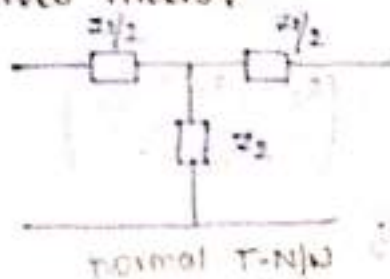
$$C_1 = \frac{f_2 - f_1}{4\pi R_0 f_1 f_2} = 5.96 \text{ nF}$$

Disadvantages of prototype filters:

1. In case of Ideal, signal passes only in pass band and attenuates the signal in stop band. But in case of prototype filters, some of the signal passes through stop band also.
2. Ideally, For any line or in case of pass band characteristic impedance Z_{OT} & Z_{ON} are always constant. But in case of proto-type filters it varies from starting to cutoff frequency varies with frequencies.

→ In order to avoid two above difficulties, m-derived filters are introduced.

Derivations of equations for T and π sections in m-derived filters:



characteristic impedances,

$$Z_{OT} = \sqrt{\left(\frac{z_1}{2}\right)^2 + z_1 z_2}, \quad Z_{OT(m-de)} = \sqrt{\left(\frac{m z_1}{2}\right)^2 + m z_1 z_2'}$$

In order to maintain same characteristic impedance,

$$Z_{OT(n)} = Z_{OT(m-de)}$$

$$(Z_{OT(n)})^2 = (Z_{OT(m)})^2$$

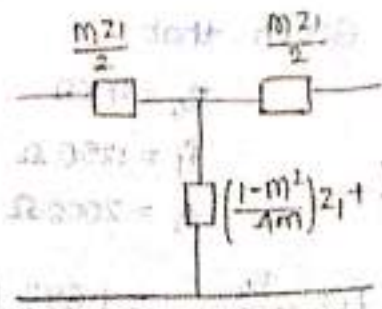
$$\left(\frac{z_1}{2}\right)^2 + z_1 z_2 = \left(\frac{m z_1}{2}\right)^2 + m z_1 z_2'$$

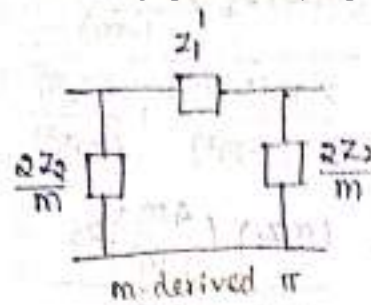
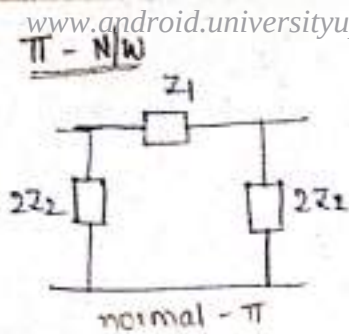
$$\frac{z_1^2}{4} + z_1 z_2 = \frac{m^2 z_1^2}{4} + m z_1 z_2'$$

$$\frac{z_1}{4} + z_2 = \frac{m^2 z_1}{4} + m z_2'$$

$$m z_2' = (1 - m^2) \frac{z_1}{4} + z_2$$

$$z_2' = \left(\frac{1 - m^2}{4m}\right) z_1 + \frac{z_2}{m}$$





Characteristic Impedance :

$$Z_{0\pi} = \frac{Z_1 Z_2}{\sqrt{\left(\frac{Z_1}{2}\right)^2 + Z_1 Z_2}}$$

$$= \frac{\sqrt{Z_1 Z_2} \sqrt{Z_1 Z_2}}{\sqrt{Z_1 Z_2} \left(\sqrt{1 + \frac{Z_1}{4Z_2}}\right)}$$

$$= \frac{\sqrt{Z_1 Z_2}}{\sqrt{1 + \frac{Z_1}{4Z_2}}}$$

$$Z_{0\pi(m)} = \frac{Z_1' \frac{Z_2}{m}}{\sqrt{\left(\frac{Z_1'}{2}\right)^2 + Z_1' \frac{Z_2}{m}}}$$

$$= \frac{Z_1' \frac{Z_2}{m}}{\sqrt{Z_1' \frac{Z_2}{m} \left(\sqrt{1 + \frac{Z_1' m}{4Z_2}}\right)}}$$

$$= \frac{\sqrt{Z_1'^2 \frac{Z_2}{m}}}{\sqrt{1 + \frac{Z_1' m}{4Z_2}}}$$

Remove square root on both sides

$$\frac{Z_1 Z_2}{1 + \frac{Z_1}{4Z_2}} = \frac{Z_1' \frac{Z_2}{m}}{1 + \frac{Z_1' m}{4Z_2}}$$

$$Z_1 Z_2 + \frac{Z_1 Z_1' m}{4} = Z_1' \frac{Z_2}{m} + \frac{Z_1 Z_1'}{4m}$$

$$Z_1' \left(\frac{Z_1}{4m} + \frac{Z_2}{m} - \frac{Z_1 m}{4} \right) = Z_1 Z_2$$

$$Z_1' \left[\frac{Z_1}{4} \left(\frac{1}{m} - m \right) + \frac{Z_2}{m} \right] = Z_1 Z_2$$

$$Z_1' = \frac{Z_1 Z_2}{Z_1 \left(\frac{1-m^2}{4m} \right) + \frac{Z_2}{m}}$$

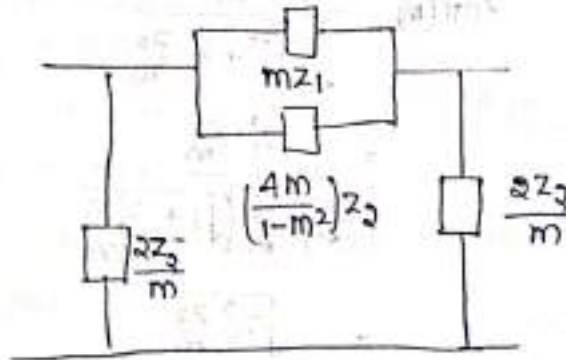
$$= \frac{Z_1 Z_2 (4m)}{Z_1 (1-m^2) + 4Z_2}$$

$$= \frac{(m Z_1) (4Z_2)}{Z_1 (1-m^2) + 4Z_2}$$

$$= \frac{(m Z_1) (4Z_2)}{Z_1 (1-m^2) + 4Z_2} \times \frac{\frac{m}{1-m^2}}{\frac{m}{1-m^2}}$$

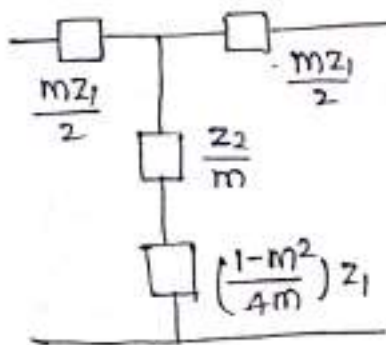
$$= \frac{(mz_1)z_2 \times \frac{4m}{1-m^2}}{\frac{z_1 m(1-m^2)}{(1-m^2)} + \frac{4z_2 m}{(1-m^2)}}$$

$$= \frac{(mz_1) \left(\frac{4m}{1-m^2}\right) z_2}{mz_1 + \left(\frac{4m}{1-m^2}\right) z_2} \Rightarrow \frac{ab}{a+b}$$

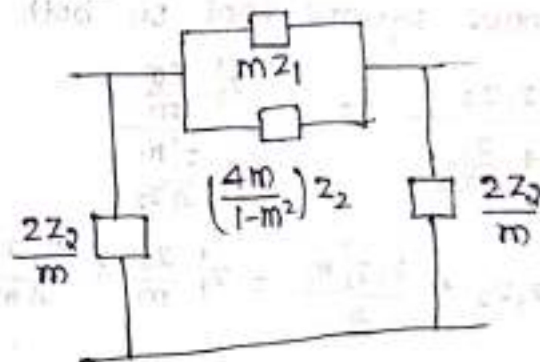


M-derived low pass filters:

Normal T π/w

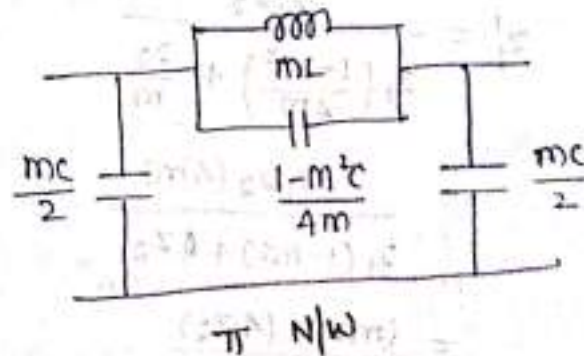
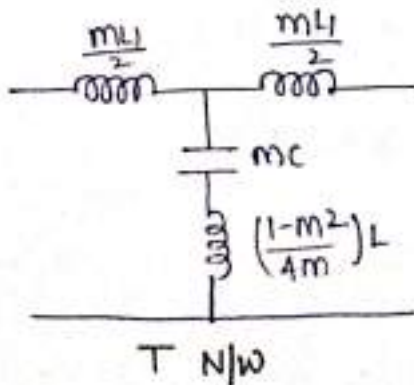


Normal π π/w



In case of LPF, we know $Z_1 = j\omega L$, $Z_2 = \frac{1}{j\omega C}$, then

$$\frac{mZ_1}{2} = j\omega \frac{mL}{2}, \quad \frac{Z_2}{m} = \frac{1}{j\omega(mC)}$$



Consider shunt arm of T section resonates on a frequency of infinite attenuation that means f_c which is selected just above cut off frequency f_c . Then the frequency of resonance is given by

$$f_{\alpha} = \frac{1}{2\pi \sqrt{(mC) \times \left(\frac{1-m^2}{4m}\right) L}}$$

$$= \frac{1}{2\pi \sqrt{LC \left(\frac{1-m^2}{4}\right)}}$$

$$= \frac{1}{\pi \sqrt{LC (1-m^2)}} \rightarrow (1)$$

We know that in case of normal LPF

$$f_c = \frac{1}{\pi \sqrt{LC}} \rightarrow (2)$$

substitute eqn(2) in eqn(1)

$$f_c = \frac{1}{\pi \sqrt{LC}} \Rightarrow f_{\alpha} = \frac{f_c}{\sqrt{1-m^2}}$$

$$\sqrt{1-m^2} = \frac{f_c}{f_{\alpha}}$$

$$(\sqrt{1-m^2})^2 = \left(\frac{f_c}{f_{\alpha}}\right)^2$$

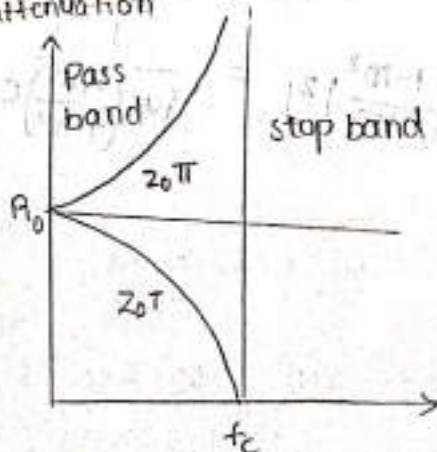
$$1-m^2 = \left(\frac{f_c}{f_{\alpha}}\right)^2$$

$$m^2 = 1 - \left(\frac{f_c}{f_{\alpha}}\right)^2$$

$$m = \sqrt{1 - \left(\frac{f_c}{f_{\alpha}}\right)^2}$$

In this case the design equations and variation Z_0 vs f are same in case of normal LPF.

Z_0 vs f : attenuation



Design Equations:

$$L = \frac{R_0}{\pi f_c}$$

$$C = \frac{1}{R_0 \pi f_c}$$

$$\left(\frac{1}{m}\right)^2 = \frac{Z_0}{Z_0}$$

1. Design an m-derived low pass filter to work into load of 500Ω and cut off frequency is 4kHz and peak attenuation at 4.5kHz .

Ans: Given data:

$$R_0 = 500\Omega$$

$$f_c = 4000\text{ Hz}$$

$$f_s = 4500\text{ Hz}$$

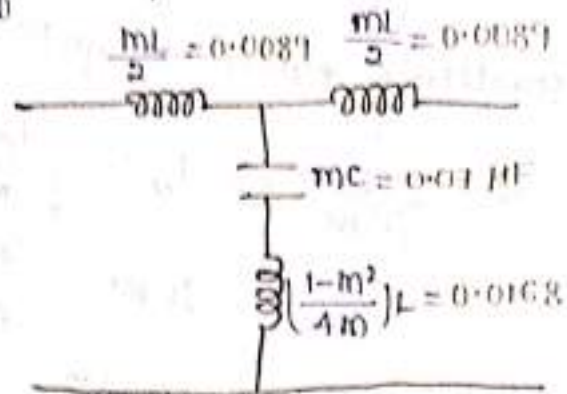
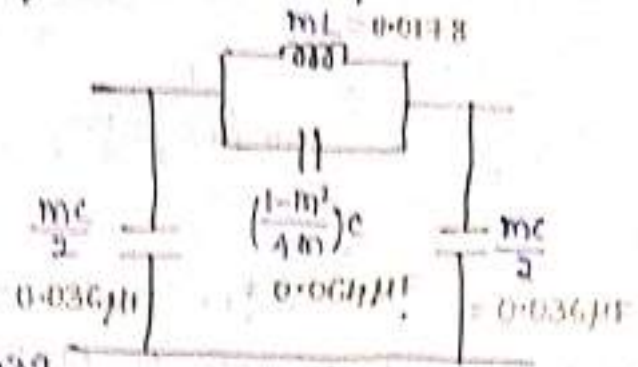
$$L = \frac{R_0}{\pi f_c} = \frac{500}{\pi \times 4000} = 0.039$$

$$C = \frac{1}{\pi R_0 f_c} = \frac{1}{\pi \times 500 \times 4000} = 0.459\text{ }\mu\text{F}$$

$$m = \sqrt{1 - \left(\frac{f_c}{f_s}\right)^2}$$

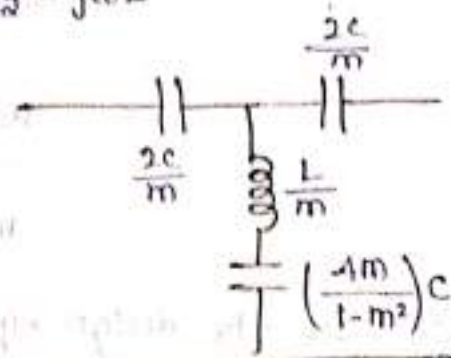
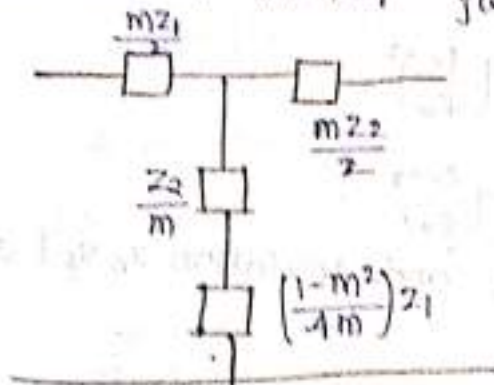
$$= \sqrt{1 - \left(\frac{4000}{4500}\right)^2}$$

$$= 0.458$$



M-derived High Pass Filter:

In case of HPF, $z_1 = \frac{1}{j\omega C}$, $z_2 = j\omega L$



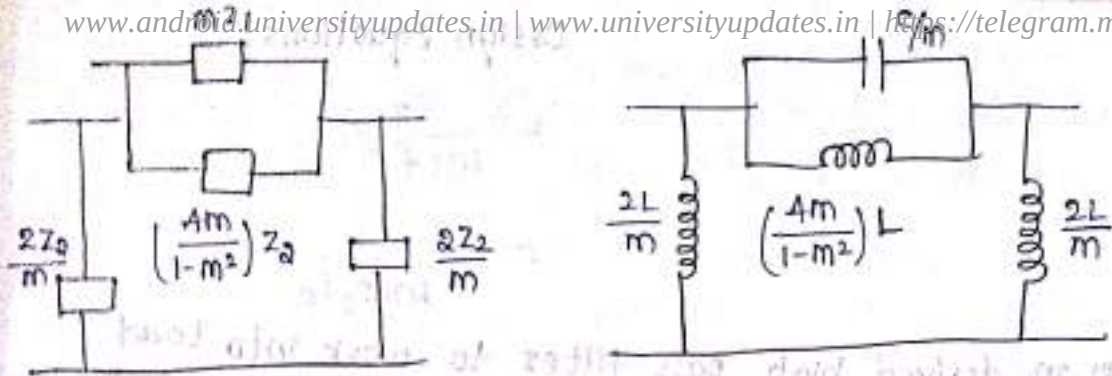
Normal T N/W

$$\frac{mZ_2}{2} = \frac{m}{2} \times \frac{1}{j\omega C}$$

$$= \frac{1}{j\omega \left(\frac{2C}{m}\right)}$$

$$\frac{Z_2}{m} = j\omega \left(\frac{L}{m}\right)$$

$$\left(\frac{1-m^2}{4m}\right)Z_1 = \frac{1}{j\omega \left(\frac{4m}{1-m^2}\right)C}$$



$$mZ_1 = m \frac{1}{j\omega C} = \frac{1}{j\omega \left(\frac{C}{m}\right)} \quad \left(\frac{4m}{1-m^2}\right)Z_2 = \left(\frac{4m}{1-m^2}\right)j\omega L$$

$$\frac{2Z_2}{m} = \frac{2}{m}j\omega L = j\omega \left(\frac{2L}{m}\right) \quad = j\omega \left(\frac{4m}{1-m^2}\right)L$$

Consider shunt arm of T section resonates at a frequency of infinite attenuation that means f_α which is selected just above cut off frequency f_c then the frequency of resonance is given by

$$f_\alpha = \frac{1}{2\pi \sqrt{\left(\frac{L}{m}\right) \times \left(\frac{4m}{1-m^2}\right)C}}$$

$$f_\alpha = \frac{\sqrt{1-m^2}}{4\pi\sqrt{LC}} \rightarrow (1)$$

But in case of HPF

$$f_c = \frac{1}{4\pi\sqrt{LC}} \rightarrow (2)$$

substitute eqn(2) in eqn(1)

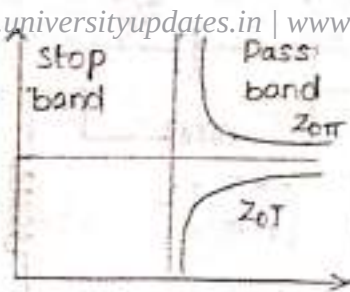
$$f_\alpha = \sqrt{1-m^2} \times f_c$$

$$\left(\frac{f_\alpha}{f_c}\right)^2 = 1-m^2$$

$$m^2 = 1 - \left(\frac{f_\alpha}{f_c}\right)^2$$

$$m = \sqrt{1 - \left(\frac{f_\alpha}{f_c}\right)^2}$$

Z_0 vs f and eqns are same in case of HPF.



Design Equations:

$$L = \frac{R_0}{4\pi f_c}$$

$$C = \frac{1}{4\pi R_0 f_c}$$

1. Design m -derived high pass filter to work into load of 600Ω with cut off frequency of $1000/\pi$ Hz and peak attenuation frequency at 300 Hz.

Sol: Given that,

$$R_0 = 600\Omega$$

$$f_c = \frac{1000}{\pi} = 318.3 \text{ Hz}$$

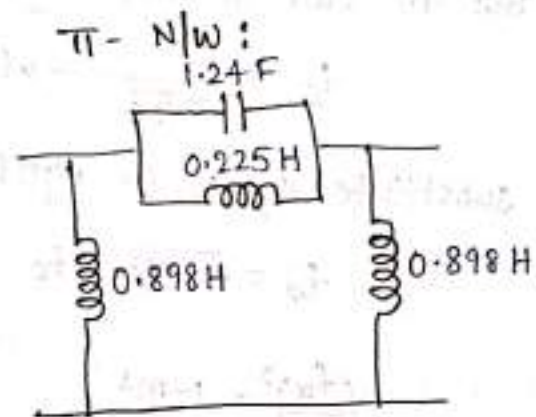
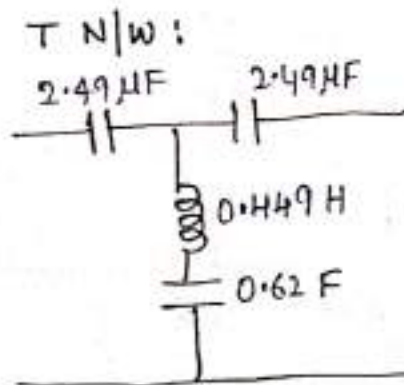
$$f_d = 300 \text{ Hz}$$

$$L = \frac{R_0}{4\pi f_c} = \frac{600}{4\pi \times 318.3} = 0.15 \text{ H}$$

$$C = \frac{1}{4\pi R_0 f_c} = \frac{1}{4\pi \times 600 \times 318.3} = 0.147 \mu\text{F}$$

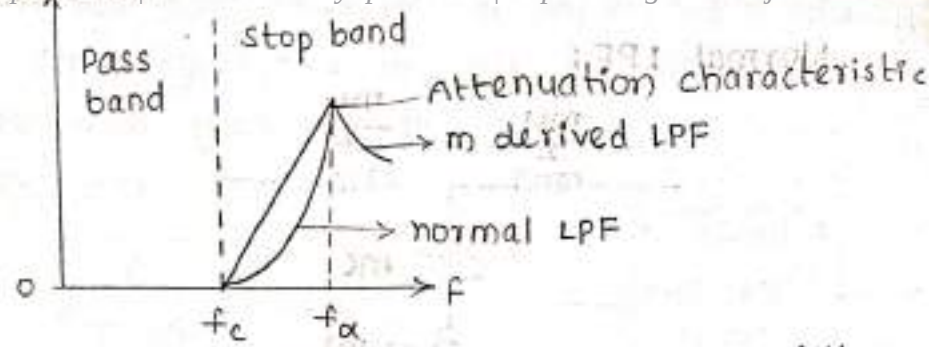
$$m = \sqrt{1 - \left(\frac{f_d}{f_c}\right)^2}$$

$$= 0.334$$

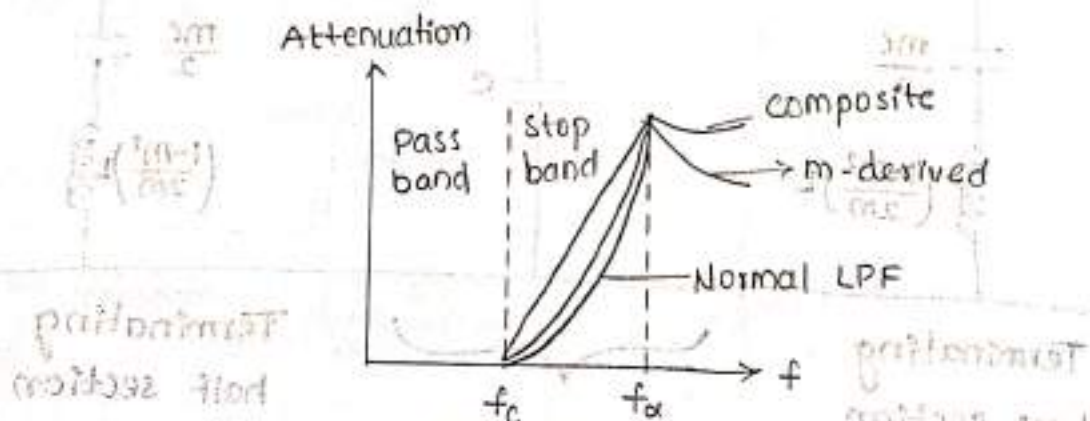
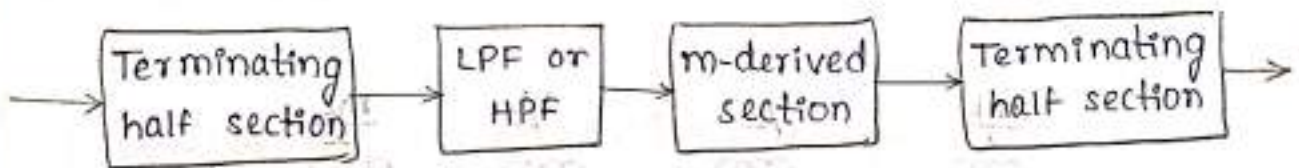


Composite Filter:

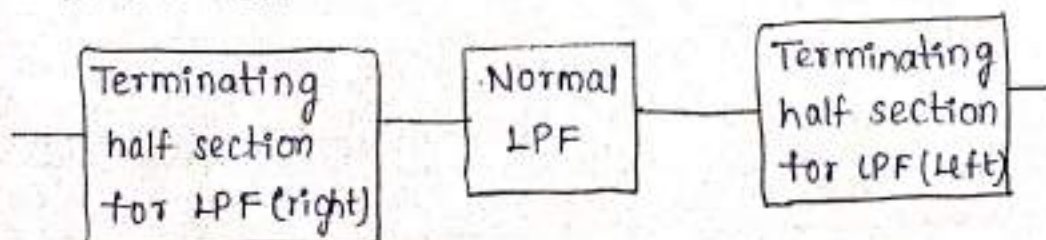
- Attenuation of prototype filter in stop band increases slowly from zero while in the m -derived filter, attenuation reaches very high value at a frequency (f_d).
- However, the attenuation decreases after f_d in the LPS and below f_d in high pass section.



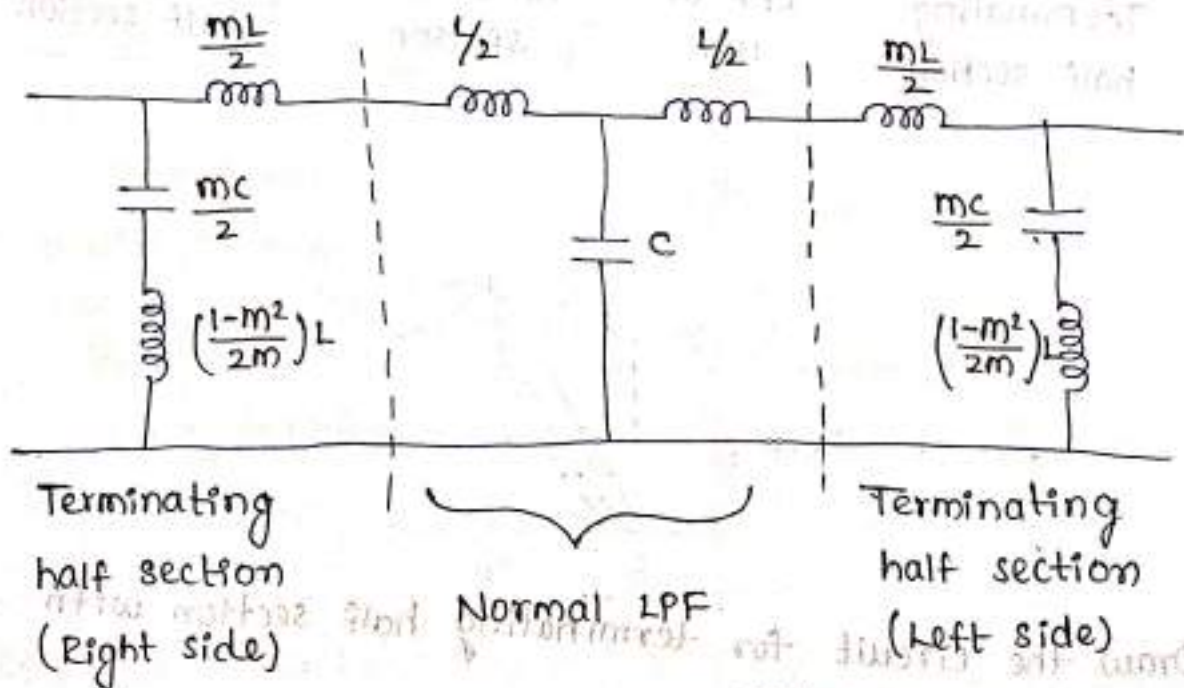
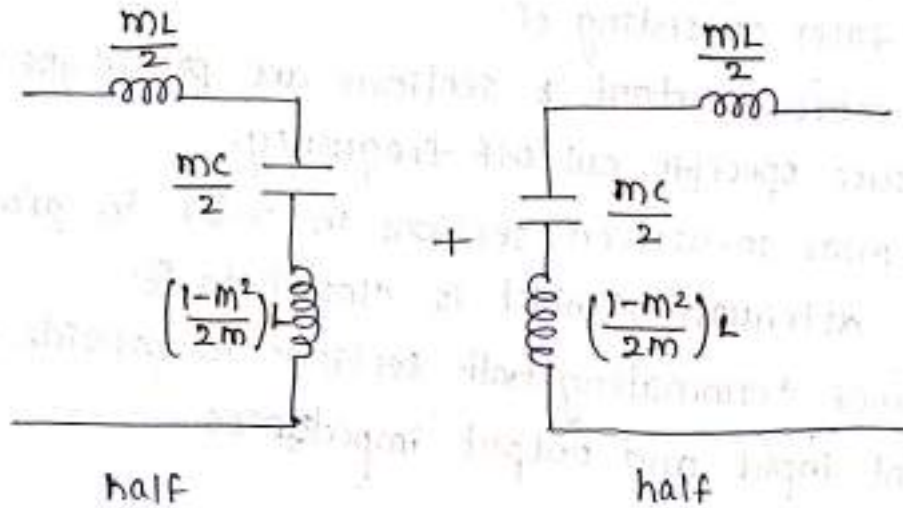
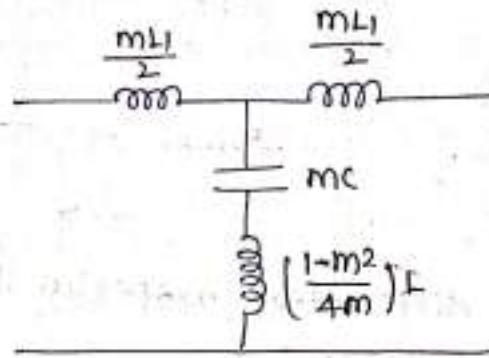
- In order to avoid these difficulties, a special type filter called composite filter is introduced.
- Composite filter consisting of
1. One or more constant k sections are prototype section to produce specific cut off frequency.
 2. One or more m -derived sections in order to produce infinite attenuation which is closed to f_c .
 3. Two no. of terminating half sections to provide almost constant input and output impedances.



1. Draw the circuit for terminating half section with normal LPF and HPF.

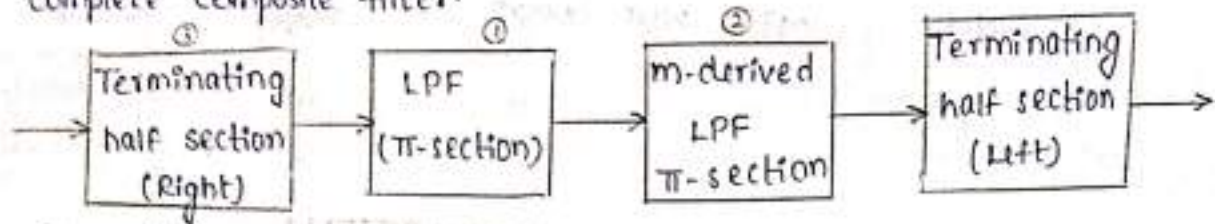


Normal LPF:



2. Design a complete low pass filter π section to work into 600Ω if cut-off frequency of 2 kHz and very high attenuation of 2.2 kHz also terminate filter properly.

sq: Complete composite filter.



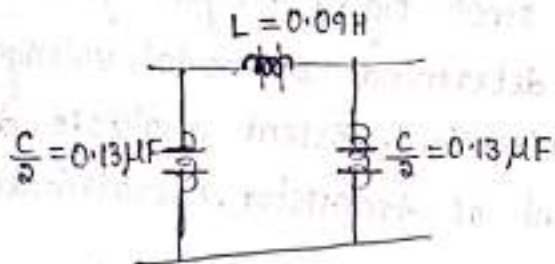
Given that,

$$R_0 = 600\Omega, f_c = 2000\text{ Hz, and } f_\infty = 2200\text{ Hz}$$

(A) Normal LPF (π -section):

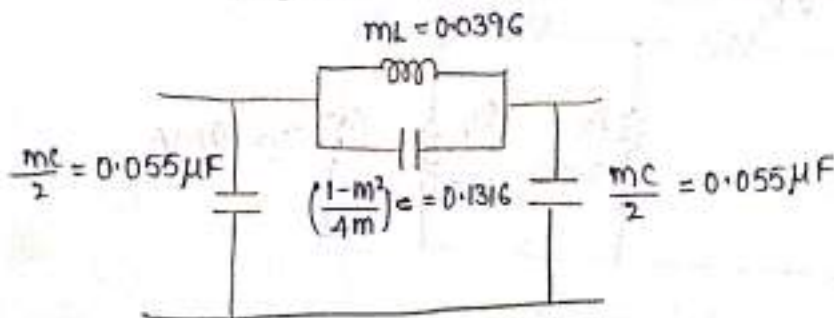
$$L = \frac{R_0}{\pi f_c} = \frac{600}{\pi \times 2000} = 0.09\text{ H}$$

$$C = \frac{1}{R_0 \pi f_c} = \frac{1}{600 \times \pi \times 2000} = 0.265\text{ }\mu\text{F}$$

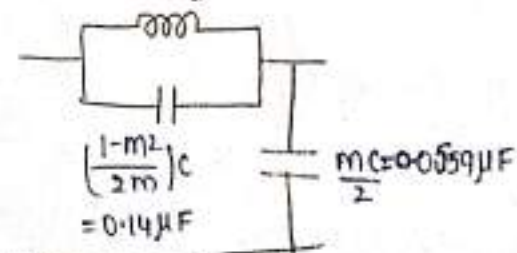
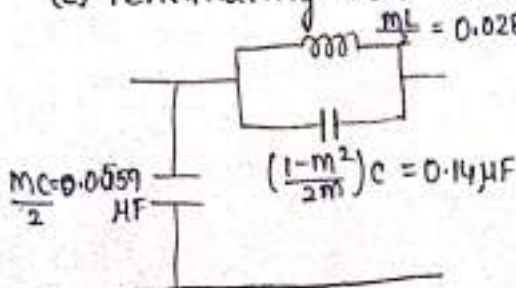


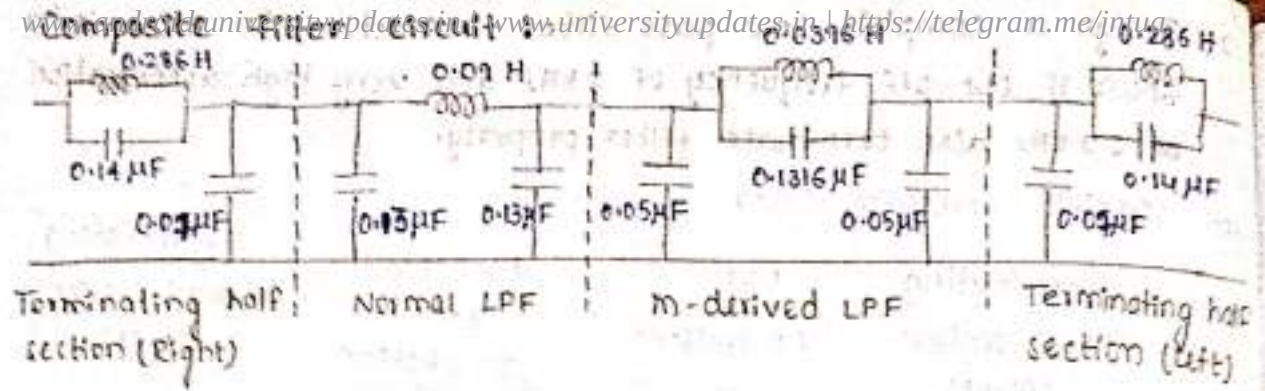
(B) m-derived low pass filter (π -section):

$$m = \sqrt{1 - \left(\frac{f_c}{f_\infty}\right)^2} = 0.416$$



(C) Terminating half section ($m=0.6$): $\frac{mL}{2} = 0.0285\text{ H}$

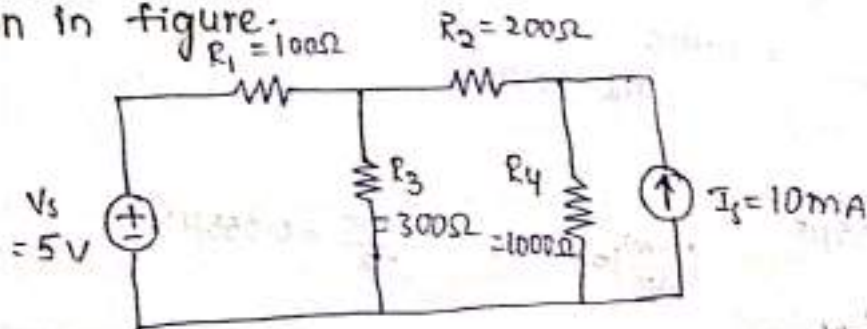




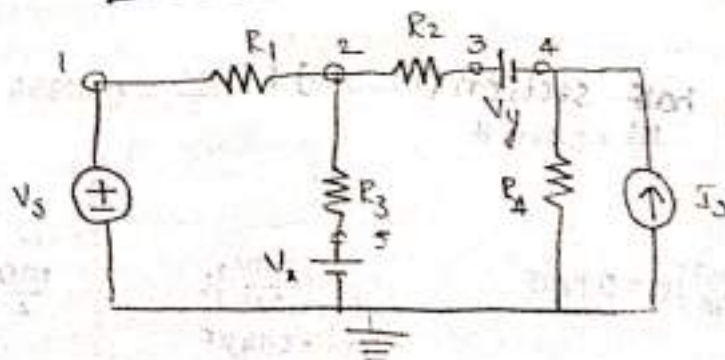
CIRCUIT SIMULATION:

- PSPICE is a personal simulated program with integrated circuit emphasis.
- PSPICE is a general purpose software that includes simulation of different circuits and can perform various analysis of electrical and electronic circuits including time domain response, small signal frequency response, total power dissipation, determination of nodal voltages and mesh analysis in the circuit, transient analysis, determination of operating point of transistor, operational amplifiers etc.

1. Write the PSPICE program and (write) also find the current flowing through resistances R_2 & R_3 as shown in figure.



Sol:

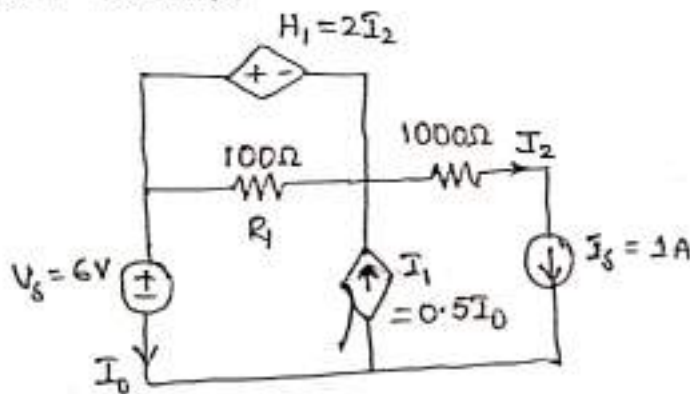


Element	From	To	Type	value
V_s	1	0	DC	5V
I_s	0	4	DC	10mA
R_1	1	2	DC	100 Ω
R_2	2	3	DC	200 Ω
R_3	2	5	DC	300 Ω
R_4	4	0	DC	1000 Ω
V_x	5	0	DC	0
V_y	3	4	DC	0

• print DC $I(R_2)$ $I(R_3)$

• End

2. Write the PSPICE program and find out current through 1000 Ω resistor.



V_s	1	0	DC	6V
I_s	4	0	DC	1A
R_1	1	2	DC	100 Ω
R_2	2	3	DC	1000 Ω
H_1	1	2	3A	2
I_1	0	2	10	0.5
V_x	3	4	DC	0

• Print DC $I(R_2)$

• End